

Chapter 8

Trigonometric Functions

Solutions

SECTION - A

Objective Type Questions (One option is correct)

[Elementary Trigonometric Functions]

1. Which of the following is correct?

- (1) $\sin 1^\circ > \sin 1^\circ$ (2) $\sin 1^\circ < \sin 1^\circ$ (3) $\sin 1^\circ = \sin 1^\circ$ (4) $\sin 1^\circ = \frac{\pi}{180^\circ} \sin 1^\circ$

Sol. Answer (2)

2. Which of the following is correct?

- (1) $\cos 1^\circ > \cos 1^\circ$ (2) $\cos 1^\circ < \cos 1^\circ$ (3) $\cos 1^\circ = \cos 1^\circ$ (4) $\sin 1^\circ = \frac{1}{180^\circ} \sin 1^\circ$

Sol. Answer (1)

3. If $\sin \theta + \operatorname{cosec} \theta = 2$, then $\sin^2 \theta + \operatorname{cosec}^2 \theta$ is equal to

- (1) 1 (2) 4 (3) 2 (4) 0

Sol. Answer (3)

4. If $\tan \theta = 3$ and θ lies in the III quadrant, then the value of $\sin \theta$ is

- (1) $\frac{1}{\sqrt{10}}$ (2) $-\frac{1}{\sqrt{10}}$ (3) $-\frac{3}{\sqrt{10}}$ (4) $\frac{3}{\sqrt{10}}$

Sol. Answer (3)

5. If $\tan \theta = \frac{-4}{3}$, then $\sin \theta$ is

- (1) $\frac{-4}{5}$ but not $\frac{4}{5}$ (2) $\frac{-4}{5}$ or $\frac{4}{5}$ (3) $\frac{4}{5}$ but not $\frac{5}{4}$ (4) $\frac{4}{5}$ and $\frac{5}{4}$ both

Sol. Answer (2)

Correct choice is (2). Since $\tan \theta = -\frac{4}{3}$ is negative, θ lies either in the II quadrant or in the IV quadrant.

Thus $\sin \theta = \frac{4}{5}$ if θ lies in the II quadrant and $\sin \theta = -\frac{4}{5}$, if θ lies in the IV quadrant.

6. If A lies in the second quadrant and $3 \tan A + 4 = 0$, then the value of $2\cot A - 5\cos A + \sin A$ is equal to

(1) $\frac{-53}{10}$

(2) $\frac{23}{10}$

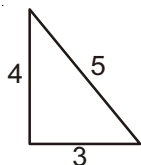
(3) $\frac{37}{10}$

(4) $\frac{7}{10}$

Sol. Answer (2)

$$3 \tan A = -4$$

$$\tan A = -\frac{4}{3}$$



$$\therefore 2\cot A - 5\cos A + \sin A$$

$$= 2\left(\frac{-3}{4}\right) - 5\left(\frac{-3}{5}\right) + \frac{4}{5}$$

$$= \frac{-6}{4} + 3 + \frac{4}{5}$$

$$= \frac{-30 + 60 + 16}{20} = \frac{46}{20} = \frac{23}{10}$$

7. The value of $(\sec A + \tan A - 1)(\sec A - \tan A + 1) - 2\tan A$ is equal to

(1) $\sec A$

(2) $2\sec A$

(3) 0

(4) 1

Sol. Answer (3)

$$= (\sec A + \tan A - 1)(\sec A - \tan A + 1) - 2\tan A$$

$$= [\sec A + (\tan A - 1)][\sec A - (\tan A - 1)] - 2\tan A$$

$$= [\sec^2 A - (\tan A - 1)^2] - 2\tan A$$

$$= \sec^2 A - \tan^2 A - 1 + 2\tan A - 2\tan A$$

$$= 1 - 1 = 0$$

8. The circular wire of diameter 10 cm is cut and placed along the circumference of a circle of diameter 1 m. The angle subtended by the wire at the centre of the circle is equal to

(1) $\frac{\pi}{4}$

(2) $\frac{\pi}{3}$

(3) $\frac{\pi}{5}$

(4) $\frac{\pi}{10}$

Sol. Answer (3)

Diameter of the wire = 10 cm

Length of wire = 10π cm

Another circle is of diameter = 1 m = 100 cm

$\therefore$ Radius = 50 cm

$$\text{Required angle} = \frac{\text{Length of arc}}{\text{Radius}} = \frac{10\pi}{50} = \frac{\pi}{5}$$

9. If $\cos \theta = \frac{-1}{2}$ and $0 < \theta < 360^\circ$, then the values of θ are

(1) $120^\circ, 210^\circ$

(2) $120^\circ, 300^\circ$

(3) $60^\circ, 240^\circ$

(4) $120^\circ, 240^\circ$

Sol. Answer (4)

$$\text{If } 0 < \theta < 360^\circ, \text{ then } \cos \theta = \frac{-1}{2} \Rightarrow \theta = \pi - \frac{\pi}{3} = \frac{2\pi}{3} \text{ and } \pi + \frac{\pi}{3} = \frac{4\pi}{3}$$

$$\theta = 120^\circ \text{ and } 240^\circ$$

10. The value of $\sin 15^\circ + \cos 105^\circ$ is

- (1) 0 (2) $2\sin 15^\circ$ (3) $\cos 15^\circ + \sin 15^\circ$ (4) $\sin 15^\circ - \cos 15^\circ$

Sol. Answer (1)

$$\sin 15^\circ + \cos 105^\circ = \sin 15^\circ + \cos(90^\circ + 15^\circ)$$

$$= \sin 15^\circ - \sin 15^\circ$$

$$= 0$$

11. If $\sin \theta_1 + \sin \theta_2 + \sin \theta_3 = 3$, then $\cos \theta_1 + \cos \theta_2 + \cos \theta_3 =$

- (1) 3 (2) 2 (3) 1 (4) 0

Sol. Answer (4)

$$\sin \theta_1 = \sin \theta_2 = \sin \theta_3 = 1.$$

$$\therefore \theta_1 = \theta_2 = \theta_3 = \frac{\pi}{2}$$

12. The value of $\cos 10^\circ - \sin 10^\circ$ is

- (1) Positive (2) Negative (3) 0 (4) 1

Sol. Answer (1)13. The value of $\cos 1^\circ \cdot \cos 2^\circ \cdots \cos 179^\circ$ is

- (1) $\frac{1}{\sqrt{2}}$ (2) 0 (3) 1 (4) -1

Sol. Answer (2)

$$\cos 1^\circ \cdot \cos 2^\circ \cdots \cos 179^\circ = 0$$

(as there will be a term $\cos 90^\circ$ in this series whose value is zero)14. If $\sec \alpha = \frac{13}{5}$, $\left(0 < \alpha < \frac{\pi}{2}\right)$, then the value of $\frac{2-3\cot \alpha}{4-9\sqrt{\sec^2 \alpha - 1}}$ is

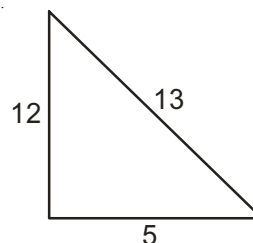
- (1) $-\frac{15}{352}$ (2) $\frac{15}{352}$ (3) $\frac{30}{352}$ (4) $\frac{5}{352}$

Sol. Answer (1)

$$\sec \alpha = \frac{13}{5}, 0 < \alpha < \frac{\pi}{2}$$

$$\therefore \frac{2-3\cot \alpha}{4-9\sqrt{\sec^2 \alpha - 1}} = \frac{2-3 \times \frac{5}{12}}{4-9 \times \frac{12}{5}} \quad (\sec^2 \alpha - 1 = \tan^2 \alpha)$$

$$= \frac{3/4}{-88/5} = -\frac{15}{352}$$



15. The value of $(\operatorname{cosec} A \cdot \operatorname{cosec} B + \cot A \cdot \cot B)^2 - (\operatorname{cosec} A \cdot \cot B + \operatorname{cosec} B \cdot \cot A)^2$ is

- (1) 1 (2) 2 (3) 0 (4) -1

Sol. Answer (1)

$$\begin{aligned} & (\operatorname{cosec} A \cdot \operatorname{cosec} B + \cot A \cdot \cot B)^2 - (\operatorname{cosec} A \cdot \cot B + \operatorname{cosec} B \cdot \cot A)^2 \\ &= \operatorname{cosec}^2 A (\operatorname{cosec}^2 B - \cot^2 B) - \cot^2 A (\operatorname{cosec}^2 B - \cot^2 B) \\ &= \operatorname{cosec}^2 A - \cot^2 A = 1 \end{aligned}$$

16. If $\tan \alpha + \cot \alpha = a$, then the value of $\tan^4 \alpha + \cot^4 \alpha$ is equal to

- (1) $a^4 + 4a^2 + 2$ (2) $a^4 - 4a^2 + 2$ (3) $a^4 - 4a^2 - 2$ (4) $-a^4 + 2a^2 + 4$

Sol. Answer (2)

$$\begin{aligned} \tan \alpha + \cot \alpha &= a, \quad \tan^4 \alpha + \cot^4 \alpha = ? \\ \tan^4 \alpha + \cot^4 \alpha &= (\tan \alpha + \cot \alpha)^4 - 4 \tan \alpha \cdot \cot \alpha (\tan^2 \alpha + \cot^2 \alpha) - 6 \tan^2 \alpha \cdot \cot^2 \alpha \\ &= a^4 - 4\{(\tan \alpha + \cot \alpha)^2 - 2 \tan \alpha \cdot \cot \alpha\} - 6 \\ &= a^4 - 4a^2 + 2 \end{aligned}$$

17. If $a \cos \theta + b \sin \theta = 3$ and $a \sin \theta - b \cos \theta = 4$, then $a^2 + b^2$ has the value

- (1) 25 (2) 14 (3) 7 (4) 15

Sol. Answer (1)

$$\begin{aligned} a \cos \theta + b \sin \theta &= 3 \\ a \sin \theta - b \cos \theta &= 4 \\ \text{Squaring and adding, } a^2 + b^2 &= 25 \end{aligned}$$

18. Suppose that $a \cos \theta = b$ and $c \sin \theta = d$, for θ and some constants a, b, c, d . Then which of the following is true?

- (1) $a^2 c^2 = b^2 c^2 + a^2 d^2$ (2) $a^2 d^2 = b^2 c^2 + a^2 c^2$ (3) $b^2 c^2 = a^2 d^2 + a^2 c^2$ (4) $a^2 b^2 = b^2 c^2 + c^2 d^2$

Sol. Answer (1)

$$\begin{aligned} a \cos \theta &= b, \quad c \sin \theta = d \\ \Rightarrow \cos \theta &= \frac{b}{a}, \sin \theta = \frac{d}{c} \end{aligned}$$

Squaring and adding,

$$\begin{aligned} \frac{b^2}{a^2} + \frac{d^2}{c^2} &= 1 \\ \Rightarrow b^2 c^2 + a^2 d^2 &= a^2 c^2 \end{aligned}$$

19. The number of intersecting points on the graph for $\sin x = \frac{x}{10}$ for $x \in [-\pi, \pi]$ is

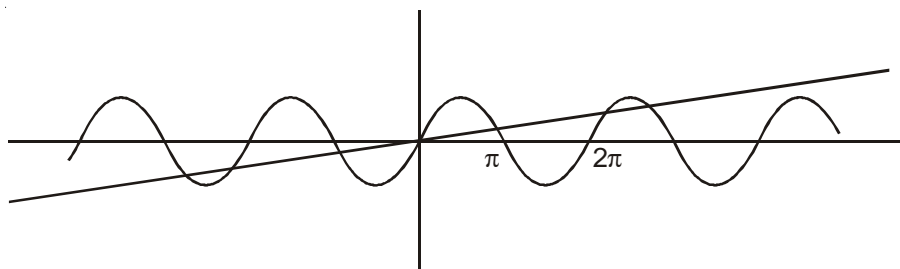
(1) 3

(2) 4

(3) 2

(4) 1

Sol. Answer (1)



$$\sin x = \frac{x}{10}, \quad x \in [-\pi, \pi]$$

Clearly the total number of points of intersection is 3.

20. $|(3 - \sec^2 x)_{\max} - (4 + \tan^2 y)_{\min}|$ equals

(1) 0

(2) 1

(3) 2

(4) 3

Sol. Answer (3)

$$|(3 - \sec^2 x)_{\max} - (4 + \tan^2 y)_{\min}| = |2 - 4| = 2$$

21. If $\tan \theta = \frac{p}{q}$, then the value of $\frac{p \sin \theta - q \cos \theta}{p \sin \theta + q \cos \theta}$ is

(1) $\frac{p^2 - q^2}{p^2 + q^2}$ (2) $\frac{p^2 + q^2}{p^2 - q^2}$

(3) 0

(4) 1

Sol. Answer (1)

$$\tan \theta = \frac{p}{q}, \quad \frac{p \sin \theta - q \cos \theta}{p \sin \theta + q \cos \theta} = ?$$

Divide numerator and denominator by $\cos \theta$

$$\frac{p \tan \theta - q}{p \tan \theta + q} = \frac{p^2 - q^2}{p^2 + q^2}$$

22. The value of $\sin 1 \cdot \cos 2 \cdot \tan 3 \cdot \cot 4 \cdot \sec 5 \cdot \operatorname{cosec} 6$ is

(1) Positive

(2) Negative

(3) Zero

(4) May be positive and Negative

Sol. Answer (2)

$$\begin{array}{cccccc} \sin 1 & \cos 2 & \tan 3 & \cot 4 & \sec 5 & \operatorname{cosec} 6 \\ + & - & - & + & + & - \end{array}$$

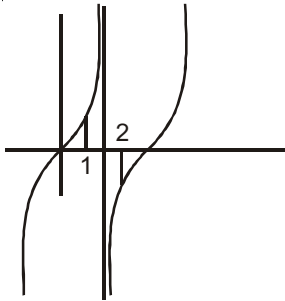
= negative

23. Which of the following is correct?

- (1) $\tan 1 > \tan 2$ (2) $\tan 2 > \tan 1$ (3) $\sin 1 < \cos 1$ (4) $\cos 3 > \cos 2$

Sol. Answer (1)

Both the options are correct



$\tan 1 > \tan 2$ (from graph)

24. The perimeter of a certain sector of a circle is equal to half that of the circle of which it is a sector. The circular measure of one angle of the sector is

- (1) $(\pi - 2)$ radian (2) $(\pi + 2)$ radian (3) π radian (4) $(\pi - 3)$ radian

Sol. Answer (1)

$2r + l = \frac{1}{2} 2\pi r$ where r is radius of the circle and l is the length of the arc i.e. $l = r\theta$

$$2r + r\theta = \pi r$$

$$\Rightarrow \theta = (\pi - 2) \text{ radian}$$

25. If $\alpha, \beta, \gamma \in [0, 2\pi]$, then the sum of all possible values of α, β, γ if

$$\sin \alpha = -\frac{1}{\sqrt{2}}, \cos \beta = -\frac{1}{2}, \tan \gamma = -\sqrt{3}, \text{ is}$$

- (1) $\frac{22\pi}{3}$ (2) $\frac{21\pi}{3}$ (3) $\frac{20\pi}{3}$ (4) 8π

Sol. Answer (1)

$$\sin \alpha = -\frac{1}{\sqrt{2}} \Rightarrow \alpha = \pi + \frac{\pi}{4}, 2\pi - \frac{\pi}{4}$$

$$\cos \beta = -\frac{1}{2} \Rightarrow \beta = \pi - \frac{\pi}{3}, \pi + \frac{\pi}{3}$$

$$\tan \gamma = -\sqrt{3} \Rightarrow \gamma = \pi - \frac{\pi}{3}, 2\pi - \frac{\pi}{3}$$

$$\therefore \text{Sum of all values} = 8\pi - \frac{2\pi}{3} = \frac{22\pi}{3}$$

26. The angles of a triangle are in A.P. and the number of degrees in the least to the number of radians in greatest is 60 to π . The angles in degree are

- (1) $60^\circ, 60^\circ, 60^\circ$ (2) $30^\circ, 60^\circ, 90^\circ$ (3) $45^\circ, 60^\circ, 75^\circ$ (4) $15^\circ, 60^\circ, 105^\circ$

Sol. Answer (2)

Let the angles be $\alpha - d, \alpha, \alpha + d$

$$\therefore 3\alpha = 180^\circ \Rightarrow \alpha = 60^\circ$$

$$\frac{\alpha - d}{\alpha + d} \times \frac{180}{\pi} = \frac{60}{\pi}$$

$$\Rightarrow d = 30^\circ$$

Hence the angles are $30^\circ, 60^\circ, 90^\circ$.

27. If $ABCD$ is a cyclic quadrilateral such that $12\tan A - 5 = 0$ and $5\cos B + 3 = 0$, then the quadratic equation whose roots are $\cos C$ and $\tan D$ is

- (1) $39x^2 - 16x - 48 = 0$ (2) $39x^2 + 88x + 48 = 0$ (3) $39x^2 - 88x + 48 = 0$ (4) $39x^2 - 13x - 46 = 0$

Sol. Answer (1)

$$12\tan A - 5 = 0, \quad 5\cos B + 3 = 0$$

$$\Rightarrow \tan A = \frac{5}{12}, \quad \cos B = -\frac{3}{5}$$

$\therefore ABCD$ is a cyclic quadrilateral

$$\therefore A + C = \pi, \quad B + D = \pi$$

$$\Rightarrow C = -A, \quad D = \pi - B$$

$$\cos C = -\cos A \quad \tan D = -\tan B$$

$$= -\frac{12}{13}$$

$$= \frac{4}{3}$$

$$\therefore \text{Equation whose roots are } \cos C \text{ and } \tan D \text{ is } x^2 - \left(-\frac{12}{13} + \frac{4}{3}\right)x + \left(-\frac{12}{13}\right) \times \frac{4}{3} = 0$$

$$39x^2 - 16x - 48 = 0$$

28. If $\cos \theta + \sec \theta = -2, \theta \in [0, 2\pi]$, then $\sin^8 \theta + \cos^8 \theta$ equal to

- (1) -2 (2) 1 (3) 2^4 (4) 2^5

Sol. Answer (2)

$$\cos \theta + \sec \theta = -2, \quad \theta \in [0, 2\pi]$$

$$\Rightarrow \cos \theta = \sec \theta = -1$$

$$\theta = \pi$$

$$\therefore \sin^8 \theta + \cos^8 \theta = 0 + 1 = 1$$

29. If the maximum value of $\cos(\cos x)$ is a and minimum value is b , then

- (1) $b = \cos a$ (2) $a = \cos b$ (3) $a = 0$ (4) $b = 0$

Sol. Answer (1)

$$y = \cos(\cos x)$$

$$y_{\max} = 1 = a, \text{ when } x = \frac{\pi}{2}$$

$$y_{\min} = \cos 1 = b \text{ when } x = 0$$

$$\therefore b = \cos a$$

30. If $\frac{2 \sin A}{1 + \sin A + \cos A} = k$, then $\frac{1 + \sin A - \cos A}{1 + \sin A}$ is

- (1) $\frac{k}{2}$ (2) k (3) $2k$ (4) $\frac{1}{k}$

Sol. Answer (2)

$$\frac{2 \sin A}{1 + \sin A + \cos A} = k, \quad \frac{1 + \sin A - \cos A}{1 + \sin A} = ?$$

$$k = \frac{2 \sin A (1 + \sin A - \cos A)}{(1 + \sin A)^2 - \cos^2 A}$$

$$\Rightarrow k = \frac{2 \sin A (1 + \sin A - \cos A)}{(1 + \sin A)^2 - (1 - \sin^2 A)}$$

$$= \frac{2 \sin A (1 + \sin A - \cos A)}{(1 + \sin A) \cdot \{(1 + \sin A) - (1 - \sin A)\}}$$

$$= \frac{1 + \sin A - \cos A}{1 + \sin A}$$

31. If $\sin x + \sin^2 x = 1$, then the value of $\cos^{12} x + 3 \cos^{10} x + 3 \cos^8 x + \cos^6 x - 1$ is equal to

- (1) 0 (2) 1 (3) -1 (4) 2

Sol. Answer (1)

$$\sin x + \sin^2 x = 1$$

$$\Rightarrow \sin x = \cos^2 x$$

$$= \cos^6 x (\cos^6 x + 3 \cos^4 x + 3 \cos^2 x + 1) - 1$$

$$= \{\cos^2 x (1 + \cos^2 x)\}^3 - 1$$

$$= \{\sin x (1 + \sin x)\}^3 - 1$$

$$= \{\sin x + \sin^2 x\}^3 - 1 = (1)^3 - 1 = 0$$

32. If $\sum_{r=15}^{29} \cos\left(\frac{r\pi}{2} + \theta\right) = S_1$ and $\sum_{r=15}^{29} \sin\left(\frac{r\pi}{2} + \theta\right) = S_2$, then $\frac{S_1}{S_2}$ equals

(1) $\cot\theta$

(2) $\tan\theta$

(3) $-\cot\theta$

(4) $-\tan\theta$

Sol. Answer (1)

$$\sum_{r=15}^{29} \cos\left(\frac{r\pi}{2} + \theta\right) = S_1$$

$$\sum_{r=15}^{29} \sin\left(\frac{r\pi}{2} + \theta\right) = S_2$$

$$S_1 = \sum_{r=15}^{29} \cos\left(\frac{r\pi}{2} + \theta\right)$$

$$= \cos\left(\frac{15\pi}{2} + \theta\right) + \cos\left(\frac{16\pi}{2} + \theta\right) + \dots + \cos\left(\frac{29\pi}{2} + \theta\right)$$

$$= (\sin\theta + \cos\theta - \sin\theta - \cos\theta) + \dots +$$

(Sum of consecutive terms is zero and these are 15 terms in all)

$$\therefore S_1 = (\sin\theta + \cos\theta - \sin\theta)$$

$$= \cos\theta$$

$$S_2 = \sum_{r=15}^{29} \sin\left(\frac{r\pi}{2} + \theta\right)$$

$$= -\cos\theta + \sin\theta + \cos\theta$$

$$= \sin\theta$$

$$\therefore \frac{S_1}{S_2} = \frac{\cos\theta}{\sin\theta} = \cot\theta$$

33. The maximum value of $\cos^2(\cos(33\pi + \theta)) + \sin^2(\sin(45\pi + \theta))$ is

(1) $1 + \sin^2 1$

(2) 2

(3) $1 + \cos^2 1$

(4) $\cos^2 2$

Sol. Answer (1)

$$\cos^2(\cos(33\pi + \theta)) + \sin^2(\sin(45\pi + \theta))$$

$$= \cos^2(-\cos\theta) + \sin^2(-\sin\theta)$$

$$= \cos^2(-\cos\theta) + \sin^2(\sin\theta)$$

$$\because \cos \leq \cos(\cos\theta) \leq 1$$

$$\cos^2 1 \leq \cos^2(\cos\theta) \leq 1$$

$$\text{and } 0 \leq \sin^2(\sin\theta) \leq \sin^2 1$$

$$\therefore \cos^2 1 \leq \cos^2(\cos\theta) + \sin^2(\sin\theta) \leq 1 + \sin^2 1$$

$$\therefore \text{Maximum value} = 1 + \sin^2 1 \left(\text{At } \theta = \frac{\pi}{2} \right)$$

34. If $a \sin^2 \theta + b \cos^2 \theta = m$, $b \sin^2 \phi + a \cos^2 \phi = n$, $a \tan \theta = b \tan \phi$, then which of the following is true?

(1) $\frac{1}{n} - \frac{1}{m} = \frac{1}{a} - \frac{1}{b}$

(2) $\frac{1}{n} + \frac{1}{m} = \frac{1}{a} + \frac{1}{b}$

(3) $m^2 + n^2 = a^2 + b^2$

(4) $m^2 - n^2 = a^2 - b^2$

Sol. Answer (2)

$$a \sin^2 \theta + b \cos^2 \theta = m \quad \dots(1)$$

$$b \sin^2 \phi + a \cos^2 \phi = n \quad \dots(2)$$

$$a \tan \theta = b \tan \phi \quad \dots(3)$$

Divide (1) by $\cos^2 \theta$ we, get

$$a \tan^2 \theta + b = m \sec^2 \theta$$

$$\Rightarrow \tan^2 \theta = \frac{m-b}{a-m} \quad \dots(4)$$

Divide (2) by $\cos^2 \phi$, we get

$$b \tan^2 \phi + a = n \sec^2 \phi$$

$$\Rightarrow \tan^2 \phi = \frac{n-a}{b-n} \quad \dots(5)$$

From (3), (4), (5)

$$a^2 \left(\frac{m-b}{a-m} \right) = b^2 \left(\frac{n-a}{b-n} \right)$$

$$\Rightarrow a^2(mb - mn - b^2 + bn) = b^2(an - a^2 - mn - am)$$

$$\Rightarrow abm(a-b) + abn(a-b) = mn(a^2 - b^2)$$

$$\Rightarrow abm + abn = mn(a+b)$$

Divide both sides by $abmn$, we get

$$\frac{1}{n} + \frac{1}{m} = \frac{1}{a} + \frac{1}{b}$$

[Transformation Formulae]

35. If $\tan \theta = \frac{1}{2}$ and $\tan \phi = \frac{1}{3}$, then the value of $\theta + \phi$ is

(1) $\frac{\pi}{6}$

(2) π

(3) 0

(4) $\frac{\pi}{4}$

Sol. Answer (4)

$$\tan(\theta + \phi) = \frac{\tan \theta + \tan \phi}{1 - \tan \theta \tan \phi}$$

36. The value of $\cot\left(\frac{\pi}{4} + \theta\right)\cot\left(\frac{\pi}{4} - \theta\right)$ is

- (1) -1 (2) 0 (3) 1 (4) Not defined

Sol. Answer (3)

37. The value of $\cos^2 48^\circ - \sin^2 12^\circ$ is

- (1) $\frac{\sqrt{5}+1}{8}$ (2) $\frac{\sqrt{5}-1}{8}$ (3) $\frac{\sqrt{5}+1}{5}$ (4) $\frac{\sqrt{5}+1}{2\sqrt{2}}$

Sol. Answer (1)

$$\cos^2 A - \sin^2 B = \cos(A+B)\cos(A-B)$$

38. The value of $\sin(45^\circ + \theta) - \cos(45^\circ - \theta)$ is

- (1) $2\cos \theta$ (2) $2\sin \theta$ (3) 1 (4) 0

Sol. Answer (4)

$$\sin \theta = \cos(90^\circ - \theta)$$

39. If $\cos(\alpha + \beta) = \frac{4}{5}$ and $\sin(\alpha - \beta) = \frac{5}{13}$ where α, β lie between 0 and $\frac{\pi}{4}$, then the value of $\tan(2\alpha)$ is

- (1) $\frac{25}{16}$ (2) $\frac{56}{33}$ (3) $\frac{19}{12}$ (4) $\frac{20}{7}$

Sol. Answer (2)

$$\cos(\alpha + \beta) = \frac{4}{5} \Rightarrow \tan(\alpha + \beta) = \frac{3}{4}$$

$$\sin(\alpha - \beta) = \frac{5}{13} \Rightarrow \tan(\alpha - \beta) = \frac{5}{12}$$

$$\tan 2\alpha = \tan[(\alpha + \beta) + (\alpha - \beta)] = \frac{\tan(\alpha + \beta) + \tan(\alpha - \beta)}{1 - \tan(\alpha + \beta)\tan(\alpha - \beta)}$$

40. The value of $\frac{1 - \tan^2 15^\circ}{1 + \tan^2 15^\circ}$ is

- (1) 1 (2) $\sqrt{3}$ (3) $\frac{\sqrt{3}}{2}$ (4) 2

Sol. Answer (3)

$$\begin{aligned} \frac{1 - \tan^2 15^\circ}{1 + \tan^2 15^\circ} &= \frac{\cos^2 15^\circ - \sin^2 15^\circ}{\cos^2 15^\circ + \sin^2 15^\circ} = \frac{\cos 30^\circ \cos 0^\circ}{1} \\ &= \frac{\sqrt{3}}{2} \end{aligned}$$

41. The value of $\cos 12^\circ + \cos 84^\circ + \cos 156^\circ + \cos 132^\circ$ is

- (1) $\frac{1}{2}$ (2) 1 (3) $-\frac{1}{2}$ (4) $\frac{1}{8}$

Sol. Answer (3)

$$\begin{aligned}
& \cos 12^\circ + \cos 132^\circ + \cos 84^\circ + \cos 156^\circ \\
&= 2\cos\left(\frac{144}{2}\right)^\circ \cos\left(\frac{120}{2}\right)^\circ + 2\cos\left(\frac{240}{2}\right)^\circ \cos\left(\frac{72}{2}\right)^\circ \\
&= 2\cos 72^\circ \times \frac{1}{2} + 2\cos 120^\circ \times \cos 36^\circ \\
&= \cos 72^\circ - \cos 36^\circ \\
&= -2\sin\left(\frac{108}{2}\right)^\circ \sin\left(\frac{36}{2}\right)^\circ \\
&= -2\sin 54^\circ \sin 18^\circ \\
&= -2\left(\frac{\sqrt{5}+1}{4}\right)\left(\frac{\sqrt{5}-1}{4}\right) \\
&= -\frac{1}{2}
\end{aligned}$$

42. The value of $\sin 50^\circ - \sin 70^\circ + \sin 10^\circ$ is equal to

- (1) 1 (2) 0 (3) $\frac{1}{2}$ (4) 2

Sol. Answer (2)

$$\begin{aligned}
& \sin 50^\circ + \sin 10^\circ - \sin 70^\circ \\
&= 2\sin\left(\frac{60}{2}\right)^\circ \cos\left(\frac{40}{2}\right)^\circ - \sin 70^\circ \\
&= (\cos 20^\circ) - \cos 20^\circ \\
&= 0
\end{aligned}$$

43. The value of $\sin \frac{\pi}{18} + \sin \frac{\pi}{9} + \sin \frac{2\pi}{9} + \sin \frac{5\pi}{18}$ is

- (1) $\sin \frac{7\pi}{18} + \sin \frac{4\pi}{9}$ (2) 1 (3) $\cos \frac{\pi}{6} + \cos \frac{3\pi}{7}$ (4) $\cos \frac{\pi}{9} + \sin \frac{\pi}{9}$

Sol. Answer (1)

Let

$$\begin{aligned}
& \therefore \sin \theta + \sin 2\theta + \sin 4\theta + \sin 5\theta \\
&= \sin \theta + \sin 5\theta + \sin 2\theta + \sin 4\theta \\
&= 2\sin 3\theta \cos 2\theta + 2\sin 3\theta \cos \theta \\
&= 2\sin 3\theta (\cos 2\theta + \cos \theta) \\
&= 2\sin \frac{3\pi}{18} \left(\cos \frac{2\pi}{18} + \cos \frac{\pi}{18} \right)
\end{aligned}$$

$$\begin{aligned}
 &= \cos \frac{\pi}{9} + \cos \frac{\pi}{18} \\
 &= \sin \left(\frac{\pi}{2} - \frac{\pi}{9} \right) + \sin \left(\frac{\pi}{2} - \frac{\pi}{18} \right) \\
 &= \sin \frac{7\pi}{18} + \sin \frac{8\pi}{18} \\
 &= \sin \frac{7\pi}{18} + \sin \frac{4\pi}{9}
 \end{aligned}$$

44. The value of $\sin \frac{\pi}{10} \sin \frac{13\pi}{10}$ is

- (1) $\frac{1}{2}$ (2) $\frac{-1}{2}$ (3) $\frac{-1}{4}$ (4) 1

Sol. Answer (3)

$$\begin{aligned}
 &\sin 18^\circ \sin \left(\pi + \frac{3\pi}{10} \right) \\
 &= -\sin 18^\circ \sin 54^\circ \\
 &= -\left(\frac{\sqrt{5}-1}{4} \right) \left(\frac{\sqrt{5}+1}{4} \right) \\
 &= -\frac{4}{16} = -\frac{1}{4}
 \end{aligned}$$

45. In a $\triangle ABC$, if $\sin A - \cos B = \cos C$, then the measure of $\angle B$ is

- (1) $\frac{\pi}{2}$ (2) $\frac{\pi}{3}$ (3) $\frac{\pi}{4}$ (4) $\frac{\pi}{6}$

Sol. Answer (1)

$$\sin A = \cos C + \cos B$$

$$\sin A = 2 \cos \frac{B+C}{2} \cdot \cos \frac{B-C}{2}$$

$$\sin A = 2 \cos \left(\frac{\pi}{2} - \frac{A}{2} \right) \cos \left(\frac{B-C}{2} \right)$$

$$\sin A = 2 \sin \frac{A}{2} \cos \left(\frac{B-C}{2} \right)$$

$$2 \sin \frac{A}{2} \cos \frac{A}{2} - 2 \sin \frac{A}{2} \cos \left(\frac{B-C}{2} \right) = 0$$

$$2 \sin \frac{A}{2} \left(\cos \frac{A}{2} - \cos \frac{B-C}{2} \right) = 0$$

$$\Rightarrow \cos \frac{A}{2} - \cos \left(\frac{B-C}{2} \right) = 0 \left[\because A \neq 0 \text{ or } \frac{A}{2} \neq 0 \Rightarrow \sin \frac{A}{2} \neq 0 \right]$$

$$\frac{A}{2} = \frac{B-C}{2}$$

$$A - B + C = 0$$

$$\Rightarrow A + C = B$$

$$\text{Also, } A + B + C = 180^\circ$$

$$2B = 180^\circ$$

$$\Rightarrow B = 90^\circ \text{ or } \frac{\pi}{2}$$

46. The value of $\cos 10^\circ \cos 20^\circ \cos 40^\circ$ is

- (1) $\frac{1}{8} \cot 10^\circ$ (2) $\frac{1}{8} \tan 10^\circ$ (3) $\frac{1}{4} \sec 10^\circ$ (4) $\frac{1}{4} \operatorname{cosec} 10^\circ$

Sol. Answer (1)

Multiply and divide by $\frac{1}{2} \sin 10^\circ$

47. The value of the expression

$$2 \cos \left(\frac{\pi}{13} \right) \cos \left(\frac{9\pi}{13} \right) + \cos \left(\frac{3\pi}{13} \right) + \cos \left(\frac{5\pi}{13} \right) \text{ is}$$

- (1) 0 (2) -1 (3) 1 (4) 2

Sol. Answer (1)

$$2 \cos \left(\frac{\pi}{13} \right) \cos \left(\frac{9\pi}{13} \right) + \cos \left(\frac{3\pi}{13} \right) + \cos \left(\frac{5\pi}{13} \right)$$

$$= \cos \left(\frac{10\pi}{13} \right) + \cos \left(\frac{8\pi}{13} \right) + \cos \left(\frac{3\pi}{13} \right) + \cos \left(\frac{5\pi}{13} \right)$$

$$= \cos \left(\pi - \frac{3\pi}{13} \right) + \cos \left(\pi - \frac{5\pi}{13} \right) + \cos \left(\frac{3\pi}{13} \right) + \cos \left(\frac{5\pi}{13} \right)$$

$$= -\cos \left(\frac{3\pi}{13} \right) - \cos \left(\frac{5\pi}{13} \right) + \cos \left(\frac{3\pi}{13} \right) + \cos \left(\frac{5\pi}{13} \right)$$

$$= 0$$

48. If $\sin \theta + \cos \theta = 1$, then the value of $\sin 2\theta$ is equal to

- (1) 1 (2) $\frac{1}{2}$ (3) 0 (4) -1

Sol. Answer (3)

On squaring both the sides, we get $(\sin \theta + \cos \theta)^2 = 1$

$$\Rightarrow \sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta = 1$$

$$\Rightarrow \sin 2\theta = 0$$

49. If $\tan A = \frac{1}{2}$ and $\tan B = \frac{1}{3}$, then $\tan(2A + B)$ is equal to

(1) 1

(2) 2

(3) 3

(4) 4

Sol. Answer (3)

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A} = \frac{2 \times \frac{1}{2}}{1 - \frac{1}{4}} = \frac{4}{3}$$

$$\tan(2A + B) = \frac{\tan 2A + \tan B}{1 - \tan 2A \tan B} = \frac{\frac{4}{3} + \frac{1}{3}}{1 - \frac{4}{3} \times \frac{1}{3}} = 3$$

50. The value of $\tan 3A - \tan 2A - \tan A$ is equal to

(1) $\tan 3A \tan 2A \tan A$ (2) $-\tan 3A \tan 2A \tan A$ (3) $\tan A \tan 2A - \tan 2A \tan 3A - \tan 3A \tan A$ (4) $\tan 6A$

Sol. Answer (1)

$$3A = 2A + A$$

$$\Rightarrow \tan 3A = \tan(2A + A)$$

51. If $\sin \theta = \frac{-4}{5}$ and θ lies in the III quadrant, then the value of $\cos \frac{\theta}{2}$ is

(1) $\frac{1}{\sqrt{5}}$ (2) $\frac{1}{\sqrt{10}}$ (3) $\frac{-1}{\sqrt{5}}$ (4) $-\frac{1}{\sqrt{10}}$

Sol. Answer (3)

$$180^\circ < \theta < 270^\circ$$

$$\Rightarrow 90^\circ < \frac{\theta}{2} < 135^\circ$$

This means $\frac{\theta}{2}$ lies in the second quadrant.

$\therefore \cos \frac{\theta}{2}$ will be negative.

$$\text{If } \sin \theta = \frac{-4}{5}, \text{ then } \cos \theta = \frac{-3}{5}$$

$$\text{Now, } \cos 2A = 2\cos^2 A - 1$$

$$\therefore \cos \theta = 2\cos^2 \frac{\theta}{2} - 1$$

$$\frac{-3}{5} = 2\cos^2 \frac{\theta}{2} - 1$$

$$2\cos^2 \frac{\theta}{2} = \frac{2}{5}$$

$$\cos^2 \frac{\theta}{2} = \frac{1}{5}$$

$$\cos \frac{\theta}{2} = -\frac{1}{\sqrt{5}} \text{ [Since } \frac{\theta}{2} \text{ lies in the II quadrant]}$$

52. The value of $\tan 75^\circ - \cot 75^\circ =$

- (1) $2\sqrt{3}$ (2) $2 + \sqrt{3}$ (3) $2 - \sqrt{3}$ (4) $-2\sqrt{3}$

Sol. Answer (1)

$$\begin{aligned}\tan 75^\circ - \cot 75^\circ &= \frac{\sin 75^\circ}{\cos 75^\circ} - \frac{\cos 75^\circ}{\sin 75^\circ} \\&= \frac{\sin^2 75^\circ - \cos^2 75^\circ}{\sin 75^\circ \cos 75^\circ} \\&= -\frac{2\cos(75^\circ \times 2)}{\sin(75^\circ \times 2)} \\&= \frac{-2\cos 150^\circ}{\sin 150^\circ} \\&= \frac{2\cos 30^\circ}{\sin 30^\circ} \\&= 2\sqrt{3}\end{aligned}$$

53. The value of $2\sin A \cos^3 A - 2\sin^3 A \cos A$ is

- (1) $\sin 4A$ (2) $\frac{1}{2}\sin 4A$ (3) $\frac{1}{4}\sin 4A$ (4) $\sin 2A$

Sol. Answer (2)

$$\begin{aligned}2\sin A \cos A (\cos^2 A - \sin^2 A) \\&= \sin 2A \cos 2A \\&= \frac{1}{2}\sin 4A\end{aligned}$$

54. If $\sin \alpha + \sin \beta = a$ and $\cos \alpha - \cos \beta = b$ then, $\tan\left(\frac{\alpha - \beta}{2}\right)$ is equal to

- (1) $-\frac{a}{b}$ (2) $-\frac{b}{a}$ (3) $\sqrt{a^2 + b^2}$ (4) $\frac{a}{b}$

Sol. Answer (2)

$$\begin{aligned}\sin \alpha + \sin \beta &= a \\ \Rightarrow 2\sin\left(\frac{\alpha + \beta}{2}\right)\cos\left(\frac{\alpha - \beta}{2}\right) &= -2\sin\left(\frac{\alpha + \beta}{2}\right)\sin\left(\frac{\alpha - \beta}{2}\right) = b \\ \Rightarrow \cot\left(\frac{\alpha - \beta}{2}\right) &= -\frac{a}{b} \\ \Rightarrow \tan\left(\frac{\alpha - \beta}{2}\right) &= -\frac{b}{a}\end{aligned}$$

55. If $\tan \alpha = \frac{\sqrt{3}+1}{\sqrt{3}-1}$, then the expression $\cos 2\alpha + (2+\sqrt{3})\sin 2\alpha$ is

- (1) $2+\sqrt{3}$ (2) -1 (3) 1 (4) $-(2+\sqrt{3})$

Sol. Answer (3)

$$\tan \alpha = \frac{\sqrt{3}+1}{\sqrt{3}-1} = 2+\sqrt{3} = \tan 75^\circ$$

$$\Rightarrow \alpha = 75^\circ$$

$$\cos 2\alpha + (2+\sqrt{3})\sin 2\alpha$$

$$= \sin 2\alpha (-\sqrt{3}+2+\sqrt{3}) = 2 \times \sin 50^\circ = 2 \times \frac{1}{2} = 1$$

56. If $0 < \theta_2 < \theta_1 < \frac{\pi}{4}$, $\cos(\theta_1 + \theta_2) = \frac{3}{5}$ and

$$\cos(\theta_1 - \theta_2) = \frac{4}{5}, \text{ then } \sin 2\theta_1 \text{ equal to}$$

- (1) -1 (2) 1 (3) 2 (4) -2

Sol. Answer (2)

$$\cos(\theta_1 + \theta_2) = \frac{3}{5}, \cos(\theta_1 - \theta_2) = \frac{4}{5}$$

$$\sin(2\theta_1) = \sin[(\theta_1 + \theta_2) + (\theta_1 - \theta_2)]$$

$$= \frac{4}{5} \cdot \frac{4}{5} + \frac{3}{5} \cdot \frac{3}{5}$$

$$= \frac{25}{25} = 1$$

57. If $\sin \theta = \frac{1}{2}$, $\cos \phi = \frac{1}{3}$, then $\theta + \phi$ belongs to, where $0 < \theta, \phi < \frac{\pi}{2}$

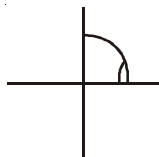
- (1) $\left(\frac{\pi}{3}, \frac{\pi}{2}\right)$ (2) $\left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$ (3) $\left(\frac{2\pi}{3}, \frac{5\pi}{6}\right)$ (4) $\left(\frac{5\pi}{6}, \pi\right)$

Sol. Answer (2)

$$\sin \theta = \frac{1}{2}, \quad \cos \phi = \frac{1}{3}$$

$$\theta = \frac{\pi}{6}, \quad \phi = \frac{\pi}{3}$$

$$\therefore \theta + \phi = \left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$$



58. The range of

$$f(\theta) = 3\cos^2\theta - 8\sqrt{3}\cos\theta\sin\theta + 5\sin^2\theta - 7 \text{ is given by}$$

- (1) $[-7, 7]$ (2) $[-10, 4]$ (3) $[-4, 4]$ (4) $[-10, 7]$

Sol. Answer (2)

$$f(\theta) = 3\cos^2\theta - 8\sqrt{3}\cos\theta\sin\theta + 5\sin^2\theta - 7$$

$$= 3(\cos^2\theta + \sin^2\theta) + 2\sin^2\theta - 4\sqrt{3}\sin 2\theta - 7$$

$$= 3 + 1 - \cos 2\theta - 4\sqrt{3}\sin 2\theta - 7$$

$$= -\cos 2\theta - 4\sqrt{3}\sin 2\theta - 3$$

$$\text{Maximum value} = \sqrt{1+48} - 3 = 7 - 3 = 4$$

$$\text{Minimum value} = -7 - 3 = -10$$

$$R_f = [-10, 4]$$

59. The maximum value of $1 + \sin\left(\frac{\pi}{4} + \theta\right) + 2\cos\left(\frac{\pi}{4} - \theta\right)$ for real values of θ is

- (1) 3 (2) 5 (3) 4 (4) 2

Sol. Answer (3)

$$1 + \sin\left(\frac{\pi}{4} + \theta\right) + 2\cos\left(\frac{\pi}{4} - \theta\right)$$

$$= 1 + \frac{1}{\sqrt{2}}(\sin\theta + \cos\theta) + 2\left(\frac{1}{2}\cos\theta + \frac{1}{\sqrt{2}}\sin\theta\right)$$

$$= 1 + \sin\theta\left(\frac{1}{\sqrt{2}} + \sqrt{2}\right) + \cos\theta\left(\frac{1}{\sqrt{2}} + \sqrt{2}\right)$$

$$= 1 + \frac{3}{\sqrt{2}}\sin\theta + \frac{3}{\sqrt{2}}\cos\theta$$

$$\text{Maximum value} = \sqrt{\frac{9}{2} + \frac{9}{2}} + 1 = 3 + 1 = 4$$

60. The value of $\cos\left(\frac{\pi}{4}\right) \cdot \cos\left(\frac{\pi}{8}\right) \cdot \cos\left(\frac{\pi}{16}\right) \cdots \cos\left(\frac{\pi}{2^n}\right)$ equals

- (1) $\frac{1}{2^n} \operatorname{cosec}\left(\frac{\pi}{2^n}\right)$ (2) $\frac{1}{2^{n-1}} \operatorname{cosec}\left(\frac{\pi}{2^{n-1}}\right)$ (3) $\frac{1}{2^n} \operatorname{cosec}\left(\frac{\pi}{2^{n-1}}\right)$ (4) $\frac{1}{2^{n-1}} \operatorname{cosec}\left(\frac{\pi}{2^n}\right)$

Sol. Answer (4)

$$\cos\left(\frac{\pi}{4}\right) \cdot \cos\left(\frac{\pi}{8}\right) \cdot \cos\left(\frac{\pi}{16}\right) \cdots \cos\left(\frac{\pi}{2^n}\right)$$

$$= \cos A \cdot \cos 2A \cdot \cos 2^2 A \cdots \cos 2^{n-2} A, \quad A = \frac{\pi}{2^n}$$

$$= \frac{1}{2^n \sin A} \times \sin 2^{n-1} A$$

$$= \frac{1}{2^n \sin \theta} \times \sin \left(\frac{2^n A}{2} \right)$$

$$= \frac{1}{2^n \sin \theta} \times \sin \left(\frac{\pi}{2} \right)$$

$$= \frac{1}{2^n \cdot \sin \left(\frac{\pi}{2^n} \right)}$$

$$= \frac{1}{2^n} \operatorname{cosec} \left(\frac{\pi}{2^n} \right)$$

61. If $\tan A + \tan B + \tan C = \tan A \cdot \tan B \cdot \tan C$, then

(1) A, B, C must be angles of a triangle

(2) The sum of any two of A, B, C is equal to third

(3) $A + B + C$ must be integral multiple of π

(4) $A + B + C$ must be odd integral multiple of π

Sol. Answer (3)

$$\text{If } A + B + C = n\pi$$

$$\Rightarrow \tan A + \tan B + \tan C = \tan A \cdot \tan B \cdot \tan C$$

62. The value of $\sin \frac{\pi}{n} + \sin \frac{3\pi}{n} + \sin \frac{5\pi}{n} + \dots$ to n terms is equal to

(1) 1

(2) 0

(3) $\frac{n}{2}$

(4) $\frac{n-1}{2}$

Sol. Answer (2)

$$\sin \frac{\pi}{n} + \sin \frac{3\pi}{n} + \sin \frac{5\pi}{n} + \dots$$

$$= \frac{\sin \left(n \cdot \frac{2\pi}{2n} \right)}{\sin \left(\frac{2\pi}{2n} \right)} \times \sin \left(\frac{\pi}{n} + (n-1) \times \frac{2\pi}{2n} \right) = 0$$

63. $\sum_{r=1}^{n-1} \cos^2 \frac{r\pi}{n}$ is equal to

(1) $\frac{n}{2}$

(2) $\frac{n-1}{2}$

(3) $\frac{n}{2} - 1$

(4) $\frac{n+1}{2}$

Sol. Answer (3)

$$\sum_{r=1}^{n-1} \cos^2 \frac{r\pi}{n}$$

$$= \cos^2 \frac{\pi}{n} + \cos^2 \frac{2\pi}{n} + \cos^2 \frac{3\pi}{n} + \dots + \cos^2 \frac{(n-1)\pi}{n}$$

$$= \frac{1}{2} \left[(n-1) + \cos \frac{2\pi}{n} + \cos \frac{4\pi}{n} + \dots \right] \text{ upto } n-1 \text{ terms}$$

$$= \frac{n-1}{2} + \frac{1}{2} \frac{\sin \left((n-1) \frac{\pi}{n} \right)}{\sin \left(\frac{\pi}{n} \right)} \cdot \cos \left(\frac{2\pi}{n} + (n-2) \cdot \frac{\pi}{n} \right)$$

$$= \frac{n-1}{2} - \frac{1}{2} = \frac{n}{2} - 1$$

64. If $\tan \theta + \tan \left(\theta + \frac{\pi}{3} \right) + \tan \left(\theta - \frac{\pi}{3} \right) = K \tan 3\theta$, then K is equal to

- (1) 1 (2) 3 (3) $\frac{1}{3}$ (4) 2

Sol. Answer (2)

$$\tan \theta + \tan \left(\theta + \frac{\pi}{3} \right) + \tan \left(\theta - \frac{\pi}{3} \right) = K \tan 3\theta$$

$$K = 3$$

65. In a triangle ABC , $\tan A + \tan B + \tan C = 6$ and $\tan A \cdot \tan B = 2$, then value of $\tan A$, $\tan B$ and $\tan C$ are

- (1) 3, 1, 2 (2) 1, 2, 4 (3) 1, 2, 3 (4) 2, 2, 2

Sol. Answer (3)

$$\tan A + \tan B + \tan C = 6$$

$$\Rightarrow \tan A \tan B \tan C = 6 \quad \Rightarrow \quad \tan C = \frac{6}{2} = 3$$

$$\tan A = 1, \tan B = 2, \tan C = 3$$

66. Let $a = \cos A + \cos B - \cos(A+B)$ and $b = 4 \sin \frac{A}{2} \cdot \sin \frac{B}{2} \cdot \cos \left(\frac{A+B}{2} \right)$, then $a - b$ is equal to

- (1) 1 (2) Zero (3) -1 (4) $\frac{1}{2}$

Sol. Answer (1)

$$a = \cos A + \cos B - \cos(A+B)$$

$$= 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2} - 2 \cos^2 \frac{A+B}{2} + 1$$

$$= 1 + 2 \cos \frac{A+B}{2} \left(\cos \frac{A-B}{2} - \cos \frac{A+B}{2} \right)$$

$$= 1 + 2 \cos \frac{A+B}{2} \times 2 \sin \frac{A}{2} \sin \frac{B}{2}$$

$$= 1 + 4 \cos \frac{A+B}{2} \times \sin \frac{A}{2} \sin \frac{B}{2}$$

$$= 1 + b$$

$$\Rightarrow a - b = 1$$

67. The value of $\tan^6 15^\circ - 15\tan^4 15^\circ + 15\tan^2 15^\circ - 3$ is

- (1) -1 (2) 1 (3) 2 (4) -2

Sol. Answer (4)

$$\text{Let } \theta = 15^\circ \Rightarrow 3\theta = 45^\circ$$

$$\tan 3\theta = 1$$

$$3 \tan \theta - \tan^3 \theta = 1 - 3 \tan^2 \theta$$

Squaring both side

$$9 \tan^2 \theta + \tan^6 \theta - 6 \tan^4 \theta = 1 + 9 \tan^4 \theta - 6 \tan^2 \theta$$

$$\Rightarrow \tan^6 \theta - 15 \tan^4 \theta + 15 \tan^2 \theta - 3 = -2$$

68. If A, B, C are in A.P., then correct relation is

- (1) $\cot A = \frac{\sin B + \sin C}{\cos C + \sin B}$ (2) $\cot B = \frac{\sin A - \sin C}{\cos C - \cos A}$
 (3) $\tan A = \frac{\sin A + \sin C}{\cos C - \cos B}$ (4) $\tan B = \frac{\sin B - \sin C}{\cos C - \cos B}$

Sol. Answer (2)

$$A, B, C \in \text{A.P.}$$

$$\Rightarrow 2B = A + C$$

$$\Rightarrow B = \frac{A+C}{2}$$

$$\cot B = \cot\left(\frac{A+C}{2}\right)$$

$$= \frac{\cos\left(\frac{A+C}{2}\right)}{\sin\left(\frac{A+C}{2}\right)}$$

$$= \frac{\cos\left(\frac{A+C}{2}\right) \cdot \sin\left(\frac{A-C}{2}\right)}{\sin\left(\frac{A+C}{2}\right) \cdot \sin\left(\frac{A-C}{2}\right)} = \frac{\sin A - \sin C}{\cos C - \cos A}$$

69. $\cos \frac{\pi}{14} + \cos \frac{3\pi}{14} + \cos \frac{5\pi}{14} = K \cot \frac{\pi}{14}$, then K is equal to

- (1) 1 (2) $\frac{1}{2}$ (3) 2 (4) -2

Sol. Answer (2)

$$\cos \frac{\pi}{14} + \cos \frac{3\pi}{14} + \cos \frac{5\pi}{14} = K \cot \frac{\pi}{14}$$

$$\Rightarrow \cos \frac{\pi}{14} + 2 \cos \frac{4\pi}{14} \cdot \cos \frac{\pi}{14} = K \cot \frac{\pi}{14}$$

$$\Rightarrow 2 \cdot \cos \frac{\pi}{14} \cos \frac{4\pi}{14} = \frac{K \cdot \cos \frac{\pi}{14}}{\sin \frac{\pi}{14}} - \cos \frac{\pi}{14}$$

$$\Rightarrow 2 \cos \frac{4\pi}{14} \cdot \sin \frac{\pi}{14} = K - \sin \frac{\pi}{14}$$

$$\Rightarrow \sin \frac{5\pi}{14} - \sin \frac{3\pi}{14} = K - \sin \frac{\pi}{14}$$

$$\Rightarrow K = \sin \frac{5\pi}{14} - \sin \frac{3\pi}{14} + \sin \frac{\pi}{14}$$

$$= 2 \sin \frac{3\pi}{14} \cos \frac{2\pi}{14} - \sin \frac{3\pi}{14}$$

$$= -2 \sin \frac{3\pi}{14} \times 2 \cos^2 \frac{\pi}{14}$$

$$= -4 \sin \frac{3\pi}{14} \cdot \cos^2 \frac{\pi}{14} \Rightarrow K = \frac{1}{2}$$

70. The minimum value of $8(\cos 2\theta + \cos \theta)$ is equal to

(1) -17

(2) -9

(3) 3

(4) -8

Sol. Answer (2)

$$8(\cos 2\theta + \cos \theta)$$

$$= 8(2\cos^2 \theta - 1 + \cos \theta)$$

$$= 8(2\cos^2 \theta + \cos \theta - 1)$$

$$= 16 \left(\cos^2 \theta + \frac{1}{2} \cos \theta - \frac{1}{2} \right)$$

$$= 16 \left(\left(\cos \theta + \frac{1}{4} \right)^2 - \frac{6}{16} \right)$$

$$\text{Minimum value} = 16 \times -\frac{9}{16} = -9$$

71. If $4n\alpha = \pi$, then $\cot \alpha \cdot \cot 2\alpha \cdot \cot 3\alpha \dots \cot (2n-1)\alpha$ $n \in \mathbb{Z}$ is equal to

(1) 1

(2) -1

(3) 2

(4) Zero

Sol. Answer (1)

$$4n\alpha = \pi \Rightarrow 2n\alpha = \frac{\pi}{2}$$

$$\cot \alpha \cdot \cot 2\alpha \cdot \cot 3\alpha \dots \cot (2n-1)\alpha$$

$$= \cot \alpha \cdot \cot 2\alpha \cdot \cot 3\alpha \dots \cot \left(\frac{\pi}{2} - 3\alpha \right) \cdot \cot \left(\frac{\pi}{2} - \alpha \right) \cdot \cot \left(\frac{\pi}{2} - \alpha \right)$$

$$= 1$$

72. The value of $\tan 7\frac{1}{2}^\circ$ is equal to

(1) $\frac{\sqrt{3}-\sqrt{2}}{\sqrt{2}+1}$

(2) $\frac{\sqrt{3}-\sqrt{2}}{\sqrt{2}-1}$

(3) $\frac{\sqrt{3}+\sqrt{2}}{\sqrt{2}+1}$

(4) $\frac{\sqrt{3}+\sqrt{2}}{\sqrt{2}-1}$

Sol. Answer (1)

$$\therefore \tan \theta = \frac{1 - \cos 2\theta}{\sin 2\theta}$$

$$\text{Put } \theta = \frac{15}{2}$$

$$\tan \frac{15^\circ}{2} = \frac{1 - \cos 15^\circ}{\sin 15^\circ} = \frac{1 - \frac{\sqrt{3}+1}{2\sqrt{2}}}{\frac{\sqrt{3}-1}{2\sqrt{2}}}$$

$$= \frac{2\sqrt{2} - \sqrt{3} - 1}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1}$$

$$= \frac{2\sqrt{6} - 3 - \sqrt{3} + 2\sqrt{2} - \sqrt{3} - 1}{3 - 1}$$

$$= \frac{2\sqrt{6} - 4 - 2\sqrt{3} + 2\sqrt{2}}{2}$$

$$= \sqrt{6} - \sqrt{4} - \sqrt{3} + \sqrt{2}$$

$$\therefore \sqrt{3} - \sqrt{2} \times \sqrt{2} - 1 = (\sqrt{3} - \sqrt{2})(\sqrt{2} + 1)$$

$$= \sqrt{6} - \sqrt{4} - \sqrt{3} + \sqrt{2} = \frac{\sqrt{3} - \sqrt{2}}{\sqrt{2} + 1}$$

73. If $\tan \alpha = \sqrt{a}$, where a is a rational number which is not a perfect square, then which of the following is a rational number?

(1) $\sin 2\alpha$

(2) $\tan 2\alpha$

(3) $\cos 2\alpha$

(4) $\operatorname{cosec} 2\alpha$

Sol. Answer (3)

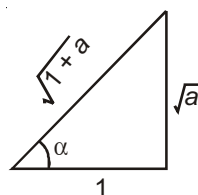
$$\tan \alpha = \sqrt{a}$$

$$\therefore \sin 2\alpha = \frac{\sqrt{a}}{\sqrt{1+a}} = \sqrt{\frac{a}{1+a}}$$

$$\sin 2\alpha = \frac{2\sqrt{a}}{1+a}$$

$$\cos 2\alpha = 2\cos^2 \alpha - 1 = 2 \times \frac{1}{1+a} - 1$$

$$= \frac{2-1-a}{1+a} = \frac{1-a}{a+1}$$



74. The value of $2\tan 18^\circ + 3\sec 18^\circ - 4\cos 18^\circ$ is

- (1) Zero (2) $\sqrt{5}$ (3) $-\sqrt{5}$ (4) $\sqrt{3}$

Sol. Answer (1)

$$2\tan 18^\circ + 3\sec 18^\circ - 4\cos 18^\circ$$

$$= \frac{2\sin 18^\circ + 3 - 4\cos^2 18^\circ}{\cos 18^\circ}$$

$$= \frac{2\sin 18^\circ + 3 - 4 + 4\sin^2 18^\circ}{\cos 18^\circ}$$

$$= \frac{4\sin^2 18^\circ + 2\sin 18^\circ - 1}{\cos 18^\circ}$$

$$= \frac{4 \times \frac{(\sqrt{5}-1)^2}{16} + 2 \cdot \frac{\sqrt{5}-1}{4} - 1}{\cos 18^\circ}$$

$$= \frac{6 - 2\sqrt{5} + 2\sqrt{5} - 2 - 4}{4\cos 18^\circ} = 0$$

75. If $\tan^2 \alpha - 2\tan^2 \beta = 1$, then which of the following is correct?

- (1) $2\cos 2\beta - \cos 2\alpha = 1$ (2) $\cos 2\alpha - 2\cos 2\beta = 1$
 (3) $2\cos 2\alpha - \cos 2\beta = -1$ (4) $\cos 2\beta - 2\cos 2\alpha = -1$

Sol. Answer (3)

$$\tan^2 \alpha - 2\tan^2 \beta = 1$$

$$\Rightarrow \tan^2 \alpha = 1 + 2\tan^2 \beta$$

$$\Rightarrow 1 + \tan^2 \alpha = 2 + 2\tan^2 \beta$$

$$\Rightarrow \sec^2 \alpha = 2\sec^2 \beta$$

$$\Rightarrow \cos^2 \beta = 2\cos^2 \alpha$$

$$\Rightarrow 2\cos^2 \beta = 4\sin^2 \alpha$$

$$\Rightarrow 1 + \cos 2\beta = 2(1 + \cos 2\alpha)$$

$$\Rightarrow 2\cos 2\alpha - \cos 2\beta = -1$$

76. The value of $\tan 10^\circ \cdot \tan 50^\circ \cdot \tan 70^\circ$ is

- (1) $\sqrt{3}$ (2) $\frac{1}{\sqrt{3}}$ (3) 1 (4) -1

Sol. Answer (2)

$$\tan 30^\circ = \frac{1}{\sqrt{3}}$$

77. The expression $2\sin 2^\circ + 4\sin 4^\circ + 6\sin 6^\circ + \dots + 180\sin 180^\circ$ equals

- (1) $\cot 1^\circ$ (2) $90\cot 1^\circ$ (3) $\sin 1^\circ$ (4) $90\cos 1^\circ$

Sol. Answer (2)

$$\begin{aligned} & 2\sin 2^\circ + 4\sin 4^\circ + 6\sin 6^\circ + \dots + 178\sin 178^\circ + 180\sin 180^\circ \\ &= (2 + 178)\sin 2^\circ + (4 + 176)\sin 4^\circ + (6 + 174)\sin 6^\circ + \dots + \sin 88^\circ \\ &= 180(\sin 2^\circ + \sin 4^\circ + \sin 6^\circ + \dots) + 90\sin 90^\circ \end{aligned}$$

$$= 180 \times \frac{\sin\left(44 \cdot \frac{2}{2}\right)}{\sin\left(\frac{2}{2}\right)} \times \sin\left(2 + (44 - 1) \cdot \frac{2}{2}\right) + 90^\circ$$

$$= 180 \times \frac{\sin 44^\circ}{\sin 1^\circ} \times \sin 45^\circ + 90$$

$$= \frac{180}{\sqrt{2}} \times \frac{\sin(45^\circ - 1^\circ)}{\sin 1^\circ} + 90$$

$$= \frac{180}{\sqrt{2}} \left[\frac{\frac{1}{\sqrt{2}} \cos 1^\circ - \frac{1}{\sqrt{2}} \sin 1^\circ}{\sin 1^\circ} \right] + 90$$

$$= \frac{180^\circ}{\sqrt{2}} \left[\frac{1}{\sqrt{2}} \cot 1^\circ - \frac{1}{\sqrt{2}} \right] + 90^\circ$$

$$= 90^\circ \cot 1^\circ - \frac{180}{2} + 90$$

$$= 90\cot 1^\circ$$

78. If $\cos 2x + 2\cos x = 1$, then $\sin^2 x(2 - \cos^2 x)$ is equal to

- (1) 1 (2) -1 (3) $-\sqrt{5}$ (4) $\sqrt{5}$

Sol. Answer (1)

$$\cos 2x + 2\cos x = 1$$

$$2\cos x = 1 - \cos 2x$$

$$2\cos x = 2\sin^2 x$$

$$\cos x = \sin^2 x$$

$$\therefore \sin^2 x(2 - \cos^2 x)$$

$$= \sin^2 x(1 + \sin^2 x)$$

$$= \sin^2 x + \sin^4 x$$

$$= \sin^2 x + \cos^2 x$$

$$= 1$$

79. The value of $\tan \frac{\theta}{2} (1 + \sec \theta)(1 + \sec 2\theta) (1 + \sec 2^2 \theta) \dots (1 + \sec 2^n \theta)$ is

- (1) $\tan 2^n \theta$ (2) $\tan 2^{n-1} \theta$ (3) $\tan 2^{n+1} \theta$ (4) $\tan 2^{n-2} \theta$

Sol. Answer (1)

$$\tan \frac{\theta}{2} (1 + \sec \theta)(1 + \sec 2\theta)(1 + \sec 4\theta) \dots (1 + \sec 2^n \theta)$$

$$= \tan \frac{\theta}{2} \times \frac{1 + \tan \theta}{\cos \theta} \times \frac{1 + \tan 2\theta}{\cos 2\theta} \times \frac{1 + \tan 4\theta}{\cos 4\theta}$$

$$= \frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} \times \frac{2 \cos^2 \frac{\theta}{2}}{\cos \theta} \times \frac{2 \cos^2 \theta}{\cos 2\theta} \times \frac{2 \cos^2 2\theta}{\cos 4\theta}$$

$$= 2^3 \cdot \sin \frac{\theta}{2} \cdot \cos \frac{\theta}{2} \cdot \frac{\cos \theta \cdot \cos 2\theta}{\cos 4\theta}$$

$$= 2^2 \cdot \sin \theta \cdot \cos \theta \cdot \frac{\cos 2\theta}{\cos 4\theta}$$

$$= 2 \frac{\sin 2\theta \cos 2\theta}{\cos 4\theta}$$

$$= \tan 4\theta$$

$$= \tan 2^2 \theta$$

$$\therefore \tan \frac{\theta}{2} (1 + \sec \theta)(1 + \sec 2\theta) \dots (1 + \sec 2^n \theta)$$

$$= \tan 2^n \theta$$

80. The value of $[100(x - 1)]$ is where $[x]$ is the greatest integer less than or equal to x and $x = \frac{\sum_{n=1}^{44^\circ} \cos n^\circ}{\sum_{n=1}^{44^\circ} \sin n^\circ}$

- (1) 140 (2) 141 (3) 142 (4) 144

Sol. Answer (2)

$$\therefore A = \sum_{n=1}^{44} \cos n^\circ$$

$$= \cos 1^\circ + \cos 2^\circ + \cos 3^\circ + \dots + \cos 44^\circ$$

$$B = \sin 1^\circ + \sin 2^\circ + \sin 3^\circ + \dots + \sin 44^\circ$$

$$x = \frac{A}{B} = \frac{\cos 1^\circ + \cos 2^\circ + \cos 3^\circ + \dots + \cos 44^\circ}{\sin 1^\circ + \sin 2^\circ + \sin 3^\circ + \dots + \sin 44^\circ}$$

$$= \frac{\sin 89^\circ + \sin 88^\circ + \sin 87^\circ + \dots + \sin 46^\circ}{\sin 1^\circ + \sin 2^\circ + \sin 3^\circ + \dots + \sin 44^\circ}$$

$$\frac{\sin\left(44 \cdot \frac{1}{2}\right)}{\sin\left(\frac{1}{2}\right)} \times \cos\left(1 + (44-1) \cdot \frac{1}{2}\right)$$

$$= \frac{\sin\left(44 \cdot \frac{1}{2}\right)}{\sin\frac{1}{2}} \times \sin\left(1 + (44-1) \cdot \frac{1}{2}\right)$$

$$= \frac{\cos\left(22 + \frac{1}{2}\right)}{\sin\left(22 + \frac{1}{2}\right)} = \cot 22 \frac{1}{2}^\circ$$

$$= \sqrt{2} + 1$$

$$\therefore [100(\sqrt{2} + 1 - 1)] = [100 \times 1.41] = 141$$

81. If $\alpha + \beta = \frac{\pi}{2}$ and $\beta + \gamma = \alpha$, then $\tan \alpha$ equals

(1) $2(\tan \beta + \tan \gamma)$

(2) $\tan \beta + 2 \tan \gamma$

(3) $\tan \beta - \tan \gamma$

(4) $\tan \gamma - \tan \beta$

Sol. Answer (2)

$$\alpha + \beta = \frac{\pi}{2}, \quad \beta + \gamma = \alpha$$

$$\alpha = \frac{\pi}{2} - \beta$$

$$\beta = \alpha - \gamma$$

$$\Rightarrow \tan \alpha = \cot \beta$$

$$\tan \beta = \frac{\tan \alpha - \tan \gamma}{1 + \tan \alpha \tan \gamma}$$

$$\Rightarrow \tan \beta + \tan \alpha \tan \beta \tan \gamma = \tan \alpha - \tan \gamma$$

$$\Rightarrow \tan \beta = \tan \alpha - 2 \tan \gamma$$

$$\Rightarrow \tan \alpha = \tan \beta + 2 \tan \gamma$$

82. If $f(\alpha, \beta) = \cos^2 \alpha + \sin^2 \alpha \cdot \cos 2\beta$, then which of the following is incorrect?

(1) $f\left(\frac{\pi}{5}, \frac{2\pi}{5}\right) \neq f\left(\frac{2\pi}{5}, \frac{\pi}{5}\right)$

(2) $f\left(\frac{\pi}{12}, \frac{\pi}{3}\right) = f\left(\frac{\pi}{3}, \frac{\pi}{12}\right)$

(3) $3f\left(\frac{\pi}{5}, \frac{\pi}{3}\right) \neq f\left(\frac{\pi}{3}, \frac{\pi}{5}\right)$

(4) $f\left(\frac{\pi}{4}, \frac{\pi}{18}\right) \neq 3f\left(\frac{\pi}{18}, \frac{\pi}{4}\right)$

Sol. Answer (1)

$$f(\alpha, \beta) = \cos^2 \alpha + \sin^2 \alpha \cdot \cos 2\beta$$

$$= 1 - \sin^2 \alpha + \sin^2 \alpha \cdot \cos 2\beta$$

$$= 1 - (1 - \cos 2\beta) \sin^2 \alpha$$

$$= 1 - (1 - \cos^2 \beta) \sin^2 \alpha$$

$$= 1 - 2\sin^2\alpha \sin^2\beta$$

$$f\left(\frac{\pi}{12}, \frac{\pi}{3}\right) = 1 - 2\sin^2\frac{\pi}{12} \sin^2\frac{\pi}{3}$$

$$= 1 - 2 \cdot \frac{(\sqrt{3}-1)^2}{8} \times \frac{3}{4}$$

$$= 1 - \frac{3(4-2\sqrt{3})}{16}$$

$$= 1 - \frac{3(2-\sqrt{3})}{8} = \frac{8-6+3\sqrt{3}}{8} = \frac{2+3\sqrt{3}}{8}$$

83. Let $f(x) = \cos 10x + \cos 8x + 3\cos 4x + 3\cos 2x$ and $g(x) = 8\cos x \cdot \cos^3 3x$, then for all x we have

(1) $f(x) = g(x)$ (2) $2f(x) = 3g(x)$ (3) $f(x) = 2g(x)$ (4) $2f(x) = g(x)$

Sol. Answer (1)

$$f(x) = \cos 10x + \cos 8x + 3\cos 4x + 3\cos 2x$$

$$g(x) = 8\cos x \cdot \cos^3 3x$$

$$= 8\cos x(4\cos^3 x - 3\cos x)^3$$

$$= 8\cos x ((4\cos^3 x)^3 - 3 \cdot (4\cos^3 x)^2 \cdot 3\cos x + 3 \cdot (4\cos^3 x)(3\cos x)^2 - (3\cos x)^3)$$

$$= 8\cos x(64\cos^9 x - 144\cos^7 x + 108\cos^5 x - 27\cos^3 x)$$

$$= 8\cos^4 x(64\cos^6 x - 144\cos^4 x + 108\cos^2 x - 27)$$

$$\therefore f(x) = \cos 10x + \cos 8x + 3\cos 4x + 3\cos 2x$$

84. $\sum_{r=1}^n \left(\frac{1}{\cos \theta + \cos(2r+1)\theta} \right)$, $n \in N$ is equal to

(1) $\frac{\sin(n+1)\theta}{\sin \theta \cdot \cos n\theta}$

(2) $\frac{\sin n\theta}{\sin 2\theta \cdot \cos(n+1)\theta}$

(3) $\frac{\tan(n+1)\theta}{\sin n\theta}$

(4) $\frac{\sin(n-1)\theta}{\sin \theta \cdot \cos n\theta}$

Sol. Answer (2)

$$\sum_{r=1}^n \left(\frac{1}{\cos \theta + \cos(2r+1)\theta} \right)$$

$$= \frac{1}{\cos \theta + \cos 3\theta} + \frac{1}{\cos \theta + \cos 5\theta} + \dots + \frac{1}{\cos \theta + \cos(2n+1)\theta}$$

$$= \frac{1}{2\cos 2\theta \cdot \cos \theta} + \frac{1}{2\cos 3\theta \cdot \cos 2\theta} + \dots + \frac{1}{2\cos(n+1)\theta \cdot \cos n\theta}$$

$$= \frac{1}{2\sin \theta} \left[\frac{\sin(2\theta - \theta)}{\cos 2\theta \cdot \cos \theta} + \frac{\sin(3\theta - 2\theta)}{\cos 3\theta \cos 2\theta} + \dots \right]$$

$$= \frac{1}{2\sin \theta} \left[\frac{\sin 2\theta \cos \theta - \cos 2\theta \sin \theta}{\cos 2\theta \cdot \cos \theta} + \frac{\sin 3\theta \cos 2\theta - \cos 3\theta \sin 2\theta}{\cos 3\theta \cdot \cos 2\theta} + \dots \right]$$

$$= \frac{1}{2\sin\theta} [\tan 2\theta - \tan\theta + \tan 3\theta - \tan 2\theta + \dots \tan(n+1)\theta - \tan n\theta]$$

$$= \frac{1}{2\sin\theta} [\tan(n+1)\theta - \tan\theta]$$

$$= \frac{1}{2\sin\theta} \times \frac{\sin(n+1-\theta)}{\cos(n+1)\theta \cos\theta}$$

$$= \frac{\sin n\theta}{\cos(n+1)\theta \cdot \sin 2\theta}$$

85. The minimum value of $27^{\cos 3x} \cdot 81^{\sin 3x}$ is

- (1) 1 (2) $\frac{1}{81}$ (3) $\frac{1}{243}$ (4) $\frac{1}{27}$

Sol. Answer (3)

$$\begin{aligned} \text{Let } y &= 27^{\cos 3x} \cdot 81^{\sin 3x} \\ &= 3^{3\cos 3x} \cdot 3^{4\sin 3x} \\ &= 3^{3\cos 3x + 4\sin 3x} \end{aligned}$$

Then minimum value of y is

$$3^{-5} = \frac{1}{243}$$

86. Given that $\sum_{k=1}^{35} \sin 5k^\circ = \tan\left(\frac{m}{n}\right)^\circ$, where m and n are relatively prime positive integers that satisfy

$\left(\frac{m}{n}\right)^\circ < 90^\circ$, then $m + n$ is equal to

- (1) 173 (2) 175 (3) 177 (4) 179

Sol. Answer (3)

$$\sum_{k=1}^{35} \sin 5k = \tan \frac{m^\circ}{n^\circ}$$

$$\Rightarrow \sin 5 + \sin 10 + \dots + \sin 175 = \tan \frac{m^\circ}{n^\circ}$$

$$\Rightarrow \frac{\sin\left(35 \cdot \frac{5}{2}\right)}{\sin\left(\frac{5}{2}\right)} \times \sin\left(5 + (35-1) \times \frac{5}{2}\right) = \tan \frac{m^\circ}{n^\circ}$$

$$\Rightarrow \frac{\sin\left(\frac{175^\circ}{2}\right)}{\sin\left(\frac{5^\circ}{2}\right)} \times \sin\left(\frac{175+5}{2}\right) = \tan \frac{m^\circ}{n^\circ}$$

$$\Rightarrow \frac{\sin\left(\frac{175^\circ}{2}\right)}{\sin\left(\frac{5^\circ}{2}\right)} = \tan \frac{m}{n}$$

$$\Rightarrow \frac{\sin\left(\frac{175}{2}\right)}{\cos\left(\frac{175}{2}\right)} = \frac{\sin\left(\frac{m}{n}\right)}{\cos\left(\frac{m}{n}\right)}$$

$$\therefore \frac{m}{n} = \frac{175}{2}$$

$$\therefore m + n = 175 + 2 = 177$$

87. For $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$, $\frac{\sin\theta + \sin 2\theta}{1 + \cos\theta + \cos 2\theta}$ lies in the interval

(1) $(-\infty, \infty)$

(2) $(-2, 2)$

(3) $(0, \infty)$

(4) $(-1, 1)$

Sol. Answer (1)

$$\frac{\sin\theta + \sin 2\theta}{1 + \cos\theta + \cos 2\theta}$$

$$= \frac{\sin\theta(1 + 2\cos\theta)}{\cos\theta(1 + 2\cos\theta)}$$

$$= \tan\theta ; \tan\theta \in (-\infty, \infty)$$

88. If $A + B + C = \pi$ and $\sin\left(A + \frac{C}{2}\right) = K \sin \frac{C}{2}$, then $\tan \frac{A}{2} \cdot \tan \frac{B}{2}$ is equal to

(1) $\frac{K-1}{K+1}$

(2) $\frac{K+1}{K-1}$

(3) $\frac{K}{K+1}$

(4) $\frac{K+1}{K}$

Sol. Answer (1)

$$A + B + C = \pi$$

$$\sin\left(A + \frac{C}{2}\right) = K \sin \frac{C}{2}$$

$$\Rightarrow \frac{\sin\left(A + \frac{C}{2}\right)}{\sin \frac{C}{2}} = \frac{K}{1}$$

$$\Rightarrow \frac{\sin\left(A + \frac{C}{2}\right) + \sin \frac{C}{2}}{\sin\left(A + \frac{C}{2}\right) - \sin \frac{C}{2}} = \frac{K+1}{K-1}$$

$$\Rightarrow \frac{2 \sin\left(\frac{A+C}{2}\right) \cos \frac{A}{2}}{2 \cos\left(\frac{A+C}{2}\right) \sin \frac{A}{2}} = \frac{K+1}{K-1}$$

$$\Rightarrow \frac{\cos \frac{B}{2}}{\sin \frac{B}{2}} \cdot \frac{\cos \frac{A}{2}}{\sin \frac{A}{2}} = \frac{K+1}{K-1}$$

$$\Rightarrow \sin \frac{A}{2} \cdot \sin \frac{B}{2} = \frac{K-1}{K+1}$$

89. If $A + B + C = \pi$ and $\cos A = \cos B \cdot \cos C$, then $\tan B \cdot \tan C$ is equal to

- (1) 1 (2) $\frac{1}{2}$ (3) 2 (4) 3

Sol. Answer (3)

$$A + B + C = \pi$$

$$\cos A = \cos B \cdot \cos C$$

$$\therefore \cos(B + C) = -\cos B \cos C$$

$$\Rightarrow \cos B \cos C - \sin B \sin C = -\cos B \cos C$$

$$\Rightarrow \sin B \sin C = 2 \cos B \cos C$$

$$\Rightarrow \tan B \tan C = 2$$

90. $\frac{\sin x}{\sin 2x \sin 3x} + \frac{\sin x}{\sin 3x \sin 4x} + \frac{\sin x}{\sin 4x \sin 5x} + \dots n$ terms is equal to

- (1) $\cot 2x - \cot nx$ (2) $\tan 2x - \tan nx$
(3) $\cot 2x - \cot (n+2)x$ (4) $\tan(nx + 3x)$

Sol. Answer (3)

$$\frac{\sin(3x-2x)}{\sin 2x \cdot \sin 3x} + \frac{\sin(4x-3x)}{\sin 3x \cdot \sin 4x} + \frac{\sin(5x-4x)}{\sin 4x \cdot \sin 5x}$$

$$= \frac{\sin 3x \cdot \cos 2x - \cos 3x \cdot \sin 2x}{\sin 2x \cdot \sin 3x} + \frac{\sin 4x \cdot \cos 3x - \cos 4x \cdot \sin 3x}{\sin 3x \cdot \sin 4x}$$

$$= (\cot 2x - \cot 3x) + (\cot 3x - \cot 4x) + \dots + \cot(n+1)x - \cot(n+2)x$$

$$= \cot 2x - \cot(n+2)x$$

[Trigonometric Equations]

91. If $2\cos^2\theta - 2\sin^2\theta = 1$, then θ is equal to

- (1) 15° (2) 30° (3) 45° (4) 60°

Sol. Answer (2)

$$2(\cos^2\theta - \sin^2\theta) = 1$$

$$2\cos 2\theta = 1$$

$$\cos 2\theta = \frac{1}{2} = \cos 60^\circ$$

$$2\theta = 60^\circ$$

$$\theta = 30^\circ$$

92. If $\sin^2 \theta = \frac{1}{4}$, then the value of θ is

- (1) $n\pi \pm \frac{\pi}{6}$ (2) $n\pi \pm \frac{\pi}{4}$ (3) $n\pi + \frac{\pi}{3}$ (4) $n\pi$

Sol. Answer (1)

93. The solution of the equation $\cos^2 \theta + \sin \theta + 1 = 0$ lies in the interval

- (1) $\left(-\frac{\pi}{4}, \frac{\pi}{4}\right)$ (2) $\left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$ (3) $\left(\frac{3\pi}{4}, \frac{5\pi}{4}\right)$ (4) $\left(\frac{5\pi}{4}, \frac{7\pi}{4}\right)$

Sol. Answer (4)

$$\cos^2 \theta + \sin \theta + 1 = 0$$

$$1 - \sin^2 \theta + \sin \theta + 1 = 0$$

$$\sin^2 \theta - \sin \theta - 2 = 0$$

$$(\sin \theta - 2)(\sin \theta + 1) = 0$$

$$\therefore \sin \theta = -1 \quad [\because \sin \theta \neq 2 \Rightarrow \sin \theta - 2 \neq 0]$$

$$\Rightarrow \theta = \pi + \frac{\pi}{2} = \frac{3\pi}{2} = \frac{6\pi}{4}$$

$$\therefore \frac{6\pi}{4} \in \left(\frac{5\pi}{4}, \frac{7\pi}{4}\right)$$

94. The general solution of $\tan 3x = 1$ is

- (1) $n\pi + \frac{\pi}{4}$ (2) $\frac{n\pi}{3} + \frac{\pi}{12}$ (3) $n\pi$ (4) $n\pi \pm \frac{\pi}{4}$

Sol. Answer (2)

95. If $\sin \theta = \sqrt{3} \cos \theta$, $-\pi < \theta < 0$, then the value of θ is

- (1) $-\frac{5\pi}{6}$ (2) $-\frac{4\pi}{6}$ (3) $\frac{2\pi}{3}$ (4) $\frac{5\pi}{6}$

Sol. Answer (2)

$$\sin \theta = \sqrt{3} \cos \theta$$

$$\Rightarrow \tan \theta = \sqrt{3} \Rightarrow \theta = n\pi + \frac{\pi}{3}$$

Taking $n = -1$, we get

$$\theta = -\pi + \frac{\pi}{3} = -\frac{2\pi}{3} = -\frac{4\pi}{6}$$

96. The total number of solutions of the equation $\tan x + \sec x = 2$ which lie in the interval $[0, 2\pi]$ is

- (1) 0 (2) 1 (3) 2 (4) 3

Sol. Answer (2)

$$\sec x = 2 - \tan x$$

$$\sec^2 x = 4 + \tan^2 x - 4 \tan x$$

$$1 + \tan^2 x = 4 + \tan^2 x - 4 \tan x$$

$$4 \tan x = 3$$

$$\tan x = \frac{3}{4}$$

Since $\tan x$ is positive in I and III quadrant only, thus $\tan x = \frac{3}{4}$ for 2 values of x .

97. If $\sqrt{3} \sec \theta + 2 = 0$, then principal value of θ may be

(1) $\frac{5\pi}{6}$

(2) $-\frac{5\pi}{6}$

(3) $\frac{7\pi}{6}$

(4) $-\frac{7\pi}{6}$

Sol. Answer (1)

$$\sec \theta = -\frac{2}{\sqrt{3}}$$

$$\Rightarrow \cos \theta = -\frac{\sqrt{3}}{2}$$

$$\Rightarrow \theta = \frac{5\pi}{6} \text{ (Principal)}$$

98. The number of solutions of the equation $2 \tan^2 \theta - 7 \sec \theta - 2 = 0$ in the interval $[0, 6\pi]$ is

(1) 8

(2) 6

(3) 4

(4) Zero

Sol. Answer (2)

$$2(1 + \sec^2 \theta) - 7 \sec \theta - 2 = 0$$

$$\Rightarrow 2 \sec^2 \theta - 7 \sec \theta = 0$$

$$\Rightarrow \sec \theta = \frac{7}{2}$$

In every interval of 2π , 2 solutions exist. In 6π interval, 6 solutions will exist.

99. The general solution of the equation $\tan x + \tan 2x + \tan x \cdot \tan 2x = 1$ is

(1) $x = n\pi + \frac{\pi}{12}, n\pi + \frac{5\pi}{12}, n \in I$

(2) $x = \frac{n\pi}{3} + \frac{\pi}{12}, n \in I$

(3) $x = n\pi - \frac{\pi}{4}, n\pi + \frac{\pi}{12}, n \in I$

(4) $x = n\pi - \frac{\pi}{4}, n\pi + \frac{5\pi}{12}, n \in I$

Sol. Answer (2)

$$\frac{\tan x + \tan 2x}{1 - \tan x \tan 2x} = 1$$

$$\Rightarrow \tan 3x = 1$$

$$\Rightarrow 3x = n\pi + \frac{\pi}{4}$$

$$\Rightarrow x = \frac{n\pi}{3} + \frac{\pi}{12}$$

100. If angles A and B satisfy $(1 + \tan A)(1 + \tan B) = 2$, then the value of $A + B$ may be

(1) $-\frac{11\pi}{4}$

(2) $\frac{15\pi}{4}$

(3) $\frac{23\pi}{4}$

(4) $-\frac{17\pi}{4}$

Sol. Answer (1)

$$1 + \tan A + \tan B + \tan A \tan B = 2$$

$$\Rightarrow \frac{\tan A + \tan B}{1 - \tan A \tan B} = 1$$

$$\Rightarrow \tan(A + B) = 1$$

$$\Rightarrow A + B = n\pi + \frac{\pi}{4}$$

For $n = -3$, $A + B = \frac{-11\pi}{4}$

101. The number of solution of the equation $\tan 3x - \tan 2x - \tan 3x \cdot \tan 2x = 1$ in $[0, 2\pi]$ is

(1) 1

(2) Zero

(3) 3

(4) 2

Sol. Answer (2)

$$\tan 3x - \tan 2x = 1 + \tan 3x \tan 2x$$

$$\Rightarrow \frac{\tan 3x - \tan 2x}{1 + \tan 3x \tan 2x} = 1$$

$$\Rightarrow \tan x = 1$$

$$\Rightarrow x = n\pi + \frac{\pi}{4}$$

$$x = \frac{\pi}{4}, \frac{5\pi}{4}, \dots$$

But for each of values $\tan 2x$, $\tan 3x$ will not be defined. Therefore no solution exists.

Trick : Always check the values by back substitution in case of $\tan x$.

102. If $x \in (0, 1)$, then greatest root of the equation $\sin 2\pi x = \sqrt{2} \cos \pi x$ is

(1) $\frac{1}{4}$

(2) $\frac{1}{2}$

(3) $\frac{3}{4}$

(4) $\frac{1}{3}$

Sol. Answer (3)

$$x \in (0, 1)$$

$$\sin 2\pi x = \sqrt{2} \cos \pi x$$

$$\Rightarrow 2 \sin \pi x \cos \pi x = \sqrt{2} \cos \pi x$$

$$\Rightarrow \sqrt{2} \cos \pi x [\sqrt{2} \sin \pi x - 1] = 0$$

$$\Rightarrow \cos \pi x = 0 \quad \text{or} \quad \sin \pi x = \frac{1}{\sqrt{2}}$$

$$\pi x = (2n+1)\frac{\pi}{2}$$

$$\pi x = n\pi + (-1)^n \frac{\pi}{4}$$

$$x = \frac{(2n+1)}{2}$$

$$x = n + \frac{(-1)^n}{4}$$

$$x = \frac{1}{2}, \frac{5}{2}, \dots$$

$$x = \frac{3}{4}, \frac{1}{4}, \frac{9}{4}, \dots$$

Greatest root in $(0, 1)$ interval is $\frac{3}{4}$.

103. The number of solutions of equation $3\cos 2\theta + 5\cos \theta = 1$ in $[0, 2\pi]$ is

(1) 8

(2) 6

(3) 4

(4) 2

Sol. Answer (4)

$$3\cos 2\theta + 5\cos \theta = 1$$

$$\Rightarrow 3(2\cos^2 \theta - 1) + 5\cos \theta = 1$$

$$\Rightarrow 6\cos^2 \theta + 5\cos \theta - 4 = 0$$

$$\Rightarrow 6\cos^2 \theta + 8\cos \theta - 3\cos \theta - 4 = 0$$

$$\Rightarrow 2\cos \theta (3\cos \theta + 4) - 1(3\cos \theta + 4) = 0$$

$$\Rightarrow (2\cos \theta - 1)(3\cos \theta + 4) = 0$$

$$\Rightarrow \cos \theta = \frac{1}{2} \quad \text{or} \quad \cos \theta = -\frac{4}{3}$$

In $[0, 2\pi]$, two

Not possible solutions exist

104. The general solution of the equations $\tan \theta = -1$ and $\cos \theta = \frac{1}{\sqrt{2}}$ is ($n \in I$)

(1) $n\pi + \frac{7\pi}{4}$

(2) $n\pi + (-1)^n \frac{7\pi}{4}$

(3) $2n\pi + \frac{7\pi}{4}$

(4) $2n\pi + \frac{5\pi}{4}$

Sol. Answer (3)

$$\tan \theta = -1, \cos \theta = \frac{1}{\sqrt{2}}$$

In $[0, 2\pi]$ interval, $\tan \theta = -1$

$$\Rightarrow \theta = \frac{3\pi}{4}, \frac{7\pi}{4}$$

$$\text{In } [0, 2\pi] \text{ interval, } \cos \theta = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \theta = \frac{\pi}{4}, \frac{7\pi}{4}$$

$$\text{Common } \theta = \frac{7\pi}{4}$$

$$\therefore \text{ General solution : } 2n\pi + \frac{7\pi}{4}$$

105. The general solution of $\sec\theta + \tan\theta = \sqrt{3}$ is

(1) $\theta = 2n\pi + \frac{\pi}{6}, 2n\pi - \frac{\pi}{2}, n \in I$

(2) $\theta = 2n\pi + \frac{\pi}{6}, n \in I$

(3) $2n\pi - \frac{\pi}{2}; n \in I$

(4) $2n\pi - \frac{\pi}{2}; n \neq 2k+1, n, k \in I$

Sol. Answer (2)

$$\sec\theta + \tan\theta = \sqrt{3}$$

$$\text{We know that } \sec^2\theta - \tan^2\theta = 1$$

$$(\sec\theta - \tan\theta)(\sec\theta + \tan\theta) = 1$$

$$\sec\theta - \tan\theta = \frac{1}{\sqrt{3}}$$

$$\sec\theta + \tan\theta = \sqrt{3}$$

$$2\sec\theta = \sqrt{3} + \frac{1}{\sqrt{3}} = \frac{4}{\sqrt{3}}$$

$$\sec\theta = \frac{2}{\sqrt{3}}$$

$$\Rightarrow \cos\theta = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \theta = 2n\pi \pm \frac{\pi}{6}$$

106. The number of the solutions of the equation $3\sin x + 4\cos x - x^2 - 16 = 0$ is

(1) 3

(2) 2

(3) 1

(4) 0

Sol. Answer (4)

$$3\sin x + 4\cos x = x^2 + 16$$

$$\text{Range of LHS} = [-5, 5]$$

$$\text{Range of RHS} = [16, \infty)$$

$\therefore$ No solution exists.

107. The solution set of $\sin^4 x - \tan^8 x = 1$ is given by

(1) $x = 2n\pi + \frac{\pi}{8}, n \in I$

(2) $x = n\pi + \frac{\pi}{12}, n \in I$

(3) $x = 2n\pi + \frac{\pi}{24}, n \in I$

(4) None of these

Sol. Answer (4)

$$\sin^4 x - \tan^8 x = 1$$

$$\Rightarrow \sin^4 x = 1 + \tan^8 x$$

$$\text{Range of LHS} : [0, 1]$$

$$\text{Range of RHS} : [1, \infty]$$

Only solution exists when LHS and RHS are both equal to 1.

$$\sin^4 x = 1, 1 + \tan^8 x = 1$$

$$\sin^2 x = 1, \tan^8 x = 0$$

$$\Rightarrow x = n\pi + \frac{\pi}{2} \text{ but at these value } \tan^8 x \neq 0.$$

108. If $\theta \in (0, 2\pi)$ and $2\sin^2\theta - 5\sin\theta + 2 > 0$, then the range of θ is

- (1) $\left(0, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, 2\pi\right)$ (2) $\left(0, \frac{\pi}{6}\right) \cup (\pi, 2\pi)$ (3) $\left(0, \frac{\pi}{6}\right) \cap (\pi, 2\pi)$ (4) ϕ

Sol. Answer (1)

$$\theta \in (0, 2\pi)$$

$$2\sin^2\theta - 5\sin\theta + 2 > 0$$

$$\Rightarrow 2\sin^2\theta - 4\sin\theta - \sin\theta + 2 > 0$$

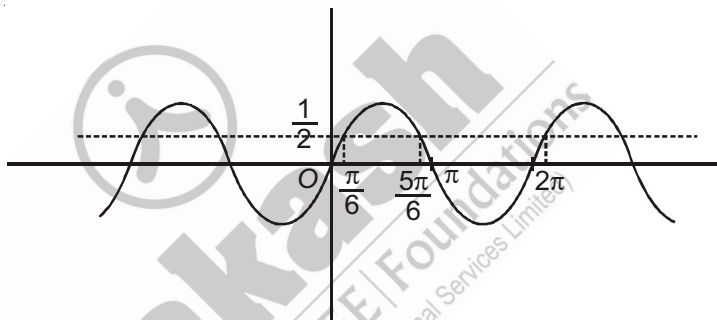
$$\Rightarrow 2\sin\theta(\sin\theta - 2) - 1(\sin\theta - 2) > 0$$

$$(2\sin\theta - 1)(\sin\theta - 2) > 0$$

$$\therefore 2\sin\theta - 1 < 0$$

$$\Rightarrow \sin\theta < \frac{1}{2}$$

$$\theta \in \left(0, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, 2\pi\right)$$



109. The general solution of the equation $\tan 2\theta \tan 3\theta = 1$ is

(1) $\theta = (4n + 1)\frac{\pi}{10}, n \neq 4k - 2, k \in I$ (2) $\theta = (2n + 3)\frac{\pi}{10}, n \neq 5k + 2, k \in I$

(3) $\theta = (4n + 3)\frac{\pi}{10}, n \neq 5k + 2, k \in I$ (4) $\theta = (2n + 1)\frac{\pi}{10}, n \neq 5k - 3, k \in I$

Sol. Answer (4)

110. If $m, n \in N (n > m)$, then number of solutions of the equation $n|\sin x| = m|\sin x|$ in $[0, 2\pi]$ is

- (1) m (2) n (3) mn (4) 3

Sol. Answer (4)

$$n|\sin x| = m|\sin x|$$

$$\Rightarrow (n - m)|\sin x| = 0$$

$$\text{Since } n > m \therefore |\sin x| = 0$$

$$\Rightarrow x = n\pi$$

$$\text{In } [0, 2\pi], x = 0, \pi, 2\pi$$

111. If $\sin A = \sin B$ and $\cos A = \cos B$, then

- (1) $A = B + n\pi, n \in I$ (2) $A = B - n\pi, n \in I$ (3) $A = 2n\pi + B, n \in I$ (4) $A = n\pi - B, n \in I$

Sol. Answer (3)

$$\sin A = \sin B, \cos A = \cos B$$

$$A = m\pi + (-1)^n B, A = 2k\pi \pm B$$

Clearly common solution is

$$A = 2n\pi + B, n \in I$$

112. The number of values of $x \in [-2\pi, 2\pi]$ satisfying $\tan x + \cot x = 2\operatorname{cosec} x$ is

- (1) 2 (2) 4 (3) 6 (4) 8

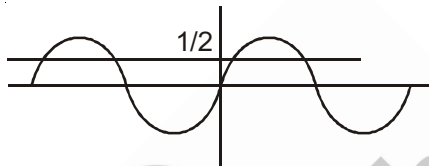
Sol. Answer (2)

$$\tan x + \cot x = 2\operatorname{cosec} x$$

$$\frac{1}{\sin x \cos x} = \frac{2}{\sin x}$$

$$\Rightarrow \cos x = \frac{1}{2} \quad (\because \sin x \neq 0)$$

$\Rightarrow$ Number of solutions in $[-2\pi, 2\pi]$ equals 4.



113. The solution set of the equation $\tan ax = \tan bx$ where $(a \neq b)$ constitutes

- (1) An A.P. with common difference $\frac{\pi}{a-b}$ or $\frac{\pi}{a+b}$ (2) An H.P. with common difference $\frac{\pi}{|a-b|}$
 (3) An A.P. with common difference $\frac{\pi}{a+b}$ (4) An A.P. with common difference $\frac{\pi}{a-b}$ or $\frac{\pi}{b-a}$

Sol. Answer (4)

$$\tan ax = \tan bx$$

$$\Rightarrow ax = n\pi + bx$$

$$\Rightarrow x = \frac{n\pi}{a-b}, n \in Z$$

$$\Rightarrow \text{A.P. with common difference} = \frac{\pi}{(a-b)} \text{ or } \frac{\pi}{(b-a)}$$

114. The general solution of the equation $\frac{\tan x}{\tan 2x} + \frac{\tan 2x}{\tan x} + 2 = 0$ is

- (1) $n\pi$ (2) $\frac{n\pi}{3}$ (3) $(2n-1)\frac{\pi}{3}, n \in I$ (4) $(3n \pm 1)\frac{\pi}{3}, n \in I$

Sol. Answer (4)

$$\frac{\tan x}{\tan 2x} + \frac{\tan 2x}{\tan x} + 2 = 0$$

$$\frac{\tan x(1 - \tan^2 x)}{2 \tan x} + \frac{2 \tan x}{(1 - \tan^2 x) \tan x} + 2 = 0$$

$$\text{If } \tan x \neq 0, \frac{1 - \tan^2 x}{2} + \frac{2}{1 - \tan^2 x} + 2 = 0$$

$$\Rightarrow \frac{(1 - \tan^2 x)^2 + 4}{2(1 - \tan^2 x)} + 2 = 0$$

$$\Rightarrow \frac{1 + \tan^4 x - 2 \tan^2 x + 4 + 4 - 4 \tan^2 x}{2(1 - \tan^2 x)} = 0$$

$$\Rightarrow \frac{\tan^4 x - 6 \tan^2 x + 9}{2(1 - \tan^2 x)} = 0$$

$$\Rightarrow \frac{(\tan^2 x - 3)^2}{2(1 - \tan^2 x)} = 0 \Rightarrow \tan^2 x = 3$$

$$\Rightarrow x = n\pi \pm \frac{\pi}{3}$$

115. The solution of the equation

 $\cos^2 \theta - 2 \cos \theta = 4 \sin \theta - \sin 2\theta$ where $\theta \in [0, \pi]$ is

$$(1) \pi - \alpha, \tan \alpha = \frac{1}{2}$$

$$(2) \pi - \alpha, \cot \alpha = -\frac{1}{2}$$

$$(3) \pi - \alpha, \tan \alpha = 2$$

$$(4) \pi + \alpha, \cot \alpha = 2$$

Sol. Answer (1)

$$\cos^2 \theta - 2 \cos \theta = 4 \sin \theta - \sin 2\theta$$

$$\Rightarrow \cos^2 \theta - 2 \cos \theta = 4 \sin \theta - 2 \sin \theta \cos \theta$$

$$\Rightarrow \cos \theta (\cos \theta - 2) + 2 \sin \theta (\cos \theta - 2) = 0$$

$$\Rightarrow (\cos \theta + 2 \sin \theta) (\cos \theta - 2) = 0$$

$$\Rightarrow \cos \theta = 2 \quad \text{or} \quad \cos \theta = -2 \sin \theta$$

Not possible

$$\tan \theta = \frac{-1}{2}$$

$$\theta = n\pi + \alpha$$

$$= n\pi + \tan^{-1} \left(\frac{-1}{2} \right)$$

$$= n\pi - \tan^{-1} \frac{1}{2}$$

$$\text{For } n = 1, \boxed{\theta = \pi - \tan^{-1} \frac{1}{2}}$$

116. The general solution of $\sin^2 \theta \sec \theta + \sqrt{3} \tan \theta = 0$ is

$$(1) \theta = n\pi + (-1)^{n+1} \frac{\pi}{3}, n \in I$$

$$(2) \theta = n\pi, n \in I$$

$$(3) \theta = n\pi + (-1)^{n+1} \frac{\pi}{3}, n \in I$$

$$(4) \theta = \frac{n\pi}{2}, n \in I$$

Sol. Answer (2)

$$\sin^2 \theta \sec \theta + \sqrt{3} \tan \theta = 0$$

$$\frac{\sin^2 \theta + \sqrt{3} \sin \theta}{\cos \theta} = 0$$

$$\Rightarrow \sin \theta (\sin \theta + \sqrt{3}) = 0, \cos \theta \neq 0$$

$$\Rightarrow \theta = n, n \in I$$

117. If $\sin x + \cos x = \sqrt{y + \frac{1}{y}}$, $x \in (0, \pi)$, then

$$(1) x = \frac{\pi}{4}, y = 1$$

$$(2) y = 0$$

$$(3) y = 2$$

$$(4) x = \frac{3\pi}{4}$$

Sol. Answer (1)

$$\sin x + \cos x = \sqrt{y + \frac{1}{y}} \quad x \in (0, \pi)$$

Comparing range on both sides,

$$\text{LHS} = [-\sqrt{2}, \sqrt{2}]$$

$$\text{RHS} = [\sqrt{2}, \infty)$$

Only solution exists $\sin x + \cos x = \sqrt{2}$

$$\sin\left(x + \frac{\pi}{4}\right) = 1$$

$$\Rightarrow x + \frac{\pi}{4} = n\pi + (-1)^n \frac{\pi}{2}$$

$$\Rightarrow x = \frac{\pi}{4}$$

$$\text{Also, } \sqrt{y + \frac{1}{y}} = \sqrt{2} \Rightarrow y = 1$$

118. The number of solutions of the equation $\cos(\pi\sqrt{x-4})\cos(\pi\sqrt{x})=1$ is

- (1) Zero (2) 1 (3) 2 (4) Infinite

Sol. Answer (2)

$$\cos\pi\sqrt{x-4}\cos\pi\sqrt{x}=1$$

Only possible when

$$\cos\pi\sqrt{x-4}=1 \quad \text{and} \quad \cos\pi\sqrt{x}=1$$

$$\pi\sqrt{x-4}=2n\pi \pm \frac{\pi}{2} \quad \pi\sqrt{x}=2m\pi \pm \frac{\pi}{2}$$

$$\sqrt{x-4}=\frac{(2n+1)}{2} \quad \sqrt{x}=\frac{(2m+1)}{2}$$

$$\sqrt{x-4}=\frac{1}{2}, \frac{5}{2}, \frac{7}{2}, \frac{9}{2} \quad \sqrt{x}=\frac{1}{2}, \frac{5}{2}, \frac{7}{2}, \dots$$

$\therefore$ No solution exists.

119. The number of solutions of $\sum_{r=1}^6 \cos(rx)=6$ in $(0, 2\pi]$ is

- (1) Zero (2) 1 (3) 6 (4) 12

Sol. Answer (2)

$$\cos x + \cos 2x + \dots + \cos 6x = 6$$

$$\Rightarrow \cos x = 1, \cos 2x = 1, \cos 3x = 1, \dots, \cos 6x = 1$$

$\cos x = 1$	$\cos 2x = 1$	$\cos 3x = 1$	$\dots$	$\cos 6x = 1$
$x = 0, 2\pi$	$x = 0, \pi, 2\pi$	$x = \frac{2\pi}{3}, \frac{4\pi}{3}, 2\pi$		$\Rightarrow x = 2\pi$

$\therefore x = 2\pi$ satisfies all conditions.

120. The number of values of x for which $\sin 2x + \cos 4x = 2$ is

- (1) Zero (2) 1 (3) 2 (4) Infinite

Sol. Answer (1)

$$\sin 2x + \cos 4x = 2$$

$$\Rightarrow \sin 2x = 1, \quad \cos 4x = 1$$

$$\Rightarrow 2x = n\pi + (-1)^n \frac{\pi}{2} \quad 4x = 2n\pi$$

$$\Rightarrow x = \frac{n\pi}{2} + (-1)^n \frac{\pi}{4} \quad x = \frac{n\pi}{2}$$

$$\Rightarrow x = \frac{\pi}{4}, \frac{5\pi}{4}, \dots \quad x = \frac{\pi}{2}, \pi, \frac{3\pi}{2}, \dots$$

$\therefore$ No solution exists. Except $x = 4$ one solution.

121. If the equation $\cos x + 3\cos(2Kx) = 4$ has exactly one solution, then

- (1) K is a rational number of the form $\frac{P}{P+1}$, $P \neq -1$
- (2) K is irrational number whose rational approximation does not exceed 2
- (3) K is irrational number
- (4) K is a rational number of the form $\frac{P}{P-1}$, $P \neq 1$

Sol. Answer (3)

$$\cos x + 3\cos 2Kx = 4$$

$$\Rightarrow \cos x = 1, \quad \cos 2Kx = 1$$

$$\Rightarrow x = 2n\pi \quad 2Kx = 2m\pi$$

$$x = \frac{m\pi}{K}$$

$$\text{Comparing } 2n\pi = \frac{m\pi}{K} \Rightarrow K = \frac{m}{2n}$$

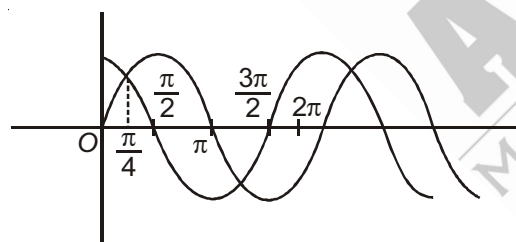
For infinite $m, n \in \text{integer}$, K will be rational but equation has only 1 solution, which is only possible when $K \in \text{irrational}$.

122. The solution of the inequality $\log_{1/2} \sin x > \log_{1/2} \cos x$ is

- (1) $x \in \left(0, \frac{\pi}{2}\right)$
- (2) $x \in \left(0, \frac{\pi}{8}\right)$
- (3) $x \in \left(0, \frac{\pi}{4}\right)$
- (4) $x \in \left[0, \frac{\pi}{4}\right]$

Sol. Answer (3)

$$\log_{1/2} \sin x > \log_{1/2} \cos x \Rightarrow \sin x < \cos x$$



From graph $\sin x < \cos x$ in $\left(0, \frac{\pi}{4}\right)$

123. The solution set of the equation $4\sin^4 x + \cos^4 x = 1$ is

- (1) $x = n\pi + \frac{\pi}{6}$
- (2) $x = n\pi \pm \theta$ where $\cos^2 \theta = \frac{3}{5}$ and $x = n\pi$, $n \in I$
- (3) $x = (2n+1)\frac{\pi}{2}$
- (4) $x = 2n\pi \pm \frac{\pi}{3}$

Sol. Answer (2)

$$4\sin^4 x + \cos^4 x = 1$$

$$\Rightarrow 4\sin^4 x + (1 - \sin^2 x)^2 = 1$$

$$\Rightarrow 4\sin^4 x + 1 + \sin^4 x - 2\sin^2 x = 1$$

$$\Rightarrow 5\sin^4 x - 2\sin^2 x = 0$$

$$\Rightarrow \sin^2 x(5\sin^2 x - 2) = 0$$

$$\Rightarrow \sin^2 x = 0, \sin^2 x = \frac{2}{5}$$

$$\Rightarrow x = n\pi \text{ or } x = n\pi \pm \theta, \text{ where } \sin^2 \theta = \frac{2}{5}, \cos^2 \theta = \frac{3}{5}, n \in I$$

124. The number of values of $\theta \in [0, 2\pi]$ satisfying $r \sin \theta = \sqrt{3}$ and $r + 4 \sin \theta = 2(\sqrt{3} + 1)$ is

(1) 4

(2) 5

(3) 6

(4) 7

Sol. Answer (1)

$$r \sin \theta = \sqrt{3}$$

$$r + 4 \sin \theta = 2(\sqrt{3} + 1)$$

$$r + \frac{4\sqrt{3}}{r} = 2\sqrt{3} + 2$$

$$\Rightarrow r^2 - 2(\sqrt{3} + 1)r + 4\sqrt{3} = 0$$

$$r(r - 2\sqrt{3}) - 2(r - 2\sqrt{3}) = 0$$

$$r = 2 \text{ or } r = 2\sqrt{3}$$

$$\sin \theta = \frac{\sqrt{3}}{2}, \sin \theta = \frac{1}{2}$$

$\therefore$ 4 solutions exist in $[0, 2\pi]$

125. The arithmetic mean of the roots of the equation $4\cos^3 x - 4\cos^2 x - \cos(315\pi + x) = 1$ in the interval $(0, 315)$ is equal to

(1) 50π

(2) 51π

(3) 100π

(4) 315π

Sol. Answer (2)

$$4\cos^3 x - 4\cos^2 x + \cos x = 1$$

$$\Rightarrow 4\cos^3 x - 4\cos^2 x + \cos x - 1 = 0$$

$$\Rightarrow (4\cos^2 x + 1)(\cos x - 1) = 0$$

$$\Rightarrow \cos x = 1 \Rightarrow x = 2n\pi, n \in I$$

$$x = 0, 2\pi, 4\pi, \dots, 100\pi$$

$$AM = \frac{2\pi + 4\pi + \dots + 100\pi}{50} = \frac{50}{2 \times 50} [102\pi] = 51\pi$$

126. How many solutions does the equation $\sec x - 1 = (\sqrt{2} - 1)\tan x$ have in the interval $(0, 6\pi]$?

- (1) 6 (2) 5 (3) 10 (4) 9

Sol. Answer (1)

$$\sec x - 1 = (\sqrt{2} - 1)\tan x$$

$$\frac{1 - \cos x}{\cos x} = \frac{(\sqrt{2} - 1)\sin x}{\cos x}$$

$$\Rightarrow (\sqrt{2} - 1)\sin x + \cos x = 1$$

Using auxiliary form, 6 solutions exist.

127. The solution of the equation $\frac{\sin 3x}{2\cos 2x + 1} = \frac{1}{2}$ is

- (1) $x = 3n\pi + \frac{\pi}{6}$ (2) $x = n\pi - \frac{\pi}{6}$ (3) $x = n\pi + (-1)^n \frac{\pi}{3}$ (4) $x = n\pi + (-1)^n \frac{\pi}{6}$

Sol. Answer (4)

$$\frac{\sin 3x}{2\cos 2x + 1} = \frac{1}{2}$$

$$\Rightarrow 2(3\sin x - 4\sin^3 x) = 2(1 - 2\sin^2 x) + 1$$

$$\Rightarrow 6\sin x - 8\sin^3 x = 2 - 4\sin^2 x + 1$$

$$\Rightarrow 8\sin^3 x - 4\sin^2 x - 6\sin x + 3 = 0$$

$$\Rightarrow 4\sin^2 x(2\sin x - 1) - 3(2\sin x - 1) = 0$$

$$\sin x = \frac{1}{2} \text{ or } \sin^2 x = \frac{3}{4}$$

$$\boxed{x = n\pi + (-1)^n \frac{\pi}{6}}$$

128. The number of values of x lying in $[-\pi, \pi]$ and satisfying $2\sin^2\theta = \cos 2\theta$ and $\sin 2\theta + 2\cos 2\theta - \cos\theta - 1 = 0$ is

- (1) 0 (2) 2 (3) 4 (4) 6

Sol. Answer (2)

Answer (2)

$$2\sin^2\theta = \cos 2\theta$$

$$\Rightarrow 2\sin^2\theta = 1 - 2\sin^2\theta$$

$$\Rightarrow \sin\theta = \pm \frac{1}{2} \quad \dots(i)$$

$$\sin 2\theta + 2\cos 2\theta - \cos\theta - 1 = 0$$

$$2\sin\theta \cos\theta - \cos\theta + 2(1 - 2\sin^2\theta) - 1 = 0$$

$$\cos\theta(2\sin\theta - 1) + (1 - 4\sin^2\theta) = 0$$

$$(2\sin\theta - 1) [\cos\theta - 2\sin\theta - 1] = 0$$

$$\sin\theta = \frac{1}{2} \text{ or } \cos\theta - 2\sin\theta = 1 \quad \dots(ii)$$

From (i) & (ii)

$$\text{Common solution is obtained at } \sin\theta = \frac{1}{2}$$

$\therefore$ Number of solutions in $[-\pi, \pi]$ is 2

129. The number of solutions of the equation $\sin x \cdot \cos x (\cos x - \sin x)^2 \cdot (\sin x + \cos x) = \lambda$, where $\lambda > \frac{1}{2\sqrt{2}}$ in the interval $[0, 4\pi]$, is

(1) 0

(2) 2

(3) 4

(4) 8

Sol. Answer (1)

$$\sin 2x \cdot (\cos x - \sin x) \cdot \cos 2x = 2\lambda$$

$$\sin 4x \cdot \cos\left(x + \frac{\pi}{4}\right) = \frac{4}{\sqrt{2}}\lambda = 2\sqrt{2}\lambda$$

L.H.S. can never exceed one while $\text{RHS} = 2\sqrt{2}\lambda > 1$

Since $\lambda > \frac{1}{2\sqrt{2}}$

$\therefore$ No solution

130. The number of solutions of x in the interval $[-\pi, \pi]$ of the equation $(1 + \cot 267^\circ)(1 + \tan 222^\circ) = \sec^2 x + \cos^2 x$ is

(1) 1

(2) 2

(3) 3

(4) 4

Sol. Answer (3)

$$\text{L.H.S.} = (1 + \cot 267^\circ)(1 + \tan 222^\circ)$$

$$= (1 + \tan 3^\circ)(1 + \tan 42^\circ)$$

$$\therefore 3 + 42 = 45^\circ$$

$$\Rightarrow \frac{\tan 3^\circ + \tan 42^\circ}{1 - \tan 3^\circ \tan 42^\circ} = \tan 45^\circ$$

$$\Rightarrow \tan 3^\circ + \tan 42^\circ + \tan 3^\circ \tan 42^\circ = 1$$

$$\Rightarrow (\tan 3^\circ + 1)(\tan 42^\circ + 1) = 2$$

$$\therefore \text{L.H.S.} = 2$$

$\therefore$ Given equation is

$$\sec^2 x + \cos^2 x = 2$$

$\therefore \sec^2 x + \cos^2 x \geq 2$, equality holds of

$$\sec^2 x = \cos^2 x \Rightarrow x = \pi, 0, \pi \text{ (As the given interval)}$$

$\therefore$ 3 solutions

131. The general solution of the equation $\tan(x + 20^\circ) \tan(x - 40^\circ) = \tan(x - 20^\circ) \tan(x + 40^\circ)$ is

- (1) $x = n\pi, n \in \mathbb{Z}$ (2) $x = n\pi + \frac{\pi}{4}, n \in \mathbb{Z}$ (3) $x = \frac{n\pi}{2}, n \in \mathbb{Z}$ (4) $x = 2n\pi + \frac{\pi}{4}, n \in \mathbb{Z}$

Sol. Answer (3)

$$\frac{\tan(x+20^\circ)}{\tan(x-20^\circ)} = \frac{\tan(x+40^\circ)}{\tan(x-40^\circ)}$$

$$\Rightarrow \frac{\sin(x+20^\circ)\cos(x-20^\circ)}{\sin(x-20^\circ)\cos(x+20^\circ)} = \frac{\sin(x+40^\circ)\cos(x-40^\circ)}{\cos(x+40^\circ)\sin(x-40^\circ)}$$

Applying components-dividends

$$\Rightarrow \frac{\sin 2x}{\sin 40^\circ} = \frac{\sin(2x)}{80^\circ}$$

$$\Rightarrow \sin 2x = 0$$

$$\Rightarrow 2x = n\pi$$

$$\Rightarrow x = \frac{n\pi}{2}, n \in \mathbb{Z}$$

132. Let $x \in [0, 2\pi]$. The curve $y = \sec x \tan x + 2 \tan x - \sec x$ and the line $y = 2$ intersect in

- (1) No point (2) 2 points (3) 3 points (4) 4 points

Sol. Answer (4)

$$\sec x \tan x + 2 \tan x - \sec x - 2 = 0$$

$$\Rightarrow (\sec x + 2)(\tan x - 1) = 0$$

$$\Rightarrow \tan x = 1, \sec x = -2$$

$$\Rightarrow x = \frac{\pi}{4}, \frac{5\pi}{4}, x = \frac{2\pi}{3}, \frac{4\pi}{3} \text{ (in } [0, 2\pi])$$

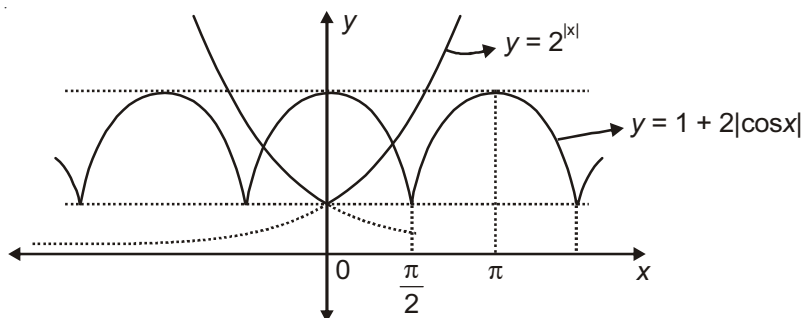
The curve and the line intersect is 4 points

133. The number of solutions of the equation $2^{|x|} = 1 + 2|\cos x|$ is

- (1) 0 (2) 2 (3) 4 (4) Infinite

Sol. Answer (2)

Clearly from graph these are two solution of the given equation



134. If the equation $1 + \sin^2 x \theta = \cos \theta$ has a non-zero solution in θ , then x must be

- (1) An integer (2) A rational number
(3) An irrational number (4) None of these

Sol. Answer (2)

$$\text{Given } 1 + \sin^2 x \theta = \cos \theta$$

$$\text{OR } \cos \theta = 1 + \sin^2 x \theta$$

It is possible iff

$$\sin^2 x \theta = 0 \text{ and } \cos \theta = 1$$

$$x \theta = n\pi \text{ and } \theta = 2m\pi \text{ where } n, m \in I$$

$$\therefore x = \frac{n\pi}{2m\pi}$$

$$x = \frac{n}{2m}$$

135. The smallest positive values of x (in degrees) such that $\tan(x + 100^\circ) = \tan(x + 50^\circ)\tan x \tan(x - 50^\circ)$ is equal to zero

- (1) 60° (2) 30° (3) 40° (4) 50°

Sol. Answer (2)

$$\frac{\tan(x + 100^\circ)}{\tan(x - 50^\circ)} = \tan(x + 50^\circ) \cdot \tan x$$

$$\Rightarrow \frac{2 \sin(x + 100^\circ) \cos(x - 50^\circ)}{2 \sin(x - 50^\circ) \cos(x + 100^\circ)} = \frac{2 \sin(x + 50^\circ) \sin x}{2 \cos(x + 50^\circ) \cos x}$$

$$\Rightarrow \frac{\sin(2x + 50^\circ) + \sin 150^\circ}{\sin(2x + 50^\circ) - \sin 150^\circ} = \frac{\cos 50^\circ - \cos(2x + 50^\circ)}{\cos(2x + 50^\circ) + \cos 50^\circ}$$

$$\Rightarrow \frac{2 \sin(2x + 50^\circ)}{1} = \frac{2 \cos 50^\circ}{-2 \cos(2x + 50^\circ)}$$

$$\Rightarrow 2 \sin(2x + 50^\circ) \cos(2x + 50^\circ) = -\cos 50^\circ$$

$$\Rightarrow \sin(4x + 100^\circ) = -\sin 40^\circ = \sin 220^\circ, \sin 320^\circ$$

$$\Rightarrow 4x + 100 = 220, 320$$

$$\Rightarrow 4x = 120, 220$$

$$\Rightarrow x = 30, 55$$

[Height and Distance]

136. An aeroplane flying at a height of 300 m above the ground passes vertically above another plane at an instant when the angles of elevation of the two planes from the same point on the ground are 60° and 45° respectively. Then the height of the lower plane from the ground is

- (1) $100\sqrt{3}$ m (2) $\frac{100}{\sqrt{3}}$ m (3) 50 m (4) $150(\sqrt{3} + 1)$ m

Sol. Answer (1)Given $BP_2 = 300$ mIn $\triangle ABP_2$

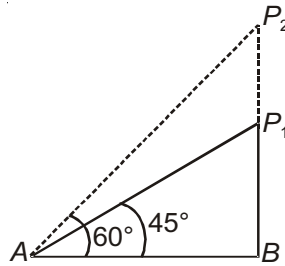
$$\frac{BP_2}{AB} = \tan 60^\circ = \sqrt{3}$$

$$\therefore AB = \frac{300}{\sqrt{3}} = 100\sqrt{3} \text{ m}$$

Also in $\triangle ABP_1$

$$\frac{P_1B}{AB} = \tan 45^\circ = 1$$

$$\therefore P_1B = 100\sqrt{3} \text{ m}$$



137. A man from the top of a 100 m high tower sees a car moving towards the tower at an angle of depression 30° . After sometime, the angle of depression becomes 60° . The distance travelled by the car during this time is

(1) $100\sqrt{3}$ m

(2) $\frac{200}{\sqrt{3}}$ m

(3) $200\sqrt{3}$ m

(4) $\frac{100}{\sqrt{3}}$ m

Sol. Answer (2)In $\triangle POA$

$$\frac{PO}{OA} = \tan 60^\circ = \sqrt{3}$$

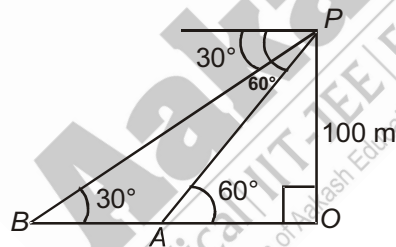
$$OA = \frac{100}{\sqrt{3}} \text{ m}$$

Also, in $\triangle POB$

$$\frac{PO}{OB} = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\therefore OB = PO\sqrt{3} = 100\sqrt{3}$$

$$\therefore AB = OB - OA = 100\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right) = \frac{200}{\sqrt{3}}$$



138. ABC is a triangular park with $AB = AC = 100$ m. A clock tower is situated at the midpoint of BC . The angles of elevation of top of the tower at A and B are $\cot^{-1}(3.2)$ and $\operatorname{cosec}^{-1}(2.6)$ respectively. The height of tower is

(1) $\frac{25}{2}$ m

(2) 25 m

(3) 50 m

(4) None of these

Sol. Answer (2)Let $OP = h$ be the tower of height given that $AB = AC = 100$ m $BO = OC$

Clearly

$$\text{So } OA = h \cot(\angle PAO)$$

$$= 3.2 h$$

$$OB = h \cot(\angle PBO)$$

Hence in right angle triangle AOB

$$AO^2 + OB^2 = AB^2$$

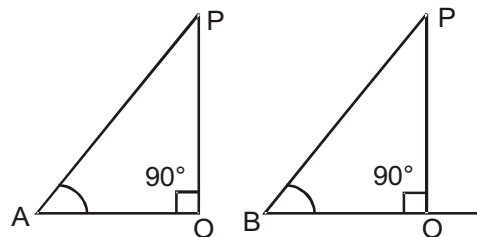
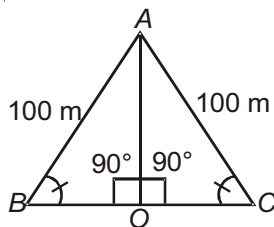
$$(3.2h)^2 + [h \cot(\angle PBO)]^2 = 100^2$$

$$h^2[3.2^2 + \operatorname{cosec}^2(\angle PBO) - 1] = 100^2$$

$$h^2[10.24 + 6.76 - 1] = 100^2$$

$$\therefore h^2[16] = 100^2$$

$$h = \frac{100}{4} = 25 \text{ m}$$



139. ABCD is a square plot. The angle of elevation of the top of a pole standing on D from A or C is 30° and that from B is θ , then $\tan \theta$ is equal to

(1) $\sqrt{6}$

(2) $\frac{1}{\sqrt{6}}$

(3) $\frac{\sqrt{3}}{\sqrt{2}}$

(4) $\frac{\sqrt{2}}{\sqrt{3}}$

Sol. Answer (2)

Let the side of square be x , then $BD = x\sqrt{2}$. Let the height of pole is h .

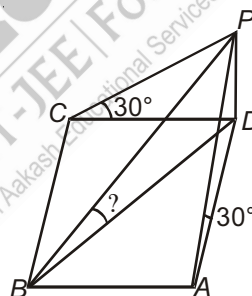
Then,

$$\frac{PD}{DA} = \frac{h}{x} = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\Rightarrow h = \frac{x}{\sqrt{3}} \quad \dots(i)$$

$$\frac{PD}{DB} = \frac{h}{x\sqrt{2}} = \tan \theta \quad \dots(ii)$$

$$\therefore \tan \theta = \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{3}} = \frac{1}{\sqrt{6}}$$



140. The upper $\frac{3}{4}$ th portion of a vertical pole subtends an angle $\tan^{-1}\left(\frac{3}{5}\right)$ at a point in the horizontal plane through its foot and at a distance 40 m from the foot. A possible height of the vertical pole is

(1) 20 m

(2) 40 m

(3) 60 m

(4) 80 m

Sol. Answer (2)

As in adjacent diagram

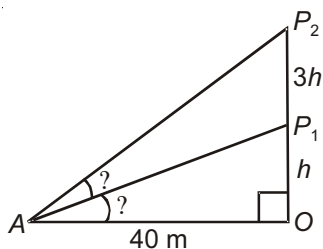
$$\tan \theta = \frac{3}{5}$$

Let angle $P_1AO = \alpha$

In $\triangle P_1AO$

$$\frac{h}{40} = \tan \alpha$$

$$h = 40 \tan \alpha$$



$$\text{Also } \tan(\theta + \alpha) = \frac{4h}{40}$$

$$\Rightarrow \frac{\tan \theta + \tan \alpha}{1 - \tan \theta \tan \alpha} = \frac{4h}{40} = \frac{h}{10}$$

$$\Rightarrow \frac{\frac{3}{5} + \frac{h}{40}}{1 - \frac{3h}{200}} = \frac{h}{10} \Rightarrow \frac{120 + 5h}{200 - 3h} = \frac{h}{10}$$

$$\Rightarrow 1200 + 50h = 200h - 3h^2$$

$$\Rightarrow 3h^2 - 150h + 1200 = 0$$

$$\Rightarrow h^2 - 50h + 400 = 0$$

$$\Rightarrow (h - 40)(h - 10) = 0$$

$$h = 10 \text{ m or } h = 40 \text{ m}$$

Hence height of pole is $4h = 40 \text{ m or } 160 \text{ m}$

[Properties of Triangle]

141. If the lengths of the sides of a triangle are 3, 5 and 7, then the largest angle of the triangle is

(1) $\frac{\pi}{2}$

(2) $\frac{5\pi}{6}$

(3) $\frac{2\pi}{3}$

(4) $\frac{3\pi}{4}$

Sol. Answer (3)

142. In $\triangle ABC$, if $\frac{b+c}{a} = \cot \frac{A}{2}$, then $\angle B =$

(1) 90°

(2) 60°

(3) 45°

(4) 30°

Sol. Answer (1)

$$\text{Given, } \cot \frac{A}{2} = \frac{b+c}{a}$$

$$\Rightarrow \frac{\cot \frac{A}{2}}{\sin \frac{A}{2}} = \frac{\sin B + \sin C}{\sin A}$$

$$\Rightarrow 2 \cos^2 \frac{A}{2} = 2 \cos \frac{A}{2} \cos \frac{B-C}{2}$$

$$\Rightarrow \cos \frac{A}{2} = \cos \frac{B-C}{2}$$

$$\text{i.e., } A = B - C$$

$$B = A + C$$

$$\text{i.e., } \angle B = 90^\circ$$

143. If $a \cos A = b \cos B$, then $\triangle ABC$ is

- (1) Isosceles (2) Right angled
(3) Equilateral (4) Either Right angled or Isosceles

Sol. Answer (4)

144. If $a = 16$, $b = 24$ and $c = 20$, then the value of $\cos\left(\frac{B}{2}\right)$ is

- (1) $\frac{3}{4}$ (2) $\frac{1}{4}$ (3) $\frac{1}{2}$ (4) $\frac{1}{3}$

Sol. Answer (1)

Find $\cos B$ and then $\cos 2\theta = 2\cos^2\theta - 1$

145. If $a = 1 + \sqrt{3}$, $b = 2$ and $c = \sqrt{2}$, then the measure of $\angle B$ is

- (1) 45° (2) 60° (3) 90° (4) 15°

Sol. Answer (1)

146. If $\angle A = 60^\circ$, $b = 2$ and $c = 4$, then the value of a is

- (1) $2\sqrt{2}$ (2) $3\sqrt{2}$ (3) $2\sqrt{3}$ (4) $\sqrt{6}$

Sol. Answer (3)

147. If $b = \sqrt{3}$, $c = 1$ and $\angle A = 30^\circ$, then the measure of $\angle B$ is

- (1) 30° (2) 120° (3) 45° (4) 150°

Sol. Answer (2)

148. If in a triangle ABC , $\frac{\sin A}{\sin C} = \frac{\sin(A-B)}{\sin(B-C)}$, then

- (1) a, b, c are in A.P. (2) a^2, b^2, c^2 are in A.P.
(3) a, b, c are in H.P. (4) a^2, b^2, c^2 are in H.P.

Sol. Answer (2)

Given that

$$\frac{\sin A}{\sin C} = \frac{\sin(A-B)}{\sin(B-C)}$$

$$\Rightarrow \sin A \cdot \sin(B - C) = \sin C \cdot \sin(A - B)$$

$$\Rightarrow \sin(B + C) \cdot \sin(B - C) = \sin(A + B) \cdot \sin(A - B) \quad [\because A + B + C = 180^\circ]$$

$$\Rightarrow \sin^2 B - \sin^2 C = \sin^2 A - \sin^2 B$$

$$\Rightarrow \frac{b^2}{4R^2} - \frac{c^2}{4R^2} = \frac{a^2}{4R^2} - \frac{b^2}{4R^2}$$

$$\Rightarrow 2b^2 = a^2 + c^2$$

$$\Rightarrow a^2, b^2, c^2 \text{ are in A.P.}$$

149. If $\frac{a}{\cos A} = \frac{b}{\cos B} = \frac{c}{\cos C}$, then $\triangle ABC$ is

- (1) Isosceles (2) Equilateral (3) Right angled (4) Scalene

Sol. Answer (2)

Given,

$$\frac{a}{\cos A} = \frac{b}{\cos B} = \frac{c}{\cos C}$$

Using sine rule,

$$\tan A = \tan B = \tan C$$

$$\text{i.e., } A = B = C$$

$\therefore$ Triangle is equilateral.

150. In triangle ABC , if $\tan\left(\frac{A-B}{2}\right) = \frac{3}{11}$ and $\frac{a}{b} = \frac{7}{4}$ then the value of angle C is

- (1) 30° (2) 60° (3) 90° (4) 45°

Sol. Answer (3)

$$\text{Given that } \tan\left(\frac{A-B}{2}\right) = \frac{3}{11} \text{ and } \frac{a}{b} = \frac{7}{4} \Rightarrow \frac{a-b}{a+b} = \frac{3}{11}$$

$$\text{We know that } \frac{A-B}{2} = \frac{a-b}{a+b} \cdot \cot \frac{C}{2}$$

$$\frac{3}{11} = \frac{3}{11} \cdot \cot \frac{C}{2} \Rightarrow \cot \frac{C}{2} = 1 \Rightarrow \boxed{C = 90^\circ}$$

151. In a triangle, the length of two larger sides are 24 and 22 respectively. If the angles are in AP, then the third side is

- (1) $12 + 2\sqrt{13}$ (2) $12 - 3\sqrt{3}$ (3) $2\sqrt{3} + 2$ (4) $2\sqrt{3} - 2$

Sol. Answer (1)

Given that, angles A , B and C are in A.P

$$\Rightarrow B = 60^\circ$$

We know that $\cos B = \frac{a^2 + c^2 - b^2}{2ac}$

$$\Rightarrow \frac{1}{2} = \frac{(24)^2 + c^2 - (22)^2}{2 \times 24 \times c} \Rightarrow c^2 + 92 = 24c$$

$$\Rightarrow c^2 - 24c + 92 = 0$$

$$\Rightarrow c = 12 + 2\sqrt{13}$$

152. In a triangle ABC , $a = 4$, $b = 3$, $\angle A = 60^\circ$, then c is the root of the equation

(1) $c^2 - 3c - 7 = 0$

(2) $c^2 + 3c + 7 = 0$

(3) $c^2 - 3c + 7 = 0$

(4) $c^2 + 3c - 7 = 0$

Sol. Answer (1)

We know that

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\Rightarrow \frac{1}{2} = \frac{9 + c^2 - 16}{2 \times 3 \times c} \Rightarrow c^2 - 3c - 7 = 0$$

153. In triangle ABC , point D lies on BC such that $\angle BAD = 45^\circ$, $\angle DAC = 30^\circ$. If $BD : DC = 2 : 1$, then $3\cot\angle ADC$ is equal to

(1) $2 + \sqrt{3}$

(2) $2 - \sqrt{3}$

(3) $\sqrt{3}$

(4) $\frac{1}{\sqrt{3}}$

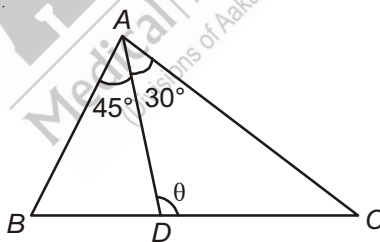
Sol. Answer (2)

Given that $BD : DC = 2 : 1$

So from (m, n) theorem

$$(2 + 1)\cot\theta = 2\cot 45^\circ - 1.\cot 30^\circ$$

$$3\cot\theta = 2 - \sqrt{3}$$



154. In $\triangle ABC$, $\tan \frac{A}{2} = \frac{5}{6}$, $\tan \frac{C}{2} = \frac{2}{5}$, then

(1) a, b, c are in A.P.

(2) a, c, b are in A.P.

(3) b, a, c are in A.P.

(4) a, b, c are in G.P.

Sol. Answer (1)

Given that $\tan \frac{A}{2} = \frac{5}{6}$, $\tan \frac{C}{2} = \frac{2}{5}$

So, $\tan \frac{A}{2}, \tan \frac{C}{2} = \frac{5}{6} \times \frac{2}{5}$

$$\sqrt{\frac{(s-b)(s-c)}{s(s-a)}} \times \frac{(s-a)(s-b)}{s(s-c)} = \frac{1}{3}$$

$$\Rightarrow \frac{s-b}{s} = \frac{1}{3} \Rightarrow \frac{a+c-b}{a+b+c} = \frac{1}{3}$$

$$\Rightarrow 2b = a + c$$

Hence a, b, c are in A.P.

155. If in a triangle ABC , $\Delta = a^2 - (b - c)^2$, then $\tan A$ is equal to

(1) $\frac{15}{16}$

(2) $\frac{8}{15}$

(3) $\frac{8}{17}$

(4) $\frac{1}{2}$

Sol. Answer (2)

Given that $\Delta = a^2 - (b - c)^2$

$$\sqrt{s(s-a)(s-b)(s-c)} = 4(s-b)(s-c)$$

$$\sqrt{\frac{(s-b)(s-c)}{s(s-a)}} = \frac{1}{4} = \tan \frac{A}{2} \quad \left[\because \tan \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} \right]$$

$$\text{Now, } \tan A = \frac{2 \tan \frac{A}{2}}{1 - \tan^2 \frac{A}{2}} = \frac{2 \times \frac{1}{4}}{1 - \frac{1}{16}} = \frac{8}{15}$$

$$\Rightarrow \tan A = \frac{8}{15}$$

156. If the angles of a triangle are 30° and 45° and the included side is $(1 + \sqrt{3})$ cm, then the area of the triangle is (in square cm)

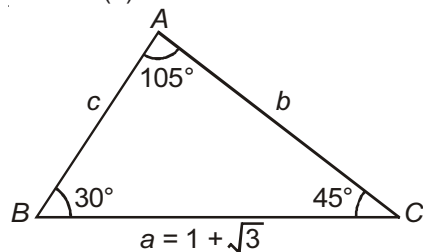
(1) $2(1 + \sqrt{3})$

(2) $\frac{1}{2}(1 + \sqrt{3})$

(3) $2(\sqrt{3} - 1)$

(4) $\frac{1}{2}(\sqrt{3} - 1)$

Sol. Answer (2)



From Sine Rule

$$\frac{1 + \sqrt{3}}{\sin 105} = \frac{c}{\sin 45} \Rightarrow c = 2$$

$$\text{Hence Area} = \frac{1}{2}ac \sin B = \frac{1}{2}(1 + \sqrt{3})$$

157. If $c^2 = a^2 + b^2$, $2s = a + b + c$, then $4s(s-a)(s-b)(s-c)$

- (1) s^4 (2) b^2c^2 (3) c^2a^2 (4) a^2b^2

Sol. Answer (4)

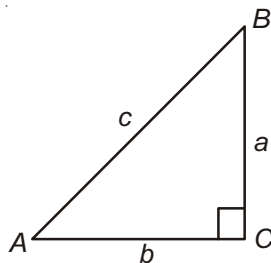
Given that $c^2 = a^2 + b^2$

$$\Rightarrow \angle C = 90^\circ$$

Now,

$$4s(s-a)(s-b)(s-c) = 4(\text{Area})^2$$

$$= 4 \times \frac{1}{4} a^2 b^2 = a^2 b^2$$



158. A triangle has its sides in the ratio 4 : 5 : 6, then the ratio of circumradius to the inradius of the triangle is

- (1) $\frac{15}{7}$ (2) $\frac{13}{6}$ (3) $\frac{16}{7}$ (4) $\frac{17}{6}$

Sol. Answer (3)

Let $a = 4\lambda$, $b = 5\lambda$, $c = 6\lambda$

We know that

$$\cos A + \cos B + \cos C = 1 + \frac{r}{R}$$

$$\frac{b^2 + c^2 - a^2}{2bc} + \frac{a^2 + c^2 - b^2}{2ac} + \frac{a^2 + b^2 - c^2}{2ab} = 1 + \frac{r}{R}$$

$$1 + \frac{r}{R} = \frac{23}{16} \Rightarrow \frac{r}{R} = \frac{7}{16} \Rightarrow \frac{R}{r} = \frac{16}{7}$$

159. In $\triangle ABC$, if $ar(\triangle ABC) = 8$. Then, $a^2 \sin(2B) + b^2 \sin(2A)$ is equal to

- (1) 2 (2) 16 (3) 32 (4) 128

Sol. Answer (3)

Given that $\Delta = 8$

$$\text{Now, } a^2 \sin^2 B + b^2 \sin^2 A = 2a^2 \sin B \cos B + 2b^2 \sin A \cos A$$

$$= 2a^2 \cdot \frac{b}{2R} \cdot \cos B + 2b^2 \cdot \frac{a}{2R} \cdot \cos A$$

$$= \frac{ab}{R} (a \cos B + b \cos A)$$

$$= \frac{abc}{R} = 4 \left(\frac{abc}{4R} \right) = 4\Delta = 32$$

$$\Rightarrow \boxed{a^2 \sin 2B + b^2 \sin 2A = 32}$$

160. In $\triangle ABC$, if $a^2 + b^2 + c^2 = ac + ab\sqrt{3}$, then

$$(1) \frac{r}{r_2} = \frac{r_3}{r_1}$$

$$(2) \frac{r}{r_1} = \frac{r_2}{r_3}$$

$$(3) r_1 + r_2 + r_3 = 2R + r$$

$$(4) r_1 r_2 r_3 = 4^3$$

Sol. Answer (1)

Given that,

$$a^2 + b^2 + c^2 = ac + ab\sqrt{3}$$

$$\Rightarrow a^2 + b^2 + c^2 - ac - ab\sqrt{3} = 0$$

$$\Rightarrow \left(\frac{a\sqrt{3}}{2} - b \right)^2 + \left(\frac{a}{2} - c \right)^2 = 0$$

Above equation is possible when

$$\frac{a\sqrt{3}}{2} - b = 0 \text{ \& } \frac{a}{2} - c = 0$$

$$\sqrt{3}a = 2b = 2\sqrt{3}c = k \text{ (let)}$$

$$\Rightarrow a = \frac{k}{\sqrt{3}}, b = \frac{k}{2}, c = \frac{k}{2\sqrt{3}}$$

$$\Rightarrow b^2 + c^2 = a^2 \Rightarrow \text{triangle is right angled triangle now taking (1) option}$$

$$\Rightarrow \frac{r}{r_2} = \frac{r_3}{r_1} \Rightarrow r_1 = r_2 r_3$$

$$\Rightarrow \frac{\Delta^2}{s(s-a)} = \frac{\Delta^2}{(s-b)(s-c)}$$

$$\Rightarrow s^2 - sa = s^2 - s(b+c) + bc$$

$$\Rightarrow (b+c-a)(b+c+a) = 2bc$$

$$\Rightarrow (b+c-a)s - bc$$

$$\Rightarrow (b+c)^2 - a^2 = 2bc \Rightarrow b^2 + c^2 - a^2 = 0$$

$$\Rightarrow \boxed{b^2 + c^2 = a^2} \text{ proved}$$

So option (1) is correct

161. In a triangle ABC , a , b and c are the sides of the triangle satisfying the relation $r_1 + r_2 = r_3 - r$ then the perimeter of the triangle

$$(1) \frac{2ab}{a+b-c}$$

$$(2) \frac{ab}{a+c-b}$$

$$(3) \frac{ab}{a-b-c}$$

$$(4) \frac{bc}{b+c-a}$$

Sol. Answer (1)Given that $r_1 + r_2 = r_3 - r$

$$\Rightarrow \frac{\Delta}{s-a} + \frac{\Delta}{s-b} = \frac{\Delta}{s-c} - \frac{\Delta}{s} \Rightarrow \frac{2s-a-b}{(s-a)(s-b)} = \frac{s-s+c}{s(s-c)}$$

$$\Rightarrow (s-a)(s-b) = s(s-c) \Rightarrow \boxed{2s = \frac{2ab}{a+b-c}}$$

Hence, option (1) is correct

162. In $\triangle ABC$, which is not right angled, if $p = \sin A \sin B \sin C$ and $q = \cos A \cos B \cos C$. Then the equation having roots $\tan A$, $\tan B$ and $\tan C$ is

(1) $qx^3 - px^2 + (1+q)x - p = 0$

(2) $qx^3 + 2px^2 + qx - p = 0$

(3) $qx^3 - px^2 + (2+q)x + pq = 0$

(4) $qx^3 - px^2 + qx + p = 0$

Sol. Answer (1)

Given that

$p = \sin A \cdot \sin B \cdot \sin C$

$q = \cos A \cdot \cos B \cdot \cos C$

Now, $\tan A \cdot \tan B \cdot \tan C = \frac{p}{q}$

And also $\tan A + \tan B + \tan C = \tan A \cdot \tan B \cdot \tan C = \frac{p}{q}$

Now, $\tan A \cdot \tan B + \tan B \cdot \tan C + \tan C \cdot \tan A$

$$= \frac{\sin A \cdot \sin B}{\cos A \cdot \cos B} + \frac{\sin B \cdot \sin C}{\cos B \cdot \cos C} + \frac{\sin C \cdot \sin A}{\cos C \cdot \cos A}$$

$$= \sin A \cdot \sin B \cdot \cos C + \sin B \sin C \cdot \cos A + \sin A \sin C \cdot \cos B / \cos A \cos B \cos C$$

$$= \sin B (\sin A \cos C + \cos A \sin C) + \sin A \cdot \sin C \cdot \cos B / \cos A \cos B \cos C$$

$$= \sin B \sin(A+C) + \sin A \cdot \sin C \cdot \cos B / \cos A \cos B \cos C$$

$$= \frac{\sin^2 B + \sin A \sin C \cdot \cos B}{\cos A \cdot \cos B \cdot \cos C}$$

$$= \frac{1 - \cos^2 B + \sin A \sin C \cdot \cos B}{\cos A \cdot \cos B \cdot \cos C}$$

$$= \frac{1 - \cos B (\cos B - \sin A \sin C)}{\cos A \cdot \cos B \cdot \cos C}$$

$$= \frac{1 - \cos B \{-\cos(A+C) - \sin A \sin C\}}{\cos A \cdot \cos B \cdot \cos C}$$

$$= \frac{1 + \cos B \cdot \cos A \cos C}{\cos A \cdot \cos B \cdot \cos C} = \frac{1+q}{q}$$

So required equation is

$$x^3 - \frac{p}{q}x^2 + \left(\frac{1+q}{q}\right)x - \frac{p}{q} = 0$$

$$\Rightarrow \boxed{qx^3 - px^2 + (1+q)x - p = 0}$$

Hence option (1) is correct

163. In $\triangle ABC$, if $\frac{R}{r} \leq 2$, and $a = 2$, then Δ is equal to ($\Delta = \text{ar}(\triangle ABC)$)

(1) 3

(2) 9

(3) $\sqrt{3}$

(4) $\frac{\sqrt{3}}{4}$

Sol. Answer (3)

Given that

$$\frac{R}{r} \leq 2 \Rightarrow \frac{r}{R} \geq \frac{1}{2}$$

$$\Rightarrow 1 + \frac{r}{R} \geq \frac{3}{2} \Rightarrow \cos A + \cos B + \cos C \geq \frac{3}{2} \quad \dots(i)$$

But we know that

$$\cos A + \cos B + \cos C \leq \frac{3}{2} \quad \dots(ii)$$

So from (i) & (ii) we get

$$\Rightarrow \cos A + \cos B + \cos C = \frac{3}{2}$$

$$\Rightarrow A = B = C$$

Hence triangle is equilateral

$$\begin{aligned} \text{So area of the triangle is} &= \frac{\sqrt{3}}{4}(\text{side})^2 \\ &= \frac{\sqrt{3}}{4} \times 4 = \sqrt{3} \end{aligned}$$

$\Rightarrow$ Option (3) is correct

164. In $\triangle ABC$, if r_1, r_2, r_3 are exradii opposites to angles A, B and C respectively. Then $\left(\frac{1}{r_1} + \frac{1}{r_2}\right)\left(\frac{1}{r_2} + \frac{1}{r_3}\right)\left(\frac{1}{r_3} + \frac{1}{r_1}\right)$ is equal to

(1) $\frac{64R^3}{abc}$

(2) $\frac{16R^3}{a^2b^2c^2}$

(3) $\frac{64R^3}{a^2b^2c^2}$

(4) $\frac{R^3}{abc}$

Sol. Answer (3)

$$\begin{aligned} &\left(\frac{1}{r_1} + \frac{1}{r_2}\right)\left(\frac{1}{r_2} + \frac{1}{r_3}\right)\left(\frac{1}{r_3} + \frac{1}{r_1}\right) \\ &= \frac{abc}{\Delta^3} \\ &= \frac{64R^3}{a^2b^2c^2} \left(\text{using } \frac{abc}{4\Delta} = R\right) \end{aligned}$$

So option (3) is correct.

165. In triangle ABC if the line joining incentre to the circumcentre is parallel to the base BC , then the value of $(\cos B + \cos C - 1)$ is equal to

- (1) 1 (2) 2 (3) 3 (4) 0

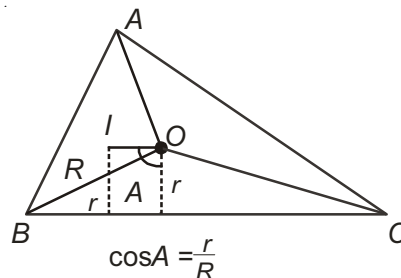
Sol. Answer (4)

$$\text{In } \triangle ABC \cos A + \cos B + \cos C = 1 + \frac{r}{R}$$

$$\Rightarrow \frac{r}{R} + \cos B + \cos C = 1 + \frac{r}{R}$$

$$\Rightarrow \cos B + \cos C - 1 = 0$$

So, option (4) is correct



166. With usual notations, if in a triangle ABC ,

$$\frac{b+c}{11} = \frac{c+a}{12} = \frac{a+b}{13}, \text{ then } \cos A : \cos B : \cos C \text{ is equal to}$$

- (1) 7 : 19 : 25 (2) 19 : 7 : 25 (3) 12 : 14 : 20 (4) 19 : 25 : 20

Sol. Answer (1)

$$\text{Let } \frac{b+c}{11} = \frac{c+a}{12} = \frac{a+b}{13} = k$$

$$\therefore b + c = 11k$$

$$c + a = 12k$$

$$a + b = 13k$$

$$2(a + b + c) = 36k$$

$$a + b + c = 18k$$

$$\therefore a = 7k, b = 6k, c = 5k$$

$$\cos A : \cos B : \cos C$$

$$\frac{b^2 + c^2 - a^2}{2bc} : \frac{a^2 + c^2 - b^2}{2ac} : \frac{a^2 + b^2 - c^2}{2ab}$$

Put the values of a, b & c

$$7 : 19 : 25$$

So option (1) is correct

167. Consider a regular polygon of 12 sides each of length one unit, then which of the following is not true?

- (1) The area of polygon is $(6 + 3\sqrt{3})$ (2) The circum-radius of polygon is $\frac{1}{2}(\sqrt{6} + \sqrt{2})$
 (3) The radius of incircle is $\frac{2 + \sqrt{3}}{2}$ (4) Each internal angle is of 135°

Sol. Answer (4)

$$\angle AOP = \frac{\pi}{12}$$

$$\text{Area of polygon} = 12 \times \text{ArAOB}$$

$$= 12 \times \frac{1}{2} \times AB \times OP$$

$$= 6 \times 1 \times OP$$

$$\tan \pi/12 = \frac{1}{2OP}$$

$$= \frac{6}{2 \tan \pi/12}$$

$$= \frac{3}{\tan \pi/12}$$

$$= \frac{3}{\sqrt{3}-1} (\sqrt{3}+1)$$

$$= 6 + 3\sqrt{3}$$

So option (1) is true.

$$(ii) \sin \frac{\pi}{12} = \frac{1}{2R}$$

$$\therefore R = \frac{1}{2 \sin \frac{\pi}{12}}$$

$$\therefore R = \frac{\sqrt{6} + \sqrt{2}}{2}$$

So option (2) is true.

$$\tan \frac{\pi}{12} = \frac{1}{2r}$$

$$\therefore r = \frac{2 + \sqrt{3}}{2}$$

So option (3) is true.

$$(iv) \text{ Each internal angle} = \frac{(2n-4)90}{n}$$

$$= \frac{20 \times 90}{12} = 150^\circ$$

Hence, option (4) is answer not true.

168. If the perimeter of a triangle is 6, then its maximum area is

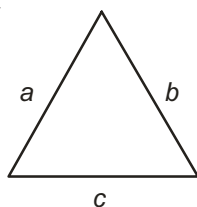
(1) $\sqrt{3}$

(2) 3

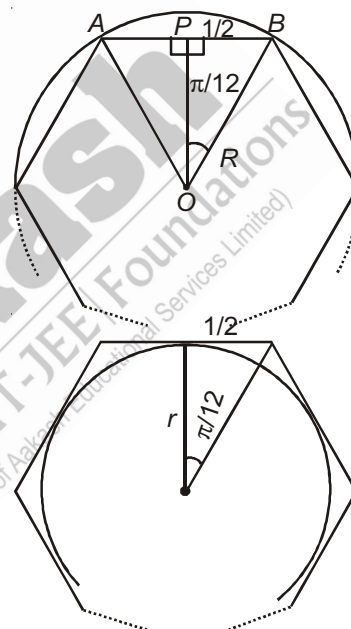
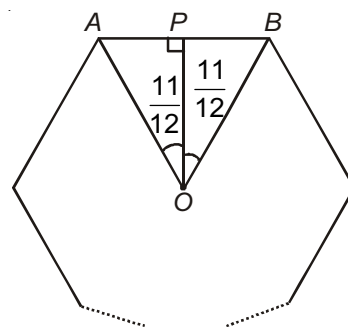
(3) 4

(4) 2

Sol. Answer (1)



$$a + b + c = 6$$



$$\frac{(s-a) + (s-b) + (s-c)}{3} \geq \{(s-a)(s-b)(s-c)\}^{1/3}$$

$$\frac{s}{3} \geq \left(\frac{s(s-a)(s-b)(s-c)}{s} \right)^{1/3}$$

$$\frac{s}{3} \geq \left(\frac{A^2}{s} \right)^{1/3}$$

$$\frac{s}{3} \geq \frac{A^{2/3}}{s^{1/3}}$$

$$\frac{s^{4/3}}{3} \geq A^{2/3}$$

$$\left(\frac{s^{4/3}}{3} \right)^{3/2} \geq A$$

$$\frac{s^2}{3\sqrt{3}} \geq A$$

$$\therefore \max A = \sqrt{3}$$

Hence, option(1) is correct.

169. In a triangle ABC, if $A - B = 120^\circ$ and $R = 8r$, then the value of $\cos C$ is

(1) $\frac{7}{8}$

(2) $\frac{3}{4}$

(3) $\frac{4}{5}$

(4) $\frac{3}{5}$

Sol. Answer (1)

$$A - B = 120^\circ$$

$$R = 8r$$

$$r = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$$

$$\frac{R}{8} = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$$

$$\frac{1}{16} = \left(\cos \frac{A-B}{2} - \cos \frac{A+B}{2} \right) \sin \frac{C}{2}$$

$$\frac{1}{16} = \left(\frac{1}{2} - \cos \frac{A+B}{2} \right) \sin \frac{C}{2}$$

$$\frac{1}{16} = \left(\frac{1}{2} - \sin \frac{C}{2} \right) \sin \frac{C}{2}$$

$$\frac{1}{16} = \left(\frac{1}{2} - x \right) x$$

Let $\sin \frac{C}{2} = x$

$$\Rightarrow 16x^2 - 8x + 1 = 0$$

$$\Rightarrow x = \frac{1}{4}$$

$$\Rightarrow \sin \frac{C}{2} = \frac{1}{4}$$

$$\Rightarrow \cos C = 1 - 2\sin^2 \frac{C}{2} = \frac{7}{8}$$

Hence option (1) is correct.

170. If in a triangle $2(a \cos B + b \cos C + c \cos A) = a + b + c$, then

- (1) The triangle is isosceles (2) The triangle is equilateral
(3) The triangle is isosceles right angled (4) The triangle is right angled

Sol. Answer (1)

$$2[a \cos B + b \cos C + c \cos A] = a + b + c$$

$$\Rightarrow 2[2R \sin A \cos B + 2R \sin B \cos C + 2R \sin C \cos A] = a + b + c$$

$$\Rightarrow 2R[2 \sin A \cos B + 2 \sin B \cos C + 2 \sin C \cos A] = 2R(\sin A + \sin B + \sin C)$$

$$\Rightarrow \sin[A + B] + \sin[A - B] + \sin[B + C] + \sin[B - C] + \sin[C + A] + \sin[C - A] = \sin A + \sin B + \sin C$$

$$\Rightarrow \sin(A - B) + \sin(B - C) + \sin(C - A) = 0$$

$$\Rightarrow \sin \frac{A - B}{2} \sin \frac{B - C}{2} \sin \frac{C - A}{2} = 0$$

which is only possible when $A = B$ or $B = C$ or $A = C$

$\therefore$ Triangle is isosceles.

Hence, option (1) is correct.

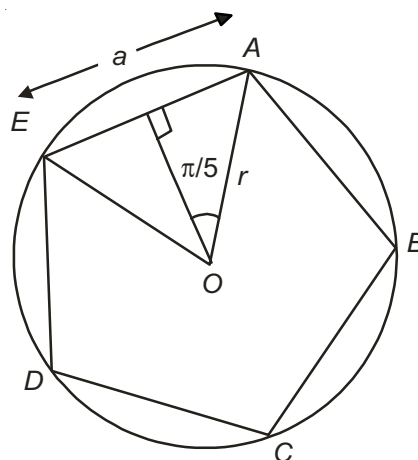
171. A regular pentagon is inscribed in a circle. If A_1 and A_2 represents the area of circle and that of regular pentagon respectively, then $A_1 : A_2$ is

- (1) $\frac{\pi}{5} \cos \frac{2\pi}{5}$ (2) $\frac{2\pi}{5} \sin \frac{2\pi}{5}$ (3) $\frac{2\pi}{5} \operatorname{cosec} \frac{2\pi}{5}$ (4) $\frac{2\pi}{5} \cos \frac{2\pi}{5}$

Sol. Answer (3)

$$\begin{aligned} \frac{A_1}{A_2} &= \frac{\pi r^2}{5 \times \frac{1}{2} \times a \times h} \\ &= \frac{\pi \times r \times r}{\frac{5}{2} \times a \times h} \\ &= \frac{2\pi}{5} \times \frac{1}{2 \sin \frac{\pi}{5}} \times \frac{1}{\cos \frac{\pi}{5}} \\ &= \frac{2\pi}{5} \times \frac{1}{\sin 2 \frac{\pi}{5}} \end{aligned}$$

$\therefore$ Option (3) is true.



172. In $\triangle ABC$, if $8R^2 - a^2 - b^2 = c^2$ then the triangle must be

- (1) Equilateral (2) Right angle (3) Scalene (4) Isosceles angle

Sol. Answer (2)

173. In $\triangle ABC$, if O is the circumcentre of $\triangle ABC$ and R_1, R_2, R_3 are the radii of circumcircles of triangles OBC ,

OCA and OAB respectively. Then the value of $\frac{a}{R_1} + \frac{b}{R_2} + \frac{c}{R_3}$ is equal to

- (1) $\frac{abc}{R}$ (2) $\frac{abc}{2R}$ (3) $\frac{abc}{4R^3}$ (4) $\frac{abc}{R^3}$

Sol. Answer (4)

Let $\angle AOB = 2\gamma$

$\angle BOC = 2\alpha$

$\angle AOC = 2\beta$

In $\triangle AOB$

$$\frac{c}{\sin 2\gamma} = 2R_3$$

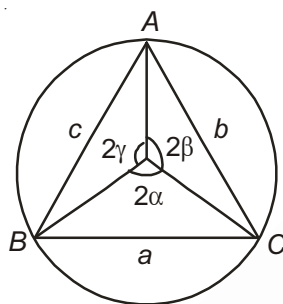
$$\frac{c}{R_3} = 2 \sin 2\gamma$$

$$\text{Similarly } \frac{a}{R_1} = 2 \sin 2\alpha$$

$$\frac{b}{R_2} = 2 \sin 2\beta$$

$$\begin{aligned} \therefore \frac{a}{R_1} + \frac{b}{R_2} + \frac{c}{R_3} &= 2 \sin 2\alpha + 2 \sin 2\beta + 2 \sin 2\gamma \\ &= 2 [\sin 2\alpha + \sin 2\beta + \sin 2\gamma] \\ &= 2 [4 \sin \alpha \sin \beta \sin \gamma] \\ &= 8 \sin \alpha \sin \beta \sin \gamma \\ &= 8 \cdot \frac{a}{2R} \cdot \frac{b}{2R} \cdot \frac{c}{2R} \\ &= \frac{abc}{R^3} \end{aligned}$$

Hence, option (4) is correct.



174. In $\triangle ABC$, the expression $\frac{b^2 - c^2}{a \sin(B-C)} + \frac{c^2 - a^2}{b \sin(C-A)} + \frac{a^2 - b^2}{c \sin(A-B)}$ is equal to

- (1) $\frac{3R}{2}$ (2) $3R$ (3) R (4) $6R$

Sol. Answer (4)

$$\begin{aligned}\frac{b^2 - c^2}{a \sin(B - C)} &= \frac{(2R \sin B)^2 - (2R \sin C)^2}{2R \sin A \sin(B - C)} \\ &= \frac{4R^2(\sin^2 B - \sin^2 C)}{2R \sin A \sin(B - C)} \\ &= \frac{4R^2 \sin(B + C) \sin(B - C)}{2R \sin A \sin(B - C)} \\ &= 2R \text{ other terms are also } 2R\end{aligned}$$

$$\therefore 2R + 2R + 2R = 6R$$

Hence, option (4) is correct.

175. In a triangle ABC , if $\Delta^2 = \frac{a^2 b^2 c^2}{2(a^2 + b^2 + c^2)}$, where Δ is the area of the triangle, then the triangle is

- (1) Isosceles but not right angled (2) Right angled
(3) Isosceles right angled (4) Equilateral

Sol. Answer (2)

$$\text{Given that } \Delta^2 = \frac{a^2 b^2 c^2}{2(a^2 + b^2 + c^2)}$$

$$\text{But we know that } \frac{abc}{4\Delta} = R \Rightarrow abc = 4\Delta R$$

$$\Delta^2 = \frac{16\Delta^2 R^2}{2(a^2 + b^2 + c^2)} \Rightarrow a^2 + b^2 + c^2 = 8R^2$$

$$\Rightarrow \sin^2 A + \sin^2 B + \sin^2 C = 2 \quad \dots(i)$$

From (i) it is clear that the triangle will be right-angled triangle.

So, option (2) is correct.

176. $ABCD$ is a trapezium such that AB and DC are parallel and BC is perpendicular to them. If $BC = 1$ cm, $CD = 2$ cm and $\angle ADB = 45^\circ$, then the length of AB is (in cm)

- (1) $\frac{12}{5}$ (2) $\frac{8}{5}$ (3) $\frac{5}{3}$ (4) $\frac{16}{3}$

Sol. Answer (3)

Using cosine formula in $\triangle DCB$

$$\cos \alpha = \frac{5 + 4 - 1}{4\sqrt{5}} = \frac{2}{\sqrt{5}}$$

Apply law of sin in $\triangle ADB$

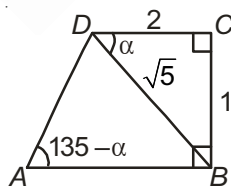
$$\frac{\sin 45^\circ}{AB} = \frac{\sin(135 - \alpha)}{\sqrt{5}}$$

$$\Rightarrow \frac{1}{\sqrt{2}AB} = \frac{\sin 135 \cos \alpha - \cos 135 \sin \alpha}{\sqrt{5}}$$

$$\Rightarrow \frac{1}{\sqrt{2}AB} = \left(\frac{2}{\sqrt{10}} + \frac{1}{\sqrt{10}} \right) \frac{1}{\sqrt{5}}$$

$$\therefore AB = \frac{5}{3}$$

Here, option (3) is correct.



177. If the median of triangle ABC through A is perpendicular AB , then the value of $\sin A \cos B + 2\sin B \cos A$ is equal to

- (1) 0 (2) 1 (3) $\frac{1}{2}$ (4) $\frac{1}{3}$

Sol. Answer (1)

From $(m_1 n)$ theorem

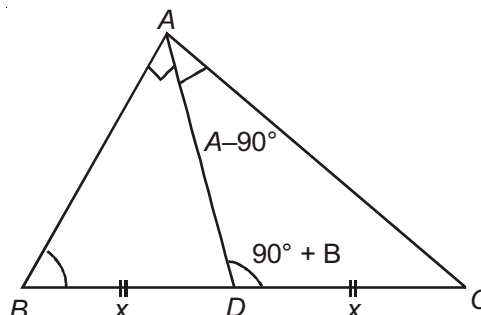
$$2x \cot(90^\circ + B) = x \cot 0 - x \cot(A - 90^\circ)$$

$$\Rightarrow -2 \tan B = \tan A$$

$$\Rightarrow -2 \sin B \cos A = \sin A \cos B$$

$$\Rightarrow \sin A \cos B + 2 \sin B \cos A = 0$$

$$\Rightarrow \text{Hence option (1) is correct}$$



178. The number of solutions of the pair of equations

$$2 \sin^2 \theta - \cos 2\theta = 0$$

$$2 \cos^2 \theta - 3 \sin \theta = 0 \text{ in the interval } [0, 2\pi] \text{ is}$$

[IIT-JEE 2007]

- (1) Zero (2) One (3) Two (4) Four

Sol. Answer (3)

$$2 \sin^2 \theta - \cos 2\theta = 0 \Rightarrow \sin^2 \theta = \frac{1}{4}$$

$$2 \cos^2 \theta - 3 \sin \theta = 0 \Rightarrow \sin \theta = \frac{1}{2} - 2$$

$$\text{common solutions are given by } \sin \theta = \frac{1}{2}$$

179. If the angles A, B and C of a triangle are in an arithmetic progression and if a, b and c denote the lengths of the sides opposite to A, B and C respectively, then the value of the expression $\frac{a}{c} \sin 2C + \frac{c}{a} \sin 2A$ is [IIT-JEE 2010]

- (1) $\frac{1}{2}$ (2) $\frac{\sqrt{3}}{2}$ (3) 1 (4) $\sqrt{3}$

Sol. Answer (4)

$$\text{Angles } A, B, C \text{ are in A.P. hence } B = 60^\circ = \frac{\pi}{3}$$

$$\frac{a}{c} \sin 2C + \frac{c}{a} \sin 2A = 2 \left(\frac{\sin C}{c} \right) a \cos C + 2 \left(\frac{\sin A}{a} \right) c \cos A$$

$$= 2(\sin A \cos C + \cos A \sin C)$$

$$= 2 \sin(A + C) = 2 \sin B = 2 \times \frac{\sqrt{3}}{2} = \sqrt{3}$$

180. Let $P = \{\theta : \sin \theta - \cos \theta = \sqrt{2} \cos \theta\}$ and $Q = \{\theta : \sin \theta + \cos \theta = \sqrt{2} \sin \theta\}$ be two sets. Then [IIT-JEE 2011]

- (1) $P \subset Q$ and $Q - P \neq \emptyset$ (2) $Q \not\subset P$ (3) $P \not\subset Q$ (4) $P = Q$

Sol. Answer (4)

We have,

$$\sin \theta - \cos \theta = \sqrt{2} \cos \theta$$

$$\Rightarrow \sin \theta = (\sqrt{2} + 1) \cos \theta$$

$$\Rightarrow \frac{1}{\sqrt{2} + 1} \sin \theta = \cos \theta$$

$$\Rightarrow (\sqrt{2} - 1) \sin \theta = \cos \theta$$

$$\Rightarrow \sqrt{2} \sin \theta = \sin \theta + \cos \theta$$

and it shows that the given equations

$$\Rightarrow \sin \theta - \cos \theta = \sqrt{2} \cos \theta$$

$$\text{and } \sin \theta + \cos \theta = \sqrt{2} \sin \theta$$

are identical. Hence their solution sets are equal

$$\Rightarrow P = Q$$

181. Let PQR be a triangle of area Δ with $a = 2$, $b = \frac{7}{2}$ and $c = \frac{5}{2}$, where a , b and c are the lengths of the sides of the triangle opposite to the angles at P , Q and R respectively. Then $\frac{2 \sin P - \sin 2P}{2 \sin P + \sin 2P}$ equals **[IIT-JEE 2012]**

(1) $\frac{3}{4\Delta}$

(2) $\frac{45}{4\Delta}$

(3) $\left(\frac{3}{4\Delta}\right)^2$

(4) $\left(\frac{45}{4\Delta}\right)^2$

Sol. Answer (3)

$$\text{As, } \frac{2 \sin P - \sin 2P}{2 \sin P + \sin 2P} = \frac{1 - \cos P}{1 + \cos P} = \tan^2 \left(\frac{P}{2} \right)$$

$$\text{Now, } a = 2, b = \frac{7}{2}, c = \frac{5}{2}$$

$$\text{So, semi perimeter} = s = \frac{a + b + c}{2} = 4$$

$$\text{Now, } \tan \frac{P}{2} = \frac{(s - b)(s - c)}{\Delta} = \frac{3}{4\Delta}$$

$$\text{So, } \tan^2 \left(\frac{P}{2} \right) = \left(\frac{3}{4\Delta} \right)^2$$

182. In a triangle the sum of two sides is x and the product of the same two sides is y . If $x^2 - c^2 = y$, where c is the third side of the triangle, then the ratio of the in radius to the circum-radius of the triangle is

[JEE(Advanced)-2014]

(1) $\frac{3y}{2x(x+c)}$

(2) $\frac{3y}{2c(x+c)}$

(3) $\frac{3y}{4x(x+c)}$

(4) $\frac{3y}{4c(x+c)}$

Sol. Answer (2)

$$a + b = x, ab = y, x^2 - c^2 = y$$

$$\text{Now, } (a + b)^2 - c^2 = ab$$

$$a^2 + b^2 - c^2 = -ab$$

$$\frac{a^2 + b^2 - c^2}{2ab} = -\frac{1}{2}$$

$$\Rightarrow \cos c = -\frac{1}{2} \quad \Rightarrow \angle c = 120^\circ$$

$$2R = \frac{c}{\sin c} = \frac{2c}{\sqrt{3}} \quad \Rightarrow R = \frac{c}{\sqrt{3}}$$

$$r = (s - c) \cdot \tan \frac{c}{2}$$

$$r = \left(\frac{x+c}{2} - c \right) \cdot \sqrt{3}$$

$$r = \frac{(x-c)\sqrt{3}}{2}$$

$$\frac{r}{R} = \frac{(x-c)\sqrt{3}}{2 \cdot c} \cdot \sqrt{3} = \frac{3(x-c)}{2c}$$

$$= \frac{3(x-c)(x+c)}{2c(x+c)}$$

$$= \frac{3(x^2 - c^2)}{2c(x+c)}$$

$$= \frac{3y}{2c(x+c)}$$

183. For $x \in (0, \pi)$, the equation $\sin x + 2\sin 2x - \sin 3x = 3$ has

- (1) Infinitely many solutions (2) Three solutions
(3) One solution (4) No solution

Sol. Answer (4)

$$\sin x + 2\sin 2x - \sin 3x = 3$$

$$2\cos 2x(-\sin x) + 4\sin x \cos x = 3$$

$$2\sin x[2\cos x - \cos 2x] = 3$$

$$2\sin x(2\cos x - (2\cos^2 x - 1)) = 3$$

$$2\sin x(1 + 2\cos x - 2\cos^2 x) = 3$$

$$2\sin x \left[\frac{3}{2} - 2 \left(\cos x - \frac{1}{2} \right)^2 \right] = 3$$

Possible only when

$$\sin x = 1 \quad \dots(i)$$

$$\text{and } \left(\cos x - \frac{1}{2} \right)^2 = 0$$

$$\Rightarrow \cos x = \frac{1}{2} \quad \dots(ii)$$

From (i) and (ii),

No solution.

[JEE(Advanced)2014]

184. Let $S = \left\{ x \in (-\pi, \pi) : x \neq 0, \pm \frac{\pi}{2} \right\}$. The sum of all distinct solutions of the equation

$$\sqrt{3} \sec x + \operatorname{cosec} x + 2(\tan x - \cot x) = 0 \text{ in the set } S \text{ is equal to}$$

[JEE(Advanced)2016]

- (1) $-\frac{7\pi}{9}$ (2) $-\frac{2\pi}{9}$ (3) 0 (4) $\frac{5\pi}{9}$

Sol. Answer (3)

$$\sqrt{3} \sec x + \operatorname{cosec} x + 2(\tan x - \cot x) = 0$$

$$\Rightarrow \sqrt{3} \sin x + \cos x + 2(\sin^2 x - \cos^2 x) = 0$$

$$\Rightarrow \frac{\sqrt{3}}{2} \sin x + \frac{1}{2} \cos x = \cos^2 x - \sin^2 x$$

$$\Rightarrow \cos 2x = \cos \left(x - \frac{\pi}{3} \right)$$

$$\Rightarrow 2x = 2n\pi \pm \left(x - \frac{\pi}{3} \right), n \in I$$

$$\Rightarrow x = 2n\pi - \frac{\pi}{3}, \frac{1}{3} \left(2n\pi + \frac{\pi}{3} \right)$$

$$\text{So, } x = -\frac{\pi}{3}, \frac{\pi}{9}, \frac{7\pi}{9}, \frac{-5\pi}{9}$$

Sum of all the solution = 0.

185. The value of $\sum_{k=1}^{13} \frac{1}{\sin\left(\frac{\pi}{4} + \frac{(k-1)\pi}{6}\right) \sin\left(\frac{\pi}{4} + \frac{k\pi}{6}\right)}$ is equal to

[JEE(Advanced)2016]

- (1) $3 - \sqrt{3}$ (2) $2(3 - \sqrt{3})$ (3) $2(\sqrt{3} - 1)$ (4) $2(2 + \sqrt{3})$

Sol. Answer (3)

$$\frac{1}{\sin \frac{\pi}{6}} \sum_{k=1}^{13} \left[\frac{\sin \left[\left(\frac{\pi}{4} + \frac{k\pi}{6} \right) - \left(\frac{\pi}{4} + \frac{(k-1)\pi}{6} \right) \right]}{\sin \left(\frac{\pi}{4} + \frac{k\pi}{6} \right) \cdot \sin \left(\frac{\pi}{4} + \frac{(k-1)\pi}{6} \right)} \right]$$

$$\frac{1}{\sin \frac{\pi}{6}} \sum_{k=1}^{13} \left[\cot \left(\frac{\pi}{4} + \frac{(k-1)\pi}{6} \right) - \cot \left(\frac{\pi}{4} + \frac{k\pi}{6} \right) \right]$$

$$= \frac{1}{\sin \frac{\pi}{6}} \left[\cot \frac{\pi}{4} - \cot \left(\frac{\pi}{4} + \frac{13 \cdot \pi}{6} \right) \right]$$

$$= \frac{1}{\sin \frac{\pi}{6}} [1 - (2 - \sqrt{3})]$$

$$= 2(\sqrt{3} - 1)$$

SECTION - B

Objective Type Questions (More than one options are correct)

[Elementary Trigonometric Functions]

1. If $x = \sec\phi - \tan\phi$ and $y = \operatorname{cosec}\phi + \cot\phi$, then

(1) $x = \frac{y+1}{y-1}$

(2) $x = \frac{y-1}{y+1}$

(3) $y = \frac{1+x}{1-x}$

(4) $xy + x - y + 1 = 0$

Sol. Answer (2, 3, 4)

$$x = \sec\phi - \tan\phi, y = \operatorname{cosec}\phi + \cot\phi$$

$$y = \cot\frac{\phi}{2}$$

$$x = \frac{1 - \sin\phi}{\cos\phi} = \frac{\left(\cos\frac{\phi}{2} - \sin\frac{\phi}{2}\right)^2}{\cos^2\frac{\phi}{2} - \sin^2\frac{\phi}{2}}$$

$$\Rightarrow x = \frac{\left(\cot\frac{\phi}{2} - 1\right)^2}{\cot^2\frac{\phi}{2} - 1} = \frac{(y-1)^2}{(y^2-1)} = \frac{y-1}{y+1}$$

2. The value of $F(x) = 6\cos x \sqrt{1+\tan^2 x} + 2\sin x \sqrt{1+\cot^2 x}$ where $x \in (0, 2\pi) - \left\{\pi, \frac{\pi}{2}, \frac{3\pi}{2}\right\}$ may be

(1) 4

(2) -4

(3) 8

(4) -8

Sol. Answer (1, 2, 3, 4)

$$f(x) = 6\cos x \sqrt{1+\tan^2 x} + 2\sin x \sqrt{1+\cot^2 x}, x \in (0, 2\pi) - \left\{\frac{\pi}{2}, \pi, \frac{3\pi}{2}\right\}$$

$$= 6\cos x \cdot |\sec x| + 2\sin x \cdot |\operatorname{cosec} x|$$

$$\left\{ \begin{array}{ll} 6+2=8 & x \in \left(0, \frac{\pi}{2}\right) \\ -6+2=-4 & x \in \left(\frac{\pi}{2}, \pi\right) \\ -6-2=-8 & x \in \left(\pi, \frac{3\pi}{2}\right) \\ 6-2=4 & x \in \left(\frac{3\pi}{2}, 2\pi\right) \end{array} \right.$$

3. If $\frac{(a+b)^2}{4ab} = \sin^2 \theta$, then

(1) $a = b \neq 0$

(2) $a \neq 0, b \neq 0$

(3) $a > b$

(4) $a < b$

Sol. Answer (1, 2)

$$\frac{(a+b)^2}{4ab} = \sin^2 \theta$$

$$\therefore 0 \leq \sin^2 \theta \leq 1$$

$$\Rightarrow \frac{(a+b)^2}{4ab} \leq 1$$

$$\Rightarrow (a-b)^2 \leq 0$$

$$\Rightarrow a = b, \text{ but not equal to zero}$$

4. If $\sec \theta = x + \frac{1}{4x}$, then $\tan \theta + \sec \theta$ is equal to

(1) $2x$

(2) $\frac{1}{2x}$

(3) $\frac{1}{x}$

(4) $4x$

Sol. Answer (1, 2)

$$\sec \theta = x + \frac{1}{4x}$$

$$\tan \theta = \sqrt{\sec^2 \theta - 1}$$

$$= \sqrt{\left(x + \frac{1}{4x}\right)^2 - 1}$$

$$= \left|x - \frac{1}{4x}\right|$$

$$\therefore \sec \theta + \tan \theta = \left(x + \frac{1}{4x}\right) + \left|x - \frac{1}{4x}\right|$$

$$= \begin{cases} 2x & \text{if } x > \frac{1}{4x} \\ \frac{1}{2x} & \text{if } x \leq \frac{1}{4x} \end{cases}$$

5. If $\tan \alpha + \cot \alpha = 2$, then

(1) $\tan^n \alpha + \cot^n \alpha = 2$

(2) $\sqrt{\tan \alpha} + \sqrt{\cot \alpha} = 2$

(3) $\sin^{2n} \alpha + \cos^{2n} \alpha = \frac{1}{2^{n-1}}$

(4) $\tan^2 \alpha + \sec^2 \alpha = 3$

Sol. Answer (1, 2, 3, 4)

$$\tan \alpha + \cot \alpha = 2$$

$$\Rightarrow \tan^2 \alpha + 1 = 2 \tan \alpha$$

$$\Rightarrow \tan \alpha = 1$$

$$(1) \tan^4 \alpha + \cot^4 \alpha = 2$$

$$(2) \sqrt{\tan \alpha} + \sqrt{\cot \alpha} = 2$$

$$(3) \sin^{2n} \alpha + \cos^{2n} \alpha = \left(\pm \frac{1}{\sqrt{2}} \right)^{2n} + \left(\pm \frac{1}{\sqrt{2}} \right)^{2n}$$

$$= \frac{1}{2^{n-1}}$$

$$(4) \tan^2 \alpha + \sec^2 \alpha = 1 + 2 = 3$$

6. Which of the following identities are correct?

$$(1) \frac{1 + \sin \theta}{\cos \theta} + \frac{\cos \theta}{1 + \sin \theta} = 2 \sec \theta$$

$$(2) (1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) = 2$$

$$(3) \sqrt{\frac{1 + \sin \theta}{1 - \sin \theta}} = |\sec \theta + \tan \theta|$$

$$(4) \frac{\cos \theta}{1 + \sin \theta} + \frac{\cos \theta}{1 - \sin \theta} = 2 \sec \theta$$

Sol. Answer (1, 2, 3, 4)

$$(1) \frac{1 + \sin \theta}{\cos \theta} + \frac{\cos \theta}{1 + \sin \theta} = 2 \sec \theta$$

$$\text{L.H.S.} = \frac{1 + \sin \theta}{\cos \theta} + \frac{\cos \theta (1 - \sin \theta)}{\cos^2 \theta}$$

$$= \frac{2}{\cos \theta} = \text{RHS}$$

$$(2) (1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) = 2$$

$$\text{L.H.S.} = \left(\frac{\sin A + \cos A - 1}{\sin A} \right) \left(\frac{\cos A + \sin A + 1}{\cos A} \right)$$

$$= \frac{1 + 2 \sin A \cos A - 1}{\sin A \cdot \cos A} = 2 = \text{R.H.S.}$$

$$(3) = \sqrt{\frac{1 + \sin \theta}{1 - \sin \theta}} = |\sec \theta + \tan \theta|$$

$$\text{L.H.S.} = \sqrt{\frac{(1 + \sin \theta)^2}{\cos^2 \theta}}$$

$$= |\sec \theta + \tan \theta| = \text{R.H.S.}$$

$$(4) \frac{\cos \theta}{1 + \sin \theta} + \frac{\cos \theta}{1 - \sin \theta} = 2 \sec \theta$$

$$\begin{aligned} \text{L.H.S.} &= \frac{\cos(1 - \sin \theta)}{\cos^2 \theta} + \frac{\cos \theta \cdot (1 + \sin \theta)}{\cos^2 \theta} \\ &= \frac{1 - \sin \theta}{\cos \theta} + \frac{1 + \sin \theta}{\cos \theta} \\ &= 2 \sec \theta = \text{R.H.S} \end{aligned}$$

7. The value of the expression $\sin x + \sin y + \sin z$ where x, y, z are real numbers satisfying $x + y + z = 180^\circ$ is

- (1) Positive (2) Zero (3) -3 (4) Negative

Sol. Answer (1, 2, 4)

$$x + y + z = 180^\circ$$

$$x = -90^\circ, y = 90^\circ, z = 360^\circ$$

8. Let $\theta \in \left(0, \frac{\pi}{4}\right)$ and $a = (\tan \theta)^{\tan \theta}$, $b = (\tan \theta)^{\cot \theta}$, $c = (\cot \theta)^{\tan \theta}$, $d = (\cot \theta)^{\cot \theta}$, then

- (1) $d > c$ (2) $b < a$ (3) $c > a$ (4) $b < a < c < d$

Sol. Answer (1, 2, 3, 4)

$$\theta \in \left(0, \frac{\pi}{4}\right)$$

$$\therefore 0 < \tan \theta < 1, 1 < \cot \theta < \infty$$

$$a = (\tan \theta)^{\tan \theta}, b = (\tan \theta)^{\cot \theta}$$

$$c = (\cot \theta)^{\tan \theta}, d = (\cot \theta)^{\cot \theta}$$

Clearly b is the smallest and d is the greatest of all

Also, Fractional value (number greater than 1) > 1

$$\therefore b < a < c < d$$

a and b both fractional values

[Transformation Formulae]

9. If $\tan\left(\frac{\pi}{4} + x\right) \cdot \tan\left(\frac{\pi}{4} - x\right) = 1$, then x equals

- (1) $\frac{\pi}{6}$ (2) $\frac{\pi}{3}$ (3) $\frac{\pi}{4}$ (4) Zero

Sol. Answer (1, 2, 4)

$$\tan\left(\frac{\pi}{4} + x\right) \cdot \tan\left(\frac{\pi}{4} - x\right) = 1$$

$$\Rightarrow \frac{1 + \tan x}{1 - \tan x} \times \frac{1 - \tan x}{1 + \tan x} = 1$$

10. The value of $\cos 22\frac{1}{2}^\circ$ is

(1) $\frac{1}{2}\sqrt{2+\sqrt{2}}$

(2) $\frac{1}{4}\{\sqrt{4+2\sqrt{2}} - \sqrt{4-2\sqrt{2}}\}$

(3) $\frac{1}{4}\{\sqrt{4+2\sqrt{2}} + \sqrt{4-2\sqrt{2}}\}$

(4) $\frac{1}{2}\sqrt{2-\sqrt{2}}$

Sol. Answer (1, 3)

$$\cos 22\frac{1}{2}^\circ = ?$$

$$2\cos^2\theta = 1 + \cos 2\theta$$

$$\Rightarrow \cos 22\frac{1}{2}^\circ = \sqrt{\frac{1 + \cos 45^\circ}{2}} = \sqrt{\frac{\sqrt{2} + 1}{2\sqrt{2}}}$$

$$= \sqrt{\frac{2 + \sqrt{2}}{4}} = \frac{\sqrt{2 + \sqrt{2}}}{2}$$

$$\therefore \cos \frac{\theta}{2} = \frac{1}{2}(\sqrt{1 + \sin \theta} + \sqrt{1 - \sin \theta})$$

$$\Rightarrow \cos 22\frac{1}{2}^\circ = \frac{1}{2}(\sqrt{1 + \sin 45^\circ} + \sqrt{1 - \sin 45^\circ})$$

$$= \frac{1}{2}\left(\sqrt{1 + \frac{1}{\sqrt{2}}} + \sqrt{1 - \frac{1}{\sqrt{2}}}\right)$$

$$= \frac{1}{2}\left(\sqrt{\frac{\sqrt{2} + 1}{\sqrt{2}}} + \sqrt{\frac{\sqrt{2} - 1}{\sqrt{2}}}\right)$$

$$= \frac{1}{2}\left(\sqrt{\frac{4 + 2\sqrt{2}}{4}} + \sqrt{\frac{4 - 2\sqrt{2}}{4}}\right)$$

$$= \frac{1}{4}(\sqrt{4 + 2\sqrt{2}} + \sqrt{4 - 2\sqrt{2}})$$

11. Let $\cos 2\alpha = \frac{3\cos 2\beta - 1}{3 - \cos 2\beta}$, then $\tan \alpha$ is equal to

(1) $\sqrt{2} \tan \beta$

(2) $-\sqrt{2} \tan \beta$

(3) $\frac{1}{\sqrt{2}} \tan \beta$

(4) $-\frac{1}{\sqrt{2}} \tan \beta$

Sol. Answer (1, 2)

$$\cos 2\alpha = \frac{3\cos 2\beta - 1}{3 - \cos 2\beta}$$

$$\frac{1 - \tan^2 \alpha}{1 + \tan^2 \alpha} = \frac{3\cos 2\beta - 1}{3 - \cos 2\beta}$$

$$\frac{1 - \tan^2 \alpha + 1 + \tan^2 \alpha}{1 - \tan^2 \alpha - 1 - \tan^2 \alpha} = \frac{3\cos 2\beta - 1 + 3 - \cos 2\beta}{3\cos 2\beta - 1 - 3 + \cos 2\beta}$$

$$\frac{2}{-2\tan^2 \alpha} = \frac{2\cos 2\beta + 2}{4\cos 2\beta - 4}$$

$$+\frac{1}{\tan^2 \alpha} = \frac{2(1 + \cos 2\beta)}{4(1 - \cos 2\beta)} = \frac{2\cos^2 \beta}{2 \times 2 \tan^2 \beta}$$

$$\tan^2 \alpha = 2 \tan^2 \beta$$

$$\tan \alpha = \pm \sqrt{2} \tan \beta$$

12. If $\sin \alpha = -\frac{3}{5}$, $\alpha \in [0, 2\pi]$ then the possible values of $\cos \frac{\alpha}{2}$ is/are

(1) $\frac{1}{\sqrt{10}}$

(2) $-\frac{1}{\sqrt{10}}$

(3) $\frac{3}{\sqrt{10}}$

(4) $-\frac{3}{\sqrt{10}}$

Sol. Answer (2, 4)

$$\sin \alpha = -\frac{3}{5}$$

$$\cos \frac{\alpha}{2} = \frac{1}{2}(\sqrt{1 + \sin \alpha} + \sqrt{1 - \sin \alpha})$$

$$= \frac{1}{2}\left(\sqrt{1 + \frac{3}{5}} + \sqrt{1 - \frac{3}{5}}\right)$$

$$= \frac{1}{2}\left(\frac{2\sqrt{2}}{\sqrt{5}} + \frac{\sqrt{2}}{\sqrt{5}}\right)$$

$$= \frac{1}{2}\left(\frac{4}{\sqrt{10}} + \frac{2}{\sqrt{10}}\right) = \frac{3}{\sqrt{10}}$$

13. The expression

$$f(\alpha, \beta) = \cos^2 \alpha + \cos^2(\alpha + \beta) - 2\cos \alpha \cos \beta \cos(\alpha + \beta)$$

(1) Is independent of β (2) Is independent of α (3) Is independent of both α and β (4) Is dependent of β

Sol. Answer (2, 4)

$$\begin{aligned}
 f(\alpha, \beta) &= \cos^2\alpha + \cos^2(\alpha + \beta) - 2\cos\alpha \cos\beta \cos(\alpha + \beta) \\
 &= \cos^2\alpha + \cos(\alpha + \beta) (\cos\alpha \cos\beta - \sin\alpha \sin\beta - 2\cos\alpha \cos\beta) \\
 &= \cos^2\alpha - \cos(\alpha + \beta) \cos(\alpha - \beta) \\
 &= \cos^2\alpha - (\cos^2\alpha - \sin^2\beta) \\
 &= \cos^2\alpha - \cos^2\alpha + \sin^2\beta \\
 &= \sin^2\beta
 \end{aligned}$$

14. The value of $\cos A \cdot \cos 2A \cdot \cos 2^2 A \dots \cos(2^{n-1}A)$, where $A \in R$ may be

- (1) 1 (2) 2 (3) -1 (4) $\frac{\sin 2^n A}{2^n \sin A}$

Sol. Answer (1, 3, 4)

$$\cos A \cdot \cos 2A \dots \cos(2^{n-1}A) = \frac{\sin 2^n A}{2^n \sin A}$$

15. If $A + B = \frac{\pi}{3}$ and $\cos A + \cos B = 1$, then

- (1) $\cos(A - B) = \frac{1}{3}$ (2) $|\cos A - \cos B| = \sqrt{\frac{2}{3}}$
 (3) $\cos(A - B) = -\frac{1}{3}$ (4) $|\cos A - \cos B| = \frac{1}{2\sqrt{3}}$

Sol. Answer (2, 3)

$$\begin{aligned}
 A + B &= \frac{\pi}{3}, \quad \cos A + \cos B = 1 \\
 \Rightarrow 2\cos \frac{A+B}{2} \cos \frac{A-B}{2} &= 1 \\
 \Rightarrow 2\cos \left(\frac{\pi}{6}\right) \cos \frac{A-B}{2} &= 1 \\
 \Rightarrow 2 \times \frac{\sqrt{3}}{2} \times \cos \left(\frac{A-B}{2}\right) &= 1 \\
 \Rightarrow \cos \left(\frac{A-B}{2}\right) &= \frac{1}{\sqrt{3}} \\
 \therefore \cos \left(\frac{A-B}{2}\right) &= \cos 2 \left(\frac{A-B}{2}\right) \\
 &= 2\cos^2 \left(\frac{A-B}{2}\right) - 1 \\
 &= 2 \times \frac{1}{3} - 1 \\
 &= \frac{2}{3} - 1 = -\frac{1}{3}
 \end{aligned}$$

$$\therefore \sin\left(\frac{A-B}{2}\right) = \sqrt{1 - \frac{1}{3}} = \sqrt{\frac{2}{3}}$$

$$|\cos A - \cos B| = \left| 2 \sin \frac{A+B}{2} \sin\left(\frac{A-B}{2}\right) \right|$$

$$= \left| 2 \times \frac{1}{2} \sin\left(\frac{A-B}{2}\right) \right|$$

$$= \sin\left(\frac{A-B}{2}\right)$$

$$= \sqrt{\frac{2}{3}}$$

16. $\cos^3 x \cdot \sin 2x = \sum_{m=1}^n a_m \sin mx$ is an identity in x , then

$$(1) \ a_3 = \frac{3}{8}, a_2 = 0$$

$$(2) \ n = 6, a_1 = \frac{1}{2}$$

$$(3) \ n = 5, a_1 = \frac{1}{4}$$

$$(4) \ \sum a_m = \frac{3}{4}$$

Sol. Answer (1, 3)

$$\cos^3 x \cdot \sin 2x = \sum_{m=1}^n a_m \sin mx$$

$$= a_1 \sin x + a_2 \sin 2x + a_3 \sin 3x + \dots + a_n \sin nx$$

$$\Rightarrow 2\cos^3 x \cdot \cos 2x - 3\cos^2 x \sin 2x \cdot \sin x = a_1 \cos x + 2a_2 \cos 2x + 3a_3 \cos 3x + \dots$$

Put $x = 0$

$$2 = a_1 + 2a_2 + 3a_3 + \dots + na_n$$

$$S_{n+1} - S_n = a_1 + 2a_2 + \dots + na_n + (n+1)a_{n+1} - a_1 - 2a_2 - \dots - na_n$$

$$\Rightarrow E_{n+1} = (n+1)a_{n+1} = \frac{1}{4}(\cos 3n + 3\cos n)\sin 2n$$

$$= \frac{1}{8}[2\sin 2x \cos 3x + 3\sin 2x \sin x]$$

$$= \frac{1}{8}[\sin 5x - \sin x + 3\sin 3x + 3\sin x]$$

$$= \frac{1}{8}[\sin 5x + 3\sin 3x + 2\sin x]$$

$$\therefore a_1 = \frac{1}{4}, a_2 = 0, a_3 = \frac{3}{8}, a_5 = \frac{1}{8}$$

17. If $0 < \alpha, \beta < \pi$ and they satisfy $\cos\alpha + \cos\beta - \cos(\alpha + \beta) = \frac{3}{2}$

- (1) $\alpha = \beta$ (2) $\alpha + \beta = \frac{2\pi}{3}$ (3) $\alpha = 2\beta$ (4) $\beta = 2\alpha$

Sol. Answer (1, 2)

$$\cos\alpha + \cos\beta - \cos(\alpha + \beta) = \frac{3}{2}$$

$$\Rightarrow \cos\alpha = \frac{1}{2}, \cos\beta = \frac{1}{2}, \cos(\alpha + \beta) = -\frac{1}{2}$$

$$\Rightarrow \alpha = \frac{\pi}{3}, \beta = \frac{\pi}{3}, \alpha + \beta = \frac{2\pi}{3}$$

18. The angles A, B, C of a triangle ABC satisfy $4\cos A \cos B + \sin 2A + \sin 2B + \sin 2C = 4$. Then which of the following statements is/are correct?

- (1) The triangle ABC is right angled
(2) The triangle ABC is isosceles
(3) The triangle ABC is neither isosceles nor right angled
(4) The triangle ABC is equilateral

Sol. Answer (1, 2)

$$4\cos A \cos B + \sin 2A + \sin 2B + \sin 2C = 4$$

$$\Rightarrow 4 \cdot \frac{1}{2} \cdot \frac{1}{2} + \frac{\sqrt{3}}{2} \times 3$$

$$2[\cos(A + B) + \cos(A - B)] + 2\cos(A + B) \cos(A - B) + 2\sin C \cos C = 2$$

$$\Rightarrow -\cos C + \cos(A - B) - \cos C \cos(A - B) + \sin C \cos C = 2$$

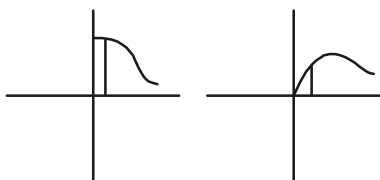
$$\Rightarrow -\cos C + \cos(A - B) + \cos C$$

19. Which of the following statement(s) is/are correct?

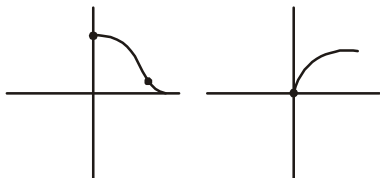
- (1) $\cos(\sin 1) > \sin(\cos 1)$ (2) $\cos(\sin 1.5) > \sin(\cos 1.5)$
(3) $\cos\left(\sin \frac{7\pi}{18}\right) > \sin\left(\cos \frac{7\pi}{18}\right)$ (4) $\cos\left(\sin \frac{5\pi}{18}\right) > \sin\left(\cos \frac{5\pi}{18}\right)$

Sol. Answer (1, 2, 3, 4)

1. $\cos(\sin 1)$ $\sin(\cos 1)$



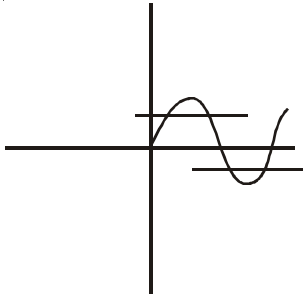
- 2.



20. If $\sin\theta = K$, $-1 \leq K \leq 1$, then number of values of θ , for same value of K in $[0, 2\pi]$ may be

- (1) 1 (2) 2 (3) 3 (4) 4

Sol. Answer (1, 2, 3)



21. If the sides of a right angled triangle are $\{\cos 2\alpha + \cos 2\beta + 2\cos(\alpha + \beta)\}$ and $\{\sin 2\alpha + \sin 2\beta + 2\sin(\alpha + \beta)\}$, then the length of hypoteneous is

- (1) $2[1 + \cos(\alpha - \beta)]$ (2) $2[1 - \cos(\alpha + \beta)]$ (3) $4\cos^2\left(\frac{\alpha - \beta}{2}\right)$ (4) $4\sin^2\left(\frac{\alpha + \beta}{2}\right)$

Sol. Answer (1, 3)

$$a = \cos 2\alpha + \cos 2\beta + 2\cos(\alpha + \beta)$$

$$= 2\cos(\alpha + \beta)\cos(\alpha - \beta) + 2\cos(\alpha + \beta)$$

$$= 2\cos(\alpha + \beta) \times 2\cos^2\left(\frac{\alpha - \beta}{2}\right)$$

$$= 4\cos(\alpha + \beta)\cos^2\left(\frac{\alpha - \beta}{2}\right)$$

$$b = \sin 2\alpha + \sin 2\beta + 2\sin(\alpha + \beta)$$

$$= 2\sin(\alpha + \beta)\cos(\alpha - \beta) + 2\sin(\alpha + \beta)$$

$$= 2\sin(\alpha + \beta) \cdot 2\cos^2\left(\frac{\alpha - \beta}{2}\right)$$

$$= 4\sin(\alpha + \beta)\cos^2\left(\frac{\alpha - \beta}{2}\right)$$

$$\therefore a^2 + b^2 = 16\cos^2(\alpha + \beta)\cos^4\left(\frac{\alpha - \beta}{2}\right) + 16\sin^2(\alpha + \beta)\cos^4\left(\frac{\alpha - \beta}{2}\right)$$

$$= 16\cos^4\left(\frac{\alpha - \beta}{2}\right)$$

$$A = 4\cos^2\left(\frac{\alpha - \beta}{2}\right)$$

22. Let a and b be real numbers such that

$$\sin a + \sin b = \frac{1}{\sqrt{2}}, \cos a + \cos b = \sqrt{\frac{3}{2}}, \text{ then}$$

$$(1) \cos(a-b) = 1 \quad (2) \cos(a-b) = 0 \quad (3) \sin(a+b) = \frac{\sqrt{3}}{2} \quad (4) \sin(a-b) = 1$$

Sol. Answer (2, 3, 4)

$$1 + 1 + 2\cos(a-b) = \frac{1}{2} + \frac{3}{2} = \frac{4}{2} = 2$$

$$\Rightarrow \cos(a-b) = 0$$

$$\therefore \sin(a-b) = \sqrt{1-0} = 1$$

23. If $\cos \alpha + \cos \beta + \cos \gamma = 0 = \sin \alpha + \sin \beta + \sin \gamma$, then

$$(1) \sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = \frac{3}{2}$$

$$(2) \sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = \frac{3}{4}$$

$$(3) \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = \frac{3}{2} \quad (4)$$

$$\cos 2\alpha + \cos 2\beta + \cos 2\gamma = -1$$

Sol. Answer (1, 3)

$$\text{Let } a = \cos \alpha + i \sin \alpha, b = \cos \beta + i \sin \beta, c = \cos \gamma + i \sin \gamma$$

$$\therefore a + b + c = 0$$

$$\text{Also, } \frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0$$

$$\Rightarrow ab + bc + ca = 0$$

$$\therefore a^2 + b^2 + c^2 = 0$$

$$\Rightarrow \cos 2\alpha + \cos 2\beta + \cos 2\gamma = 0 = \sin 2\alpha + \sin 2\beta + \sin 2\gamma$$

$$\therefore 2\cos^2 \alpha - 1 + 2\cos^2 \beta - 1 + 2\cos^2 \gamma - 1 = 0$$

$$\therefore 2\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = \frac{3}{2}$$

$$1 - 2\sin^2 \alpha + 1 - 2\sin^2 \beta + 1 - 2\sin^2 \gamma = 0$$

$$\Rightarrow \sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = \frac{3}{2}$$

[Trigonometric Equations]

24. If $4\cos \theta - 3\sec \theta = 2\tan \theta$, then θ equals to

$$(1) n\pi + (-1)^n \left(\frac{\pi}{10} \right)$$

$$(2) n\pi + (-1)^n \frac{\pi}{6}$$

$$(3) n\pi - (-1)^n \frac{3\pi}{10}$$

$$(4) n\pi$$

Sol. Answer (1, 3)

$$4 \cos \theta - 3 \sec \theta = 2 \tan \theta$$

$$4 \cos^2 \theta - 3 = 2 \sin \theta, \cos \theta \neq 0$$

$$4 \sin^2 \theta + 2 \sin \theta - 1 = 0$$

$$\sin \theta = \frac{-1 + \sqrt{5}}{2 \times 4}$$

$$\sin \theta = -1 + \frac{\sqrt{5}}{4}, \sin \theta = \frac{-1 - \sqrt{5}}{4}$$

$$\theta = n\pi(-1)^n \frac{\pi}{10} \text{ or } \theta = n\pi - (-1)^n \frac{3\pi}{10}$$

25. If $\alpha \in [-2\pi, 2\pi]$ and

$$\cos\left(\frac{\alpha}{2}\right) + \sin\left(\frac{\alpha}{2}\right) = \sqrt{2}(\cos 36^\circ - \sin 18^\circ), \text{ then the value of } \alpha \text{ is}$$

(1) $\frac{7\pi}{6}$

(2) $\frac{\pi}{6}$

(3) $-\frac{5\pi}{6}$

(4) $-\frac{\pi}{6}$

Sol. Answer (1, 4)

$$\cos \frac{\alpha}{2} + \sin \frac{\alpha}{2} = \sqrt{2} (\cos 36^\circ - \sin 18^\circ), \alpha \in [-2\pi, 2\pi]$$

$$\sin\left(\frac{\alpha}{2} + \frac{\pi}{4}\right) = \left(\frac{\sqrt{5}+1}{4}\right) - \left(\frac{\sqrt{5}-1}{4}\right) = \frac{1}{2}$$

$$\Rightarrow \frac{\alpha}{2} + \frac{\pi}{4} = n\pi + (-1)^n \frac{\pi}{6}$$

$$\alpha = 2n\pi + (-1)^n \cdot \frac{\pi}{3} - \frac{\pi}{2}$$

$$\Rightarrow \alpha = \frac{-\pi}{6}, \frac{7\pi}{6}$$

26. If $\sin \theta = a$ for exactly one value of $\theta \in \left[0, \frac{7\pi}{3}\right]$, then the value of a is

(1) $\frac{\sqrt{3}}{2}$

(2) 1

(3) Zero

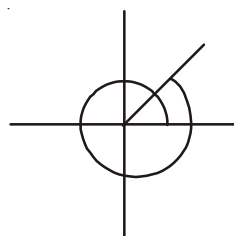
(4) -1

Sol. Answer (2, 4)

$$\sin \theta = a \text{ for exactly one value of } \theta \in \left[0, \frac{7\pi}{3}\right]$$

$\therefore$ in $[0, 2\pi]$ $\sin \theta$ takes every value twice except ± 1 .

$$\therefore a = \pm 1$$



27. If $\sin\theta + \sqrt{3}\cos\theta = 6y - y^2 - 11$, $\theta \in [0, 4\pi]$, $y \in \mathbb{R}$ holds for

- (1) No value of y and two values of θ (2) One value of y and two values of θ
 (3) Two values of y and one value of θ (4) Two pairs of (y, θ)

Sol. Answer (2, 4)

$$\sin\theta + \sqrt{3}\cos\theta = 6y - y^2 - 11$$

$$2\left(\sin\left(\theta + \frac{\pi}{3}\right)\right) = -(y^2 - 6y + 9) + 9 - 11$$

$$2\sin\left(\theta + \frac{\pi}{3}\right) = -(y - 3)^2 - 2$$

$$\text{LHS} \in [-2, 2],$$

$$\text{RHS} \geq -2$$

$\therefore$ only solution exists if $\text{LHS} = \text{RHS} = -2$

$$\sin\left(\theta + \frac{\pi}{3}\right) = -1 \quad \text{and} \quad (y - 3) = 0$$

$$\therefore \theta \in [0, 4\pi], y = 3$$

$$\Rightarrow \theta + \frac{\pi}{3} = \frac{3\pi}{2}, \frac{7\pi}{2}$$

$\therefore$ Two values of θ and one value of y .

28. The equation $\sin x + \sqrt{3}\cos x = \sqrt{2}$ is satisfied by value of x which is equal to

- (1) $x = n\pi - \frac{\pi}{12}$, $n \in \mathbb{I}$ (2) $x = 2n\pi + \frac{5\pi}{12}$, $n \in \mathbb{I}$ (3) $x = n\pi + \frac{5\pi}{12}$, $n \in \mathbb{I}$ (4) $x = 2n\pi - \frac{\pi}{12}$, $n \in \mathbb{I}$

Sol. Answer (2, 4)

$$\sin x + \sqrt{3}\cos x = \sqrt{2}$$

$$\text{Or, } \frac{\sin x}{2} + \frac{\sqrt{3}}{2}\cos x = \frac{1}{\sqrt{2}}$$

$$\text{Or, } \sin\left(x + \frac{\pi}{3}\right) = \sin\frac{\pi}{4}$$

$$\therefore x + \frac{\pi}{3} = n\pi + (-1)^n \frac{\pi}{4}$$

$$n = \text{even}; x + \frac{\pi}{3} = 2k\pi + \frac{\pi}{4}$$

$$\therefore x = 2k\pi + \left(\frac{\pi}{4} - \frac{\pi}{3}\right)$$

$$x = 2k\pi - \frac{\pi}{12}; k \in I$$

$$n = \text{odd}; x + \frac{\pi}{3} = (2k+1)\pi - \frac{\pi}{4}$$

$$\therefore x = 2k\pi + \pi - \frac{\pi}{4} - \frac{\pi}{3}$$

$$= 2k\pi + \frac{3\pi}{4} - \frac{\pi}{3}$$

$$= 2k\pi + \frac{5\pi}{12}; k \in I$$

29. The possible value of $\theta \in [-\pi, \pi]$ satisfying the equation $2(\cos\theta + \cos 2\theta) + (1 + 2\cos\theta)\sin 2\theta = 2\sin\theta$ are

(1) $-\frac{\pi}{2}$

(2) $-\frac{\pi}{3}$

(3) $\frac{\pi}{3}$

(4) π

Sol. Answer (1, 2, 3, 4)

$$\theta \in [-\pi, \pi]$$

$$2(\cos\theta + \cos 2\theta) + (1 + 2\cos\theta)\sin 2\theta = 2\sin\theta$$

$$\text{or, } 2(\cos\theta + \cos 2\theta) + 2(\cos\theta + 2\cos^2\theta)\sin\theta = 2\sin\theta$$

$$\text{or, } 2(\cos\theta + \cos 2\theta) + 2\sin[\cos\theta + 2\cos^2\theta - 1] = 0$$

$$\text{or, } 2(\cos\theta + \cos 2\theta) + 2\sin\theta(\cos\theta + \cos 2\theta) = 0$$

$$\text{or, } 2(\cos\theta + \cos 2\theta)(1 + \sin\theta) = 0$$

30. The number of all possible triplets (p, q, r) such that $p + q\cos 2\theta + r\sin^2\theta = 0$ for all θ , is

(1) $(k, -k, -2k)$

(2) $\left(\frac{k}{2}, -\frac{k}{2}, -k\right)$

(3) $\left(-\frac{k}{2}, \frac{k}{2}, k\right)$

(4) $(-k, k, 2k)$

Sol. Answer (1, 2, 3, 4)

$$p + q\cos 2\theta + r\sin^2\theta = 0 \quad \forall \theta \in R$$

$$\text{or, } p + q(1 - 2\sin^2\theta) + r\sin^2\theta = 0$$

$$\forall \theta \in R$$

$$\text{or, } (r - 2q)\sin^2\theta + (q + p) = 0 \quad \forall \theta \in R$$

$$\Rightarrow r - 2q = q + p = 0$$

$$\Rightarrow r = 2q, q = -p$$

$$\Rightarrow \frac{r}{2} = \frac{q}{1} = \frac{-p}{1}$$

$$\text{Or, } \frac{p}{-1} = \frac{q}{1} = \frac{r}{2}$$

31. The co-ordinates of points of intersection of the curves $y = \cos x$ and $y = \sin 3x$, $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ are

- (1) $\left(\frac{\pi}{4}, \frac{1}{\sqrt{2}}\right)$ (2) $\left(\frac{\pi}{8}, \cos \frac{\pi}{8}\right)$ (3) $\left(-\frac{3\pi}{8}, \cos \frac{3\pi}{8}\right)$ (4) $\left(\frac{3\pi}{8}, \cos \frac{3\pi}{8}\right)$

Sol. Answer (1, 2, 3)

$$\cos x = \sin 3x$$

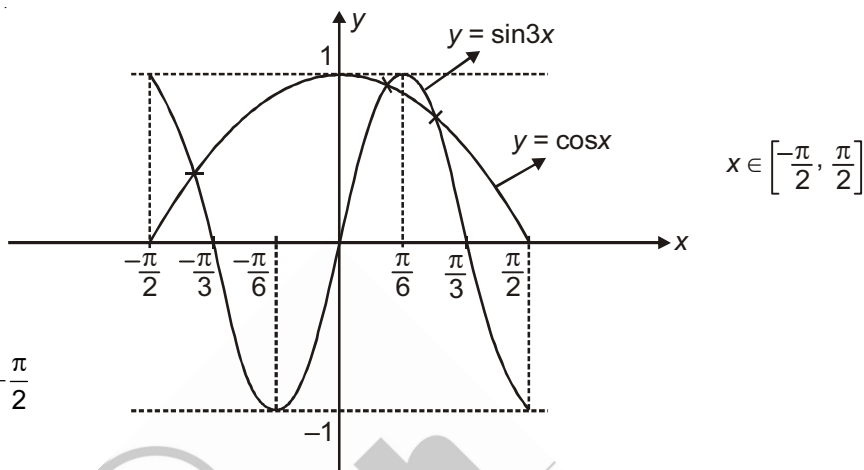
$$\text{or, } \cos x = \cos\left(\frac{\pi}{2} - 3x\right)$$

$$\therefore x = 2n\pi \pm \left(\frac{\pi}{2} - 3x\right)$$

$$\Rightarrow 4x = 2n\pi + \frac{\pi}{2} \text{ or, } -2x = 2n\pi - \frac{\pi}{2}$$

$$\therefore x = \frac{2n\pi}{4} + \frac{\pi}{8} \text{ or, } x = -n\pi + \frac{\pi}{4}$$

So, the values of x in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is $\frac{\pi}{4}, \frac{\pi}{8}$ and $-\frac{3\pi}{8}$



32. A value of θ satisfying $4\cos^2\theta\sin\theta - 2\sin^2\theta = 3\sin\theta$ is

- (1) $\frac{9\pi}{10}$ (2) $\frac{\pi}{10}$ (3) $-\frac{13\pi}{10}$ (4) $-\frac{17\pi}{10}$

Sol. Answer (1, 2)

$$\sin\theta[4\cos^2\theta - 2\sin\theta] = 3\sin\theta$$

$$\text{or, } \sin\theta[4\cos^2\theta - 2\sin\theta - 3] = 0$$

$$\text{or, } \sin\theta[-4\sin^2\theta - 2\sin\theta + 1] = 0$$

$$\text{or, } \sin\theta[4\sin^2\theta + 2\sin\theta - 1] = 0$$

$$\therefore \sin\theta = 0, \frac{\sqrt{5}-1}{4}, \frac{-\sqrt{5}-1}{4}$$

$$\sin\theta = 0$$

$$\therefore \theta = n\pi$$

$$\sin\theta = \frac{\sqrt{5}-1}{4}$$

$$\sin\theta = \sin \frac{\pi}{10}$$

$$\therefore \theta = n\pi + (-1)^n \left(\frac{\pi}{10}\right)$$

$$\sin\theta = -\frac{\sqrt{5}+1}{4}$$

$$\sin\theta = -\sin \frac{3\pi}{10}$$

$$\therefore \theta = n\pi + (-1)^{n+1} \frac{3\pi}{10}$$

33. The equation $\sin x + \sin 2x + 2\sin x \sin 2x = 2\cos x + \cos 2x$ is satisfied by values of x for which

(1) $x = n\pi + (-1)^n \frac{\pi}{6}, n \in I$

(2) $x = 2n\pi + \frac{2\pi}{3}, n \in I$

(3) $x = 2n\pi - \frac{2\pi}{3}, n \in I$

(4) $x = 2n\pi - \frac{\pi}{2}, n \in I$

Sol. Answer (1, 2, 3, 4)

$$\sin x + \sin 2x + 2\sin x \sin 2x = 2\cos x + \cos 2x$$

$$\text{or, } (\sin x + \sin 2x) - \cos 2x + (2\sin x \sin 2x - 2\cos x) = 0$$

$$\text{or, } \sin x(1 + 2\cos x) - \cos 2x + 2\cos x(2\sin^2 x - 1) = 0$$

$$\text{or, } \sin x(1 + 2\cos x) - \cos 2x - 2\cos x(1 - 2\sin^2 x) = 0$$

$$\text{or, } \sin x(1 + 2\cos x) - \cos 2x - 2\cos x \cos 2x = 0$$

$$\text{or, } \sin x(1 + 2\cos x) - \cos 2x(1 + 2\cos x) = 0$$

$$\text{or, } (1 + 2\cos x)(\sin x - \cos 2x) = 0$$

$$\Rightarrow \text{Either, } \cos x = \frac{-1}{2} \quad \text{or, } \cos 2x = \sin x$$

$$\Rightarrow x = 2n\pi \pm \frac{2\pi}{3}$$

$$\text{or, } 1 - 2\sin^2 x = \sin x$$

$$\text{or, } 2\sin^2 x + \sin x - 1 = 0$$

$$\therefore \sin x = -1, \frac{1}{2}$$

$$\Rightarrow x = 2n\pi - \frac{\pi}{2}, n\pi + (-1)^n \frac{\pi}{6}, n \in I$$

So, finally the general solution is

$$x = 2n\pi \pm \frac{2\pi}{3}, 2n\pi - \frac{\pi}{2}, n\pi + (-1)^n \frac{\pi}{6}, n \in I$$

34. The solution set of $\sin 3\theta + \cos 2\theta = -2$ is

(1) $\left\{x \mid x = \frac{n\pi}{3} + (-1)^{n+1} \frac{\pi}{6}\right\} \cap \left\{x \mid x = n\pi \pm \frac{\pi}{2}\right\} \quad n \in I$

(2) $\left\{x \mid x = \frac{n\pi}{3} + (-1)^{n+1} \frac{\pi}{6}\right\} \cup \left\{x \mid x = n\pi \pm \frac{\pi}{2}\right\} \quad n \in I$

(3) $\left\{x \mid x = \frac{n\pi}{3} + (-1)^{n+1} \frac{\pi}{6}\right\} \quad n \in I$

(4) $\left\{x \mid x = n\pi \pm \frac{\pi}{2}\right\} \quad n \in I$

Sol. Answer (1, 2, 3, 4)

$$\sin 3\theta + \cos 2\theta = -2$$

$$\text{or, } 3\sin\theta - 4\sin^3\theta + 1 - 2\sin^2\theta = -2$$

$$\text{or, } -4\sin^3\theta - 2\sin^2\theta + 3\sin\theta + 3 = 0$$

$$\text{or, } 4\sin^3\theta + 2\sin^2\theta - 3\sin\theta - 3 = 0$$

$$\text{or, } 4\sin^3\theta - 4\sin^2\theta + 6\sin^2\theta - 6\sin\theta + 3\sin\theta - 3 = 0$$

$$\text{or, } 4\sin^2\theta(\sin\theta - 1) + 6\sin\theta(\sin\theta - 1) + 3(\sin\theta - 1) = 0$$

$$\therefore (\sin\theta - 1) [4\sin^2\theta + 6\sin\theta + 3] = 0$$

$$\therefore \sin\theta = 1 \quad (\because \text{Another equation will not give real solutions})$$

$$\Rightarrow \sin\theta = 1 \text{ is the only real solution}$$

$$\therefore \theta = 2n\pi + \frac{\pi}{2}; n \in I$$

35. The general solution of the equation $2^{\cos 2x} + 1 = 3 \cdot (2)^{-\sin^2 x}$ is

$$(1) \quad n\pi, n \in I$$

$$(2) \quad n\pi + \frac{\pi}{2}, n \in I$$

$$(3) \quad n\pi - \frac{\pi}{2}, n \in I$$

$$(4) \quad n\pi + \frac{\pi}{3}, n \in I$$

Sol. Answer (1, 2, 3)

$$2^{1-2\sin^2 x} + 1 = 3 \cdot 2^{-\sin^2 x}$$

$$\frac{2 + 2^{2\sin^2 x}}{2^{2\sin^2 x}} = \frac{3}{2^{\sin^2 x}}$$

$$2 + 2^{2\sin^2 x} = 3 \cdot 2^{\sin^2 x}$$

$$2^{\sin^2 x} = y$$

$$\Rightarrow y^2 - 3y + 2 = 0$$

$$\Rightarrow (y - 1)(y - 2) = 0$$

$$y = 2^{\sin^2 x} = 1 \text{ or } 2^{\sin^2 x} = 2$$

$$\Rightarrow \sin^2 x = 0 \text{ or } \sin^2 x = 1$$

$$x = n\pi, x = n\pi \pm \frac{\pi}{2}$$

36. $(m+2)\sin\theta + (2m-1)\cos\theta = 2m+1$, if

$$(1) \quad \tan\theta = \frac{3}{4}$$

$$(2) \quad \tan\theta = \frac{4}{3}$$

$$(3) \quad \tan\theta = \frac{2m}{m^2 - 1}$$

$$(4) \quad \tan\theta = \frac{2m}{m^2 + 1}$$

Sol. Answer (2, 3)

$$(m+2)\sin\theta + (2m-1)\cos\theta = 2m+1$$

$$\Rightarrow (m+2)\tan\theta + (2m-1) = (2m+1)\sec\theta$$

$$\Rightarrow (m+2)^2\tan^2\theta + (2m-1)^2 + 2(m+2)(2m-1)\tan\theta = (2m+1)^2(1+\tan^2\theta)$$

$$\Rightarrow (3m^2 - 3)\tan^2\theta - 2(2m^2 + 3m - 2)\tan\theta + 8m = 0$$

$$3(m^2 - 1)\tan^2\theta - \{4(m^2 - 1) + 6m\}\tan\theta + 8m = 0$$

$$\Rightarrow (m^2 - 1)\tan\theta \cdot (3\tan\theta - 4) - 2m(3\tan\theta - 4) = 0$$

$$\tan\theta = \frac{4}{3} \text{ or } \tan\theta = \frac{2m}{m^2 - 1}$$

[Properties of Triangle]

37. In a triangle ABC , $\frac{a}{b} = 2 + \sqrt{3}$, $\angle C = 60^\circ$ then in the triangle

- (1) One angle is 105° (2) One angle is four times another angle
 (3) One angle is 25° (4) One angle is five times another angle

Sol. Answer (1, 2)

In $\triangle ABC$

Using $\sin\theta$ formula $\frac{\sin A}{\sin B} = \frac{a}{b} = \frac{2 + \sqrt{3}}{1}$

Apply components and dividends $\frac{a+b}{a-b} = \frac{3 + \sqrt{3}}{1 + \sqrt{3}}$

$$\Rightarrow \frac{\sin A + \sin B}{\sin A - \sin B} = \frac{3 + \sqrt{3}}{1 + \sqrt{3}}$$

$$\Rightarrow \frac{2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}}{2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}} = \frac{3 + \sqrt{3}}{1 + \sqrt{3}}$$

$$\Rightarrow \cot \frac{A-B}{2} = 1$$

$$\therefore A - B = 90^\circ$$

$$\text{and } A + B = 120^\circ$$

$$\angle A = 105^\circ$$

$$\therefore \angle B = 15^\circ$$

$\therefore$ Option (1) and (2) are correct but option (3) & (4) are not correct.

38. In a triangle ABC

$$(1) \sin A \cdot \sin B \cdot \sin C = \frac{\Delta}{2R^2}$$

$$(2) \sin A \cdot \sin B \cdot \sin C = \frac{r}{2R} (\sin A + \sin B + \sin C)$$

$$(3) a \cos A + b \cos B + c \cos C = \frac{abc}{2R^2}$$

$$(4) \sin A \cdot \sin B \cdot \sin C = \frac{R}{2r} (\sin A + \sin B + \sin C)$$

Sol. Answer (1, 2)

$$(1) \sin A \cdot \sin B \cdot \sin C$$

$$= \frac{2\Delta}{bc} \cdot \frac{2\Delta}{ac} \cdot \frac{2\Delta}{ab} \quad \Delta = \frac{1}{2} ab \sin C$$

$$= \frac{8\Delta^3}{a^2 b^2 c^2} \quad \Delta = \frac{abc}{4R}$$

$$= \frac{8\Delta^3}{16\Delta^2 R^2}$$

$$= \frac{\Delta}{2R^2} \text{ So option (1) is correct}$$

$$(2) \text{ RHS } \frac{r}{2R}(\sin A + \sin B + \sin C)$$

$$= \frac{r}{2R} \left(\frac{2\Delta}{bc} + \frac{2\Delta}{ac} + \frac{2\Delta}{ab} \right)$$

$$= \frac{r}{2R} \cdot 2\Delta \left(\frac{a+b+c}{abc} \right)$$

$$= \frac{r\Delta}{R} \cdot \frac{2s}{abc}$$

$$= \frac{2\Delta^2}{R \cdot abc}$$

$$= \frac{2\Delta^2}{R \cdot 4\Delta R}$$

$$= \frac{\Delta}{2R^2} = \text{LHS}$$

So option 2 is correct

(3) Since option (2) is correct $\therefore$ option 4 is not correct

Hence correct answers are (1, 2)

39. If $\cos A + \cos B = 4\sin^2\left(\frac{C}{2}\right)$, then

(1) $2\sin B = \sin A + \sin C$

(2) $\cos C = 1 - \frac{r}{R}$

(3) $\cos A + \cos B = \frac{2r}{R}$

(4) a, c, b are in G.P.

Sol. Answer (2, 3)

$$\cos A + \cos B = 4\sin^2 \frac{C}{2}$$

$$\Rightarrow 2\cos \frac{A+B}{2} \cos \frac{A-B}{2} = 4\sin^2 \frac{C}{2}$$

$$\Rightarrow \cos \frac{A-B}{2} = 2\sin \frac{C}{2} \quad \text{Multiply by } 2\cos \frac{C}{2} \text{ both sides}$$

$$\Rightarrow 2\cos \frac{C}{2} \cos \frac{A-B}{2} = 2\sin C$$

$$\Rightarrow \sin A + \sin B = 2\sin C$$

$\therefore a, b, c$ are in A.P.

$\therefore$ Option 1 is not correct

$$\text{So we know that } \cos A + \cos B + \cos C = 1 + \frac{r}{R}$$

$$\therefore 4 \sin^2 \frac{C}{2} + \cos C = 1 + \frac{r}{R}$$

$$(1 - \cos C) + \cos C = 1 + \frac{r}{R}$$

$$1 - \frac{r}{R} = \cos C$$

$\therefore$ Option (2) is correct

$$\text{Again } \cos A + \cos B + \cos C = 1 + \frac{r}{R}$$

$$\text{Put } \cos C = 1 + \frac{r}{R}$$

$$\therefore \cos A + \cos B + 1 - \frac{r}{R} = 1 + \frac{r}{R}$$

$$\therefore \cos A + \cos B = \frac{2r}{R}$$

$\therefore$ Option (3) is correct

As a, c, b are in A.P. and all three are not equal

$\therefore a, c, b$ are not in G.P

40. In a triangle ABC , point D and E are taken on side BC such that $BD = DE = EC$. If angle $ADE =$ angle $AED = \theta$, then

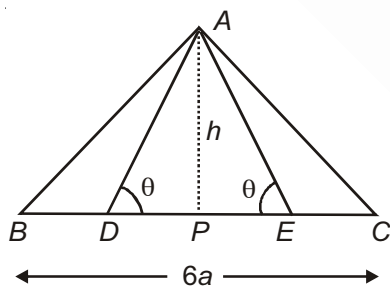
(1) $\tan \theta = 3 \tan B$

(2) $3 \tan \theta = \tan C$

(3) $\frac{6 \tan \theta}{\tan^2 \theta - 9} = \tan A$

(4) $\angle B = \angle C$

Sol. Answer (1, 3, 4)



Let $BC = 6a$

$$\therefore BD = DE = EC = 2a$$

$$\text{And } DP = PE = a$$

$$\tan B = \frac{h}{3a}$$

$$\tan \theta = \frac{h}{a}$$

$$\therefore \tan \theta = 3 \tan B$$

$\therefore$ Option (1) is correct

By figure it can be seen that $\angle B = \angle C$

$\therefore$ Option (4) is correct

And option (2) is incorrect

$$\text{Now option (3); LHS} \Rightarrow \frac{6 \tan \theta}{\tan^2 \theta - 9} = \frac{6 \frac{h}{a}}{\left(\frac{h}{a}\right)^2 - 9} = \frac{6ah}{h^2 - 9a^2}$$

$$\text{L.H.S. } \tan A = -\tan(B + C)$$

$$= \frac{[\tan B + \tan C]}{1 - \tan B \tan C}$$

$$= \frac{\tan B + \tan C}{\tan B \tan C - 1} = \frac{\frac{h}{3a} + \frac{h}{3a}}{\frac{h^2}{9a^2} - 1} = \frac{6ah}{h^2 - 9a^2}$$

$\therefore$ Option (3) is correct

41. In a triangle, with usual notations, the length of the bisector of angle A is

(1) $\frac{2bc \cos \frac{A}{2}}{b+c}$

(2) $\frac{2bc \sin \frac{A}{2}}{b+c}$

(3) $\frac{abc \operatorname{cosec} \frac{A}{2}}{2R(b+c)}$

(4) $\frac{2\Delta}{b+c} \operatorname{cosec} \frac{A}{2}$

Sol. Answer (1, 3, 4)

Let $AD = l$

$$\text{Ar } \triangle ABD + \text{Ar } \triangle ADC = \text{Ar } \triangle ABC$$

$$\frac{1}{2}cl \sin \theta + \frac{1}{2}bl \sin \theta = \frac{1}{2}bc \sin 2\theta$$

$$\frac{1}{2}l \sin \theta [b+c] = \frac{1}{2}bc 2 \sin \theta \cos \theta$$

$$l = \frac{2bc \cos \theta}{b+c} = \frac{2bc \cos A/2}{b+c}$$

So option (1) is correct and option (2) is incorrect

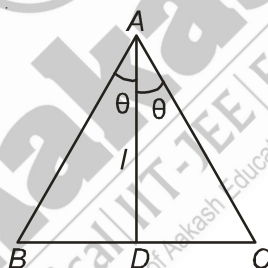
For option (3)

$$= \frac{a}{a} \frac{2bc \cos A/2}{b+c} = \frac{2abc \cos A/2}{(b+c) \cdot 2R \sin A} = \frac{2abc \cos A/2}{(b+c) \cdot 2R \cdot 2 \sin A/2 \cos A/2}$$

$$= \frac{abc \operatorname{cosec} A/2}{2R(b+c)}$$

For option (4)

$$\frac{abc \operatorname{cosec} A/2}{2R(b+c)} = \frac{4R \Delta \operatorname{cosec} A/2}{2R(b+c)} = \frac{2\Delta \operatorname{cosec} A/2}{b+c}$$



42. An ordered triplet solution (x, y, z) with $x, y, z \in (0, 1)$ and satisfying $x^2 + y^2 + z^2 + 2xyz = 1$ is

(1) $\left(\cos \frac{\pi}{6}, \cos \frac{7\pi}{18}, \cos \frac{4\pi}{9}\right)$

(2) $\left(\cos \frac{2\pi}{5}, \cos \frac{\pi}{3}, \cos \frac{\pi}{10}\right)$

(3) $\left(\cos \frac{7\pi}{12}, \cos \frac{\pi}{4}, \cos \frac{\pi}{6}\right)$

(4) $\left(\cos \frac{\pi}{12}, \cos \frac{4\pi}{9}, \cos \frac{17\pi}{36}\right)$

Sol. Answer (1, 4)

Let $A = \cos^{-1} x, B = \cos^{-1} y, C = \cos^{-1} z$, where $A + B + C = \pi$

$$A + B + C = \pi$$

$$\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = \pi$$

$$\cos^{-1} x + \cos^{-1} y = \pi - \cos^{-1} z$$

$$\cos^{-1}[xy - \sqrt{1-x^2}\sqrt{1-y^2}] = \cos^{-1}(-z)$$

$$xy + z = \sqrt{1-x^2}\sqrt{1-y^2}$$

$$(xy + z^2) = (1-x^2)(1-y^2)$$

$$x^2y^2 + z^2 + 2xyz = 1 - x^2 - y^2 + x^2y^2$$

$$x^2y^2 + z^2 + 2xyz = 1 \text{ which is the given identity}$$

$$\therefore A + B + C = \pi$$

which is satisfied by options (1) and (4) and A, B, C should be acute.

43. In a $\triangle ABC$, $a \cos B + b \cos C + c \cos A = \frac{a+b+c}{2}$, then

(1) Triangle is isosceles

(2) Triangle may be equilateral

(3) $\sin(A-B) + \sin(B-C) + \sin(C-A) = \frac{3}{2}$

(4) $4 \sin\left(\frac{A-B}{2}\right) \sin\left(\frac{B-C}{2}\right) \sin\left(\frac{C-A}{2}\right) = 1$

Sol. Answer (1, 2)

In $\triangle ABC$

$$a \cos B + b \cos C + c \cos A = \frac{a+b+c}{2}$$

$$2R \sin A \cos B + 2R \sin B \cos C + 2R \sin C \cos A = \frac{2R \sin A + 2R \sin B + 2R \sin C}{2}$$

$$\sin(A+B) + \sin(A-B) + \sin(B+C) + \sin(B-C) + \sin(C+A) + \sin(C-A) = \sin A + \sin B + \sin C$$

$$\Rightarrow \sin(B+C) + \sin(C-A) + \sin(A+B) = 0$$

$$\Rightarrow 4 \sin \frac{B-A}{2} \sin \frac{A-C}{2} \sin \frac{C-B}{2} = 0$$

$$\therefore \sin \frac{B-A}{2} = 0 \text{ or } \sin \frac{A-C}{2} = 0 \text{ or } \sin \frac{C-B}{2} = 0$$

$$\therefore B = A \text{ or } A = C \text{ or } C = B$$

Option (1) is correct

Option (2) is correct (If $A = B = C$)

Option (3) is not correct

Option (4) is not correct

44. In a triangle ABC with fixed base BC , the vertex A moves such that

$$\cos B + \cos C = 4 \sin^2 \frac{A}{2}$$

If a , b and c denote the lengths of the sides of the triangle opposite to the angles A , B and C respectively, then
[IIT-JEE 2009]

(1) $b + c = 4a$

(2) $b + c = 2a$

(3) Locus of point A is an ellipse

(4) Locus of point A is a pair of straight lines

Sol. Answer (2, 3)

Given $\cos B + \cos C = 4 \sin^2 \frac{A}{2}$

$$\Rightarrow 2 \cos\left(\frac{B+C}{2}\right) \cdot \cos\left(\frac{B-C}{2}\right) = 4 \sin^2 \frac{A}{2}$$

$$\Rightarrow 2 \sin\left(\frac{A}{2}\right) \cdot \cos\left(\frac{B-C}{2}\right) = 4 \sin^2 \frac{A}{2}$$

$$\Rightarrow \cos\left(\frac{B-C}{2}\right) = 2 \sin \frac{A}{2}, \text{ as } \sin \frac{A}{2} \neq 0$$

$$\Rightarrow \frac{\cos\left(\frac{B-C}{2}\right)}{\cos\left(\frac{B+C}{2}\right)} = 2$$

$$\Rightarrow \frac{1 + \tan \frac{B}{2} \cdot \tan \frac{C}{2}}{1 - \tan \frac{B}{2} \cdot \tan \frac{C}{2}} = 2$$

$$\Rightarrow \frac{1 + \sqrt{\frac{(s-a)(s-c)}{s(s-b)}} \cdot \sqrt{\frac{(s-a)(s-b)}{s(s-c)}}}{1 - \sqrt{\frac{(s-a)(s-c)}{s(s-b)}} \cdot \sqrt{\frac{(s-a)(s-b)}{s(s-c)}}} = 2$$

$$\Rightarrow \frac{1 + \frac{(s-a)}{s}}{1 - \frac{s-a}{s}} = 2$$

$$\Rightarrow \frac{2s-a}{a} = 2$$

$$\Rightarrow \frac{b+c}{a} = 2$$

$$\boxed{b+c=2a}$$

and also $b + c = 2a$

$$\Rightarrow AC + AB = 2BC$$

$\Rightarrow AC + AB > BC$, which shows that the locus of point A is an ellipse.

45. If $\frac{\sin^4 x}{2} + \frac{\cos^4 x}{3} = \frac{1}{5}$, then

[IIT-JEE 2009]

(1) $\tan^2 x = \frac{2}{3}$

(2) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{1}{125}$

(3) $\tan^2 x = \frac{1}{3}$

(4) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{2}{125}$

Sol. Answer (1, 2)

We have,

$$\frac{\sin^4 x}{2} + \frac{\cos^4 x}{3} = \frac{1}{5}$$

$$\Rightarrow 3 \sin^4 x + 2 \cos^4 x = \frac{6}{5}$$

$$\Rightarrow 3(1 - \cos^2 x)^2 + 2 \cos^4 x = \frac{6}{5}$$

$$\Rightarrow 3(1 + \cos^4 x - 2 \cos^2 x) + 2 \cos^4 x = \frac{6}{5}$$

$$\Rightarrow 3 + 3 \cos^4 x - 6 \cos^2 x + 2 \cos^4 x = \frac{6}{5}$$

$$\Rightarrow 5 \cos^4 x - 6 \cos^2 x + 3 - \frac{6}{5} = 0$$

$$\Rightarrow 5 \cos^4 x - 6 \cos^2 x + \frac{9}{5} = 0$$

$$\Rightarrow 25 \cos^4 x - 30 \cos^2 x + 9 = 0$$

$$\Rightarrow (5 \cos^2 x - 3)^2 = 0$$

$$\Rightarrow \cos^2 x = \frac{3}{5}$$

$$\Rightarrow \sec^2 x = \frac{5}{3}$$

$$\tan^2 x = \sec^2 x - 1 = \frac{5}{3} - 1 = \frac{2}{3}$$

Hence option (1) is true.

$$\sin^2 x = 1 - \cos^2 x = 1 - \frac{3}{5} = \frac{2}{5}$$

$$\text{Now, } \frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{\left(\frac{2}{5}\right)^4}{8} + \frac{\left(\frac{3}{5}\right)^4}{27} = \frac{2}{5^4} + \frac{3}{5^4} = \frac{5}{5^4} = \frac{1}{125} \text{ option (2).}$$

46. For $0 < \theta < \frac{\pi}{2}$, the solution(s) of $\sum_{m=1}^6 \operatorname{cosec}\left(\theta + \frac{(m-1)\pi}{4}\right) \operatorname{cosec}\left(\theta + \frac{m\pi}{4}\right) = 4\sqrt{2}$ is(are)

[IIT-JEE 2009]

(1) $\frac{\pi}{4}$

(2) $\frac{\pi}{6}$

(3) $\frac{\pi}{12}$

(4) $\frac{5\pi}{12}$

Sol. Answer (3, 4)

We have

$$T_m = \operatorname{cosec}\left(\theta + \frac{m\pi - \pi}{4}\right) \operatorname{cosec}\left(\theta + \frac{m\pi}{4}\right)$$

$$= \frac{1}{\sin\left(\theta + \frac{m\pi}{4} - \frac{\pi}{4}\right) \sin\left(\theta + \frac{m\pi}{4}\right)}$$

$$= \frac{1}{\sin\frac{\pi}{4}} \left[\frac{\sin\left(\left(\theta + \frac{m\pi}{4}\right) - \left(\theta + \frac{m\pi}{4} - \frac{\pi}{4}\right)\right)}{\sin\left(\theta + \frac{m\pi}{4}\right) \sin\left(\theta + \frac{m\pi}{4} - \frac{\pi}{4}\right)} \right]$$

$$= \sqrt{2} \left[\frac{\sin\left(\theta + \frac{m\pi}{4}\right) \cos\left(\theta + \frac{m\pi}{4} - \frac{\pi}{4}\right) - \cos\left(\theta + \frac{m\pi}{4}\right) \sin\left(\theta + \frac{m\pi}{4} - \frac{\pi}{4}\right)}{\sin\left(\theta + \frac{m\pi}{4}\right) \sin\left(\theta + \frac{m\pi}{4} - \frac{\pi}{4}\right)} \right]$$

$$= \sqrt{2} \left[\cot\left(\theta + \frac{m\pi}{4} - \frac{\pi}{4}\right) - \cot\left(\theta + \frac{m\pi}{4}\right) \right]$$

$$\Rightarrow T_1 = \sqrt{2} \left(\cot(\theta) - \cot\left(\theta + \frac{\pi}{4}\right) \right)$$

$$T_2 = \sqrt{2} \left(\cot\left(\theta + \frac{\pi}{4}\right) - \cot\left(\theta + \frac{2\pi}{4}\right) \right)$$

$$T_6 = \sqrt{2} \left(\cot\left(\theta + \frac{5\pi}{4}\right) - \cot\left(\theta + \frac{6\pi}{4}\right) \right)$$

$$\text{Sum } T_1 + T_2 + T_3 + T_4 + T_5 + T_6 = 4\sqrt{2}$$

$$\Rightarrow \sqrt{2} \left[\cot\theta - \cot\left(\theta + \frac{6\pi}{4}\right) \right] = 4\sqrt{2}$$

$$\cot\theta + \tan\theta = 4$$

$$\Rightarrow \frac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta} = 4$$

$$\Rightarrow \sin \theta \cos \theta = \frac{1}{4}$$

$$2 \sin \theta \cos \theta = \frac{1}{2}$$

$$\sin 2\theta = \frac{1}{2}$$

$$\Rightarrow 2\theta = \frac{\pi}{6}, \frac{5\pi}{6} \Rightarrow \theta = \frac{\pi}{12}, \frac{5\pi}{12}$$

47. Let ABC be a triangle such that $\angle ACB = \frac{\pi}{6}$ and let a, b and c denote the lengths of the sides opposite to A, B and C respectively. The value(s) of x for which $a = x^2 + x + 1$, $b = x^2 - 1$ and $c = 2x + 1$ is/are [IIT-JEE 2010]

- (1) $-(2 + \sqrt{3})$ (2) $1 + \sqrt{3}$ (3) $2 + \sqrt{3}$ (4) $4\sqrt{3}$

Sol. Answer (2)

$$\cos C = \frac{b^2 + a^2 - c^2}{2ab}$$

$$\frac{\sqrt{3}}{2} = \frac{(x^2 - 1)^2 + (x^2 + x + 1)^2 - (2x + 1)^2}{2(x^2 - 1)(x^2 + x + 1)}$$

$$\Rightarrow (x + 2)(x + 1)x(x - 1) + (x^2 - 1)^2 = \sqrt{3}(x^2 + x + 1)(x^2 - 1)$$

$$\Rightarrow \sqrt{3}(x^2 + x + 1) = x^2 + 2x + x^2 - 1$$

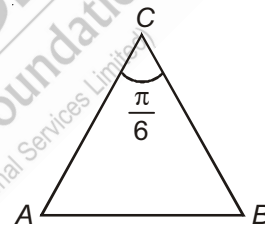
$$\Rightarrow (2 - \sqrt{3})x^2 + (2 - \sqrt{3})x - (\sqrt{3} + 1) = 0$$

$$\Rightarrow x = -(2 + \sqrt{3}) \text{ \& } x = 1 + \sqrt{3}$$

$$\text{But } x = -(2 + \sqrt{3})$$

shows that c is negative, which is not possible in a triangle.

Hence $x = 1 + \sqrt{3}$ is only possible value of x . $\Rightarrow x = 1 + \sqrt{3}$



48. Let $\theta, \phi \in [0, 2\pi]$ be such that

$$2\cos\theta(1 - \sin\phi) = \sin^2\theta \left(\tan\frac{\theta}{2} + \cot\frac{\theta}{2} \right) \cos\phi - 1, \tan(2\pi - \theta) > 0 \text{ and } -1 < \sin\theta < -\frac{\sqrt{3}}{2}.$$

Then ϕ cannot satisfy

[IIT-JEE 2012]

- (1) $0 < \phi < \frac{\pi}{2}$ (2) $\frac{\pi}{2} < \phi < \frac{4\pi}{3}$ (3) $\frac{4\pi}{3} < \phi < \frac{3\pi}{2}$ (4) $\frac{3\pi}{2} < \phi < 2\pi$

Sol. Answer (1, 3, 4)

$$2\cos\theta(1 - \sin\phi) = \sin^2\theta \left(\frac{1}{\sin\frac{\theta}{2} \cdot \cos\frac{\theta}{2}} \right) \cos\phi - 1$$

$$\Rightarrow 2\cos\theta(1 - \sin\phi) = 2\sin\theta \cdot \cos\phi - 1$$

$$\Rightarrow 2\cos\theta + 1 = 2\sin(\theta + \phi) \quad \dots(i)$$

Now, $\tan\theta < 0$ and $-1 < \sin\theta < \frac{-\sqrt{3}}{2}$ gives

$$\theta \in \left(\frac{3\pi}{2}, \frac{5\pi}{3}\right)$$

$$\text{Using (i) ; } \sin(\theta + \phi) = \cos\theta + \frac{1}{2}$$

$$\Rightarrow \frac{1}{2} < \sin(\theta + \phi) < 1$$

$$\text{i.e., } \frac{\pi}{6} < \theta + \phi < \frac{5\pi}{6} \text{ or, } \frac{13\pi}{6} < \theta + \phi < \frac{17\pi}{6}$$

$$\Rightarrow \frac{13\pi}{6} - \theta < \phi < \frac{17\pi}{6} - \theta$$

$$\Rightarrow \frac{2\pi}{3} < \phi < \frac{7\pi}{6}$$

So, options (1), (3) and (4) are correct.

49. In a triangle PQR , P is the largest angle and $\cos P = \frac{1}{3}$. Further the incircle of the triangle touches the sides PQ , QR and RP at N , L and M respectively, such that the lengths of PN , QL and RM are consecutive even integers. Then possible length(s) of the side(s) of the triangle is (are) **[JEE(Advanced)-2013]**

(1) 16

(2) 18

(3) 24

(4) 22

Sol. Answer (2, 4)

QR is largest side

$$PM = PN = 2k$$

$$RM = RL = 2k + 4$$

$$QL = QN = 2k + 2$$

$$\therefore QR = 4k + 6$$

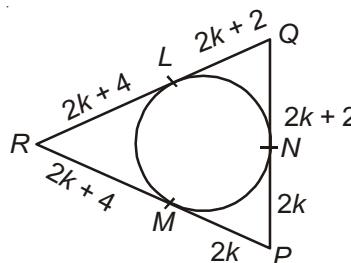
$$RP = 4k + 4$$

$$QP = 4k + 2$$

$$\cos P = \frac{(PQ)^2 + (PR)^2 - (QR)^2}{2(PQ)(PR)}$$

$$\frac{1}{3} = \frac{(4k+2)^2 + (4k+4)^2 - (4k+6)^2}{2(4k+2)(4k+4)}$$

$$\frac{1}{3} = \frac{(2k+1)^2 + 4(k+1)^2 - (2k+3)^2}{4(2k+1)(k+1)}$$



$$\frac{1}{3} = \frac{4(k+1)^2 - (4k+4)(2)}{4(2k+1)(k+1)}$$

$$\frac{1}{3} = \frac{(k+1)^2 - 2(k+1)}{(k+1)(2k+1)}$$

$$(k+1)(2k+1) = 3(k-1)(k+1)$$

$$k = -1 \text{ or } 2k+1 = 3k-3$$

$$4 = k$$

$$\text{So, } PQ = 18, QR = 22, RP = 20$$

50. If $\alpha = 3 \sin^{-1}\left(\frac{6}{11}\right)$ and $\beta = 3 \cos^{-1}\left(\frac{4}{9}\right)$, where the inverse trigonometric functions take only the principal values, then the correct option(s) is (are) [JEE(Advanced)-2015]

(1) $\cos \beta > 0$

(2) $\sin \beta < 0$

(3) $\cos(\alpha + \beta) > 0$

(4) $\cos \alpha < 0$

Sol. Answer (2, 3, 4)

$$\alpha = 3 \sin^{-1} \frac{6}{11}$$

$$\text{Since } \frac{\pi}{4} > \sin^{-1} \frac{6}{11} > \frac{\pi}{6}$$

$$\frac{3\pi}{4} > 3 \sin^{-1} \frac{6}{11} > \frac{\pi}{2}$$

$$\cos \alpha < 0$$

$$\cos\left(3 \cos^{-1} \frac{4}{9}\right) = 4 \times \left(\frac{4}{9}\right)^3 - 3 \times \frac{4}{9} = \frac{4 \times 64}{729} - \frac{12}{9} = \frac{256 - 12 \times 81}{72} < 0$$

$$\frac{5\pi}{12} > \cos^{-1} \frac{4}{9} > \frac{\pi}{3}$$

$$\frac{5\pi}{4} > 3 \cos^{-1} \frac{4}{9} > \pi$$

$$\sin\left(3 \cos^{-1} \frac{4}{9}\right) < 0$$

$$\frac{3\pi}{4} > 3 \sin^{-1} \frac{6}{11} > \frac{\pi}{2}$$

$$\frac{5\pi}{4} > 3 \cos^{-1} \frac{4}{9} > \pi$$

$$2\pi > 3 \sin^{-1} \frac{6}{11} + 3 \cos^{-1} \frac{4}{9} > \frac{3\pi}{2}$$

$$\cos\left(3 \sin^{-1} \frac{6}{11} + 3 \cos^{-1} \frac{4}{9}\right) > 0$$

51. In a triangle XYZ , let x, y, z be the lengths of sides opposite to the angles X, Y, Z , respectively, and $2s = x + y + z$.

If $\frac{s-x}{4} = \frac{s-y}{3} = \frac{s-z}{2}$ and area of incircle of the triangle XYZ is $\frac{8\pi}{3}$, then

[JEE(Advanced)-2016]

(1) Area of the triangle XYZ is $6\sqrt{6}$

(2) The radius of circumcircle of the triangle XYZ is $\frac{35}{6}\sqrt{6}$

(3) $\sin \frac{X}{2} \sin \frac{Y}{2} \sin \frac{Z}{2} = \frac{4}{35}$

(4) $\sin^2 \left(\frac{X+Y}{2} \right) = \frac{3}{5}$

Sol. Answer (1, 3, 4)

$$\frac{s-x}{4} = \frac{s-y}{3} = \frac{s-z}{2} = k$$

$$s-x = 4k \quad \dots(i)$$

$$s-y = 3k \quad \dots(ii)$$

$$s-z = 2k \quad \dots(iii)$$

$$3s - (x+y+z) = 9k$$

$$3s - 2s = 9k$$

$$s = 9k$$

From (i), (ii), (iii)

$$x = 5k, y = 6k, z = 7k$$

$$\text{Area of incircle} = \pi r^2 = \frac{8\pi}{3}$$

$$r^2 = \frac{8}{3}$$

$$\Rightarrow \frac{\Delta^2}{s^2} = \frac{8}{3}$$

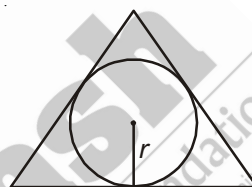
$$\Rightarrow \frac{s(s-x)(s-y)(s-z)}{s^2} = \frac{8}{3}$$

$$\frac{(9k)(4k)(3k)(2k)}{81k^2} = \frac{8}{3}$$

$$k^2 = 1$$

$$\Rightarrow k = +1$$

Now side length



$$x = 5, y = 6, z = 7 \text{ and } s = 9$$

$$(1) \text{ Area of triangle } XYZ = \sqrt{s(s-x)(s-y)(s-z)}$$

$$= \sqrt{9 \cdot 4 \cdot 3 \cdot 2}$$

$$= 3 \cdot 2\sqrt{6}$$

$$= 6\sqrt{6}$$

$$(2) \text{ Radius of circumcircle of } \triangle XYZ$$

$$R = \frac{xyz}{4\Delta} = \frac{5 \cdot 6 \cdot 7}{4 \cdot 6\sqrt{6}} = \frac{35}{4\sqrt{6}}$$

$$(3) \sin \frac{X}{2} \cdot \sin \frac{Y}{2} \cdot \sin \frac{Z}{2} = \sqrt{\frac{(s-x)(s-z)}{xz} \cdot \frac{(s-y)(s-z)}{yz} \cdot \frac{(s-x)(s-y)}{xy}}$$

$$= \frac{(s-x)(s-y)(s-z)}{xyz}$$

$$= \frac{4 \cdot 3 \cdot 2}{5 \cdot 6 \cdot 7} = \frac{4}{35}$$

$$(4) \sin^2 \left(\frac{X+Y}{2} \right) = \cos^2 \frac{Z}{2} = \frac{s(s-z)}{xy} = \frac{9 \cdot 2}{5 \cdot 6} = \frac{3}{5}$$

52. Let α and β be non-zero real numbers such that $2(\cos\beta - \cos\alpha) + \cos\alpha \cos\beta = 1$. Then which of the following is/are true? [JEE(Advanced)-2017]

$$(1) \tan\left(\frac{\alpha}{2}\right) + \sqrt{3} \tan\left(\frac{\beta}{2}\right) = 0$$

$$(2) \sqrt{3} \tan\left(\frac{\alpha}{2}\right) - \tan\left(\frac{\beta}{2}\right) = 0$$

$$(3) \tan\left(\frac{\alpha}{2}\right) - \sqrt{3} \tan\left(\frac{\beta}{2}\right) = 0$$

$$(4) \sqrt{3} \tan\left(\frac{\alpha}{2}\right) + \tan\left(\frac{\beta}{2}\right) = 0$$

Sol. Answer (1, 3)

$$\text{As } 2(\cos\beta - \cos\alpha) = 1 - \cos\alpha \cdot \cos\beta$$

$$\Rightarrow \cos\alpha = \frac{2\cos\beta - 1}{2 - \cos\beta}$$

Using componendo and dividendo

$$\Rightarrow \frac{1 - \cos\alpha}{1 + \cos\alpha} = 3 \left(\frac{1 - \cos\beta}{1 + \cos\beta} \right)$$

$$\Rightarrow \tan^2 \frac{\alpha}{2} - 3 \tan^2 \frac{\beta}{2} = 0$$

$$\text{So, } \tan \frac{\alpha}{2} + \sqrt{3} \tan \frac{\beta}{2} = 0$$

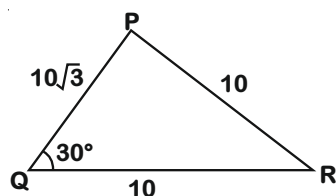
Or

$$\tan \frac{\alpha}{2} - \sqrt{3} \tan \frac{\beta}{2} = 0$$

53. In a triangle PQR , let $\angle PQR = 30^\circ$ and the sides PQ and QR have lengths $10\sqrt{3}$ and 10 , respectively. Then, which of the following statement(s) is (are) TRUE? [JEE(Advanced)-2018]

- (1) $\angle QPR = 45^\circ$
 (2) The area of the triangle PQR is $25\sqrt{3}$ and $\angle QRP = 120^\circ$
 (3) The radius of the incircle of the triangle PQR is $10\sqrt{3} - 15$
 (4) The area of the circumcircle of the triangle PQR is 100π

Sol. Answer (2, 3, 4)



$$\cos 30^\circ = \frac{(10\sqrt{3})^2 + 10^2 - PR^2}{2(10\sqrt{3})(10)}$$

$$\Rightarrow PR = 10$$

$$\angle QPR = 30^\circ$$

$$\text{Area of triangle } PQR = \frac{1}{2}(10)(10\sqrt{3})(\sin 30^\circ) = 25\sqrt{3}$$

$$\text{Also, } r = \frac{\Delta}{s} = \frac{25\sqrt{3}}{10 + 5\sqrt{3}} = 5\sqrt{3}(2 - \sqrt{3}) = 10\sqrt{3} - 15$$

$$R = \frac{abc}{4\Delta} = \frac{(10)(10)(10\sqrt{3})}{4(25\sqrt{3})} = 10$$

$$\text{Area of the circumcircle} = \pi(10)^2 = 100\pi$$

SECTION - C

Linked Comprehension Type Questions

Comprehension - I

$$\text{If } P_n = \cos^n \theta + \sin^n \theta,$$

1. $6P_{10} - 15P_8 + 10P_6$ is equal to

- (1) -1 (2) 0 (3) 1 (4) 8

2. Maximum value of P_{1000} will be

- (1) 2 (2) 1 (3) 4 (4) $\frac{1}{2}$

3. $\theta \in \left[0, \frac{\pi}{2}\right], n \in (-\infty, 2)$, then minimum of P_n will be

- (1) 1 (2) $\frac{1}{2}$ (3) $\sqrt{2}$ (4) $\frac{1}{\sqrt{2}}$

Solution of Comprehension - I

- Answer (3)
- Answer (2)
- Answer (1)

$$P_n = \cos^n \theta + \sin^n \theta$$

$$P_n = \cos^n \theta + \sin^n \theta$$

$$P_{n-2} = \cos^{n-2} \theta + \sin^{n-2} \theta$$

$$P_n - P_{n-2} = (\cos^n \theta + \sin^n \theta) - (\cos^{n-2} \theta + \sin^{n-2} \theta)$$

$$= \cos^{n-2} \theta \cdot (-\sin^2 \theta) + \sin^{n-2} \theta \cdot (-\cos^2 \theta)$$

$$= -\sin^2 \theta \cos^2 \theta (P_{n-4})$$

$$\begin{aligned} 1. \quad \therefore 6P_{10} - 15P_8 + 10P_6 \\ &= 6P_{10} - 6P_8 - 9P_8 + 9P_6 + P_6 - P_4 + P_4 - P_2 + P_2 \\ &= 6(-\sin^2 \theta \cos^2 \theta \cdot P_6) - 9(-\sin^2 \theta \cos^2 \theta \cdot P_4) + (-\sin^2 \theta \cos^2 \theta \cdot P_2) + (-\sin^2 \theta \cos^2 \theta \cdot P_0) + 1 \\ &= \sin^2 \theta \cos^2 \theta \cdot [-6P_6 + 9P_4 - P_2] - 2] + 1 \\ &= \sin^2 \theta \cos^2 \theta \cdot [-6(P_6 - P_4) + 3(P_4 - P_2) - 2P_2 - 2] + 1 \\ &= \sin^2 \theta \cos^2 \theta \cdot [-6 \cdot (-\sin^2 \theta \cos^2 \theta \cdot P_2) + 3(\sin^2 \theta \cos^2 \theta \cdot P_0) + 2 - 2] + 1 \\ &= \sin^4 \theta \cdot \cos^4 \theta [-6P_2 + 3P_0] + 1 \\ &= \sin^4 \theta \cdot \cos^4 \theta [-6 + 6] + 1 = 1 \end{aligned}$$

2. Maximum value of $P_{1000} = ?$

$$\cos^{1000} \theta + \sin^{1000} \theta$$

$$= \cos^2 \theta \cdot \cos^{1998} \theta + \sin^2 \theta \cdot \sin^{1998} \theta \leq \cos^2 \theta + \sin^2 \theta = 1$$

$$\therefore (P_{1000})_{\max} = 1$$

3. $P_n = \cos^n \theta + \sin^n \theta, \theta \in \left[0, \frac{\pi}{2}\right], n \in (-\infty, 2)$

Minimum value of $P_n = ?$

$$\sin^2 \theta \leq (\sin \theta)^n \quad \forall n \in (-\infty, 2) \& \theta \in \left(0, \frac{\pi}{2}\right)$$

$$\text{and } \cos^2 \theta \leq (\cos \theta)^4 \quad \forall n \in (-\infty, 2) \& \theta \in \left(0, \frac{\pi}{2}\right)$$

$$\therefore \sin^n \theta + \cos^n \theta \geq 1$$

$$\therefore (P_n)_{\min} = 1$$

Comprehension-II

From algebra we know that if $ax^2 + bx + c = 0$; $a(\neq 0)$, $b, c \in R$ has roots α and β then $\alpha + \beta = -\frac{b}{a}$ and $\alpha\beta = \frac{c}{a}$.

Trigonometric functions $\sin\theta$ and $\cos\theta$; $\tan\theta$ and $\sec\theta$; $\csc\theta$ and $\cot\theta$ obey $\sin^2\theta + \cos^2\theta = 1$. A linear relation in $\sin\theta$ and $\cos\theta$, $\sec\theta$ and $\tan\theta$ or $\csc\theta$ and $\cot\theta$ can be transformed into a quadratic equation in, say, $\sin\theta$, $\tan\theta$ or $\cot\theta$ respectively. And then one can apply sum and product of roots to find the desired result. Let $a\cos\theta + b\sin\theta = c$ have two roots θ_1 and θ_2 , $\theta_1 \neq \theta_2$.

1. The value of $\cos(\theta_1 + \theta_2)$ is (a and b not being simultaneously zero)

(1) $\frac{a^2 - b^2}{a^2 + b^2}$ (2) $\frac{b^2 - a^2}{a^2 + b^2}$ (3) $\frac{a^2 + c^2}{a^2 - b^2}$ (4) $\frac{c^2 - a^2}{a^2 - b^2}$

Sol. Answer (1)

$$a\cos\theta + b\sin\theta = c$$

$$(a\cos\theta - c)^2 = ab^2\sin^2\theta = b^2 - b^2\cos^2\theta$$

$$a^2\cos^2\theta - 2accos\theta + c^2 - b^2 + b^2\cos^2\theta = 0$$

$$(a^2 + b^2)\cos^2\theta - 2accos\theta + (c^2 - b^2) = 0$$

$$\cos\theta_1 + \cos\theta_2 = \frac{2ac}{a^2 + b^2}, \cos\theta_1 \cos\theta_2 = \frac{c^2 - b^2}{a^2 + b^2}$$

$$a^2\cos^2\theta = c^2 + b^2\sin^2\theta - 2bcsin\theta$$

$$a^2 - b^2\sin^2\theta - b^2\sin^2\theta + 2bcsin\theta - c^2 = 0$$

$$\Rightarrow (a^2 + b^2) - 2bcsin\theta + (c^2 - a^2) = 0$$

$$\therefore \sin\theta_1 + \sin\theta_2 = \frac{2bc}{a^2 + b^2}$$

$$\sin\theta_1 \sin\theta_2 = \frac{c^2 - a^2}{a^2 + b^2}$$

$$\therefore \cos(\theta_1 + \theta_2) = \cos\theta_1 \cos\theta_2 - \sin\theta_1 \sin\theta_2$$

$$= \frac{c^2 - b^2}{a^2 + b^2} - \frac{c^2 - a^2}{a^2 + b^2}$$

$$= \frac{a^2 - b^2}{a^2 + b^2}$$

2. The value of $\cos(\theta_1 - \theta_2)$ is (a and b not being simultaneously zero)

(1) $-1 - \frac{2c^2}{a^2 + b^2}$ (2) $\frac{2c^2}{a^2 + b^2} + 1$ (3) $1 - \frac{2c^2}{a^2 + b^2}$ (4) $\frac{2c^2}{a^2 + b^2} - 1$

Sol. Answer (4)

$$\cos(\theta_1 - \theta_2) = \cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2$$

$$= \frac{c^2 - b^2 + c^2 - a^2}{a^2 + b^2}$$

$$= \frac{2c^2}{a^2 + b^2} - 1$$

3. The values of $\tan\theta_1\tan\theta_2$ is (given $|b| \neq |c|$)

(1) $\frac{a^2 - c^2}{a^2 + b^2}$

(2) $\frac{a^2 + c^2}{a^2 + b^2}$

(3) $\frac{a^2 - c^2}{c^2 - b^2}$

(4) $\frac{c^2 - a^2}{c^2 - b^2}$

Sol. Answer (4)

$$\tan\theta_1\tan\theta_2 = \frac{c^2 - a^2}{c^2 + b^2} \times \frac{a^2 + b^2}{c^2 - b^2}$$

$$= \frac{c^2 - a^2}{c^2 - b^2}$$

Comprehension-III

In reducing a given trigonometric equation to the standard form ($\sin x = \sin \alpha$) or ($\cos x = \cos \alpha$) etc. we apply several trigonometric or Algebraic transformation. As a result of which final form so obtained may not be equivalent to the original equation resulting either in loss of solutions or appearance of extraneous solutions.

1. The solution set of the equation

$$\sqrt{5 - 2\sin x} = 6\sin x - 1 \text{ is}$$

(1) $n\pi + (-1)^n$ where $\sin\theta = \frac{-2}{9}$ or $x = n\pi + (-1)^n \frac{\pi}{6}, n \in I$

(2) $x = n\pi + (-1)^n \frac{\pi}{6}, n \in I$

(3) $x = n\pi + (-1)^n \theta$ where $\sin\theta = \frac{2}{9}$

(4) Null set

Sol. Answer (2)

$$\sqrt{5 - 2\sin x} = 6\sin x - 1$$

$$5 - 2\sin x = 36\sin^2 x - 12\sin x + 1$$

$$36\sin^2 x - 10\sin x - 4 = 0$$

$$18\sin^2 x - 5\sin x - 2 = 0$$

$$18\sin^2 x - 9\sin x + 4\sin x - 2 = 0$$

$$(9\sin x + 2)(2\sin x - 1) = 0$$

Putting $\sin x = \frac{1}{2}$, $\sin x = -\frac{2}{9}$

$\sin x = \frac{1}{2}$ in given equation

$$\text{LHS} = \sqrt{5 - 2 \times \frac{1}{2}} = 2$$

$$\text{RHS} = 6 \times \frac{1}{2} - 1 = 2$$

Putting $\sin x = -\frac{2}{9}$

$$\text{LHS} = \sqrt{5 + \frac{4}{9}} = \frac{7}{3}$$

$$\text{RHS} = 6 \times -\frac{2}{9} - 1 < 0$$

$\Rightarrow$ No solution

$\therefore \sin x = \frac{1}{2}$ is a solution

$$\therefore x = n\pi + (-1)^n \frac{\pi}{6}; n \in I$$

2. The equation $2\cot 2x - 3\cot 3x = \tan 2x$ has

(1) Two solutions in $\left(0, \frac{\pi}{3}\right)$

(3) No solution in $(-\infty, \infty)$

(2) One solution in $\left(0, \frac{\pi}{3}\right)$

(4) Three solution in $(0, \pi)$

Sol. Answer (3)

$$2\cot 2x - 3\cot 3x = \tan 2x$$

$$\Rightarrow 2(\cot 2x - \cot 3x) = \tan 2x + \cot 3x$$

$$\Rightarrow \frac{2(\cos 2x \sin 3x - \cos 3x \sin 2x)}{\sin 2x \sin 3x} = \frac{\sin 2x \sin 3x + \cos 2x \cos 3x}{\cos 2x \sin 3x}$$

$$\Rightarrow \frac{2 \sin x}{\sin 2x \sin 3x} = \frac{\cos x}{\cos 2x \sin 3x}$$

$$\Rightarrow 2 \tan x = \tan 2x$$

$$\Rightarrow 1 - \tan 2x = 1$$

$$\Rightarrow \tan 2x = 0$$

But $\tan x \neq 0$ as given equation involves terms of cotangent.

$\therefore$ No solution

3. The solution of $\sqrt{13 - 18 \tan x} = 6 \tan x - 3$ is

(1) $\tan x = -\frac{1}{6}$

(2) $\tan x = \frac{2}{3}$

(3) $\tan x \in \left(-\infty, -\frac{1}{6}\right) \cup \left(\frac{2}{3}, \infty\right)$

(4) $\tan x = \frac{3}{4}$

Sol. Answer (2)

$$\sqrt{13 - 18 \tan x} = 6 \tan x - 3$$

$$\Rightarrow 13 - 18 \tan x = 36 \tan^2 x - 36 \tan x + 9$$

$$\Rightarrow 36 \tan^2 x - 18 \tan x - 4 = 0$$

$$\Rightarrow 18 \tan^2 x - 9 \tan x - 2 = 0$$

$$18 \tan^2 x - 12 \tan x + 3 \tan x - 2 = 0$$

$$(6 \tan x + 1)(3 \tan x - 2) = 0$$

$$\tan x = -\frac{1}{6}, \tan x = \frac{2}{3}$$

Putting $\tan x = -\frac{1}{6}$ in given equation,

$$\text{L.H.S.} = \sqrt{13 + 3} = 4 \quad \text{R.H.S.} = -1 - 3 = -4$$

$\Rightarrow$ No solution

Putting $\tan x = \frac{2}{3}$ in given equation

$$\text{L.H.S.} = \sqrt{13 - 12} = 1$$

$$\text{R.H.S.} = 4 - 3 = 1$$

$\Rightarrow \tan x = \frac{2}{3}$ is a solution.

Comprehension-IV

There are trigonometrical equations that are nonstandard in aspect in that the equation may contain dissimilar terms, say algebraic as well as trigonometric expressions, or there may be only one equation in more than one unknowns. In such cases we make use of the extreme values of trigonometric functions or algebraic functions. In particular $|\sin x| \leq 1$, $|\cos x| \leq 1$, and maxima/minima of quadratic expressions are widely made use of.

1. The equation $\sin\left(\frac{\pi x}{6}\right) = x^2 - 6x + 10$ holds for

(1) Infinitely many values of x

(2) Finitely many values of x

(3) Just one value of x

(4) No value of x

Sol. Answer (3)

$$\sin\left(\frac{x\pi}{6}\right) = (x-3)^2 + 1 \geq 1, \forall x \in R$$

We know $\sin\left(\frac{\pi x}{6}\right) \leq 1$

Equality holds at $x = 3$.

Hence the given equation has only one solution.

2. The equation $8\cos^4 \frac{x}{2} \sin^2 \frac{x}{2} = x^2 + \frac{1}{x^2}, x \in (0, 4\pi]$ holds for

- (1) No value of x
- (2) Exactly two values of x , both greater than π
- (3) Exactly two values of x , one smaller than π and the other greater than π
- (4) Just one value of x

Sol. Answer (1)

$$2\cos^2 \frac{x}{2} \sin^2 x = x^2 + \frac{1}{x^2} \geq 2 \quad \forall x \in R, \text{ except } x = 0$$

$$\Rightarrow \cos^2 \frac{x}{2} \sin^2 x \geq 1 \text{ (which is not possible)}$$

Because $0 \leq \cos^2 \frac{x}{2} \leq 1$ and $0 \leq \sin^2 x \leq 1$

Hence $\cos^2 \frac{x}{2} \sin^2 x \nless 1$

But if $\cos^2 \frac{x}{2} \sin^2 x = 1$

Then $\cos^2 \frac{x}{2} = 1$ and $\sin^2 x = 1$

$$\Rightarrow \frac{x}{2} = n\pi \text{ and } x = k\pi \pm \frac{\pi}{2}, n, k \in Z, \text{ which is not possible simultaneously. Hence no solution exists.}$$

3. The equation $\operatorname{cosec} \frac{x}{2} + \operatorname{cosec} \frac{y}{2} + \operatorname{cosec} \frac{z}{2} = 6$, where $0 < x, y, z < \frac{\pi}{2}$ and $x + y + z = \pi$, have

- | | |
|---|---|
| (1) Three ordered triplet (x, y, z) solutions | (2) Two ordered triplet (x, y, z) solutions |
| (3) Just one ordered triplet (x, y, z) solution | (4) No ordered triplet (x, y, z) solution |

Sol. Answer (3)

$$\operatorname{cosec} \frac{x}{2} + \operatorname{cosec} \frac{y}{2} + \operatorname{cosec} \frac{z}{2} = 2 + 2 + 2$$

$$\Rightarrow x = y = z = \frac{\pi}{3}$$

Comprehension-V

Recall that $\sin x + \cos x = u$ (say)

and $\sin x \cos x = v$ (say) are connected by

$$(\sin x + \cos x)^2 = \sin^2 x + \cos^2 x + 2\sin x \cos x$$

$$\Rightarrow u^2 = 1 + 2v$$

$$\Rightarrow v = \frac{u^2 - 1}{2}$$

It follows that any rational integral function of $\sin x + \cos x$, and $\sin x \cos x$ i.e., $R(\sin x + \cos x, \sin x \cos x)$, or in our

notation $R(u, v)$ can be transformed to $R\left(u, \frac{u^2 - 1}{2}\right)$. Thus to solve an equation of the form $R(u, v) = 0$, we form

a polynomial equation in u and then look for solutions.

1. The solution set of $\sin x + \cos x - 2\sqrt{2} \sin x \cos x = 0$ is completely described by

$$(1) \quad x = 2n\pi + \frac{\pi}{4}, 2n\pi - \frac{5\pi}{12}, 2n\pi + \frac{11\pi}{12}, n \in \mathbb{Z}$$

$$(2) \quad x = 2n\pi - \frac{\pi}{4}, 2n\pi + \frac{\pi}{12}, 2n\pi + \frac{7\pi}{12}, n \in \mathbb{Z}$$

$$(3) \quad x = 2n\pi + \frac{\pi}{4}, 2n\pi - \frac{\pi}{12}, 2n\pi - \frac{7\pi}{12}, n \in \mathbb{Z}$$

$$(4) \quad x = 2n\pi - \frac{\pi}{4}, 2n\pi + \frac{\pi}{12}, 2n\pi - \frac{7\pi}{12}, n \in \mathbb{Z}$$

Sol. Answer (1)

$$\sin x + \cos x = 2\sqrt{2} \sin x \cos x$$

$$\Rightarrow u = 2\sqrt{2}v = 2\sqrt{2} \left(\frac{u^2 - 1}{2} \right) \Rightarrow \sqrt{2}u^2 - u - \sqrt{2} = 0$$

$$\Rightarrow u = \frac{1 \pm \sqrt{1+8}}{2\sqrt{2}} = \sqrt{2}, -\frac{1}{\sqrt{2}} \Rightarrow \sin x + \cos x = \sqrt{2}, -\frac{1}{\sqrt{2}}$$

$$\Rightarrow \sin\left(x + \frac{\pi}{4}\right) = -\frac{1}{2} = \sin\left(-\frac{\pi}{6}\right)$$

$$\Rightarrow x + \sin \frac{\pi}{4} = n\pi + (-1)^{n-1} \cdot \frac{\pi}{6}$$

$$\text{and } \sin\left(x + \frac{\pi}{4}\right) = 1 = \sin \frac{\pi}{2}$$

$$\Rightarrow x + \frac{\pi}{4} = n\pi + (-1)^n \frac{\pi}{2}$$

$$\Rightarrow x = n\pi - \frac{\pi}{4} + (-1)^n \frac{\pi}{2} = n\pi + \frac{\pi}{4} \text{ for } n \text{ is even or odd.}$$

2. The complete solution of the equation

$\sin 2x - 12(\sin x - \cos x) + 12 = 0$ is given by

$$(1) \quad x = 2n\pi + \frac{\pi}{2}, (2n-1)\frac{\pi}{4}, n \in \mathbb{Z}$$

$$(2) \quad x = n\pi + \frac{\pi}{2}, (2n+1)\pi, n \in \mathbb{Z}$$

$$(3) \quad x = 2n\pi + \frac{\pi}{2}, (2n+1)\pi, n \in \mathbb{Z}$$

$$(4) \quad x = n\pi + \frac{\pi}{2}, (2n-1)\pi, n \in \mathbb{Z}$$

Sol. Answer (3)

$$\text{Let } u = \cos x - \sin x, v = \sin x \cos x = \frac{1-u^2}{2}$$

Thus the given equation reduces to $2v + 12u + 12 = 0$

$$\Rightarrow 1 - u^2 + 12u + 12 = 0$$

$$\Rightarrow u^2 - 12u - 13 = 0$$

$$\Rightarrow (u+1)(u-13) = 0$$

$$\Rightarrow \cos x - \sin x = -1 \text{ as } u \neq 13$$

$$\Rightarrow \cos \left(x + \frac{\pi}{4} \right) = \cos \frac{3\pi}{4}$$

$$\Rightarrow x + \frac{\pi}{4} = 2n\pi \pm \frac{3\pi}{4}$$

$$\Rightarrow x = 2n\pi + \frac{\pi}{2}, (2n-1)\pi, n \in \mathbb{Z} \text{ (by taking positive and negative respectively)}$$

3. The number of solutions of the equation $\sin \theta + \cos \theta = 1 + \sin \theta \cos \theta$ in the interval $[0, 4\pi]$ is

(1) Four

(2) Six

(3) Eight

(4) Five

Sol. Answer (4)

$$\text{Let } y = \sin x + \cos x$$

Then the given equation can be reduced to

$$y = 1 + \frac{y^2 - 1}{2}$$

$$\Rightarrow y^2 - 2y + 1 = 0$$

$$\Rightarrow y = 1$$

$$\Rightarrow \sin x + \cos x = 1 \quad \Rightarrow \sin \left(x + \frac{\pi}{4} \right) = \frac{1}{\sqrt{2}}$$

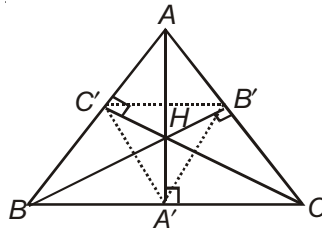
$$\Rightarrow x + \frac{\pi}{4} = \frac{\pi}{4}, \pi - \frac{\pi}{4}, 2\pi + \frac{\pi}{4}, 3\pi - \frac{\pi}{4}, 4\pi + \frac{\pi}{4}, 5\pi - \frac{\pi}{4}$$

$$\Rightarrow x = 0, \frac{\pi}{2}, 2\pi, \frac{5\pi}{2}, 4\pi, \frac{9\pi}{2}$$

Hence in the given interval $x = 0, \frac{\pi}{2}, 2\pi, \frac{5\pi}{2}, 4\pi$ five solution exists.

Comprehension-VI

Consider a triangle ABC , and that AA' , BB' , CC' be the perpendicular from A , B and C upon the sides opposite to them. These three perpendiculars meet in H , called the orthocentre of the triangle. The triangle $A'B'C'$ formed by the feet of the perpendicular is called the pedal triangle of ABC . (Assume $\angle A, B, C \neq 90^\circ$)



1. Suppose the triangle ABC have angles 60° , 70° and 50° . Then the pedal triangle $A'B'C'$ have angles given by
 (1) $80^\circ, 60^\circ, 40^\circ$ (2) $120^\circ, 40^\circ, 20^\circ$ (3) $30^\circ, 65^\circ, 85^\circ$ (4) $45^\circ, 55^\circ, 80^\circ$

Sol. Answer (1)

$$\angle A' = \pi - 2A = \pi - 120 = 60^\circ$$

$$\angle B' = \pi - 2B = \pi - 140 = 40^\circ$$

$$\angle C' = \pi - 2C = \pi - 100 = 80^\circ$$

2. Suppose a triangle ABC has its sides 13, 14 and 15 cm. Then the circumradius of the pedal triangle is (in cm)
 (1) $\frac{65}{24}$ (2) $\frac{65}{8}$ (3) $\frac{26}{3}$ (4) $\frac{65}{16}$

Sol. Answer (4)

$$\text{Given } a = 13, b = 14, c = 15$$

Let R is circumradius of $\triangle ABC$, then

$$R = \frac{abc}{4\Delta}$$

$$\text{Now, } 2s = 13 + 14 + 15 = 42$$

$$s = 21$$

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{21 \cdot 8 \cdot 7 \cdot 6} = 84$$

$$R = \frac{13 \cdot 14 \cdot 15}{4 \cdot 21 \cdot 4} = \frac{65}{8}$$

$$\text{Now, circumradius of pedal triangle} = \frac{R}{2} = \frac{65}{16}$$

3. Suppose ABC is an acute angled triangle, then the area of the pedal triangle is (R being the circum radius of triangle ABC)

(1) $\frac{R^2}{2} \sin 2A \sin 2B \sin 2C$

(2) $R^2 \sin 2A \sin 2B \sin 2C$

(3) $\frac{R^2}{2} \sin A \sin B \sin C$

(4) $R^2 \sin A \sin B \sin C$

Sol. Answer (1)

$$\text{Area of } \triangle DEF = \frac{DE \cdot EF \cdot FD}{4 \cdot R'}$$

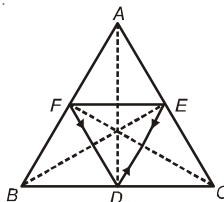
where R' is the circumradius of $\triangle DEF$

$$= \frac{(a \cos A)(b \cos B)(c \cos C)}{4 \cdot \left(\frac{R}{2}\right)}$$

$$= \frac{(2R \sin A \cos A)(2R \sin B \cos B)(2R \sin C \cos C)}{2R}$$

$$= \frac{R^3}{2R} \sin 2A \cdot \sin 2B \cdot \sin 2C$$

$$= \frac{1}{2} R^2 \sin 2A \cdot \sin 2B \cdot \sin 2C$$



Comprehension-VII

Let us consider a triangle ABC having $BC = 5$ cm, $CA = 4$ cm, $AB = 3$ cm D, E are points on BC such that $BD = DE = EC$, $\angle CAE = \theta$, then

1. AE^2 is equal to

(1) $\frac{73}{3}$

(2) $\frac{73}{5}$

(3) $\frac{73}{7}$

(4) $\frac{73}{9}$

Sol. Answer (4)

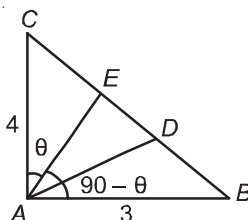
Given $BD = DE = EC = x$ (let)

And $\angle ACE = \theta$

In $\triangle ACE$, by using cosine rule

$$\cos \theta = \frac{AE^2 + 16 - CE^2}{2AE \cdot 4} \text{ and } CE = \frac{5}{3}$$

$$\cos \theta = \frac{AE^2 + 16 - \frac{25}{9}}{2 \cdot AE \cdot 4} \dots (i)$$



In $\triangle AEB$, by using sine rule

$$\frac{\sin(90^\circ - \theta)}{BE} = \frac{\sin B}{AE}$$

$$\cos \theta = \frac{BE}{AE} \sin B = \frac{10}{3 \cdot AE} \cdot \frac{4}{5} \quad \dots (ii)$$

From equation (i) and (ii)

$$\frac{10}{3 \cdot AE} \cdot \frac{4}{5} = \frac{AE^2 + 16 - \frac{25}{9}}{2 \cdot AE \cdot 4}$$

$$\frac{64}{3} = AE^2 + \frac{119}{9}$$

$$\therefore AE^2 = \frac{73}{9}$$

2. $\tan \theta$ is equal to

(1) $\frac{3}{4}$

(2) $\frac{1}{2}$

(3) $\frac{3}{8}$

(4) $\frac{5}{8}$

Sol. Answer (3)

From equation (ii)

$$\cos \theta = \frac{8}{3 \cdot AE}$$

$$\cos \theta = \frac{8}{3 \left(\frac{\sqrt{73}}{3} \right)} = \frac{8}{\sqrt{73}}$$

$$\therefore \tan \theta = \frac{3}{8}$$

3. AD^2 is equal to

(1) $\frac{52}{3}$

(2) $\frac{52}{9}$

(3) $\frac{52}{7}$

(4) 52

Sol. Answer (2)

In $\triangle ABD$, by using cosine rule

$$\cos B = \frac{BA^2 + BD^2 - AD^2}{2BA \cdot BD}$$

$$\frac{3}{5} = \frac{9 + \frac{25}{9} - AD^2}{2 \cdot 3 \cdot \frac{5}{3}}$$

$$6 = 9 + \frac{25}{9} - AD^2$$

$$\therefore AD^2 = \frac{25}{9} + 3$$

$$AD^2 = \frac{52}{9}$$

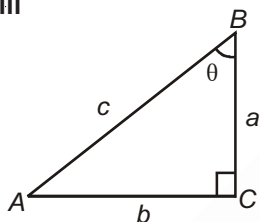
Comprehension-VIII

Given a right angled triangle ABC with right angled at C . $\angle C = 90^\circ$ and a, b, c are length of corresponding sides. ($b > a$).

Solution of Comprehension-VIII

$$a^2 + b^2 = c^2$$

$$\angle(A + B) = 90^\circ$$



1. $\cos(A - B)$ will be equal to

- (1) $\cos 2B$ (2) $\sin 2A$ (3) $\cos 2A$ (4) $\sin B$

Sol. Answer (2)

$$\begin{aligned}\therefore \cos(90^\circ - B - B) &= \cos(90^\circ - 2B) \\ &= \sin 2B\end{aligned}$$

$$\begin{aligned}\text{Also } \cos(A - (90^\circ - A)) &= \cos(2A - 90^\circ) \\ &= \sin(2A)\end{aligned}$$

2. $\sin 2A$ will be equal to

- (1) $\frac{2bc}{a^2}$ (2) $\frac{b^2 - a^2}{c^2}$ (3) $\frac{2ab}{c^2}$ (4) $\frac{b^2 - c^2}{2bc}$

Sol. Answer (3)

$$\therefore \sin 2A = 2 \sin A \cos A$$

$$= 2 \cdot \frac{a}{c} \times \frac{b}{c}$$

$$= \frac{2ab}{c^2}$$

$$c^2 - b^2 = a^2$$

3. $\tan \frac{A}{2}$ will be equal to

- (1) $\frac{2ab}{a^2 - b^2}$ (2) $\frac{2ab}{b^2 - a^2}$ (3) $\frac{c - b}{a}$ (4) $\frac{a}{c - b}$

Sol. Answer (3)

$$\begin{aligned}\tan^2 \frac{A}{2} &= \frac{1 - \cos A}{1 + \cos A} = \frac{1 - \frac{b}{c}}{1 + \frac{b}{c}} = \frac{c - b}{c + b} \\ &= \frac{(c - b)^2}{c^2 - b^2} = \frac{(c - b)^2}{a^2} \\ \Rightarrow \tan \frac{A}{2} &= \frac{c - b}{a}\end{aligned}$$

SECTION - D**Assertion-Reason Type Questions**

1. STATEMENT-1 : Minimum value of $\sin x + \operatorname{cosec} x$ is 2 for all $x \in (0, \pi)$.

andSTATEMENT-2 : $\operatorname{Min}(\sin x + \operatorname{cosec} x) = 2, \forall x \in R$.**Sol.** Answer (3)Statement (2): $\operatorname{Min}(\sin x + \operatorname{cosec} x) = 2, \forall x \in R$

False

 $\therefore (\sin x + \operatorname{cosec} x)_{\min} = 2$ If $\sin x, \operatorname{cosec} x$ are positiveStatement (1) minimum value of $(\sin x + \operatorname{cosec} x) = 2$ when $x \in R(0, \pi)$ is true

2. STATEMENT-1 : If $\cos(A + B)\sin(C + D) = \cos(A - B)\sin(C - D)$, then the value of $\cot A \cot B \cot C$ is $\cot D$.

andSTATEMENT-2 : If $\cos(A + B)\sin(C + D) = \cos(A - B)\sin(C - D)$, then the value of $\tan A \tan B \tan C$ is $\tan D$.**Sol.** Answer (1)

From Statement-2

$$\frac{\cos(A + B)}{\cos(A - B)} = \frac{\sin(C - D)}{\sin(C + D)}$$

$$\frac{1 - \tan A \tan B}{1 + \tan A \tan B} = \frac{\tan C - \tan D}{\tan C + \tan D}$$

$$\Rightarrow \tan A \cdot \tan B \cdot \tan C = \tan D$$

$$\Rightarrow \cot A \cdot \cot B \cdot \cot C = \cot D$$

3. STATEMENT-1 : If $\cos \theta = \frac{1}{7}$ and $\cos \phi = \frac{13}{14}$ where θ and ϕ both are acute angles, then the value of $\theta - \phi$ is

$$\frac{\pi}{3}$$

and

$$\text{STATEMENT-2 : } \cos \frac{\pi}{3} = \frac{1}{2}$$

Sol. Answer (4)

$$\cos(\theta - \phi) = \cos\theta \cdot \cos\phi + \sin\theta \cdot \sin\phi$$

$$= \frac{1}{7} \cdot \frac{13}{14} + \frac{\sqrt{48}}{7} \cdot \frac{\sqrt{27}}{14}$$

$$= \frac{13+36}{91} = \frac{49}{91}$$

4. STATEMENT-1 : The value of k for which $(\cos\theta - \sin\theta)^2 + k\sin\theta\cos\theta - 1 = 0$ is an identity is -2 .

andSTATEMENT-2 : An identity in θ is satisfied by all real values of θ .**Sol.** Answer (4)

$$\sin^2\theta + \cos^2\theta - 2\sin\theta \cdot \cos\theta + k\sin\theta \cdot \cos\theta - 1$$

$$(k-2)\sin\theta \cdot \cos\theta = 0 \Rightarrow k = 2$$

Statement-1 is false, Statement-2 is true.

5. STATEMENT-1 : If θ is acute and $1 + \cos\theta = k$, then $\sin\frac{\theta}{2}$ is $\sqrt{\frac{2-k}{2}}$.

and

$$\text{STATEMENT-2 : } 2\sin^2\frac{\theta}{2} = 1 - \cos\theta.$$

Sol. Answer (1)

$$1 + \cos\theta = k$$

$$1 + 1 - 2\sin^2\frac{\theta}{2} = k$$

$$\frac{2-k}{2} = \sin^2\frac{\theta}{2}$$

$$\Rightarrow \sin\frac{\theta}{2} = \sqrt{\frac{2-k}{2}}$$

Statement-1 is true, Statement-2 is true and Statement-2 is correct explanation of (1).

6. STATEMENT-1 : The general solution of $2^{\sin x} + 2^{\cos x} = 2^{1-\frac{1}{\sqrt{2}}}$ is $n\pi + \frac{\pi}{4}$.

andSTATEMENT-2 : A.M. $\geq$ G.M.**Sol.** Answer (1)

Statement-1

$$2^{\sin x} + 2^{\cos x} = 2^{1-\frac{1}{\sqrt{2}}}$$

$$\begin{aligned} \text{L.H.S.} &= 2^{\sin x} + 2^{\cos x} \geq 2\sqrt{2^{\sin x} 2^{\cos x}} \\ &\geq 2\sqrt{2^{\sin x + \cos x}} \end{aligned}$$

$$\therefore \text{LHS} \geq 2^{1-\frac{1}{\sqrt{2}}}$$

$$\text{RHS} = 2^{1-\frac{1}{\sqrt{2}}}$$

$$\therefore \text{LHS} = \text{RHS}$$

$$\therefore \text{Equality holds if } 2^{\sin x} = 2^{\cos x}$$

$$\Rightarrow \sin x = \cos x$$

$$\therefore x = n\pi + \frac{\pi}{4}$$

7. STATEMENT-1 : The equation $3\cos x + 4\sin x = 6$ has no solution.

and

STATEMENT-2 : Due to periodic nature of sine and cosine functions, the equation $3\cos x + 4\sin x = 6$ has infinitely many solutions.

Sol. Answer (3)

$$-5 \leq 3\cos x + 4\sin x \leq 5$$

$\therefore$ Given equation has no solution & statement 2 is false.

8. STATEMENT-1 : If $\sum_{r=1}^n \sin(x_r) = n$, then $\sum_{r=1}^n \cot(x_r) = n$.

and

STATEMENT-2 : The number of solutions of the equation $\cos x = x$ is 1.

Sol. Answer (4)

Statement (2) $\cos x = x$

Only one solution true

Statement-1

$$\text{If } \sum_{r=1}^n \sin x_r = n, \text{ then } \cot x_r = n$$

$$\sin x_1 + \sin x_1 + \dots + \sin x_n = n$$

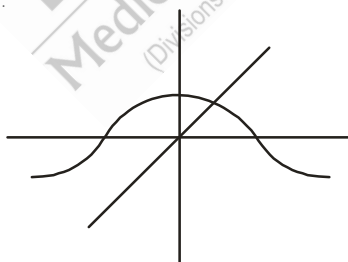
$$\Rightarrow \sin x_1 = \sin x_2 = \dots = \sin x_n = 1$$

$$\Rightarrow x_1 = x_2 = \dots = x_n = \frac{\pi}{2}$$

$$\text{i.e., } \cot(x_r) = 0 \quad \forall r \in [0, n]$$

$$\Rightarrow \cot x_1 + \cot x_2 + \dots + \cot x_n = 0$$

$\therefore$ Statement-1 is false.



9. STATEMENT-1 : The number of real solution of the equation $\sin x = 4^x + 4^{-x}$ is zero.

and

STATEMENT-2 : $|\sin x| \leq 1 \quad \forall x \in R$.

Sol. Answer (1)

Statement (2) : $|\sin x| \leq 1 \quad \forall x \in R$ is true.

Statement (1) : $\sin x + 4^x + 4^{-x}$.

L.H.S. $\in [-1, 1]$ (as from statement-2)

R.H.S. $\Rightarrow$ No solution

$\therefore$ Both statements are correct and statement -2 is the correct explanation of statement -1.

10. STATEMENT-1 : The general solution of $\tan 5\theta = \cot 2\theta$ is $\theta = \frac{n\pi}{7} + \frac{\pi}{14}, n \in Z$.

and

STATEMENT-2 : The equation $\tan \theta = k$ has exactly one solution in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ for all $k \in R$.

Sol. Answer (4)

$\theta = \frac{n+1}{7} + \frac{\pi}{4}$ does not hold good for $n = 3$.

Statement-1 is false.

Statement-2 is true.

11. STATEMENT-1 : $\sin x = \sin y \Rightarrow x = y$.

and

STATEMENT-2 : $\sin x = \sin y$ has infinitely many solutions for real values of x and y .

Sol. Answer (4)

Statement (2) is correct thus clearly

Statement (1) is wrong

$\sin x = \sin y$

$\Rightarrow (x = y, \pi - y, \dots)$

12. STATEMENT-1 : $f(x) = \log_{\cos x} \sin x$ is well defined in $\left(0, \frac{\pi}{2}\right)$.

and

STATEMENT-2 : $\sin x$ and $\cos x$ are positive in $\left(0, \frac{\pi}{2}\right)$.

Sol. Answer (2)

Statement (2) is correct as both $\sin x$ and $\cos x$ are positive in I quadrant

Statement (1)

$$f(x) = \log_{\cos x}(\sin x)$$

$$= \frac{\ln \sin x}{\ln \cos x}$$

Is defined if $\cos x > 0$ and $\cos x \neq 1$

$\therefore$ It is defined in $\left(0, \frac{\pi}{2}\right)$ (as given)

$\therefore$ Statement (1) is true but (2) is not correct explanation for (1)

13. STATEMENT-1 : If in $\triangle ABC$, $3bc = (a - b + c)(a + b - c)$ then $A = 120^\circ$.

and

$$\text{STATEMENT-2 : } \cos 120^\circ = -\frac{1}{2}$$

Sol. Answer (2)

$$3bc = (a - b + c)(a + b - c)$$

$$= [a - (b - c)][a + (b - c)]$$

$$\Rightarrow 3bc = a^2 - (b - c)^2$$

$$\Rightarrow 3bc = a^2 - b^2 - c^2 + 2bc$$

$$\Rightarrow b^2 + c^2 - a^2 = -bc$$

$$\Rightarrow \frac{b^2 + c^2 - a^2}{2bc} = -\frac{1}{2}$$

$$\Rightarrow \cos A = -\frac{1}{2}$$

$$A = 120^\circ$$

14. ABCD is a quadrilateral in which a circle is inscribed.

STATEMENT-1 : The length of the sides of the quadrilateral can be A.P.

and

STATEMENT-2 : The length of tangents from an external point to a circle are equal.

Sol. Answer (1)

If sum of opposite sides of a quadrilateral is equal, then and only then a circle can be inscribed in a quadrilateral

Statement-1 : Length of sides of quadrilateral can be in A.P. if common difference is zero.

Statement-2 : True.

15. STATEMENT-1 : In $\triangle ABC$, if $a < b \sin A$, then the triangle is possible.

and

$$\text{STATEMENT-2 : In } \triangle ABC \quad \frac{a}{\sin A} = \frac{b}{\sin B}$$

Sol. Answer (4)

In $\triangle ABC$

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

$$\sin A > \frac{a}{b}$$

$$\sin A > \frac{\sin A}{\sin B}$$

$$\boxed{\sin B > 1}$$

16. Let $ABCD$ be a cyclic quadrilateral then

$$\text{STATEMENT-1 : } \sin A + \sin B + \sin C + \sin D = 0$$

and

$$\text{STATEMENT-2 : } \cos A + \cos B + \cos C + \cos D = 0$$

Sol. Answer (4)

In cyclic quadrilateral $ABCD$

$$A + C = 180 \text{ and } B + D = 180$$

$$A = 180 - C \text{ and } B = 180 - D$$

$$\text{Then, } \sin A = \sin C \quad \text{and } \sin B = \sin D$$

$$\text{Or } \cos A = -\cos C \quad \text{and } \cos B = -\cos D$$

$$\text{Hence } \cos A + \cos B + \cos C + \cos D = 0$$

17. STATEMENT-1 : In a triangle ABC if $\tan A : \tan B : \tan C = 1 : 2 : 3$, then $A = 45^\circ$.

and

$$\text{STATEMENT-2 : If } p : q : r = 1 : 2 : 3, \text{ then } p = 1$$

Sol. Answer (3)

Given,

$$\tan A : \tan B : \tan C = 1 : 2 : 3$$

$$\tan A = k$$

$$\tan B = 2k$$

$$\tan C = 3k$$

In $\triangle ABC$

$$\tan A + \tan B + \tan C = \tan A \tan B \tan C$$

$$6K = 6K^3$$

$$K \neq 0 \Rightarrow K = \pm 1 \quad K \neq -1$$

$$\Rightarrow K = 1$$

$$\tan A = 1 \Rightarrow A = 45^\circ$$

18. STATEMENT-1 : In an acute angled triangle minimum value of $\tan\alpha + \tan\beta + \tan\gamma$ is $3\sqrt{3}$.

and

STATEMENT-2 : If a, b, c are three positive real numbers then $\frac{a+b+c}{3} \geq \sqrt[3]{abc}$ also in a $\triangle ABC$,

$$\tan A + \tan B + \tan C = \tan A \cdot \tan B \cdot \tan C.$$

Sol. Answer (1)

$$\tan\alpha + \tan\beta + \tan\gamma \geq (\tan\alpha \cdot \tan\beta \cdot \tan\gamma)^{1/3}$$

Equality will hold when $\tan\alpha = \tan\beta = \tan\gamma = \sqrt{3}$

SECTION - E

Matrix-Match Type Questions

1. Match the following

Column-I

(A) The minimum value of $\sec^2 x + \operatorname{cosec}^2 x - 4$ is

(B) The maximum value of $||\sin x| - 4| - 3$ is

(C) $10^{\log_{10} 3} + \log_{10} \tan 1^\circ + \log_{10} \tan 2^\circ + \dots + \log_{10} \tan 89^\circ$
equals

(D) If $\cos x + \cos^2 x = 1$, then the value of $4\sin^2 x(2 - \cos^2 x)$

Column-II

(p) 3

(q) 4

(r) 1

(s) 0

Sol. Answer A(s), B(r), C(p), D(q)

(A) $f(x) = \sec^2 x + \operatorname{cosec}^2 x - 4$

$$= \frac{\sin^2 x + \cos^2 x}{\sin^2 x \cos^2 x} - 4$$

$$= \frac{4}{\sin^2 2x} - 4$$

$$f_{\min} = 4 - 4 = 0$$

(B) $f(x) = ||\sin x| - 4| - 3$

$$f_{\max} = |0 - 4| - 3$$

$$= 1$$

(C) $10^{\log_{10} 3} + \log_{10} \tan 1^\circ + \log_{10} \tan 2^\circ + \dots + \log_{10} \tan 89^\circ$

$$= 3 + \log_{10} \{(\tan 1^\circ \cdot \tan 89^\circ) (\tan 2^\circ \cdot \tan 88^\circ) \dots (\tan 44^\circ \tan 46^\circ) \tan 45^\circ\}$$

$$= 3$$

(D) $\cos x + \cos^2 x = 1 \Rightarrow \cos x = \sin^2 x$

$$\therefore 4\sin^2 x \cdot (2 - \cos^2 x)$$

$$= 4\sin^2 x \cdot (1 + \sin^2 x) (\because \sin^2 x + \cos^2 x = 1)$$

$$= 4(\sin^2 x + \sin^4 x)$$

$$= 4(\sin^2 x + \cos^2 x)$$

$$= 4$$

2. Match the expression on column I with the expression on column II that is equal to it.

Column-I

(A) $\sqrt{\frac{1+\cos\theta}{1-\cos\theta}}$

(B) $\frac{1+\tan^2\theta}{1+\cot^2\theta}$

(C) $1+\frac{2\tan^2\theta}{\cos^2\theta}$

(D) $(1-\sin\theta-\cos\theta)^2$

Column-II

(p) $\tan^4\theta + \sec^4\theta$

(q) $\operatorname{cosec}\theta + \cot\theta$

(r) $\left(\frac{1-\tan\theta}{1-\cot\theta}\right)^2$

(s) $2(1-\sin\theta)(1-\cos\theta)$

Sol. Answer A(q), B(r), C(p), D(s)

(A) $\sqrt{\frac{1+\cos\theta}{1-\cos\theta}}$

$$= \sqrt{\frac{(1+\cos\theta)^2}{1-\cos^2\theta}}$$

$$= \left| \frac{1+\cos\theta}{\sin\theta} \right|$$

$$= |\operatorname{cosec}\theta + \cot\theta|$$

(B) $\frac{1+\tan^2\theta}{1+\cot^2\theta}$

$$= \frac{\sec^2\theta}{\operatorname{cosec}^2\theta}$$

$$= \tan^2\theta$$

$$\text{Also, } \left(\frac{1-\tan\theta}{1-\cot\theta}\right)^2 = \left\{ \frac{1-\tan\theta}{\tan\theta-1} \times \tan\theta \right\}^2$$

$$= \tan^2\theta$$

(C) $1+\frac{2\tan^2\theta}{\cot^2\theta}$

$$= (\sec^2\theta - \tan^2\theta)^2 + 2\tan^2\theta \cdot \sec^2\theta$$

$$= \sec^4\theta + \tan^4\theta$$

(D) $(1-\sin\theta-\cos\theta)^2$

$$= 2 - 2\sin\theta - 2\cos\theta + 2\sin\theta \cos\theta$$

$$= 2(1-\sin\theta)(1-\cos\theta)$$

3. Match the expression given in column I with its extreme values given in column II.

Column-I**Column-II**

- (A) If $\sin^2 x + \operatorname{cosec}^2 x = 2$, then $2\sin x + 3\operatorname{cosec} x$ is (p) 5
- (B) If $\sin x_1 + \sin x_2 + \sin x_3 + \sin x_4 + \sin x_5 = \pm 5$, then
 $\cos x_1 + \cos x_2 + \cos x_3 + \cos x_4 + \cos x_5$ is equal to (q) -5
- (C) If $\cos^2 x_1 + \cos^2 x_2 + \cos^2 x_3 + \cos^2 x_4 + \cos^2 x_5 = 5$, then
 $\sin x_1 + 2\sin x_2 + 3\sin x_3 + 4\sin x_4 + 5\sin x_5$ is
less than or equal to (r) 0
- (D) If $\cos^3 x_1 + \cos^3 x_2 + \cos^3 x_3 + \cos^3 x_4 + \cos^3 x_5 + 5 = 0$,
then $\sin^3 x_1 + 2^3 \sin^3 x_2 + 3^3 \sin^3 x_3 + 4^3 \sin^3 x_4 + 5^3 \sin^3 x_5$
is equal to (s) 15
- (t) 225

Sol. Answer A(p, q), B(r), C(p, r, s, t), D(r)

(A) $\sin^2 x + \operatorname{cosec}^2 x = 2 \Rightarrow \sin x = \pm 1 = \operatorname{cosec} x = \pm 1$

$\therefore 2 \sin x + 3 \operatorname{cosec} x = \pm 5$

(B) $\sin x_1 + \sin x_2 + \dots + \sin x_5 = \pm 5$

$\Rightarrow$ Each of $x_1, x_2, \dots, x_5$ is $= \pm \frac{\pi}{2}$

$\therefore \cos x_1 = \cos x_2 = \dots = \cos x_5 = 0$

$\therefore \cos x_1 + \cos x_2 + \dots + \cos x_5 = 0$

(C) $\cos^2 x_1 + \cos^2 x_2 + \dots + \cos^2 x_5 = 5$

$\cos^2 x_1 = \cos^2 x_2 = \dots = \cos^2 x_5 = 1$

$\therefore \cos x_1 = \cos x_2 = \dots = \cos x_5 = \pm 1$

Then $\sin x_1 + 2\sin x_2 + \dots + 5 \sin x_5 = 0$

$\cos^3 x_1 + \cos^3 x_2 + \dots + \cos^3 x_5 = -5$

$\cos x_1 = \cos x_2 = \dots = \cos x_5 = -1$

$\Rightarrow \sin x_1 = \sin x_2 = \dots = \sin x_5 = 0$

$\therefore \sin x_1 + 2\sin x_2 + \dots + 5\sin x_5 = 0$

4. Match the following :

Column-I**Column-II**

- (A) $\sqrt{3} \operatorname{cosec} 20^\circ - \sec 20^\circ$ (p) $-\frac{4}{\sqrt{3}}$
- (B) $2 \left(\sin^4 \frac{\pi}{8} + \sin^4 \frac{3\pi}{8} + \sin^4 \frac{5\pi}{8} + \sin^4 \frac{7\pi}{8} \right)$ (q) 3
- (C) $\cos \frac{2\pi}{7} \cos \frac{4\pi}{7} \cos \frac{6\pi}{7}$ (r) 4
- (D) $\sec 10^\circ - \frac{1}{\sqrt{3}} \operatorname{cosec} 10^\circ$ (s) $\frac{1}{8}$

Sol. Answer A(r), B(q), C(s) D(p)

(A) $\sqrt{3} \operatorname{cosec} 20^\circ - \sec 20^\circ$

$$= 4 \left(\frac{\frac{\sqrt{3}}{2} \cos 20^\circ - \frac{1}{2} \sin 20^\circ}{2 \sin 20^\circ \cos 20^\circ} \right)$$

$$= 4 \cdot \frac{\sin 40^\circ}{\sin 40^\circ}$$

$$= 4$$

(B) $2 \left(\sin^4 \frac{\pi}{8} + \sin^4 \frac{3\pi}{8} + \sin^4 \frac{5\pi}{8} + \sin^4 \frac{7\pi}{8} \right)$

$$= 4 \left(\sin^4 \frac{\pi}{8} + \sin^4 \frac{3\pi}{8} \right)$$

$$= \left(\left(2 \sin^2 \frac{\pi}{8} \right)^2 + \left(2 \sin^2 \frac{3\pi}{8} \right)^2 \right)$$

$$= \left(1 - \cos \frac{\pi}{4} \right)^2 + \left(1 + \cos \frac{3\pi}{4} \right)^2$$

$$= \left(1 - \frac{1}{\sqrt{2}} \right)^2 + \left(1 + \frac{1}{\sqrt{2}} \right)^2$$

$$= 1 + \frac{1}{2} + 1 + \frac{1}{2}$$

$$= 3$$

(C) $\cos \frac{2\pi}{7} \cdot \cos \frac{4\pi}{7} \cdot \cos \frac{6\pi}{7}$

$$= -\cos \frac{\pi}{7} \cdot \cos \frac{2\pi}{7} \cdot \cos \frac{4\pi}{7}$$

$$= -\frac{1}{2^3 \sin \frac{\pi}{7}} \cdot \sin \frac{8\pi}{7}$$

$$= \frac{1}{8}$$

(D) $\sec 10^\circ - \frac{1}{\sqrt{3}} \operatorname{cosec} 10^\circ$

$$= \frac{1}{\sqrt{3}} (\sqrt{3} \sec 10^\circ - \operatorname{cosec} 10^\circ)$$

$$\begin{aligned}
 &= \frac{1}{\sqrt{3}} \times \left(\frac{\sqrt{3} \sec 10^\circ - \cos 10^\circ}{\sin 10^\circ \cos 10^\circ} \right) \\
 &= \frac{1}{\sqrt{3}} \times 4 \left(\frac{\frac{\sqrt{3}}{2} \sec 10^\circ - \frac{1}{2} \cos 10^\circ}{\sin 20^\circ} \right) \\
 &= \frac{4}{\sqrt{3}} \times \frac{\sin(10^\circ - 30^\circ)}{\sin 20^\circ} \\
 &= -\frac{4}{\sqrt{3}}
 \end{aligned}$$

5. Match the value of the expression given in column-I with value greater than or equal to the number given in column-II.

Column-I

- (A) $2^{\sin x} + 2^{\cos x}$
 (B) $2^{\sin^2 x} + 2^{\cos^2 x}$
 (C) $2^{\sin^6 x} + 2^{\cos^6 x}$
 (D) $2^{\sin^4 x} + 2^{\cos^4 x}$

Column-II

- (p) 2
 (q) $2^{3/2}$
 (r) $2^{1-\sqrt{2}}$
 (s) $2^{5/4}$
 (t) $2^{9/8}$

Sol. Answer A(r), B(p, q, r, s, t), C(p, r, t), D(p, r, s, t)

$$\begin{aligned}
 \text{(A)} \quad & \frac{2^{\sin x} + 2^{\cos x}}{2} \geq 2^{\frac{\sin x + \cos x}{2}} \\
 \Rightarrow & 2^{\sin x} + 2^{\cos x} \geq 2^{\frac{\sin x + \cos x + 2}{2}} \\
 &= 2^{\frac{2 - \sqrt{2}}{2}} \\
 &= 2^{1 - \frac{1}{\sqrt{2}}}
 \end{aligned}$$

$$\begin{aligned}
 \text{(B)} \quad & \frac{2^{\sin^2 x} + 2^{\cos^2 x}}{2} \geq 2^{\sin x \cos x} \\
 \Rightarrow & 2^{\sin^2 x} + 2^{\cos^2 x} \geq 2^{\sin x \cos x + 1} \\
 &= 2^{\frac{1}{2} \sin 2x + 1} \\
 &= 2^{\frac{1}{2} + 1} = 2^{\frac{3}{2}}
 \end{aligned}$$

$$(C) \quad 2^{\sin^6 x} + 2^{\cos^6 x} \geq 2^{\sin^3 x \cos^3 x + 1}$$

$$= 2^{\frac{1}{8}(\sin 2x)^3 + 1} = 2^{\frac{9}{8}}$$

$$(D) \quad 2^{\sin^4 x} + 2^{\cos^4 x} \geq 2^{\sin^2 x \cos^2 x + 1}$$

$$= 2^{\frac{1}{4}(\sin 2x)^2 + 1} = 2^{\frac{5}{4}}$$

6. Match the following :

Column-I

Column-II

(A) If $\frac{\sin \theta}{\sin \phi} = \frac{1}{2}$ and $\frac{\cos \theta}{\cos \phi} = \frac{3}{2}$; θ and ϕ are acute angles,

(p) 12

then $\tan \theta$ is equal to

(B) If $\frac{\sin \theta}{\sin \phi} = \frac{1}{2}$ and $\frac{\cos \theta}{\cos \phi} = \frac{3}{2}$; θ and ϕ are acute angles,

(q) $\frac{1}{3}\sqrt{\frac{5}{3}}$

then $\tan \phi$ is equal to

(C) The numerical value of $4 \sin 50^\circ - \sqrt{3} \tan 50^\circ$ is equal to

(r) $\sqrt{\frac{5}{3}}$

(D) The minimum value of $9 \tan^2 \theta + 4 \cot^2 \theta$ is greater than or equal to

(s) 1

(t) 2

Sol. Answer A(q), B(r), C(s), D(p)

(A) $\frac{\sin \theta}{\sin \phi} = \frac{1}{2}, \frac{\cos \theta}{\cos \phi} = \frac{3}{2}$

$$\sin \phi = 2 \sin \theta$$

$$\cos \phi = \frac{2}{3} \cos \theta$$

$$1 \pm 4 \sin \theta + \frac{4}{9} \cos^2 \theta$$

$$\Rightarrow \sec^2 \theta = 4 \sin^2 \theta + \frac{4}{9}$$

$$\Rightarrow 1 + \sin^2 \theta = 4 \sin^2 \theta + \frac{4}{9}$$

$$\Rightarrow 3 \sin^2 \theta = 1 - \frac{4}{9} = \frac{5}{9}$$

$$\Rightarrow \sin^2 \theta = \frac{5}{27}$$

$$\Rightarrow \tan \theta = \frac{\sqrt{5}}{3\sqrt{3}} = \frac{1}{3}\sqrt{\frac{5}{3}}$$

$$(B) \frac{\sin \theta}{\sin \phi} = \frac{1}{2}, \frac{\cos \theta}{\cos \phi} = \frac{3}{2}$$

$$\Rightarrow \sin \theta = \frac{1}{2} \sin \phi, \cos \theta = \frac{3}{2} \cos \phi$$

$$1 = \frac{1}{4} \sin^2 \theta + \frac{9}{4} \cos^2 \phi$$

$$\Rightarrow \sec^2 \phi = \frac{1}{4} \sin^2 \phi + \frac{9}{4}$$

$$\Rightarrow 1 + \sin^2 \phi = \frac{1}{4} \sin^2 \phi + \frac{9}{4}$$

$$\Rightarrow \frac{3}{4} \sin^2 \phi = \frac{5}{4}$$

$$\Rightarrow \sin^2 \phi = \frac{5}{3}$$

$$\Rightarrow \sin \phi = \sqrt{\frac{5}{3}}$$

$$(C) 4 \sin 50^\circ - \sqrt{3} \cdot \sin 50^\circ$$

$$= 4 \sin 50^\circ - \sqrt{3} \frac{\sin 50^\circ}{\cos 50^\circ}$$

$$= \frac{4 \sin 50^\circ \cos 50^\circ - \sqrt{3} \sin 50^\circ}{\cos 50^\circ}$$

$$= \frac{2 \sin 100^\circ - \sqrt{3} \sin 50^\circ}{\cos 50^\circ}$$

$$= \frac{2 \sin 100^\circ - 2 \sin 50^\circ}{\cos 50^\circ}$$

$$= \frac{2 \sin 100^\circ - \sin 80^\circ - \sin 20^\circ}{\cos 50^\circ}$$

$$= \frac{\sin 100^\circ - \sin 20^\circ}{\cos 50^\circ} = \frac{2 \cos 60^\circ \cdot \sin 40^\circ}{\cos 50^\circ}$$

$$= 1$$


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7. Match the following

Column-I**(Trigonometric equation)****Column-II****(Family of solutions)**

(A) $\sin 9\theta = \cos\left(\frac{\pi}{2} - \theta\right)$

(p) $(2n+1)\frac{\pi}{10}, n \in \mathbb{Z}$

(B) $\sin 5\theta = \sin\left(\frac{\pi}{2} + 2\theta\right)$

(q) $\frac{n\pi}{2} + (-1)^n \frac{\pi}{4}, n \in \mathbb{Z}$

(C) $\cos 11\theta = \cos 3\theta$

(r) $\frac{n\pi}{7}, n \in \mathbb{Z}$

(D) $3\tan(\theta - 15^\circ) = \tan(\theta + 15^\circ)$

(s) $(4n+1)\frac{\pi}{14}, n \in \mathbb{Z}$

Sol. Answer A(p, q), B(s), C(q, r), D(q)

(A) $\sin 9\theta = \cos\left(\frac{\pi}{2} - \theta\right)$

$$\Rightarrow \sin 9\theta - \sin \theta = 0 \Rightarrow 2\cos 5\theta \sin 4\theta = 0$$

$$\Rightarrow \cos 5\theta = \cos \frac{\pi}{2} \text{ or } \sin 4\theta = 0$$

$$\Rightarrow 5\theta = n\pi + \frac{\pi}{2} \Rightarrow \theta = (2n+1)\frac{\pi}{10} \text{ or } \theta = \frac{n\pi}{4}, n \in \mathbb{Z}$$

(B) $\sin 5\theta - \sin\left(\frac{\pi}{2} + 2\theta\right) = 0$

$$\Rightarrow 2\cos\left(\frac{7\theta + \pi/2}{2}\right) \sin\left(\frac{3\theta - \pi/2}{2}\right) = 0$$

$$\Rightarrow \frac{7\theta + \pi/2}{2} = n\pi + \frac{\pi}{2}$$

$$\Rightarrow \theta = (4n+1)\frac{\pi}{14}$$

(C) $\cos 11\theta - \cos 3\theta = -2 \sin 7\theta \sin 4\theta = 0$

$$\Rightarrow \theta = \frac{n\pi}{7} \text{ or } \theta = \frac{n\pi}{4}$$

(D) $3 \left[\frac{\tan \theta - \tan 15^\circ}{1 + \tan \theta \tan 15^\circ} \right] = \frac{\tan \theta + \tan 15^\circ}{1 - \tan \theta \tan 15^\circ}$

$$2[\tan\theta + \tan^2 15^\circ \tan\theta] = 4[\tan^2\theta \tan 15^\circ + \tan 15^\circ]$$

$$\Rightarrow \tan\theta \sec^2 15^\circ = 2\tan 15^\circ \cdot \sec^2\theta$$

$$\Rightarrow \sin\theta \cos\theta = 2 \sin 15 \cos 15$$

$$\Rightarrow \frac{\sin 2\theta}{2} = \frac{1}{2}$$

$$\Rightarrow 2\theta = n\pi + (-1)^n \frac{\pi}{2}$$

$$\Rightarrow \theta = \frac{n\pi}{2} + (-1)^n \frac{\pi}{4}, n \in \mathbb{Z}$$

8. Match the following

Column-I

Column-II

(A) The number of solutions of $\sin 3x + \cos 2x = 0$ in $\left[0, \frac{3\pi}{2}\right]$ is (p) 1

(B) The number of solutions of the equation (q) 4

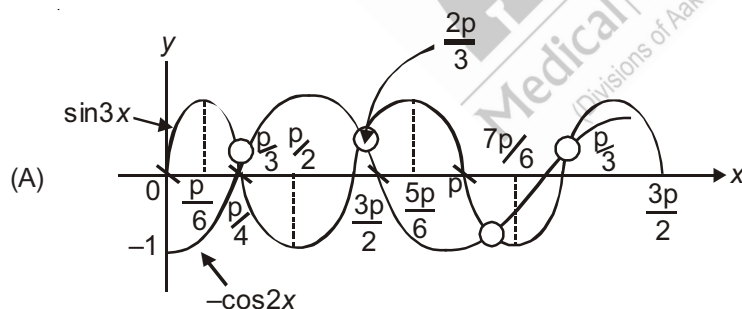
$$(1 - \tan x)(1 + \sin 2x) = 1 + \tan x \text{ in } \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \text{ is}$$

(C) The number of solutions of the equation $\sin(\pi^x) = e^x + e^{-x}$ is (r) 2

(D) The number of solutions of the equation (s) 0

$$\sin^3 x - 3\sin x \cos^2 x + 2\cos^3 x = 0 \text{ in } \left[-\frac{\pi}{4}, \frac{\pi}{4}\right] \text{ is}$$

Sol. Answer A(q), B(r), C(s), D(p)



It is clear from the graph that 4 points satisfy the given equation

(B) $(\cos x - \sin x)(1 + \sin 2x) = \cos x + \sin x$

$$\Rightarrow -\sin x + \cos x - \sin x \sin 2x + \cos x \sin 2x = \sin x + \cos x$$

$$\Rightarrow -1 + \sin 2x + 2\cos^2 x = 1 \text{ or } \sin x = 0$$

$$\Rightarrow -\sin x \cos x = 1 - \cos^2 x$$

$$\Rightarrow \sin x [\cos x + \sin x] = 0$$

$$\Rightarrow x = 0 \text{ or } -\frac{\pi}{4}$$

(C) $-1 \leq \sin(\pi x) \leq 1$

But $e^x + e^{-x} \geq 2$

Hence no solution exists.

(D) $\sin^3 x - 3\sin x \cos^2 x + 2\cos^3 x = 0$

$$\Rightarrow \sin^2 x (\sin x - \cos x) + \sin x \cos x (\sin x - \cos x) - 2 \cos^2 x (\sin x - \cos x) = 0.$$

$$\Rightarrow (\sin x - \cos x) (\sin x - \cos x) (\sin x + 2\cos x) = 0$$

$$\Rightarrow (\sin x - \cos x)^2 (\sin x + 2\cos x) = 0$$

$$\Rightarrow x = \frac{\pi}{4} \text{ or } \tan x = -2 \Rightarrow x < -\frac{\pi}{4}$$

$$\Rightarrow x = \frac{\pi}{4} \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$$

9. Match the following

Column-I

Column-II

(A) The range of k such that the equation $\sin^8 x + \cos^8 x = k$

(p) $k \in \left[\frac{1}{2}, 1\right]$

admits of a solution is given by

(B) The range of k such that the equation $k \sin x - 4 \cos x = k + 2$

(q) $k \in \left[\frac{1}{4}, 1\right]$

admits of a solution is given by

(C) The range of k such that the equation $\sin^4 x + \cos^4 x = k$

(r) $k \in (-\infty, 3]$

admits of a solution is given by

(D) The range of k such that the equation $\sin^6 x + \cos^6 x = k$

(s) $k \in \left[\frac{1}{8}, 1\right]$

admits of a solution is given by

Sol. Answer A(s), B(r), C(p), D(q)

(A) $\frac{\sin^8 x + \cos^8 x}{2} \geq (\sin^8 x \cos^8 x)^{\frac{1}{2}}$

$$\Rightarrow \frac{k}{2} \geq \frac{(\sin^4 2x)}{16} = \frac{1}{16}$$

$$\Rightarrow k \geq \frac{1}{8}$$

$$\Rightarrow k \in \left[\frac{1}{8}, 1\right]$$

(B) In order that the given equation admits of a solution we must have

$$(k + 2)^2 \leq k^2 + 16$$

$$\Rightarrow k^2 + 4k + 4 \leq k^2 + 16$$

$$\Rightarrow k \leq 3 \Rightarrow k \in (-\infty, 3]$$

(C) Do as 'a' part (A.M. $\geq$ G.M)

(D) Do as 'a' part (A.M. $\geq$ G.M)

10. Match the following

Column-I

(A) $\cos x \sin^3 x < \sin x \cos^3 x$, $0 \leq x \leq 2\pi$ is satisfied

for x lying in

(B) The value of x for which $4 \sin^2 x - 8 \sin x + 3 \leq 0$,

where $x \in [0, 2\pi]$ lies in

(C) $|\tan x| \leq 1$ and $x \in [-\pi, \pi]$ is

(D) $\cos x - \sin x \geq 1$ and $0 \leq x \leq 2\pi$ is

Column-II

(p) $\left[\frac{\pi}{6}, \frac{5\pi}{6}\right]$

(q) $\left[-\pi, -\frac{3\pi}{4}\right] \cup \left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$

(r) $\left[\frac{3\pi}{2}, 2\pi\right] \cup \{0\}$

(s) $\left[0, \frac{\pi}{4}\right]$

(t) $\left[\frac{3\pi}{4}, \pi\right]$

Sol. Answer A(s), B(p), C(q, s, t), D(r)

(A) $\cos x \sin^3 x - \sin x \cos^3 x \leq 0$

$$\Rightarrow \sin x \cos x (\sin^2 x - \cos^2 x) \leq 0$$

$$\Rightarrow \sin x \cdot \cos x (\cos^2 x - \sin^2 x) \geq 0$$

$$\Rightarrow \sin x \cdot \cos x (\cos 2x) \geq 0$$

$$\Rightarrow 2 \sin x \cdot \cos x (\cos 2x) \geq 0$$

$$\Rightarrow \sin 2x \cdot \cos 2x \geq 0$$

$$\sin 4x \geq 0$$

$$\Rightarrow 4x \in [0, \pi]$$

$$x \in \left[0, \frac{\pi}{4}\right]$$

(B) $4 \sin^2 x - 6 \sin x - 2 \sin x + 3 \leq 0$

$$\Rightarrow 2 \sin x (2 \sin x - 3) - 1(2 \sin x - 3) \leq 0$$

$$(2 \sin x - 3) (2 \sin x - 1) \leq 0$$

$$\text{But } 2 \sin x - 3 \leq 0$$

$$\Rightarrow 2 \sin x - 1 \leq 0$$

$$\sin x \geq \frac{1}{2}$$

$$\Rightarrow x \in \left[\frac{\pi}{6}, \frac{5\pi}{6}\right]$$

$$(C) |\tan x| \leq 1 \Rightarrow -1 \leq \tan x \leq 1$$

$$x \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right] \cup \left[-\pi, -\frac{3\pi}{4}\right]$$

$$(D) \cos x - \sin x \geq 1$$

$$\Rightarrow \sqrt{2} \left(\cos x \sin \frac{\pi}{4} - \sin x \cos \frac{\pi}{4} \right) \geq 1$$

$$\sin \left(\frac{\pi}{4} - x \right) \geq \frac{1}{\sqrt{2}} \Rightarrow x \in \left[\frac{3\pi}{2}, 2\pi \right] \cup [0]$$

11. Based on the relation between the variables in column I match the type of the triangle in column II.

Column-I

$$(A) r_1 = r_2 + r_3 + r$$

$$(B) (a^2 + b^2) \sin(A - B) = (a^2 - b^2) \sin(A + B)$$

$$(C) \tan A \tan B < 1$$

$$(D) R = 2r$$

Column-II

(p) Isosceles or right angled

(q) Obtuse but not necessarily isosceles

(r) Right angled but not necessary isosceles

(s) Equilateral

Sol. Answer A(r), B(p), C(q), D(s)

$$(A) r_1 = r_2 + r_3 + r$$

$$\Rightarrow r_1 - r = r_2 + r_3$$

$$\frac{\Delta}{s-a} - \frac{\Delta}{s} = \frac{\Delta}{s-b} + \frac{\Delta}{s-c}$$

$$\frac{a}{s(s-a)} = \frac{2s-(s-c)}{(s-b)(s-c)}$$

$$\Rightarrow s(s-a) = (s-b)(s-c)$$

$$\Rightarrow \frac{(s-b)(s-c)}{s(s-a)} = 1$$

$$\tan^2 \frac{A}{2} = 1$$

$$\frac{A}{2} = \frac{\pi}{4} \Rightarrow \angle A = 90^\circ$$

Then triangle is right angled

$$(B) (a^2 + b^2) \sin(A - B) = (a^2 - b^2) \sin(A + B)$$

$$\Rightarrow \frac{\sin(A - B)}{\sin(A + B)} = \frac{a^2 - b^2}{a^2 + b^2}$$

By using componendo and dividendo

$$\Rightarrow \frac{\sin(A-B) + \sin(A+B)}{\sin(A-B) - \sin(A+B)} = \frac{2a^2}{-2b^2}$$

$$\Rightarrow \frac{2\sin A \cos B}{-2\cos A \sin B} = \frac{a^2}{+b^2}$$

$$\Rightarrow \frac{\sin A}{\cos A} - \frac{\cos B}{\sin B} = \frac{\sin^2 A}{\sin^2 B}$$

$$\Rightarrow \frac{\cos B}{\cos A} = \frac{\sin A}{\sin B}$$

$$\Rightarrow \sin^2 A = \sin^2 B$$

$$\Rightarrow \boxed{A = B} \text{ or } 2A = \pi - 2B$$

$$A + B = \frac{\pi}{2}$$

Then $\triangle ABC$ is isosceles or right angled triangle

(C) $\tan A \cdot \tan B < 1$

Let B is acute angle then

$$\Rightarrow \tan A < \cot B$$

$$\Rightarrow \tan A < \cot B (\pi/2 - B)$$

$$A + B < \frac{\pi}{2}$$

$$\therefore \angle C > \frac{\pi}{2}$$

(D) $R = 2r$

$$\therefore \cos A + \cos B + \cos C = 1 + \frac{r}{R} = \frac{3}{2}$$

Then it is possible

$$\angle A = \angle B = \angle C$$

12. Match the following

Column-I

(A) In triangle ABC , the angle $A = 60^\circ$ and the sides $b = 3$ cm, $c = 5$ cm, then the length of median through A is (in cm)

(B) In triangle ABC , the angle $A = 120^\circ$ and the sides $b = 4$ cm, $c = 6$ cm, then the length of angle bisector through A is (in cm)

Column-II

(p) $\frac{12}{5}$

(q) $\frac{7}{2}$

(C) In triangle ABC , the medians through A and B

(r) 8

are perpendicular, then the value of $\frac{a^2 + b^2}{c^2}$

(D) If the lengths of median of a triangle are 3, 4, 5 cm,
then the area of the triangle is (in cm^2)

(s) 5

Sol. Answer A(q), B(p), C(s), D(r)

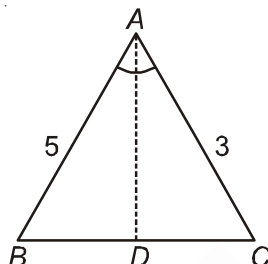
(A) $\angle A = 60^\circ$, $b = 3$ and $c = 5$

$$\therefore \cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{1}{2}$$

$$b^2 + c^2 - a^2 = bc$$

$$25 + 9 - a^2 = 15$$

$$a^2 = 19$$



$$\text{Length of median } AD = \frac{1}{2} \sqrt{2b^2 + 2c^2 - a^2}$$

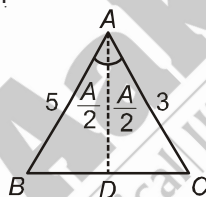
$$= \frac{1}{2} \sqrt{16 + 50 - 19}$$

$$= \frac{7}{2}$$

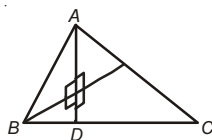
(B) $\angle A = 120^\circ$, $b = 4$ and $c = 6$ cm

$$AD = \left(\frac{2bc}{b+c} \right) \cos \frac{A}{2}$$

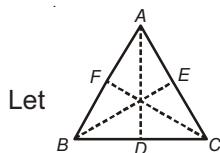
$$\frac{48}{10} \cdot \frac{1}{2} = \frac{12}{5}$$



(C)



(D) Let AD , BE and CF are the median of triangle ABC



Let

$$\therefore AD = 3$$

$$BE = 4$$

$$CF = 5$$

13. Match the items of Column I with Column II.

Column-I

(A) If in $\triangle ABC$, $\sin^2 A + \sin^2 B = \sin^2(A + B)$,
then the triangle must be

(B) In a $\triangle ABC$, $\frac{bc}{2\cos A} = b^2 + c^2 - 2bc \cos A$,
then the $\triangle ABC$ must be

(C) In $\triangle ABC$, $\tan \frac{A}{2} + \tan \frac{B}{2} + \tan \frac{C}{2} = \sqrt{3}$,
then $\triangle$ must be

(D) In a $\triangle ABC$, the sides and the altitudes
are in A.P. then $\triangle$ must be

Column-II

(p) Right angled triangle

(q) Equilateral

(r) Isosceles

(s) Obtuse angled

(t) Acute angled

Sol. Answer A(p), B(r), C(q, r, t), D(q, r, t)

(A) In $\triangle ABC$

$$\sin^2 A + \sin^2 B = \sin^2(A + B)$$

$$\Rightarrow \sin^2 A + \sin^2 B = \sin^2 C$$

$\Rightarrow$ By using Sine Rule

$$a^2 + b^2 = c^2$$

Triangle must be right angled triangle

(B) $\frac{bc}{2\cos A} = b^2 + c^2 - 2bc \cos A$

By using Cosine Rule

$$\frac{bc(2bc)}{2(b^2 + c^2 - a^2)} = b^2 + c^2 - 2bc \left(\frac{b^2 + c^2 - a^2}{2bc} \right)$$

$$\Rightarrow \frac{b^2 c^2}{b^2 + c^2 - a^2} = a^2$$

$$\Rightarrow b^2 c^2 = a^2 b^2 + a^2 c^2 - a^4$$

$$\Rightarrow (b^2 c^2 - a^2 b^2) + a^4 - a^2 c^2 = 0$$

$$\Rightarrow b^2(c^2 - a^2) + a^2(a^2 - c^2) = 0$$

$$\Rightarrow (a^2 - b^2)(a^2 - c^2) = 0$$

$$a^2 - b^2 = 0 \text{ or } a^2 - c^2 = 0$$

$$a = b \text{ or } a = c$$

$$(C) \tan \frac{A}{2} + \tan \frac{B}{2} + \tan \frac{C}{2} = \sqrt{3} \quad \dots(i)$$

$$\therefore \tan \frac{A}{2} \cdot \tan \frac{B}{2} + \tan \frac{B}{2} \cdot \tan \frac{C}{2} + \tan \frac{C}{2} \cdot \tan \frac{A}{2} = 1$$

From equation (i)

By squaring both side

$$\Rightarrow \tan^2 \frac{A}{2} + \tan^2 \frac{B}{2} + \tan^2 \frac{C}{2} + 2 \left(\tan \frac{A}{2} \tan \frac{B}{2} + \tan \frac{B}{2} \tan \frac{C}{2} + \tan \frac{C}{2} \tan \frac{A}{2} \right) = 3$$

$$\Rightarrow \tan^2 \frac{A}{2} + \tan^2 \frac{B}{2} + \tan^2 \frac{C}{2} = 1$$

$$\text{Now, } \left(\tan \frac{A}{2} - \tan \frac{B}{2} \right)^2 + \left(\tan \frac{B}{2} - \tan \frac{C}{2} \right)^2 + \left(\tan \frac{C}{2} - \tan \frac{A}{2} \right)^2$$

$$= 2 \left[\tan^2 \frac{A}{2} + \tan^2 \frac{B}{2} + \tan^2 \frac{C}{2} - \left(\tan \frac{A}{2} \tan \frac{B}{2} + \tan \frac{B}{2} \tan \frac{C}{2} + \tan \frac{C}{2} \tan \frac{A}{2} \right) \right]$$

$$= 2[1 - 1] = 0$$

It is possible if

$$\tan \frac{A}{2} = \tan \frac{B}{2} = \tan \frac{C}{2}$$

$$\angle A = \angle B = \angle C$$

(D) Let a, b, c are the sides with usual notations given a, b, c are in A.P.

$$2b = a + c \quad \dots(i)$$

Let P_1, P_2, P_3 are the length of altitudes AP, BE, CE respectively

$$\text{Now, } \Delta = \frac{1}{2}aP_1 = \frac{1}{2}bP_2 = \frac{1}{2}cP_3$$

$$\text{Or } P_1 = \frac{2\Delta}{a}, P_2 = \frac{2\Delta}{b}, P_3 = \frac{2\Delta}{c}$$

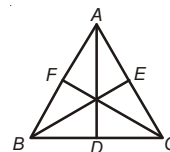
And given P_1, P_2, P_3 are in A.P.

$$\text{Then } 2P_2 = P_1 + P_3$$

$$2 \left(\frac{2\Delta}{b} \right) = \frac{2\Delta}{a} + \frac{2\Delta}{c}$$

$$\frac{2}{b} = \frac{1}{a} + \frac{1}{c} \quad \dots(ii)$$

From (i) and (ii) a, b, c are in A.P. as well as in H.P. then it is possible if $a = b = c$



SECTION - F

Integer Answer Type Questions

1. If $5\cos \theta = 3$, then, $\frac{\operatorname{cosec} \theta + \cot \theta}{\operatorname{cosec} \theta - \cot \theta}$ is equal to

Sol. Answer (4)

$$\text{Given, } 5\cos \theta = 3$$

$$\frac{\operatorname{cosec} \theta + \cot \theta}{\operatorname{cosec} \theta - \cot \theta}$$

$$= (\operatorname{cosec} \theta + \cot \theta)^2$$

$$= \frac{(1 + \cos \theta)^2}{\sin^2 \theta}$$

$$= \frac{\left(1 + \frac{3}{5}\right)^2}{\frac{16}{25}} = 4$$

2. Least integral value of $3(\tan^2 \theta + \cot^2 \theta) - 8(\tan \theta + \cot \theta) + 10$, $\theta \in \left(0, \frac{\pi}{2}\right)$ can be

Sol. Answer (0)

$$f(\theta) = 3(\tan^2 \theta + \cot^2 \theta) - 8(\tan \theta + \cot \theta) + 10, \theta \in \left(0, \frac{\pi}{2}\right)$$

$$= 3(\tan \theta + \cot \theta)^2 - 6 - 8(\tan \theta + \cot \theta) + 10$$

$$= 3\left\{\frac{1}{\sin \theta \cos \theta}\right\}^2 - 8\left\{\frac{1}{\sin \theta \cos \theta}\right\} + 4$$

$$= \frac{12}{\sin^2 2\theta} - \frac{16}{\sin 2\theta} + 4$$

$$= 4(3\operatorname{cosec}^2 2\theta - 4\operatorname{cosec} 2\theta + 1)$$

$$= 12\left[\operatorname{cosec}^2 2\theta - \frac{4}{3}\operatorname{cosec} 2\theta + \frac{4}{9}\right] - \frac{16}{3} + 4$$

$$= 12\left(\operatorname{cosec} 2\theta - \frac{2}{3}\right)^2 - \frac{4}{3}$$

$$f_{\min} = 0$$

3. $x^2 - 2x + 2\cos^2\theta + \sin^2\theta = 0$, then maximum number of ordered pair (x, θ) such that $x \in R, \theta \in [0, 2\pi]$.

Sol. Answer (2)

$$x^2 - 2x + 2\cos^2\theta + \sin^2\theta = 0, x, \theta \in R$$

$$x^2 - 2x + \cos^2\theta + 1 = 0$$

$$D = 4 - 4(\cos^2\theta + 1)$$

$$= -4 \cos^2\theta$$

$$\therefore x \text{ is real} \Rightarrow \Delta \geq 0$$

$$\therefore \cos\theta = 0$$

$$\text{In } [0, 2\pi], \theta = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$\therefore x^2 - 2x + 1 = 0 \Rightarrow x = 1$$

$$\therefore (x, \theta) = \left(1, \frac{\pi}{2}\right), \left(1, \frac{3\pi}{2}\right)$$

4. The smallest positive value of $\frac{8\sqrt{2}}{\pi}p$ for which $\sin(p \cos\theta) = \cos(p \sin\theta)$ has a solution in $[0, 2\pi]$ is

Sol. Answer (4)

$$\sin(p \cos\theta) = \sin\left(\frac{\pi}{2} - p \sin\theta\right)$$

$$\Rightarrow p \cos\theta = \frac{\pi}{2} - p \sin\theta$$

$$\Rightarrow p(\sin\theta + \cos\theta) = \frac{\pi}{2}$$

$$\Rightarrow \sin\theta + \cos\theta = \frac{\pi}{2p}$$

$$\Rightarrow \sin\left(\theta + \frac{\pi}{4}\right) = \frac{\pi}{2\sqrt{2}p}$$

$$\left|\frac{\pi}{2\sqrt{2}p}\right| \leq 1$$

$$\Rightarrow p \geq \frac{\pi}{2\sqrt{2}}$$

$$\therefore \frac{8\sqrt{2}}{\pi} \times \frac{\pi}{2\sqrt{2}} = 4$$

$$\Rightarrow \text{Smallest value of } p \text{ is } 4.$$

5. If $\sec(\theta - \phi)$, $\sec\theta$ and $\sec(\theta + \phi)$ are in A.P. then $\cos^2\theta \sec^2\frac{\phi}{2} + 3$, then is equal to

Sol. Answer (5)

$$\sec(\theta - \phi), \sec\theta, \sec(\theta + \phi) \in \text{A.P.}$$

$$\Rightarrow 2\sec\theta = \sec(\theta - \phi) + \sec(\theta + \phi)$$

$$\Rightarrow \frac{2}{\cos\theta} = \frac{1}{\cos(\theta - \phi)} + \frac{1}{\cos(\theta + \phi)}$$

$$\Rightarrow \frac{2}{\cos\theta} = \frac{\cos(\theta + \phi) + \cos(\theta - \phi)}{\cos(\theta - \phi)\cos(\theta + \phi)}$$

$$\Rightarrow \frac{2}{\cos\theta} = \frac{2\sin\theta\cos\phi}{\cos^2\theta - \sin^2\phi}$$

$$\Rightarrow \cos^2\theta - \sin^2\phi = \cos^2\theta\sin^2\phi$$

$$\Rightarrow \cos^2\theta(1 - \cos\phi) = \sin^2\phi$$

$$\Rightarrow \sin^2\theta = \frac{\sin^2\phi}{1 - \cos\phi}$$

$$\Rightarrow \cos^2\theta = \frac{4\sin^2\frac{\phi}{2}\cos^2\frac{\phi}{2}}{2\sin^2\frac{\phi}{2}}$$

$$\Rightarrow \cos^2\theta \cdot \sec^2\frac{\phi}{2} = 2$$

$$\Rightarrow \cos^2\theta \cdot \sec^2\frac{\phi}{2} + 3 = 2 + 3 = 5$$

6. The sum of maximum and minimum values of the expression $5\cos x + 3\sin\left(\frac{\pi}{6} - x\right) + 4$ is

Sol. Answer (8)

$$5\cos x + 3\sin\left(\frac{\pi}{6} - x\right) + 4$$

$$= 5\cos x + 3\left(\frac{1}{2}\cos x - \frac{\sqrt{3}}{2}\sin x\right) + 4$$

$$= \frac{13}{2}\cos x - \frac{3\sqrt{3}}{2}\sin x + 4$$

$$\text{Maximum value} = \sqrt{\frac{169}{4} + \frac{27}{4}} + 4$$

$$= \sqrt{\frac{196}{4}} + 4$$

$$= 7 + 4 = 11$$

$$\text{Minimum value} = -7 + 4 = -3$$

$$\text{Sum of maximum and minimum value} = 11 - 3 = 8$$

7. Let PQ and RS be two parallel chords of a given circle of radius 6 cm lying on the same side of the centre. If the chords subtend angles of 72° and 144° at the centre and the distance between the chords is d , then d^2 is equal to

Sol. Answer (9)

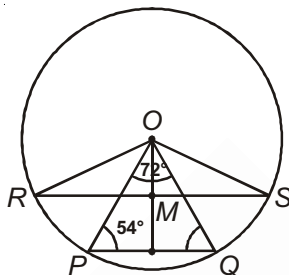
$$OM = 6 \sin 18^\circ, ON = 6 \sin 54^\circ$$

$$d = 6 \sin 54^\circ - 6 \sin 18^\circ$$

$$= 6(\cos 36^\circ - \sin 18^\circ)$$

$$= 6 \left(\frac{\sqrt{5}+1}{4} - \frac{\sqrt{5}-1}{4} \right) = 6 \left(\frac{1}{2} \right) = 3$$

$$\therefore d^2 = 9$$



8. If $4 \tan \theta \tan \phi = 3$, then $\frac{\cos(\theta - \phi)}{\cos(\theta + \phi)}$ is equal to

Sol. Answer (7)

$$\frac{\cos(\theta - \phi)}{\cos(\theta + \phi)} = k$$

$$\Rightarrow \frac{\cos(\theta - \phi) - \cos(\theta + \phi)}{\cos(\theta - \phi) + \cos(\theta + \phi)} = \frac{k-1}{k+1}$$

$$\Rightarrow \frac{2 \sin \theta \sin \phi}{2 \cos \theta \cos \phi} = \frac{k-1}{k+1}$$

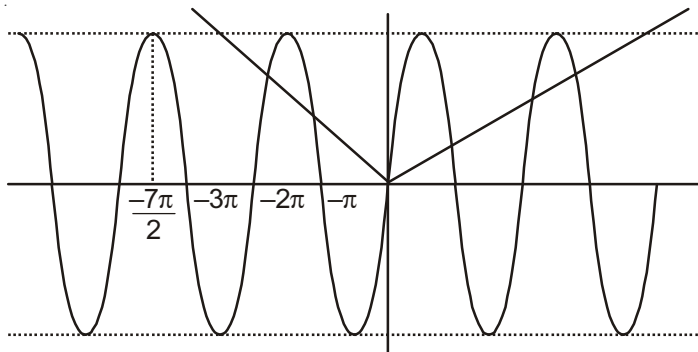
$$\Rightarrow \tan \theta \tan \phi = \frac{k-1}{k+1}$$

$$\Rightarrow \frac{3}{4} = \frac{k-1}{k+1}$$

$$\Rightarrow k = 7$$

9. The number of solution of $10 \sin x = |x|$ is _____.

Sol. Answer (6)



10. The number of solutions of $(81)^{\sin^2 x} + (81)^{\cos^2 x} = 30$ for $x \in [0, 2\pi]$ is equal to _____.

Sol. Answer (8)

$$81^{\sin^2 x} + 81^{\cos^2 x} = 30$$

$$81^{\sin^2 x} = y$$

$$y + \frac{81}{y} = 30$$

$$\Rightarrow y^2 - 30y + 81 = 0$$

$$y = 3, y = 27$$

$$3^{4\sin^2 x} = 3^1, 3^{4\sin^2 x} = 3^3$$

$$\Rightarrow \sin^2 x = \frac{1}{4} \text{ or } \sin^2 x = \frac{3}{4}$$

$$\therefore x \in [0, 2\pi]$$

$$\therefore x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6} \text{ or } x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$$

Number of solution = 8

11. If $a, b \in [0, 2\pi]$ and equation $x^2 - 2x + 4 = 3\sin(ax + b)$ has at least one solution and the least positive value of

$a + b$ is k , then $\frac{2k}{\pi}$ is equal to

Sol. Answer (1)

$$x^2 - 2x + 4 = 3\sin(ax + b)$$

$$(x - 1)^2 + 3 = 3\sin(ax + b)$$

$$\text{LHS} \geq 3$$

$$\text{RHS} \in [-3, 3]$$

$n = 1$ is only solution.

$$3 = 3\sin(a + b)$$

$$\Rightarrow a + b = \frac{\pi}{2}$$

12. If $\sin^2 x - 2\sin x = 1$ has exactly four solutions in $[0, n\pi]$, then minimum value of n is equal to.

Sol. Answer (4)

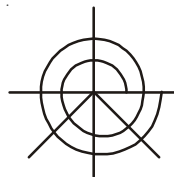
$\sin^2 x - 2\sin x = 1$ has exactly four solutions in $[0, n\pi]$

$$\sin^2 x - 2\sin x - 1 = 0$$

$$\sin x = \frac{2 \pm \sqrt{4+4}}{2} = 1 \pm \sqrt{2}$$

$$\sin x \neq 1 + \sqrt{2}, \sin x = (1 - \sqrt{2})$$

Minimum value of $n = 4$ for which this equation has exactly four solutions in $(0, n\pi)$



13. If $\tan \theta = \frac{a}{b}$ and $0 < \theta < \pi$, then the value of $\frac{\sqrt{a^2 + b^2}}{|a|} \sin \theta$ is

(Assume $b \neq 0, a \neq 0, a, b \in R$)

Sol. Answer (1)

$$\tan \theta = \frac{a}{b}, \frac{\sqrt{a^2 + b^2}}{|a|} \sin \theta = ?, \theta \in (0, \pi)$$

$\therefore \theta$ lies in $(0, \pi)$

$\therefore \tan \theta$ will be positive or negative but $\sin \theta$ will be positive

$$\therefore \sin \theta = \frac{|a|}{\sqrt{a^2 + b^2}}$$

$$\therefore \frac{\sqrt{a^2 + b^2}}{|a|} \sin \theta = 1$$

14. Let $\cos(\alpha - \beta) + \cos(\beta - \gamma) + \cos(\gamma - \alpha) = \frac{-3}{2}$, then the value of $\cos \alpha + \cos \beta + \cos \gamma$ is

Sol. Answer (0)

$$\cos(\alpha - \beta) + \cos(\beta - \gamma) + \cos(\gamma - \alpha) = -\frac{3}{2}$$

$$2\cos(\alpha - \beta) + 2\cos(\beta - \gamma) + 2\cos(\gamma - \alpha) + 3 = 0$$

$$2\cos(\alpha - \beta) + 2\cos(\beta - \gamma) + 2\cos(\gamma - \alpha) + \cos^2 \alpha + \sin^2 \alpha + \cos^2 \beta + \sin^2 \beta + \cos^2 \gamma + \sin^2 \gamma = 0$$

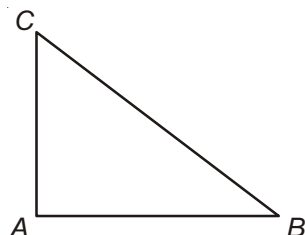
$$(\cos \alpha + \cos \beta + \cos \gamma)^2 + (\sin \alpha + \sin \beta + \sin \gamma)^2 = 0$$

$$\Rightarrow \cos \alpha + \cos \beta + \cos \gamma = \sin \alpha + \sin \beta + \sin \gamma = 0$$

15. In right angled $\triangle ABC$, if $AB = AC$, then value of $\left[\frac{R}{r} \right]$ (value $[x]$ denote the greatest integer of x)

Sol. Answer (2)

In right angled $\triangle ABC$



$$AB = AC$$

$$\therefore \angle A = 90^\circ, \angle B = 45^\circ, \angle C = 45^\circ$$

$$\text{And } r = 4R \sin \frac{A}{2} \cdot \sin \frac{B}{2} \cdot \sin \frac{C}{2}$$

$$r = 4R \sin 45^\circ \sin \frac{45^\circ}{2} \cdot \sin \frac{45^\circ}{2} = \frac{4R}{\sqrt{2}} \sin^2 \frac{45^\circ}{2} = \frac{4R}{\sqrt{2}} \left[\frac{1 - \cos 45^\circ}{2} \right] = \frac{2R}{\sqrt{2}} \left[1 - \frac{1}{\sqrt{2}} \right]$$

$$\frac{r}{R} = \frac{2}{\sqrt{2}} - 1 = (\sqrt{2} - 1)$$

$$\frac{R}{r} = \frac{1}{\sqrt{2} - 1} = 2.414$$

$$\therefore \left[\frac{R}{r} \right] = 2$$

16. In a $\triangle ABC$, $bc = a$, $a = 2$, the value of $2\Delta R$ is equal to

Sol. Answer (2)

In a $\triangle ABC$

$$bc = a \text{ and } a = 2$$

$$\therefore \Delta = \frac{abc}{4R}$$

$$\Rightarrow \Delta R = \frac{abc}{4}$$

$$\Rightarrow \Delta R = \frac{a(a)}{4} = \frac{4}{4}$$

$$\Rightarrow \Delta R = 1$$

$$\therefore 2\Delta R = 2$$

17. In a triangle ABC with usual notation $b \operatorname{cosec} B = a$, then value of $\left(\frac{b+c}{r+R} \right)$ is

Sol. Answer (2)

$$\text{Given } b \operatorname{cosec} B = a$$

$$b = a \sin B$$

$$\therefore \frac{a}{1} = \frac{b}{\sin B}$$

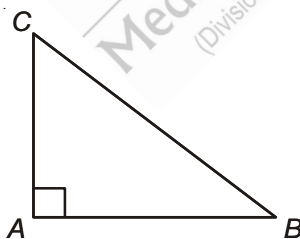
$$\therefore \angle A = 90^\circ$$

$\therefore \triangle ABC$ is a right angled triangle then

$$\therefore R = \frac{a}{2} \text{ and } r = (s-a) \tan \frac{A}{2}$$

$$\text{Now, } \frac{b+c}{r+R} = \frac{b+c}{(s-a) + \frac{a}{2}} = \frac{b+c}{s - \frac{a}{2}} = \frac{2(b+c)}{a+b+c-a}$$

$$\frac{b+c}{r+R} = 2$$



18. In a triangle ABC , $2B = A + C$ and $b^2 = ac$, then $\frac{a(a+b+c)}{3bc}$ is

Sol. Answer (1)

Clearly $\angle B = 60^\circ$

$$\frac{1}{2} = \frac{c^2 + a^2 - b^2}{2ac}$$

$$\Rightarrow c^2 + a^2 - b^2 = ac$$

But a, b, c are in G.P.

$$\Rightarrow b^2 = ac$$

$$\Rightarrow c^2 + a^2 - 2ac = 0$$

$$\Rightarrow c = a = b$$

$$\text{Hence } \frac{a(a+b+c)}{3bc} = \frac{3a^2}{3a^2} = 1$$

19. In an equilateral triangle with usual notations the value of $\frac{27r^2R}{r_1r_2r_3}$ is equal to

Sol. Answer (4)

In an equilateral triangle

$$r = \frac{\Delta}{s}$$

$$\therefore \text{Area of equilateral triangle} = \frac{\sqrt{3}}{4}a^2$$

$$s = \frac{3a}{2}$$

$$r = \frac{\sqrt{3}a^2(2)}{4(3a)} = \frac{1}{2\sqrt{3}}a$$

$$\text{And } r_1 = \frac{\Delta}{s-a} = \frac{\sqrt{3}a^2}{4\left(\frac{3a}{2}-a\right)} = \frac{\sqrt{3}a^2}{4\left(\frac{a}{2}\right)} = \frac{\sqrt{3}}{2}a = r_2 = r_3$$

$$R = \frac{a}{\sin A} = \frac{2a}{\sqrt{3}}$$

$$\text{Now } \frac{27r^2R}{r_1r_2r_3} = \frac{27a^2}{4 \cdot 3} \cdot \frac{2a}{\sqrt{3}} \cdot \frac{8}{3\sqrt{3}a^3} = 4$$

20. In a triangle ABC , $\frac{c \cos(A-\theta) + a \cos(C+\theta)}{b \cos \theta}$ is equal to

Sol. Answer (1)

In $\triangle ABC$

$$\therefore c \cos(A-\theta) + a \cos(C+\theta)$$

$$= c[\cos A \cos \theta + \sin A \sin \theta] + a[\cos C \cos \theta - \sin C \sin \theta]$$

$$= \cos\theta [c \cdot \cos A + a \cos C] + \sin\theta [c \cdot \sin A - a \cdot \sin C]$$

By using projection formula and sine rule

$$= \cos\theta (b) + 0$$

$$\therefore \frac{c \cos(A - \theta) + a \cos(C + \theta)}{b \cos\theta} = \frac{b \cos\theta}{b \cos\theta} = 1$$

21. In a $\triangle ABC$ if $\frac{1}{r^2} + \frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} = \frac{a^2 + b^2 + c^2}{\Delta^n}$, then n is equal to _____.

Sol. Answer (2)

Given that

$$\frac{1}{r^2} + \frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} = \frac{a^2 + b^2 + c^2}{\Delta^n}$$

Taking L.H.S.

$$\frac{1}{r^2} + \frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2}$$

$$= \frac{s^2 + (s-a)^2 + (s-b)^2 + (s-c)^2}{\Delta^2}$$

$$= \frac{(a+b+c)^2 + (b+c-a)^2 + (a-b+c)^2 + (a+b-c)^2}{4\Delta^2}$$

$$= \frac{4(a^2 + b^2 + c^2)}{4\Delta^2} = \frac{a^2 + b^2 + c^2}{\Delta^2}$$

So $n = 2$

22. In $\triangle ABC$, medians AD , CE are drawn such that $AD = 5$, $\angle DAC = \frac{\pi}{8}$, $\angle ACE = \frac{\pi}{4}$. If area of $\triangle ABC$ is P then

$$\frac{3P}{5} =$$

Sol. Answer (5)

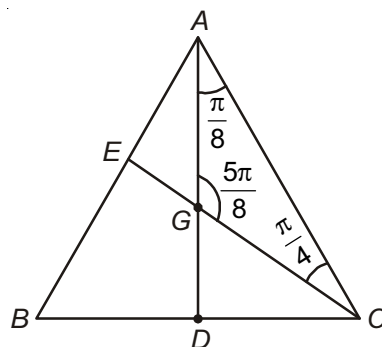
From sine rule

$$\frac{AG}{\sin \frac{\pi}{4}} = \frac{CG}{\sin \frac{\pi}{8}}$$

$$AG = \frac{10}{3}$$

$$CG = \frac{10}{3} \sqrt{2} \times \sin \frac{\pi}{8}$$

Area of $\triangle ABC (P)$



$$= 3 \text{ Area } \triangle AGC$$

$$= 3 \times \frac{1}{2} \times AG \cdot CG \sin\left(\frac{5\pi}{8}\right)$$

$$= \frac{3}{2} \times \frac{10}{3} \times \frac{10\sqrt{2}}{3} \sin \frac{\pi}{8} \cdot \sin\left(\frac{5\pi}{8}\right)$$

$$= \frac{25}{3} \sqrt{2} \left(\cos \frac{3\pi}{4} \right)$$

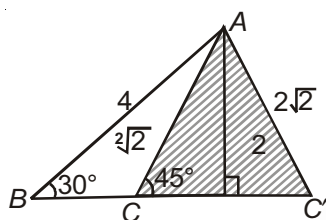
$$P = \frac{25}{3}$$

$$\text{Then, } \frac{3P}{5} = 5$$

23. Let ABC and ABC' be two non-congruent triangles with sides $AB = 4$, $AC = AC' = 2\sqrt{2}$ and angle $B = 30^\circ$. The absolute value of the difference between the areas of these triangles is [IIT-JEE 2009]

Sol. Answer (4)

Required area will be the shaded region in the figure



$$\text{Area of } \triangle ACC' = \frac{1}{2} \times 2 \times 4 = 4$$

24. The number of values of θ in the interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ such that $\theta \neq \frac{n\pi}{5}$ for $n = 0, \pm 1, \pm 2$ and $\tan \theta = \cot 5\theta$ as well as $\sin 2\theta = \cos 4\theta$ is [IIT-JEE 2010]

Sol. Answer (3)

$$\tan \theta = \cot 5\theta$$

we have,

$$\Rightarrow \tan \theta = \tan\left(\frac{\pi}{2} - 5\theta\right)$$

$$\Rightarrow \frac{\pi}{2} - 5\theta = n\pi + \theta$$

$$\Rightarrow 6\theta = \frac{\pi}{2} - n\pi$$

$$\Rightarrow \theta = \frac{\pi}{12} - \frac{n\pi}{6}$$

Also,

$$\cos 4\theta = \sin 2\theta = \cos\left(\frac{\pi}{2} - 2\theta\right)$$

$$\Rightarrow 4\theta = 2n\pi \pm \left(\frac{\pi}{2} - 2\theta\right)$$

Taking positive

$$6\theta = 2n\pi + \frac{\pi}{2}$$

$$\theta = \frac{n\pi}{3} + \frac{\pi}{12}$$

Taking negative

$$2\theta = 2n\pi - \frac{\pi}{2}$$

$$\theta = n\pi - \frac{\pi}{4}$$

Above values of θ suggests that there are only 3 common solutions.

25. The maximum value of the expression $\frac{1}{\sin^2 \theta + 3\sin \theta \cos \theta + 5\cos^2 \theta}$ is

[IIT-JEE 2010]

Sol. Answer (2)

We have,

$$f(\theta) = \frac{1}{\sin^2 \theta + 3\sin \theta \cos \theta + 5\cos^2 \theta}$$

$$\text{Let } g(\theta) = \sin^2 \theta + 3\sin \theta \cos \theta + 5\cos^2 \theta$$

$$= \frac{1 - \cos 2\theta}{2} + 5\left(\frac{1 + \cos 2\theta}{2}\right) + \frac{3}{2}\sin 2\theta$$

$$= 3 + 2\cos 2\theta + \frac{3}{2}\sin 2\theta$$

$$g(\theta)_{\min} = 3 - \sqrt{4 + \frac{9}{4}} = 3 - \frac{5}{2} = \frac{1}{2}$$

$$f(\theta)_{\max} = \frac{1}{g(\theta)_{\min}} = 2$$

26. Consider a triangle ABC and let a , b and c denote the lengths of the sides opposite to vertices A , B and C respectively. Suppose $a = 6$, $b = 10$ and the area of the triangle is $15\sqrt{3}$. If $\angle ACB$ is obtuse and if r denotes the radius of the incircle of the triangle, then r^2 is equal to

[IIT-JEE 2010]

Sol. Answer (3)

$$\sin C = \frac{\sqrt{3}}{2} \text{ and } C \text{ is given to be obtuse.}$$

$$\Rightarrow C = \frac{2\pi}{3}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab} \Rightarrow -\frac{1}{2} = \frac{36 + 100 - c^2}{2 \times 6 \times 10}$$

$$c = 14$$

$$r^2 = \left(\frac{15\sqrt{3} \times 2}{6 + 8 + 14} \right)^2 = 3$$

27. The positive integer value of $n > 3$ satisfying the equation $\frac{1}{\sin\left(\frac{\pi}{n}\right)} = \frac{1}{\sin\left(\frac{2\pi}{n}\right)} + \frac{1}{\sin\left(\frac{3\pi}{n}\right)}$ is [IIT-JEE 2011]

Sol. Answer (7)

According to the question,

$$\frac{1}{\sin \frac{\pi}{n}} = \frac{1}{\sin \frac{2\pi}{n}} + \frac{1}{\sin \frac{3\pi}{n}}$$

$$\Rightarrow \sin \frac{2\pi}{n} \sin \frac{3\pi}{n} = \sin \frac{\pi}{n} \sin \frac{3\pi}{n} + \sin \frac{\pi}{n} \sin \frac{2\pi}{n}$$

$$\Rightarrow \cos \frac{\pi}{n} - \cos \frac{5\pi}{n} = \cos \frac{2\pi}{n} - \frac{4\pi}{n} + \cos \frac{\pi}{n} - \cos \frac{3\pi}{n}$$

$$\Rightarrow \sin \frac{9\pi}{2n} \times \sin \frac{\pi}{2n} = \sin \frac{5\pi}{2n} \cdot \sin \frac{\pi}{n}$$

$$\Rightarrow \sin \frac{9\pi}{2n} = \sin \frac{5\pi}{2n}$$

$$\Rightarrow \frac{9\pi + 5\pi}{2n} = (2k + 1)\pi, \text{ say}$$

$$\Rightarrow n = \frac{7}{2k + 1}, k \text{ is an integer}$$

Let us put $k = 0$

Hence $n = 7$

28. The number of distinct solutions of the equation $\frac{5}{4} \cos^2 2x + \cos^4 x + \sin^4 x + \cos^6 x + \sin^6 x = 2$ in the interval $[0, 2\pi]$ is [JEE(Advanced)-2015]

Sol. Answer (8)

$$\frac{5}{4} \cos^2 2x + (\sin^2 x + \cos^2 x)^2 + (\sin^2 x + \cos^2 x)^3 - 5 \sin^2 x \cdot \cos^2 x = 2$$

$$\Rightarrow \frac{5}{4} (1 - \sin^2 2x) - \frac{5}{4} \sin^2 2x + 2 = 2$$

$$\Rightarrow 1 - 2 \sin^2 2x = 0$$

$$\Rightarrow \sin^2 2x = \frac{1}{2}$$

$$\Rightarrow 2x = n\pi \pm \frac{\pi}{4}$$

$$\Rightarrow x = \frac{n\pi}{2} \pm \frac{\pi}{8}$$

Number of solution = 8

29. Let a, b, c be three non-zero real numbers such that the equation

$$\sqrt{3} a \cos x + 2b \sin x = c, x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right],$$

has two distinct real roots α and β with $\alpha + \beta = \frac{\pi}{3}$. Then, the value of $\frac{b}{a}$ is _____. [JEE(Advanced)-2018]

Sol. Answer (0.5)

$$\text{Let } \tan \frac{x}{2} = t$$

$$\sqrt{3}a \left(\frac{1-t^2}{1+t^2} \right) + \frac{2b \cdot 2t}{1+t^2} = c$$

$$\Rightarrow t^2(c + \sqrt{3}a) + t(-4b) + (c - \sqrt{3}a) = 0$$

$$\tan \frac{\alpha}{2} + \tan \frac{\beta}{2} = \frac{4b}{c + \sqrt{3}a}$$

$$\tan \frac{\alpha}{2} \cdot \tan \frac{\beta}{2} = \frac{c - \sqrt{3}a}{c + \sqrt{3}a}$$

$$\frac{\alpha}{2} + \frac{\beta}{2} = \frac{\pi}{6}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{4b}{c + \sqrt{3}a - (c - \sqrt{3}a)}$$

$$\Rightarrow \frac{b}{a} = 0.5$$

SECTION - G

Multiple True-False Type Questions

1. STATEMENT-1 : $\sin 1 > \sin 2 > \sin 3$

STATEMENT-2 : $\sin 3 < \sin 1 < \sin 2$

STATEMENT-3 : $\sin x_1 < \sin x_2, x_1 < x_2, x_1, x_2 \in \left(0, \frac{\pi}{2}\right)$

(1) T F T

(2) F T T

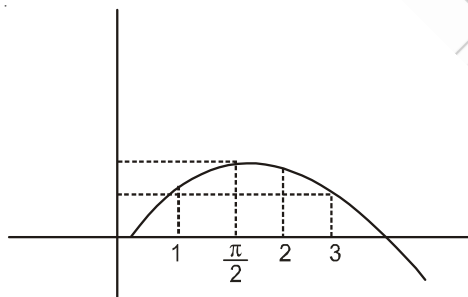
(3) F T F

(4) T F F

Sol. Answer (2)

Statement-I :

$\sin 1 > \sin 2 > \sin 3$ (false)



From graph $\sin 2 > \sin 1 > \sin 3$

Statement-II :

True as explained above

Statement- III :

True clearly from graph $\sin x$ increases in $\left(0, \frac{\pi}{2}\right)$

$\sin x, \sin x_2$ if $x_1 < x_2$ is $\left(0, \frac{\pi}{2}\right)$

$\therefore$ F T T

2. STATEMENT-1 : $\sin^2 6^\circ + \sin^2 12^\circ + \sin^2 18^\circ + \dots + \sin^2 84^\circ = 7$

STATEMENT-2 : $\tan 9^\circ \tan 27^\circ \tan 45^\circ \tan 36^\circ \tan 81^\circ = 1$

STATEMENT-3 : $\left(\tan \frac{\pi}{4} + \cot \frac{\pi}{4} + \operatorname{cosec} \frac{\pi}{4} \right) \left(\tan \frac{\pi}{4} + \cot \frac{\pi}{4} - \operatorname{cosec} \frac{\pi}{4} \right) = \sec^2 \frac{\pi}{3}$

(1) T T T

(2) F F T

(3) T T F

(4) T F F

Sol. Answer (4)

Statement-I :

$$\sin^2 6^\circ + \sin^2 12^\circ + \sin^2 18^\circ + \dots + \sin^2 84^\circ = 7$$

$$\text{L.H.S.} = (\sin^2 6^\circ + \cos^2 6^\circ) + (\sin^2 12^\circ + \cos^2 12^\circ) + \dots + (\sin^2 42^\circ + \cos^2 42^\circ)$$

$$(\sin^2 84^\circ = \cos^2 6^\circ) \text{ \& so on}$$

$$= 1 + 1 + \dots + 1 \text{ (7 times)}$$

$$= 7 = \text{R.H.S. True}$$

Statement-II :

$$\tan 9^\circ \cdot \tan 27^\circ \cdot \tan 45^\circ \cdot \tan 36^\circ \cdot \tan 81^\circ = 1$$

$$\text{L.H.S.} = \tan 9^\circ \cdot \tan 27^\circ \cdot \tan 36^\circ \cdot \cot 9^\circ$$

$$= \tan 27^\circ \cdot \tan 36^\circ \neq 1 \text{ (false)}$$

Statement-III :

$$\left(\tan \frac{\pi}{4} + \cot \frac{\pi}{4} + \operatorname{cosec} \frac{\pi}{4} \right) \left(\tan \frac{\pi}{4} + \cot \frac{\pi}{4} - \operatorname{cosec} \frac{\pi}{4} \right) = \sec^2 \frac{\pi}{3}$$

$$\text{L.H.S.} = (1+1+\sqrt{2})(1+1-\sqrt{2})$$

$$= 4 - 2 = 2 \neq \sec^2 \frac{\pi}{3} \text{ false}$$

3. STATEMENT-1 : $f(\theta) = \sin^2 \theta + \cos^2 \theta$ then $f(\theta) = 1$ for every real value of θ .

STATEMENT-2 : $g(\theta) = \sec^2 \theta - \tan^2 \theta$ then $g(\theta) = 1$ for every real value of θ .

STATEMENT-3 : $f(\theta) = g(\theta)$ for every real value of θ .

(1) T F F

(2) T T T

(3) T F T

(4) T T F

Sol. Answer (1)

Statement -I :

$$f(\theta) = \sin^2 \theta + \cos^2 \theta = 1, \theta \in R, \text{ true}$$

Statement -II :

$$f(\theta) = \sec^2 \theta - \tan^2 \theta = 1, \theta \in R, \text{ (given statement is false)}$$

$$\therefore \text{ it is true for } \theta \in R - \left\{ (2n+1)\frac{\pi}{2} \right\}$$

Statement-III :

$$f(\theta) = g(\theta) \text{ for all } \theta$$

(false) $\therefore g(\theta)$ doesn't contain odd multiples of $\frac{\pi}{2}$ in its domain

$\therefore$ Domains of f and g are different

$\therefore f$ & g are not equal T F F

4. STATEMENT-1 : $\cos \theta = x + \frac{1}{x}$ is possible for some real values of x .

STATEMENT-2 : $\sin 1^\circ < \sin 1$.

STATEMENT-3 : The minimum value of $5\cos x - 12\sin x + 13$ is 0.

(1) T T T

(2) F T T

(3) F F T

(4) F T F

Sol. Answer (2)

$$5\cos x - 12\sin x + 13$$

$$= -\sqrt{5^2 + 12^2} + 13$$

$$= 0$$

5. STATEMENT-1 : $\tan 22\frac{1}{2}^\circ$ is a root of the equation $\frac{1+x^2}{1-x^2} = \sqrt{2}$.

STATEMENT-2 : $\sin \alpha = x + \frac{k}{x}$ is possible for real value x if $k \leq \frac{1}{4}$.

STATEMENT-3 : $|\sin nx| \leq n|\sin x|$ is valid for all natural numbers n .

(1) T T T

(2) T F T

(3) T F F

(4) F F F

Sol. Answer (1)

$$x = \sqrt{2} - 1$$

$$\Rightarrow x^2 = 2 + 1 - 2\sqrt{2}$$

$$\Rightarrow \frac{x^2}{1} = \frac{3 - 2\sqrt{2}}{1}$$

$$\Rightarrow \frac{x^2 + 1}{x^2 - 1} = \frac{3 - 2\sqrt{2} + 1}{3 - 2\sqrt{2} - 1} = \frac{4 - 2\sqrt{2}}{2 - 2\sqrt{2}} = \frac{2 - \sqrt{2}}{1 - \sqrt{2}} = -\sqrt{2} \left(\frac{1 - \sqrt{2}}{1 - \sqrt{2}} \right) = -\sqrt{2}$$

$$\Rightarrow \frac{x^2 + 1}{1 - x^2} = \sqrt{2}$$

$$\frac{x + \frac{k}{x}}{2} \geq \sqrt{k}$$

$$k \leq \frac{1}{4}$$

$$\frac{|\sin nx| \leq n |\sin x|}{|\sin nx| \leq 10}$$

6. STATEMENT-1 : $\sin x = \frac{1}{p} \Rightarrow \tan x = \frac{1}{\sqrt{p^2-1}}, p \neq 1$

STATEMENT-2 : $\cos x = x^3 + \frac{3}{x}, x > 0$ has no solution in R

STATEMENT-3 : $\tan^2 x - \sin^2 x = \tan^2 x \cdot \sin^2 x, x \in R.$

(1) F F F

(2) F T F

(3) T T T

(4) T T F

Sol. Answer (2)

Statement-1 : $\tan x = \pm \frac{1}{\sqrt{p^2-1}}$

Statement-2 : As $x^3 + \frac{3}{x} \geq x$, hence no solution exist

Statement-3 : false for $x = (2x+1)\frac{\pi}{2}, x \in Z$

7. STATEMENT-1 : If α, β are different values of x satisfying $a \cos x + b \sin x = c, (b \neq 0)$, then $\tan\left(\frac{\alpha+\beta}{2}\right) = \frac{a}{b}$.

STATEMENT-2 : $\sin 2x + \cos 2x + \sin x + \cos x + 1 = 0$ has no solution in $\left(0, \frac{\pi}{2}\right)$.

STATEMENT-3 : $\sin \theta + \sqrt{3} \cos \theta \geq 1$ when $\theta \in \left[-\frac{\pi}{6}, \frac{\pi}{2}\right]$.

(1) F F F

(2) T T T

(3) T T F

(4) F T T

Sol. Answer (4)

Statement-1 :

$a \sin x + b \cos x = c$

$\therefore \alpha, \beta$ are two different values of x satisfying this equation

$$\begin{aligned} a \cos \alpha + b \sin \alpha &= c \\ a \cos \beta + b \sin \beta &= c \\ \therefore \frac{a \cos \alpha + b \sin \alpha - a \cos \beta - b \sin \beta}{a(\cos \alpha - \cos \beta) + b(\sin \alpha - \sin \beta)} &= 0 \end{aligned}$$

$$\Rightarrow -2a \sin \frac{\alpha+\beta}{2} \cdot \sin \frac{\alpha-\beta}{2} + 2b \cos \frac{\alpha+\beta}{2} \cdot \sin \frac{\alpha-\beta}{2} = 0$$

$$\Rightarrow \tan \frac{\alpha+\beta}{2} = \frac{b}{a} \quad (\because \alpha \neq \beta) \Rightarrow \quad (\text{False})$$

Statement-2 :

$\sin \theta + \sqrt{3} \cos \theta \geq 1$

L.H.S. = $2 \left(\sin \theta \cdot \frac{1}{2} + \frac{\sqrt{3}}{2} \cdot \cos \theta \right)$

$$= 2\sin\left(\theta + \frac{\pi}{3}\right)$$

$$\Rightarrow 2\sin\left(\theta + \frac{\pi}{3}\right) \geq 1$$

$$\sin\left(\theta + \frac{\pi}{3}\right) \geq \frac{1}{2}$$

$$\frac{\pi}{6} \leq \theta + \frac{\pi}{3} \leq \frac{5\pi}{6}$$

$$-\frac{\pi}{6} \leq \theta \leq \frac{\pi}{2} \quad (\text{True})$$

Statement-3 :

$$\sin 2x + \cos 2x + \sin x + \cos x + 1 = 0, \quad x \in \left(0, \frac{\pi}{2}\right)$$

$$2\sin x \cdot \cos x + 2\cos^2 x - 1 + \sin x + \cos x + 1 = 0$$

$$\sin x(2\cos x + 1) + \cos x(2\cos x + 1) = 0$$

$$\therefore x \in \left(0, \frac{\pi}{2}\right)$$

$\therefore$ Each term is positive.

$\therefore$ No solution (True)

8. STATEMENT-1 : The number of solutions of $|\cos x| = \sin x$ $x \in [0, 4\pi]$ is 4.

STATEMENT-2 : The equation $\sin^2 x - \frac{1}{\sqrt{2}}(\sqrt{3}+1)\sin x + \frac{\sqrt{3}}{4} = 0$ has two roots in $\left[0, \frac{\pi}{2}\right]$.

STATEMENT-3 : The number of solutions of $\sin \theta + \sin 5\theta = \sin 3\theta$, $\theta \in [0, \pi]$ is 5.

(1) T T T

(2) T F F

(3) T T F

(4) F T T

Sol. Answer (2)

Statement-1 :

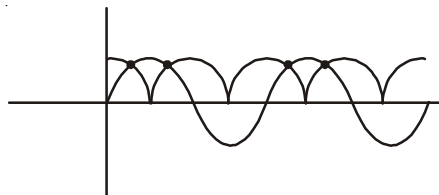
$$|\cos x| = \sin x$$

Number of solutions = 4 (True)

Statement-2 :

$$\sin^2 x - \frac{1}{\sqrt{2}}(\sqrt{3}+1)\sin x + \frac{\sqrt{3}}{4} = 0$$

$$\sin x = \frac{\frac{\sqrt{3}+1}{\sqrt{2}} \pm \sqrt{\frac{3+1+2\sqrt{3}}{2} - \frac{\sqrt{3}}{2}}}{2}$$



$$= \frac{\frac{\sqrt{3}+1}{\sqrt{2}} \pm \sqrt{2}}{2}$$

$$= \frac{\sqrt{3}-1}{2\sqrt{2}}, \sin x \neq \frac{\sqrt{3}+3}{2\sqrt{2}}$$

$\therefore$ One solution in $\left(0, \frac{\pi}{2}\right)$ (False)

Statement-3 :

$$\sin \theta + \sin 5\theta = \sin 3\theta, \theta \in [0, \pi]$$

$$2 \sin 3\theta \cdot \cos 2\theta = \sin 3\theta$$

$$\sin 3\theta = 0, \quad \cos 2\theta = \frac{1}{2}$$

$$\theta = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \quad \theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$\Rightarrow$ 6 Solution (False)

9. STATEMENT-1 : In a $\triangle ABC$, maximum value of $\sin A \sin B \sin C$ is $\frac{3\sqrt{3}}{8}$

STATEMENT-2 : In a $\triangle ABC$, $\sin^2 A + \sin^2 B + \sin^2 C = 2 - \cos^2 C + \cos C \cos(A - B)$

STATEMENT-3 : In a $\triangle ABC$, $\sin^2 A + \sin^2 B + \sin^2 C \leq 9/4$

(1) T T T

(2) T F T

(3) T F F

(4) F T F

Sol. Answer (1)

Statement-1 :

In a $\triangle ABC$

$$\therefore \sin A \cdot \sin B \cdot \sin C \leq \frac{3\sqrt{3}}{8}$$

Statement-2 :

$$\sin^2 A + \sin^2 B + \sin^2 C = 2 - \cos^2 C + \cos C \cdot \cos(A - B)$$

$$\text{R.H.S.} = 1 + (1 - \cos^2 C) + \cos[\pi - (A + B)] \cos(A - B)$$

$$= 1 + \sin^2 C - \cos(A + B) \cdot \cos(A - B)$$

$$= 1 + \sin^2 C - [\cos^2 A - \sin^2 B]$$

$$= (1 - \cos^2 A) + \sin^2 B + \sin^2 C$$

$$= \sin^2 A + \sin^2 B + \sin^2 C$$

Statement-3 :

In a $\triangle ABC$

$$\sin^2 A + \sin^2 B + \sin^2 C \leq \frac{9}{4}$$

10. STATEMENT-1 : In a $\triangle ABC$, if $2a^2 + 4b^2 + c^2 = 4ab + 2ac$, then $\cos A = \frac{1}{4}$

STATEMENT-2 : In a $\triangle ABC$ if $\cos A = \frac{1}{4}$, then $(a + b + c)(b + c - a) = \frac{5}{2}bc$

STATEMENT-3 : For any $\triangle ABC$, the expression $(b + c - a)(c + a - b)(a + b - c) - abc$ is negative.

(1) T T T

(2) F T T

(3) T F T

(4) T T F

Sol. Answer (1)

Statement-1 :

In a $\triangle ABC$

$$\text{If } 2a^2 + 4b^2 + c^2 = 4ab + 2ac$$

$$\Rightarrow a^2 + c^2 - 2ac + a^2 - 4ab + 4b^2 = 0$$

$$\Rightarrow (a - c)^2 + (a - 2b)^2 = 0$$

It is possible of if

$$a - c = 0 \text{ and } a - 2b = 0$$

$$\Rightarrow c = a \text{ and } b = \frac{a}{2}$$

$$\text{Let } a = x, b = \frac{x}{2}, \text{ and } c = x$$

$$\therefore \cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$= \frac{\frac{x^2}{4} + x^2 - x^2}{2\left(\frac{x}{2}\right)(x)}$$

$$\cos A = \frac{1}{4}$$

Statement-2 :

In a $\triangle ABC$

$$\text{Given } \cos A = \frac{1}{4} = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\Rightarrow \frac{bc}{2} = b^2 + c^2 - a^2$$

$$\text{Now, } (b + c + a)(b + c - a) = (b + c)^2 - a^2$$

$$= b^2 + c^2 - a^2 + 2bc$$

$$= \frac{bc}{2} + 2bc$$

$$= \frac{5bc}{2}$$

Statement-3 :In a $\triangle ABC$

$$(b + c - a)(c + a - b)(a + b - c) \leq abc$$

$$\text{Let } x = b + c - a$$

$$y = c + a - b$$

$$z = a + b - c$$

$$\text{Now, } \frac{x+y}{2} = c, \frac{y+z}{2} = a, \frac{z+x}{2} = b$$

By using A.M. $\geq$ G.M

$$x + y \geq 2\sqrt{xy}$$

$$y + z \geq 2\sqrt{yz}$$

$$z + x \geq 2\sqrt{zx}$$

$$\text{And } (x + y)(y + z)(z + x) \geq 8xyz$$

$$(2a)(2b)(2c) \geq 8(b + c - a)(c + a - b)(a + b - c)$$

$$\Rightarrow (b + c - a)(c + a - b)(a + b - c) \leq abc$$

SECTION - H**Aakash Changers Questions**

1. If $a \cos^3 \alpha + 3a \cos \alpha \sin^2 \alpha = m$, $a \sin^3 \alpha + 3a \cos^2 \alpha \sin \alpha = n$, then $(m+n)^{2/3} + (m-n)^{2/3}$ is equal to

(1) $2a^2$

(2) $2a^{2/3}$

(3) $a^{2/3}$

(4) $2a^3$

Sol. Answer (2)

$$m + n = a\{(\sin^3 \alpha + \cos^3 \alpha) + 3\sin \alpha \cos \alpha (\sin \alpha + \cos \alpha)\}$$

$$= a(\sin \alpha + \cos \alpha) [1 - \sin \alpha \cos \alpha + 3\sin \alpha \cos \alpha]$$

$$= a(\sin \alpha + \cos \alpha)^3$$

$$\text{Also } m - n = a[(\cos^3 \alpha - \sin^3 \alpha)] + 3 \sin \alpha \cos \alpha - (\sin \alpha - \cos \alpha)$$

$$= a(\cos \alpha - \sin \alpha) [1 + \sin \alpha \cos \alpha - 3\sin \alpha \cos \alpha]$$

$$= a(\cos \alpha - \sin \alpha)^3 \quad (1)$$

$$\therefore (m + n)^{2/3} + (m - n)^{2/3} = a^{2/3}[(\sin \alpha + \cos \alpha)^2 + (\cos \alpha - \sin \alpha)^2]$$

$$= 2a^{2/3}$$

2. Let f and g be function defined by $f(\theta) = \cos^2 \theta$ and $g(\theta) = \tan^2 \theta$. Suppose α and β satisfy $2f(\alpha) - g(\beta) = 1$, then the value of $2f(\beta) - g(\alpha)$ is

Sol. $f(\theta) = \cos^2 \theta$, $g(\theta) = \sin^2 \theta$

$$2f(\alpha) - g(\beta) = 1$$

$$2\cos^2 \alpha - \sin^2 \beta = 1$$

$$2\cos^2 \alpha = 1 + \sin^2 \beta = \sin^2 \beta$$

$$2\cos^2 \beta = \sec^2 \alpha$$

$$2f(\beta) - g(\alpha)$$

$$= 2\cos^2 \beta - \sin^2 \alpha$$

$$= \sec^2 \alpha - \sin^2 \alpha$$

$$= 1$$

3. The value of $\cot 16^\circ \cot 44^\circ + \cot 44^\circ \cot 76^\circ - \cot 76^\circ \cot 16^\circ$ is

Sol. $\cot 16^\circ \cdot \cot 44^\circ + \cot 44^\circ \cdot \cot 76^\circ - \cot 76^\circ \cot 16^\circ$

$$= \cot 16^\circ \cdot \cot(60^\circ - 16^\circ) + \cot(60^\circ - 16^\circ) \cdot \cot(60^\circ + 16^\circ) - \cot(60^\circ + 16^\circ) \cdot \cot 16^\circ$$

$$= \cot 16^\circ \cdot \left(\frac{\cot 60^\circ \cot 16^\circ + 1}{\cot 60^\circ - \cot 16^\circ} \right) + \frac{\cos^2 60^\circ - \sin^2 16^\circ}{\sin^2 60^\circ - \sin^2 16^\circ} - \frac{\cot 60^\circ \cdot \cot 16^\circ - 1}{\cot 60^\circ + \cot 16^\circ} \cdot \cot 16^\circ$$

$$= \cot 16^\circ \left(\frac{\cot 16^\circ + \sqrt{3}}{1 - \sqrt{3} \cot 16^\circ} \right) + \frac{1 - 4 \sin^2 16^\circ}{3 - 4 \sin^2 16^\circ} - \frac{\cot 16^\circ - \sqrt{3}}{1 + \sqrt{3} \cot 16^\circ} \cdot \cot 16^\circ - (1 - \sqrt{3} \cot 16^\circ)(\cot 16^\circ - \sqrt{3})$$

$$= (\cot 16^\circ) \left(\frac{(\cot 16^\circ + \sqrt{3})(1 + \sqrt{3} \cot 16^\circ)}{1 - 3 \cot^2 16^\circ} \right) + \frac{1 - 4 \sin^2 16^\circ}{3 - 4 \sin^2 16^\circ}$$

$$= (\cot 16^\circ) \left(\frac{\cot 16^\circ + \sqrt{3} \cot^2 16^\circ + \sqrt{3} + 3 \cot 16^\circ - \cot 16^\circ + \sqrt{3} + \sqrt{3} \cot^2 16^\circ - 3 \cot 16^\circ}{1 - 3 \cot^2 16^\circ} \right) + \frac{1 - 4 \sin^2 16^\circ}{3 - 4 \sin^2 16^\circ}$$

$$= \frac{\cot^2 16^\circ \cdot 2\sqrt{3}(1 + \cot 16^\circ)}{1 - 3 \cot^2 16^\circ} + \frac{1 - 4 \sin^2 16^\circ}{3 - 4 \sin^2 16^\circ}$$

$$= 3$$

4. If $\cos(\theta - \theta_1) = \frac{1}{2\sqrt{3}}$ and $\sin(\theta - \theta_2) = \frac{1}{3\sqrt{2}}$, where $0 < \theta < \theta_1$, $\theta - \theta_2 < \frac{\pi}{2}$, then the value of

$$\frac{108}{5} \left\{ \cos^2(\theta_1 - \theta_2) + \frac{\sqrt{6}}{18} \sin(\theta_1 - \theta_2) \right\} \text{ is$$

Sol. $\cos(\theta - \theta_1) = \frac{1}{2\sqrt{3}}, \quad \sin(\theta - \theta_2) = \frac{1}{3\sqrt{2}}$

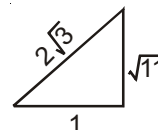
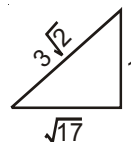
$$\therefore \cos(\theta_1 - \theta_2) = \cos((\theta_1 - \theta) + (\theta - \theta_2))$$

$$= \cos(\theta - \theta_2) \cos(\theta - \theta_1)$$

$$= \cos(\theta - \theta_2) \cos(\theta - \theta_1) + \sin(\theta - \theta_1) \sin(\theta - \theta_2)$$

$$= \frac{\sqrt{17}}{3\sqrt{2}} \cdot \frac{1}{2\sqrt{3}} + \frac{\sqrt{11}}{2\sqrt{3}} \cdot \frac{1}{3\sqrt{2}}$$

$$= \frac{\sqrt{17} + \sqrt{11}}{6\sqrt{6}}$$



$$\begin{aligned}
 \sin(\theta_1 - \theta_2) &= \sin((\theta - \theta_2) - (\theta - \theta_1)) \\
 &= \sin(\theta - \theta_2)\cos(\theta - \theta_1) - \cos(\theta - \theta_2)\sin(\theta - \theta_1) \\
 &= \frac{1}{3\sqrt{2}} \cdot \frac{1}{2\sqrt{3}} - \frac{\sqrt{17}}{3\sqrt{2}} \cdot \frac{\sqrt{11}}{2\sqrt{3}} \\
 &= \frac{1 - \sqrt{17} \times \sqrt{11}}{6\sqrt{6}}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \frac{108}{5} &\left\{ \frac{17 + 11 + 2\sqrt{17}\sqrt{11}}{216} + \frac{\sqrt{6}}{18} \times \frac{(1 - \sqrt{7} \cdot \sqrt{11})}{6\sqrt{6}} \right\} \\
 &= \frac{1}{10} (28 + 2\sqrt{17}\sqrt{11}) + \frac{2}{10} (1 - \sqrt{7} - \sqrt{11}) \\
 &= \frac{1}{10} (28 + 2\sqrt{17}\sqrt{11} + 2 - 2\sqrt{7} - 2\sqrt{11}) \\
 &= \frac{1}{10} \times 30 = 3
 \end{aligned}$$

5. If $A = \tan 270 - \tan \theta$ and $B = \frac{\sin \theta}{\cos 30} + \frac{\sin 30}{\cos 90} + \frac{\sin 90}{\cos 270}$, then the value of $\frac{A+B}{A-B} + \frac{A-B}{A+B} - \frac{10}{3}$ is.....

Sol. Given $A = \tan 270 - \tan \theta$

$$= \tan 270 - \tan 90 + \tan 90 - \tan 30 + \tan 30 - \tan \theta$$

$$\therefore \tan 30 - \tan \theta = \frac{\sin 2\theta}{\cos 30 \cos \theta} = \frac{2 \sin \theta}{\sin 30}$$

$$\tan 90 - \tan 30 = \frac{2 \sin 30}{\cos 90}$$

$$\tan 270 - \tan 90 = \frac{2 \sin 90}{\cos 270}$$

$$\Rightarrow A = 2B$$

$$\Rightarrow \frac{A}{B} = 2$$

$$\Rightarrow \frac{A+B}{A-B} = \frac{2+2}{2-1} = 3, \frac{A-B}{A+B} = \frac{2-1}{2+1} = \frac{1}{3}$$

$$\Rightarrow \frac{A+B}{A-B} + \frac{A-B}{A+B} = \frac{3+1}{3} = \frac{10}{3}$$

$$\Rightarrow \frac{A+B}{A-B} + \frac{A-B}{A+B} - \frac{10}{3} = 0$$

6. If $A = \sqrt{3} \cot 20^\circ - 4 \cos 20^\circ$ and $B = \sin 12^\circ \sin 48^\circ \sin 54^\circ$ be such that $A + \lambda B = 2$, then the value of $\lambda^3 + 2000$ is.....

Sol. Given $A = \sqrt{3} \cot 20^\circ - 4 \cos 20^\circ$

$$\begin{aligned}
 &= \sqrt{3} \frac{\cos 20^\circ}{\sin 20^\circ} - 4 \cos 20^\circ \\
 &= \frac{\sqrt{3} \cos 20^\circ - 4 \sin 20^\circ \cos 20^\circ}{\sin 20^\circ} \\
 &= \frac{2(\sin 60^\circ + \cos 20^\circ) - 2 \sin 40^\circ}{\sin 20^\circ} \\
 &= \frac{2 \sin 60^\circ \cos 20^\circ - 2 \sin 40^\circ}{\sin 20^\circ} \\
 &= \frac{\sin 80^\circ + \sin 40^\circ - 2 \sin 40^\circ}{\sin 20^\circ} \\
 &= \frac{\sin 80^\circ - \sin 40^\circ}{\sin 20^\circ} \\
 &= \frac{2 \cos 60^\circ \sin 20^\circ}{\sin 20^\circ} \\
 &= 2 \times \frac{1}{2} = 1
 \end{aligned}$$

$$B = \sin 12^\circ \sin 48^\circ \sin 54^\circ$$

$$\begin{aligned}
 &= \frac{1}{2} (2 \sin 48^\circ \sin 12^\circ) \cos 36^\circ \\
 &= \frac{1}{2} (\cos 36^\circ - \cos 60^\circ) \cos 36^\circ \\
 &= \frac{1}{2} \left(\frac{\sqrt{5}+1}{4} - \frac{1}{2} \right) \cdot \frac{\sqrt{5}+1}{4} \\
 &= \frac{1}{2} \times \frac{\sqrt{5}-1}{4} \times \frac{\sqrt{5}+1}{4} \\
 &= \frac{1}{2} \times \frac{4}{16} \\
 &= \frac{1}{8}
 \end{aligned}$$

$$\text{Now, } A + 8B = 1 + 1 = 2$$

$$A + 8B = 2$$

$$\lambda = 8$$

$$\therefore \lambda^3 + 2000 = 8^3 + 2000$$

$$= 512 + 2000$$

$$= 2512$$

7. If $\tan \alpha = \frac{1}{\sqrt{x(x^2+x+1)}}$, $\tan \beta = \frac{\sqrt{x}}{\sqrt{x^2+x+1}}$ and $\tan \gamma = \sqrt{\frac{1}{x^3} + \frac{1}{x^2} + \frac{1}{x}}$ be such that $l\alpha + m\beta + n\gamma = 0$, then the value of $l^3 + m^3 + n^3 - 3lmn$ is (where $l, m, n \in \mathbb{Z}$ having no common factor except one)

Sol. Given $\tan \alpha = \frac{1}{\sqrt{x(x^2+x+1)}}$, $\tan \beta = \frac{\sqrt{x}}{\sqrt{x^2+x+1}}$

$$\tan \gamma = \frac{1}{\sqrt{\frac{1+x+x^2}{x^3}}} = \frac{x\sqrt{x}}{\sqrt{x^2+x+1}}$$

Now, $\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$

$$= \frac{\frac{1}{\sqrt{x(x^2+x+1)}} + \frac{\sqrt{x}}{\sqrt{x^2+x+1}}}{1 - \frac{\sqrt{x}}{\sqrt{x(x^2+x+1)}}}$$

$$= \frac{\frac{(1+x)}{\sqrt{x^2+x+1}}\sqrt{x}}{\frac{x^2+x+1-1}{(x^2+x+1)}}$$

$$= \frac{(x+1)}{\sqrt{x}\sqrt{x^2+x+1}} \times \frac{\sqrt{(x^2+x+1)}}{x(x+1)}$$

$$= \frac{\sqrt{x^2+x+1}}{x\sqrt{x}}$$

$$= \sqrt{\frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3}}$$

$$= \tan \gamma$$

$$\Rightarrow \alpha + \beta = \gamma$$

$$\Rightarrow \alpha + \beta - \gamma = 0$$

$$\Rightarrow l = 1, m = 1, n = -1$$

$$\text{Now, } l^3 + m^3 + n^3 = 3lmn$$

$$= 1 + 1 - 1 + 3$$

$$= 4$$

8. $A = \{\theta | 2\cos^2\theta + \sin\theta \leq 2\}$

$B = \left\{\theta \left| \frac{\pi}{2} \leq \theta \leq \frac{3\pi}{2} \right.\right\}$. Find $A \cap B$

Sol. $A = \{\theta | 2\cos^2\theta + \sin\theta \leq 2\}$

$$2\sin^2\theta - \sin\theta \geq 0$$

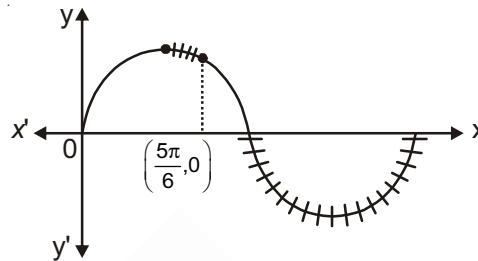
$$\sin\theta(2\sin\theta - 1) \geq 0 \quad \dots(1)$$

Since

$$\theta \in \left[\frac{\pi}{2}, \frac{3\pi}{2}\right] \quad \dots(2)$$

Hence (1) and (2) will represent

$$\theta \in \left[\frac{\pi}{2}, \frac{5\pi}{6}\right] \cup \left[\pi, \frac{3\pi}{2}\right]$$



9. Solve the equation, $\sin^2 n\theta - \sin^2[(n-1)\theta] = \sin^2\theta$.

Sol. $\sin^2 n\theta - \sin^2(n-1)\theta = \sin^2\theta$

$$\Rightarrow \sin(2n\theta - \theta) \cdot \sin\theta = \sin^2\theta$$

$$\Rightarrow \sin\theta = 0 \text{ or } \sin(2n-1)\theta = \sin\theta$$

$$\sin\theta = 0$$

$$\Rightarrow \theta = k\pi$$

$$\sin(2n-1)\theta = \sin\theta$$

$$\Rightarrow (2n-1)\theta = k\pi + (-1)^k\theta$$

$$k = \text{even} = 2m$$

$$\Rightarrow (2n-1)\theta = 2m\pi + \theta$$

$$\theta = \frac{m\pi}{n-2}$$

$$k = \text{odd} = 2m+1$$

$$(2n-1)\theta = (2m+1)\pi - \theta$$

$$\theta = \frac{(2m+1)\pi}{2n}$$

$$\therefore \theta = k\pi \text{ or } \left(\frac{2m+1}{2n}\right)\pi$$

$$\text{or } \frac{m\pi}{n-2}$$

$$\text{where } k, m \in \mathbb{Z}$$

10. Solve the equation $\tan x + \frac{\cos x}{\sqrt{1+\sin 2x}} = 2$.

Sol. $\tan x + \frac{\cos x}{\sqrt{1+\sin 2x}} = 2$

$$\tan x + \frac{\cos x}{\sin x + \cos x} = 2, \quad \text{if } \sin x + \cos x > 0$$

$$\tan x + \frac{1}{1 + \tan x} = 2$$

$$(1 + \tan x) + \frac{1}{1 + \tan x} = 3$$

Let $1 + \tan x = y$

$$\Rightarrow y^2 - 3y + 1 = 0$$

$$\Rightarrow y = \frac{3 \pm \sqrt{5}}{2}$$

$$\therefore \tan x = \frac{1 + \sqrt{5}}{2}$$

$$\tan x = \frac{\sqrt{5} + 1}{2}, \quad \tan x = \frac{-\sqrt{5} + 1}{2}$$

$$x = n\pi + \tan^{-1}\left(\frac{\sqrt{5} + 1}{2}\right) \quad \text{or} \quad n\pi + \tan^{-1}\left(\frac{-\sqrt{5} + 1}{2}\right)$$

Case II

$$\sin x + \cos x < 0$$

$$\tan x - \frac{\cos x}{\sin x + \cos x} = 2$$

$$\tan x - \frac{1}{1 + \tan x} = 2$$

$$1 + \tan x - \frac{1}{1 + \tan x} = 3$$

$$y^2 - 3y - 1 = 0 \quad (1 + \tan x = y)$$

$$y = \frac{3 \pm \sqrt{13}}{2}$$

$$\tan x = \frac{1 \pm \sqrt{13}}{2}$$

$$\frac{1 + \sqrt{13}}{2}, \frac{1 - \sqrt{13}}{2}$$

11. Let x and y , $(0 < x, y < \frac{\pi}{2})$ satisfy $3\sin^2x + 2\sin^2y = 1$ and $3\sin 2x = 2\sin 2y$, then the value of $\frac{232}{\pi}(x+2y)$ is _____.

Sol. We have,

$$3\sin^2x + 2\sin^2y = 1$$

$$\Rightarrow 3\sin^2x = \cos 2y \text{ and } 3\sin 2x = 2\sin 2y$$

$$\Rightarrow \tan x = \cot 2y = \tan\left(\frac{\pi}{2} - 2y\right)$$

$$\Rightarrow x + 2y = n\pi + \frac{\pi}{2}, \quad n \in \mathbb{Z}$$

$$= (2n+1)\frac{\pi}{2}$$

$$\text{Since } 0 < x, y < \frac{\pi}{2}$$

$$\Rightarrow 0 < x + 2y < \frac{3\pi}{2} \Rightarrow x + 2y = \frac{\pi}{2}$$

$$\Rightarrow \frac{232}{\pi}(x+2y) = \left(\frac{232}{\pi}\right)\left(\frac{\pi}{2}\right) = 116$$

12. If angles A and B satisfy $\sqrt{2}\cos A = \cos B + \cos^3 B$ and $\sqrt{2}\sin A = \sin B - \sin^3 B$, then the value of $1620\sin^2(A-B)$ is _____.

Sol. $\sqrt{2}\cos A = \cos B(1 + \cos^2 B) = \cos B\left(\frac{3 + \cos 2B}{2}\right)$

$$\text{and } \sqrt{2}\sin A = \sin B(1 - \sin^2 B) = \sin B\left(\frac{1 + \cos 2B}{2}\right)$$

$$\text{Hence, } \sqrt{2}\sin(A-B) = -\sin B \cos B$$

$$\Rightarrow 2\sqrt{2}\sin(A-B) = -\sin 2B$$

$$\text{and } 8 = \cos^2 B \left(\frac{3 + \cos 2B}{2}\right)^2 + \sin^2 B \left(\frac{1 + \cos 2B}{2}\right)^2$$

After simplification we shall get

$$\cos 2B = \frac{1}{3} \text{ and } \sin 2B = \frac{-2\sqrt{2}}{3}$$

$$\text{Hence } 1620\sin^2(A-B) = 1620 \times \frac{\sin^2 2B}{8} = 1620 \times \frac{8}{8 \times 9} = 180$$

13. Let the sum of all solutions of the equation $3\sqrt{3}\sin^3 x + \cos^3 x + 3\sqrt{3}\sin x \cos x = 1$ in the interval $[0, 10\pi]$ be denoted by S . Then $\frac{3S}{\pi}$ is _____.

Sol. We have,

$$3\sqrt{3}\sin^3 x + \cos^3 x + 3\sqrt{3}\sin x \cos x = 1$$

$$\Rightarrow (\sqrt{3}\sin x)^3 + (\cos x)^3 + (-1)^3 - 3(\sqrt{3}\sin x)(\cos x)(-1) = 0$$

$$\Rightarrow \sqrt{3}\sin x + \cos x - 1 = 0$$

$$\Rightarrow \cos x + \sqrt{3}\sin x = 1$$

$$\Rightarrow (\cos x)\frac{1}{2} + \frac{\sqrt{3}}{2}\sin x = \frac{1}{2}$$

$$\Rightarrow \sin\left(x + \frac{\pi}{6}\right) = \sin \frac{\pi}{6}$$

$$\text{or } \cos\left(x - \frac{\pi}{3}\right) = \cos \frac{\pi}{3}$$

$$\Rightarrow x - \frac{\pi}{3} = 2n\pi \pm \frac{\pi}{3}$$

$$\Rightarrow x = 2n\pi, 2n\pi + \frac{2\pi}{3}, n \in \mathbb{Z}$$

The solutions of the given equation lying in $[0, 10\pi]$ are

$$0, 2\pi, 4\pi, 6\pi, 8\pi, 10\pi; \frac{2\pi}{3}, \frac{8\pi}{3}, \frac{14\pi}{3}, \frac{20\pi}{3}, \frac{26\pi}{3}$$

Thus the sum of all solutions of the given equation is $\frac{160\pi}{3} = s$

$$\Rightarrow \frac{3s}{\pi} = \frac{3}{\pi} \times \frac{160\pi}{3} = 160$$

14. If the equation $\sin 6x + \cos 4x = -2$ have a family of nonnegative solutions x_k 's, where $0 \leq x_1 < x_2 < x_3$ _____.

$< x_k < x_{k+1}$, then the value of $\frac{1}{\pi} \sum_{k=1}^{1000} |x_{k+1} - x_k|$ is _____.

Sol. We have,

$$\sin 6x + \cos 4x = -2$$

Which will be possible only when $\sin 6x = -1$, $\cos 4x = -1$

$$\Rightarrow 6x = 2k\pi - \frac{\pi}{2}, 4x = 2m\pi - \pi, k, m \in \mathbb{Z}$$

$$x = n\pi + \frac{\pi}{4} \text{ where } n \in \mathbb{Z}.$$

$$\text{So } x_{k+1} - x_k = \pi$$

$$\Rightarrow \frac{1}{\pi} \sum_{k=1}^{1000} |x_{k+1} - x_k| = 1000$$

15. Find all values of x , y and k for which the system of equations

$$\sin x \cos 2y = k^4 - 2k^2 + 2$$

$$\cos x \sin 2y = k + 1$$

has a solution.

Sol. $\sin x \cos 2y = k^4 - 2k^2 + 2$

$$\cos x \sin 2y = k + 1$$

$$\sin x \cos 2y = (k^2 - 1)^2 + 1$$

$$\text{and } \cos x \sin 2y = k + 1$$

But L.H.S. is less than or equal to 1.

$$\Rightarrow k = \pm 1$$

So,

Case-I, $k = 1$

$$\Rightarrow \sin x \cos 2y = 1$$

$$\Rightarrow \sin x = 1 \text{ and } \cos 2y = 1$$

or $\cos x \sin 2y = 2$ which is not possible

$$\Rightarrow \text{Case-II,}$$

When $k = -1$

$$\sin x \cos 2y = 1$$

$$\Rightarrow \sin x = 1 \text{ and } \cos 2y = 1$$

$$\text{or, } \sin x = -1 \text{ or } \cos 2y = -1$$

$$\text{and } \cos x \sin 2y = 0$$

$$x = 2n\pi \pm \frac{\pi}{2} \text{ and } y = \frac{n\pi}{2}$$

16. The number of solutions of the equation $\sin^5 \theta + \frac{1}{\sin \theta} = \frac{1}{\cos \theta} + \cos^5 \theta$ where $\theta \in \left(0, \frac{\pi}{2}\right)$, is

(1) 0

(2) 1

(3) 2

(4) 3

Sol. Answer (2)

$$\sin^5 \theta - \cos^5 \theta = \frac{1}{\cos \theta} - \frac{1}{\sin \theta}$$

$$(\sin \theta - \cos \theta) [\sin^4 \theta + \sin^3 \theta \cos \theta + \sin^2 \theta \cos^2 \theta + \sin \theta \cos^3 \theta + \cos^4 \theta] = \frac{\sin \theta - \cos \theta}{\sin \theta \cos \theta}$$

$$\Rightarrow \sin\theta - \cos\theta = 0 \quad \dots(i)$$

$$\text{OR } \sin^4\theta + \cos^4\theta + \sin\theta\cos\theta (\sin^2\theta + \cos^2\theta) + \sin^2\theta \cos^2\theta - \frac{1}{\sin\theta\cos\theta} = 0 \quad \dots(ii)$$

$$\text{From (i) } \sin\theta = \cos\theta \Rightarrow \theta = \frac{\pi}{4}$$

From (ii)

$$1 - \frac{1}{2}\sin^2 2\theta + \frac{1}{2}\sin 2\theta + \frac{1}{4}\sin^2 2\theta - \frac{2}{\sin 2\theta} = 0$$

$$\Rightarrow \sin^3 2\theta - 2\sin^2 2\theta - 4\sin 2\theta + 8 = 0$$

$$\Rightarrow (\sin 2\theta - 2)(\sin^2 2\theta - 4) = 0$$

$\Rightarrow$ No solution

$$\therefore \text{ Only solution is } \theta = \frac{\pi}{4}$$

17. Let the number of solutions of the equation $\sin x = x^2 - 4x + 5$ be λ . The general solution of the equation

$$1 + \tan((\lambda^2 - 2\lambda + 1)\theta) + \frac{2}{1 - \tan((\lambda^2 + 2\lambda + 1)\theta)} = 0 \text{ is}$$

$$(1) \ n\pi + \frac{\pi}{4} \text{ only } n \in \mathbb{Z} \quad (2) \ n\pi \pm \frac{\pi}{4}, n \in \mathbb{Z} \quad (3) \ n\pi + \frac{\pi}{3} \text{ only } n \in \mathbb{Z} \quad (4) \ n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$$

Sol. Answer (4)

$$\sin x = (x - 2)^2 + 1$$

$$-1 \leq \text{L.H.S} \leq 1 \text{ and } \text{RHS} \geq 1$$

$$\text{L.H.S.} = 1 \text{ at } x = \frac{\pi}{2} \text{ while } \text{R.H.S.} = 1 \text{ at } x = 2$$

No solution

$$\therefore \lambda = 0$$

Putting $\lambda = 0$ in given equation, we get

$$1 + \tan\theta + \frac{2}{1 - \tan\theta} = 0$$

$$1 - \tan^2\theta + 2 = 0$$

$$\Rightarrow \tan^2\theta = 3$$

$$\therefore \text{ General solution is } \theta = n\pi \pm \frac{\pi}{3}$$

18. Let three sets be defined as

$$A = \{x | \cos 2x + \cos x + 1 = 0\}$$

$$B = \{x | \cos 2x + 3\sin x = 2\}$$

$$C = \{x | \sec x + \tan x = \cos x\}$$

The number of elements in $A \cap B \cap C$ is

$$(1) \ 0 \quad (2) \ 4 \quad (3) \ 8 \quad (4) \ \text{Infinitely many}$$

Sol. Answer (1)

(i) For set A

$$\cos 2x + \cos x + 1 = 0$$

$$2\cos^2 x + \cos x = 0$$

$$\cos x(2\cos x + 1) = 0$$

$$\cos x = 0 \text{ or } \cos x = -\frac{1}{2}$$

$$x = (2n+1)\frac{\pi}{2} \text{ or } x = 2n\pi \pm \frac{2\pi}{3}$$

(ii) For set B

$$\cos 2x = 3\sin x = 2$$

$$1 - 2\sin^2 x + 3\sin x = 2$$

$$\Rightarrow 2\sin^2 x - 3\sin x + 1 = 0$$

$$(2\sin x - 1)(\sin x - 1) = 0$$

$$\sin x = \frac{1}{2} \text{ or } \sin x = 1$$

$$x = n\pi + (-1)^n \cdot \frac{\pi}{6} \text{ or } x = 2n\pi + \frac{\pi}{2}$$

(iii) For set C

$$\sec x + \tan x = \cos x$$

$$1 + \sin x = \cos^2 x$$

$$\Rightarrow \sin^2 x + \sin x = 0$$

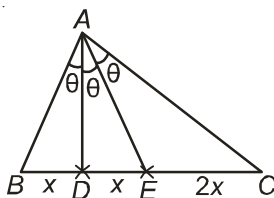
$$\sin x = 0, \sin x \neq -1 \text{ as } \cos x \neq 0$$

$$\Rightarrow x = n\pi$$

$\therefore$ From (i), (ii), (iii) it is clear that there exists no x satisfying all 3 conditions

$\therefore$ No. of elements in $A \cap B \cap C = 0$

19. In a triangle ABC, if the median and altitude from A trisect angle A, then $(B - C)$ is (in degrees) equal to _____.

Sol.

From the question it is clear

$$BD = DE \text{ and } BE = EC$$

$$\text{Let } BD = x$$

$$\text{So, } BD = DE = x$$

$$\& EC = 2x$$

$\Rightarrow AE$ is the angle bisector of $\triangle ADC$

$$\text{So, } \frac{AD}{AC} = \frac{1}{2}, \text{ But } \frac{AD}{DC} = \sin C$$

$$\Rightarrow \sin C = \sin 30^\circ \Rightarrow C = 30^\circ$$

$$\Rightarrow \frac{AD}{DC} = \tan 30 \Rightarrow \frac{AD}{3x} = \frac{1}{\sqrt{3}} \Rightarrow AD = \sqrt{3}x$$

In $\triangle ABD$

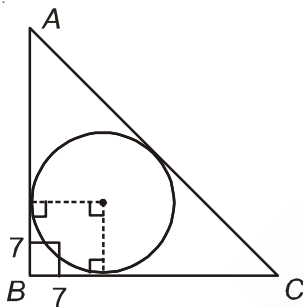
$$\tan B = \frac{AD}{x} = \frac{\sqrt{3}x}{x}$$

$$\Rightarrow B = 60$$

So, $\angle B - \angle C = 30^\circ$

20. In a right angled triangle ABC , if r (inradius) = 7 cm and R (circumradius) = 32.5 cm, then the area of the triangle (in square cm) is _____

Sol.



$$r = 7 \text{ cm}$$

$$R = 32.5 \text{ cm}$$

$$A = ?$$

Let $C\theta = x$

$$C_{\theta} = C_5$$

$$AP = A5$$

[Length of tangent drawn from internal point are equal]

Apply Pythagoras theorem in w.r.t. $\triangle ABC$

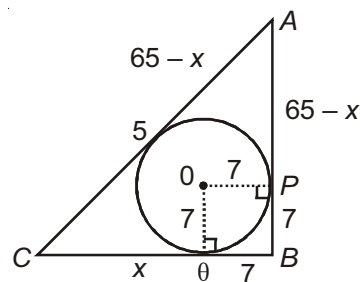
$$AB^2 + BC^2 = AC^2$$

$$(72 - x)^2 + (7 + x)^2 = 65^2$$

Simplifying we get $x = 56$ or 9

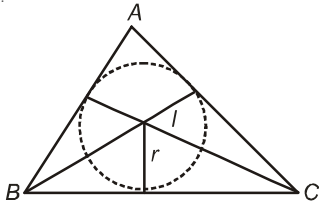
$$\text{Ar } \triangle ABC = \frac{1}{2} \times BC \times AB = \frac{1}{2} \times 16 \times 63$$

= 504 sq. units



21. In a triangle ABC , perpendicular distance of BC from the point of intersection of angle bisector of $\angle B$ and $\angle C$ is _____.

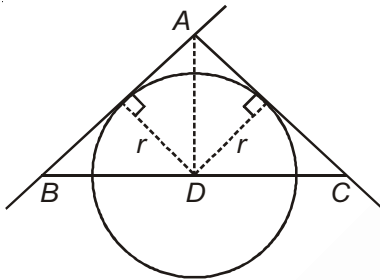
Sol.



From figure it is clear that the perpendicular distance of BC from the point of intersection of angle bisector of $\angle B$ and $\angle C$ is
 = inradius
 = r

22. The radius of the circle touching two sides AB and AC of a triangle ABC and having its centre on BC is equal to _____.

Sol.



Let the radius of the circle be r

$$\text{ar}(\triangle ABC) = \text{ar}(\triangle ABD) + \text{ar}(\triangle ACD)$$

$$\Rightarrow \Delta = \frac{1}{2} \cdot r \cdot b + \frac{1}{2} \cdot r \cdot c$$

$$\Rightarrow \boxed{r = \frac{2\Delta}{b+c}}$$

23. Find the cubic equation whose roots are the radius of three inscribed circles in term of inradius, circumradius and perimeter.

Sol. Let roots be r_1, r_2 and r_3

$$\therefore \text{Equation } x^3 - (r_1 + r_2 + r_3)x^2 + (r_1r_2 + r_2r_3 + r_1r_3)x - r_1r_2r_3 = 0$$

$$r_1 + r_2 + r_3 = \frac{\Delta^2}{(s-a)(s-b)} + \frac{\Delta^2}{(s-b)(s-c)} + \frac{\Delta^2}{(s-c)(s-a)}$$

$$= s\Delta^2 \left[\frac{s-a+s-b+s-c}{s(s-a)(s-b)(s-c)} \right]$$

$$= \frac{s\Delta^2 \cdot s}{\Delta^2} = s^2$$

$$\text{Also, } \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} = \frac{r_2 r_3 + r_1 r_3}{r_1 r_2 r_3} + r_1 r_2 = \frac{s^2}{r_1 r_2 r_3} = \frac{1}{r}$$

$$\therefore r_1 r_2 r_3 = s^2 r$$

$$\text{and } r_1 + r_2 + r_3 = 4R + r$$

$$\therefore \text{Equation } x^3 - (4R + r)x^2 + s^2 x - s^2 r = 0$$

24. $ABCD$ is a cyclic quadrilateral inscribed in circle of radius 21 cm, having diagonals AC and BD at right angles, the point of intersection of AC and BD being E . Then the sum of squares of E from the four vertices (in square cm) is _____.

Sol. We have to find

$$EB^2 + ED^2 + EA^2 + EC^2$$

$$= \frac{1}{2} \{ (EB^2 + EA^2) + (EB^2 + EC^2) + (EC^2 + ED^2) + (ED^2 + EA^2) \}$$

$$= \frac{1}{2} \{ AB^2 + BC^2 + CD^2 + AD^2 \}$$

Now, let R be the radius of the circumscribed circle using sine rule

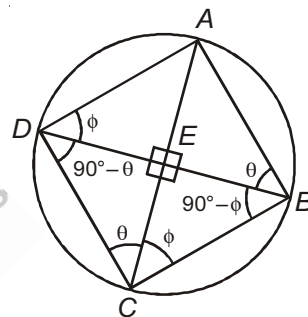
$$AD^2 = 4R^2 \sin^2 \theta ; AB^2 = 4R^2 \sin^2 \phi$$

$$BC^2 = 4R^2 \cos^2 \theta ; CD^2 = 4R^2 \cos^2 \phi$$

$$\text{Hence } \frac{1}{2} \{ AB^2 + BC^2 + CD^2 + AD^2 \} = \frac{1}{2} \{ 8R^2 \}$$

$$= 4R^2$$

$$= 1764$$



25. In $\triangle ABC$, $3\sin A + 4\cos B = 6$ and $4\sin B + 3\cos A = 1$. Find the measure of angle C .

$$3\sin A + 4\cos B = 6$$

$$3\cos A + 4\sin B = 1$$

$$\text{Sol. } \frac{3(\sin A - \cos A) + 4(\cos B - \sin B)}{3(\sin A - \cos A) + 4(\cos B - \sin B)} = 5$$

$$4\sin B + 3\cos A = 1$$

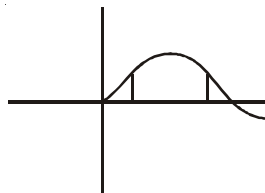
$$9\sin^2 A + 16\cos^2 B + 24\sin A \cos B = 36$$

$$9\cos^2 A + 16\sin^2 B + 24\cos A \sin B = 1$$

$$9 + 16 + 24(\sin A \cos B + \cos A \sin B) = 37$$

$$24\sin(A + B) = 37 - 25 = 12$$

$$\sin(A + B) = \frac{1}{2}$$



$$\Rightarrow \sin(\pi - C) = \frac{1}{2}$$

$$\Rightarrow \sin C = \frac{1}{2}$$

$$\Rightarrow C = \frac{\pi}{6}, \frac{5\pi}{6}$$

26. If the angle A and angle B of a triangle ($A > B$) satisfy the equation $2\tan x - \lambda(1 + \tan^2 x) = 0$, then determine the angle C where $0 < \lambda < 1$.

Sol. $2\tan x - \lambda(1 + \tan^2 x) = 0$

A, B satisfy this equation

$$\frac{2 \sin x}{\cos x} = \frac{\lambda}{\cos^2 x}$$

$$\Rightarrow \sin 2x = \lambda$$

$$\Rightarrow \sin 2A = \lambda = \sin 2B$$

$$\Rightarrow 2A = \pi - 2B \quad (\because A > B)$$

$$\Rightarrow A + B = \frac{\pi}{2}$$

$$\therefore C = \frac{\pi}{2}$$

27. Given that the angles α, β, γ are connected by relation

$$2\tan^2 \alpha \tan^2 \beta \tan^2 \gamma + \tan^2 \alpha \tan^2 \beta + \tan^2 \beta \tan^2 \gamma + \tan^2 \gamma \tan^2 \alpha = 1$$

find the value of $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma$.

Sol. Let $\sin^2 \alpha = a$, $\sin^2 \beta = b$, $\sin^2 \gamma = c$

$$\tan^2 \alpha = \frac{a}{1-a}$$

$$\tan^2 \beta = \frac{b}{1-b}$$

$$\tan^2 \gamma = \frac{c}{1-c}$$

Then from question

$$2\left(\frac{a}{1-a}\right)\left(\frac{b}{1-b}\right)\left(\frac{c}{1-c}\right) + \frac{a}{1-a} \cdot \frac{b}{1-b} + \frac{b}{1-b} \cdot \frac{c}{1-c} + \frac{c}{1-c} \cdot \frac{a}{1-a} = 1$$

$$\Rightarrow 2abc + ab + bc + ca - 3abc = (1-a)(1-b)(1-c)$$

$$-abc + ab + bc + ca = 1 - a - b - c + ab + bc + ca - abc$$

$$a + b + c = 1$$

$$\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 1$$

28. Solve the equation, $2(\sin x + \sin y) - 2\cos(x - y) = 3$ for smallest positive values of x and y .

Sol. $2 \cdot 2 \sin\left(\frac{x+y}{2}\right) \cdot \cos\left(\frac{x-y}{2}\right) - 2\left[2\cos^2\left(\frac{x-y}{2}\right) - 1\right] = 3$

$$4 \sin\left(\frac{x+y}{2}\right) \cdot \cos\left(\frac{x-y}{2}\right) - 4\cos^2\left(\frac{x-y}{2}\right) - \sin^2\left(\frac{x+y}{2}\right) = \cos^2\left(\frac{x+y}{2}\right)$$

$$\left[\sin\left(\frac{x+y}{2}\right) - 2\cos\left(\frac{x-y}{2}\right)\right]^2 + \cos^2\left(\frac{x+y}{2}\right) = 0$$

$$\cos\left(\frac{x+y}{2}\right) = 0, \quad x+y = \pi \quad \dots(i)$$

$$1 - 2\cos\left(\frac{x-y}{2}\right) = 0, \quad x-y = \frac{2\pi}{3} \quad \dots(ii)$$

$$2x = \frac{5\pi}{3} \Rightarrow x = \frac{5\pi}{6}$$

$$y = \frac{\pi}{6}$$

